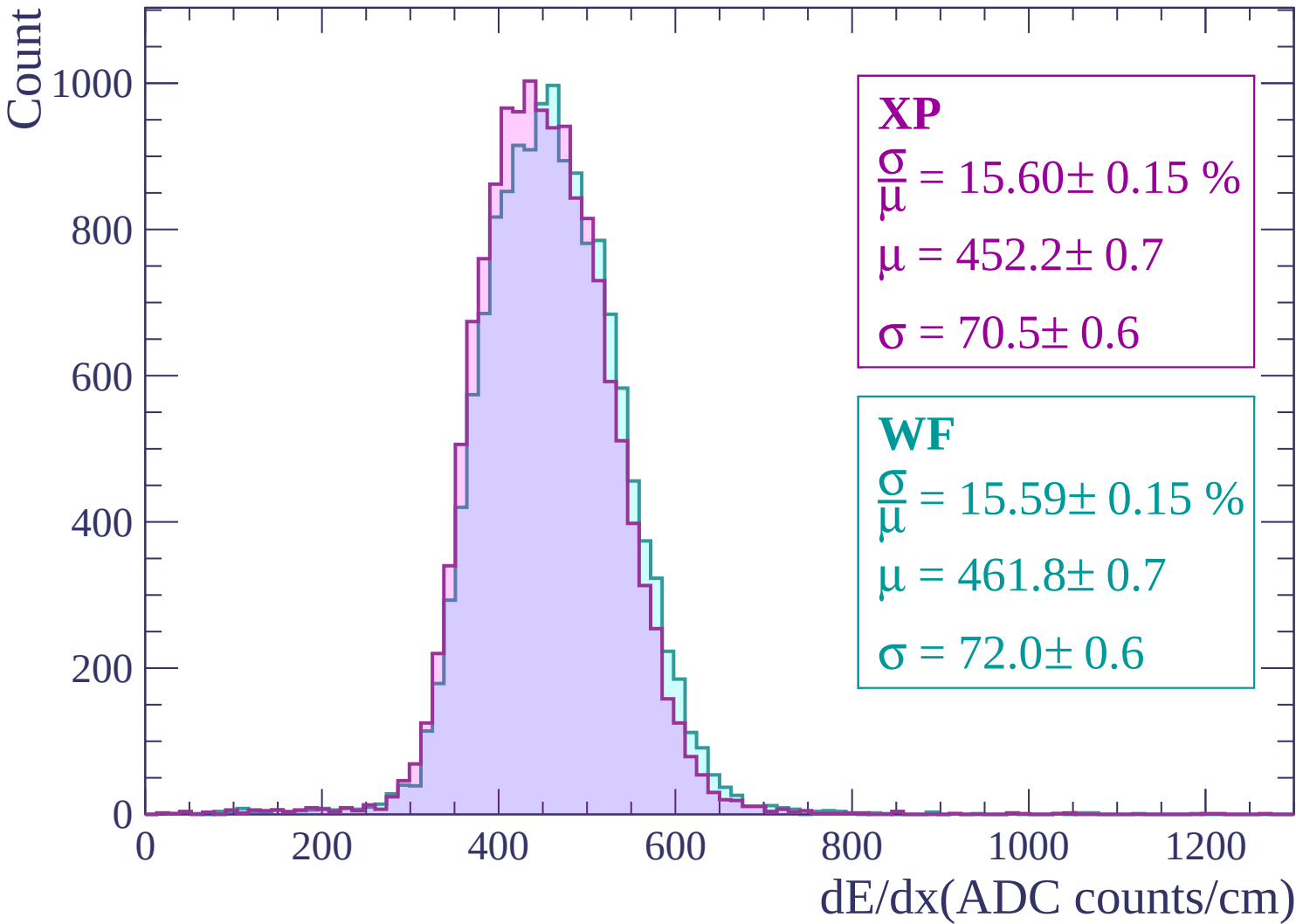
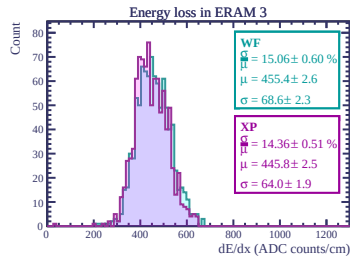
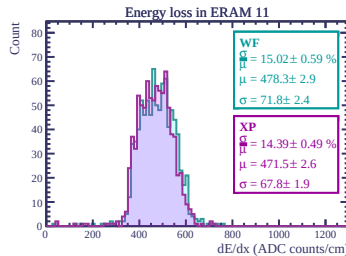
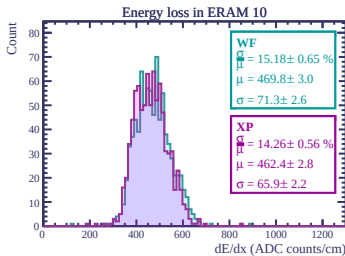
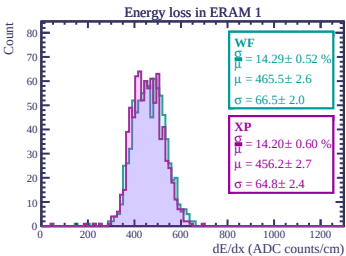
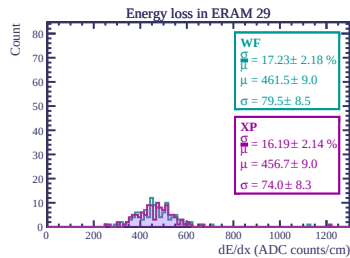
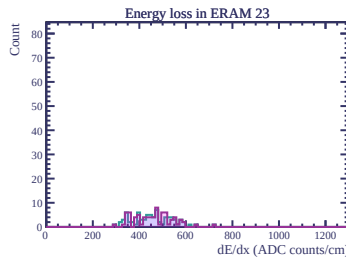
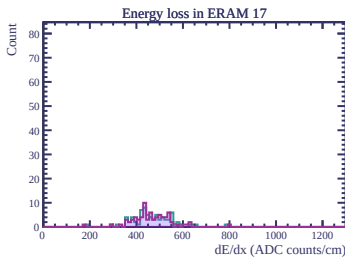
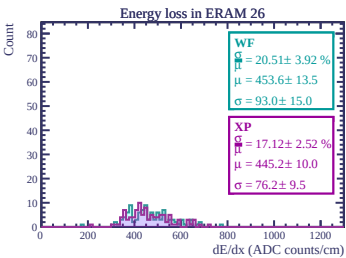
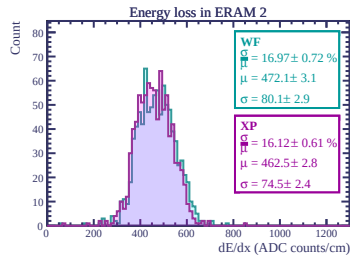
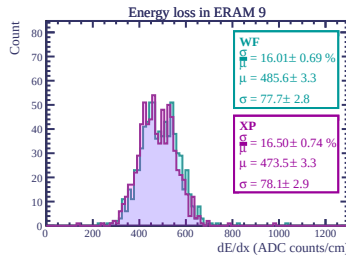
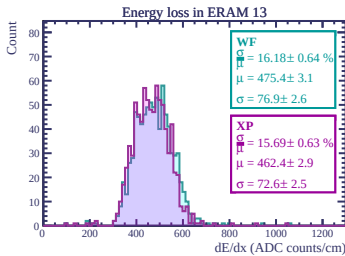
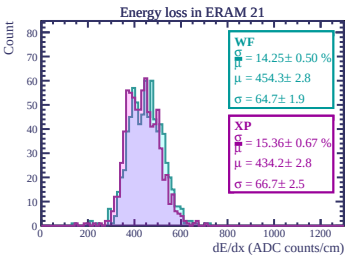
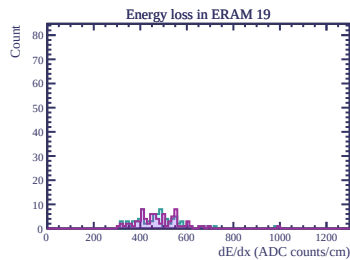
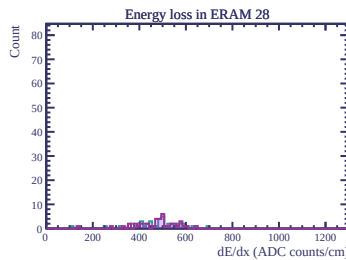
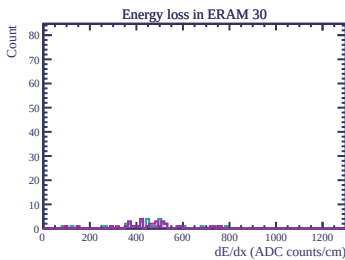
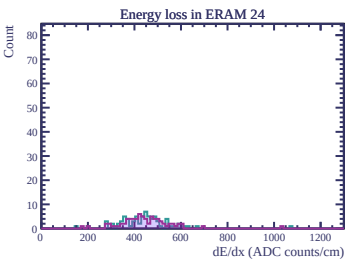
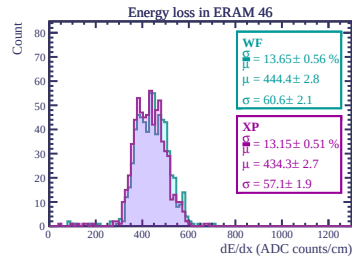
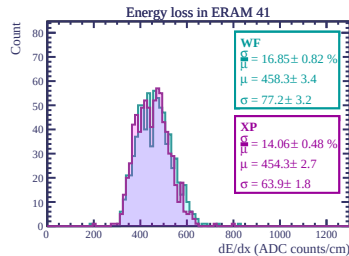
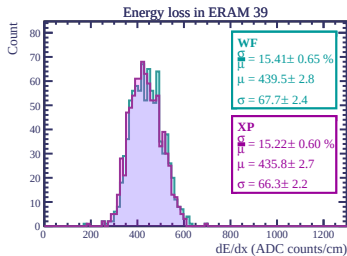
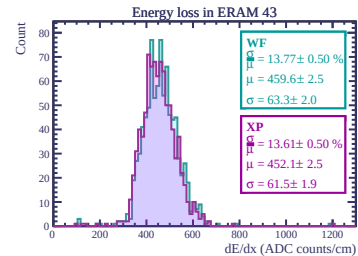
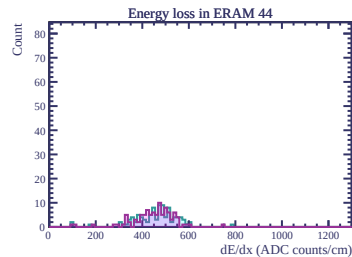
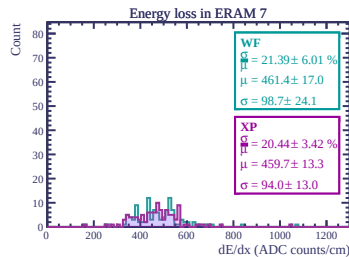
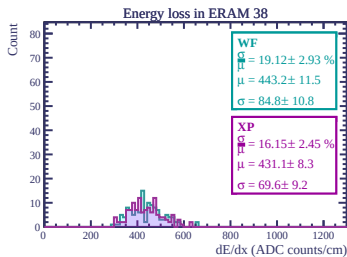
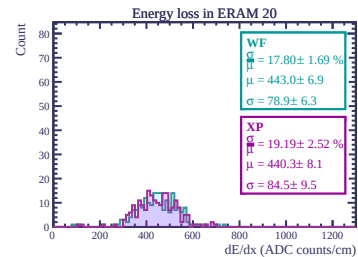
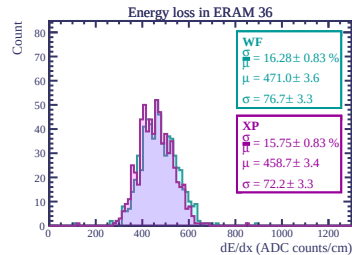
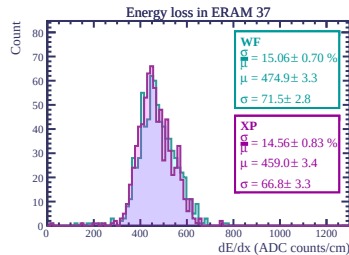
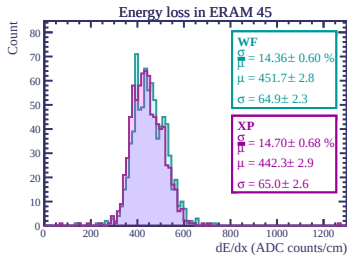
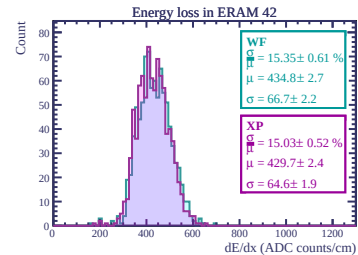
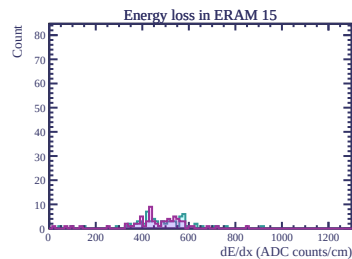
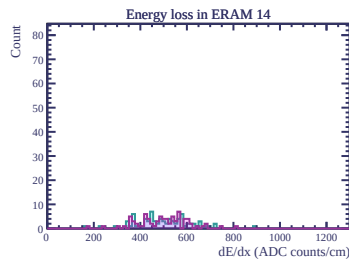
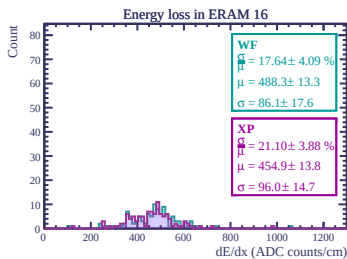
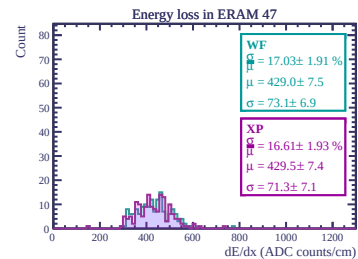
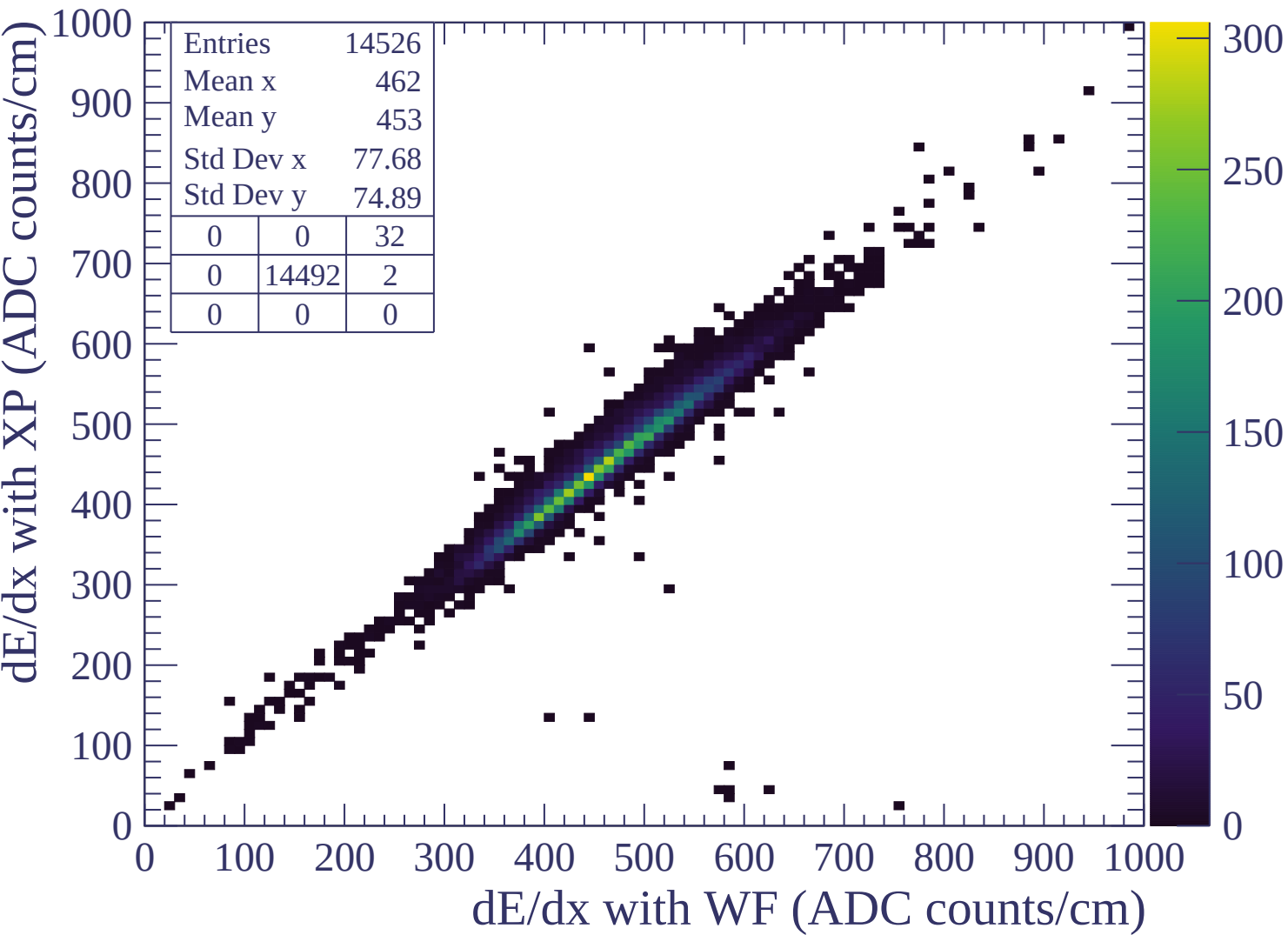
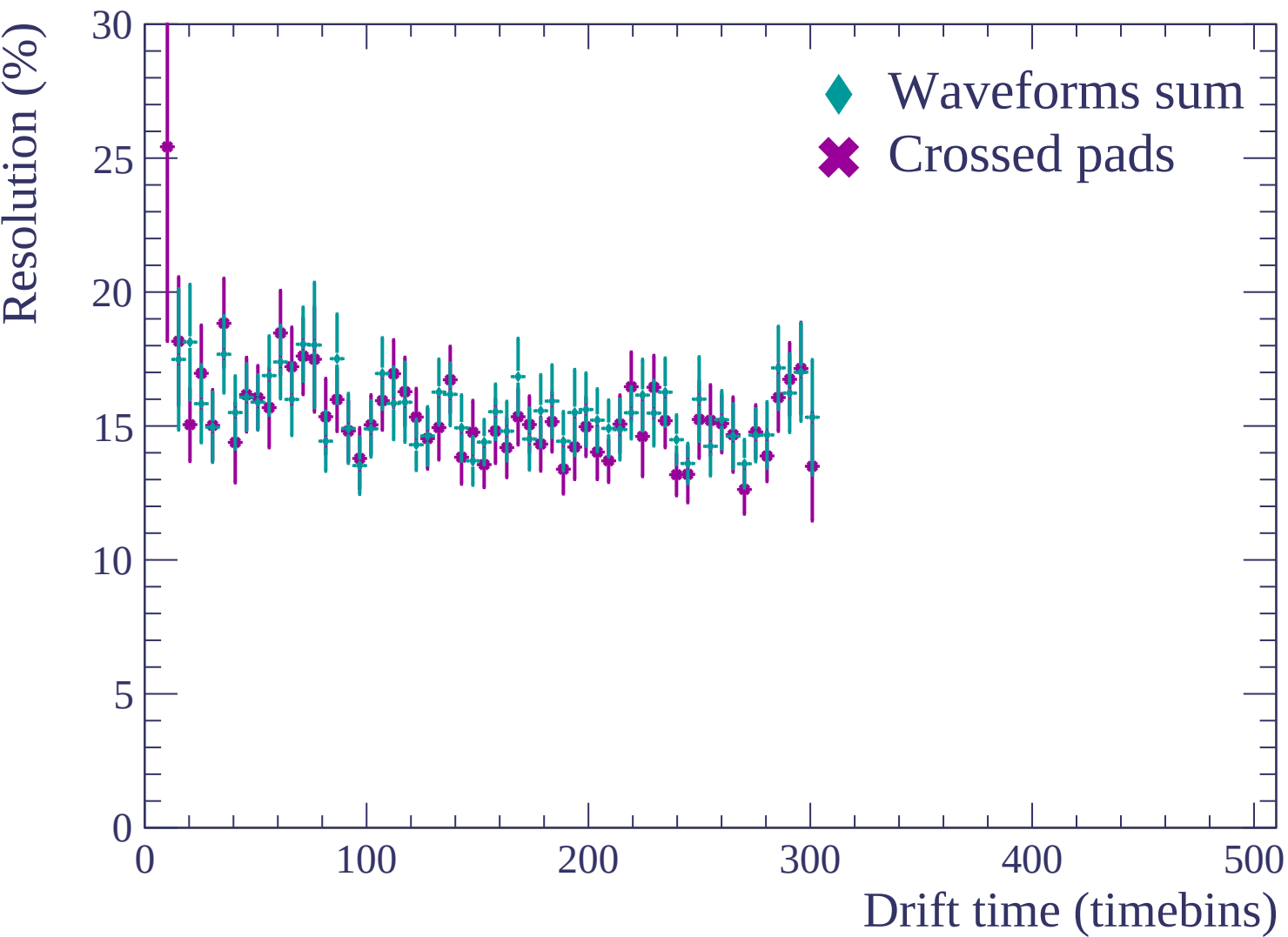


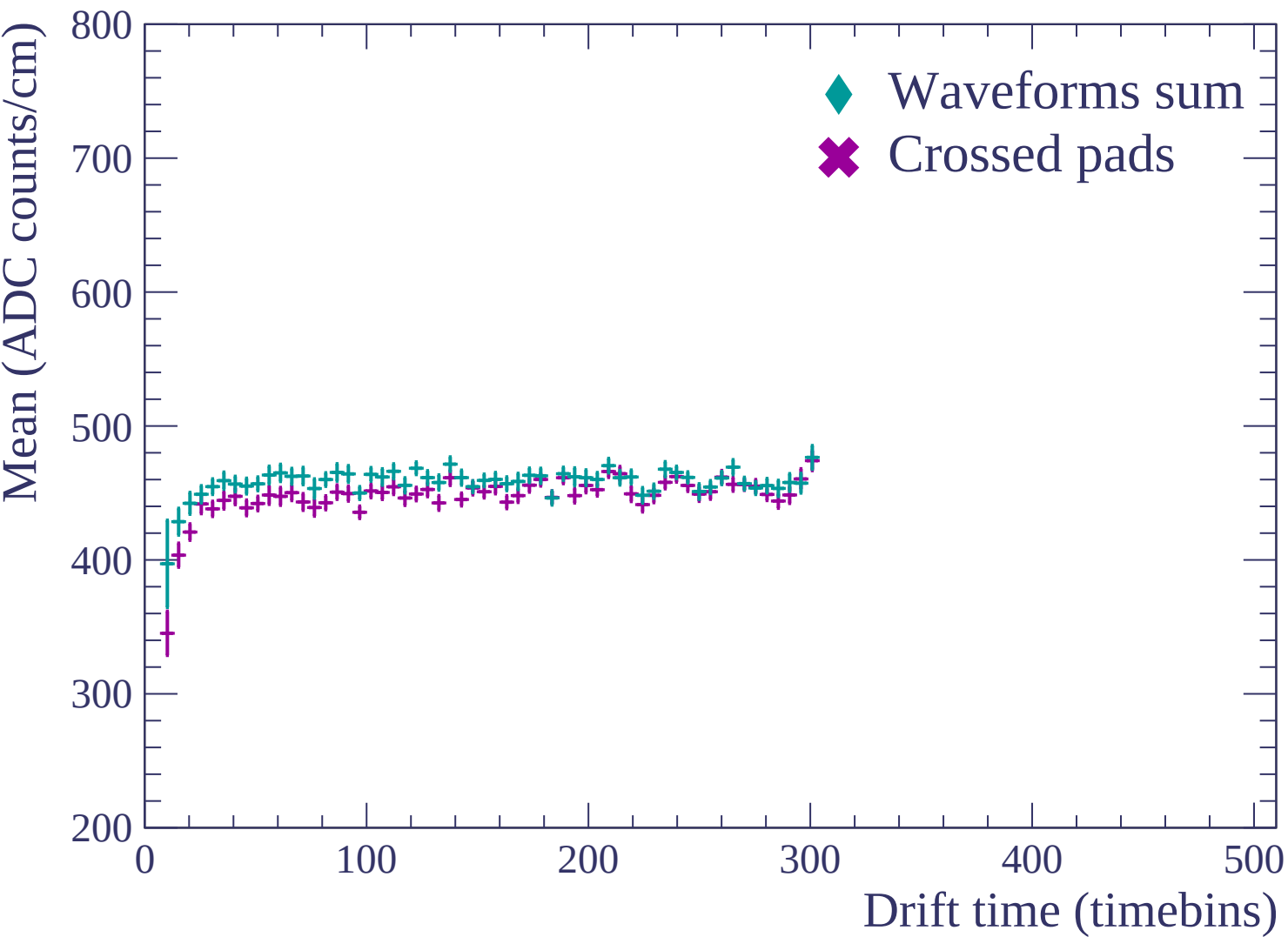


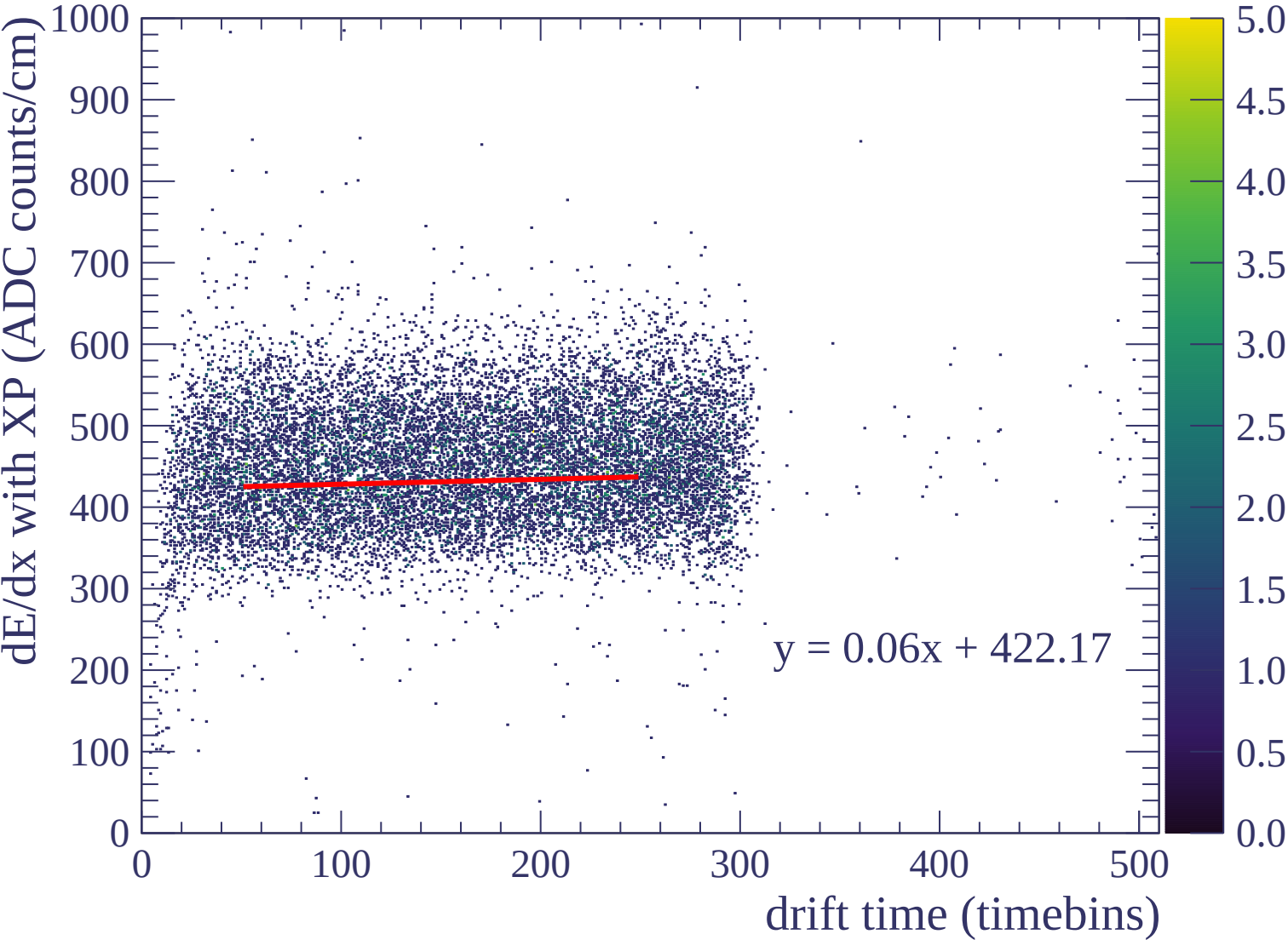


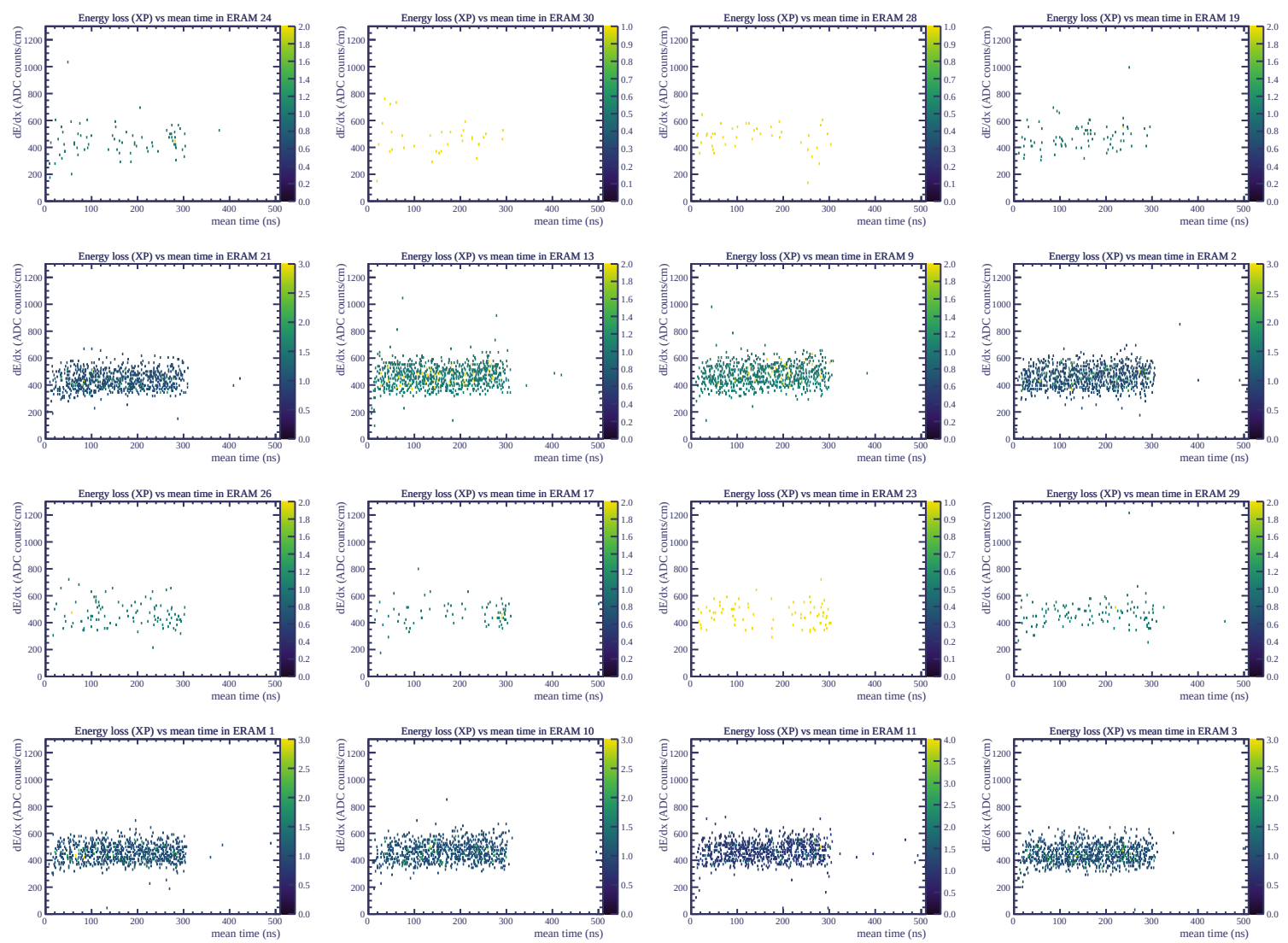




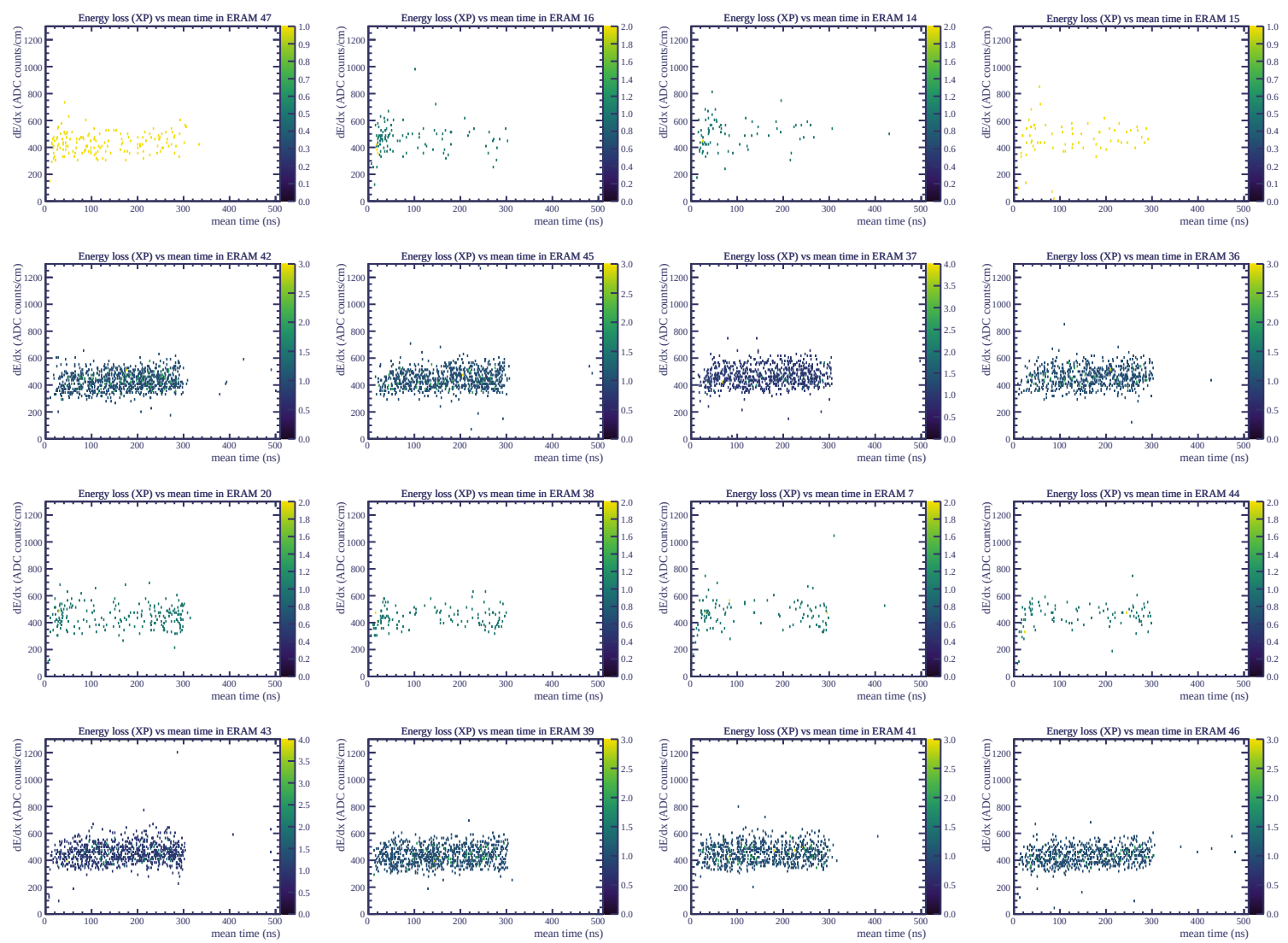


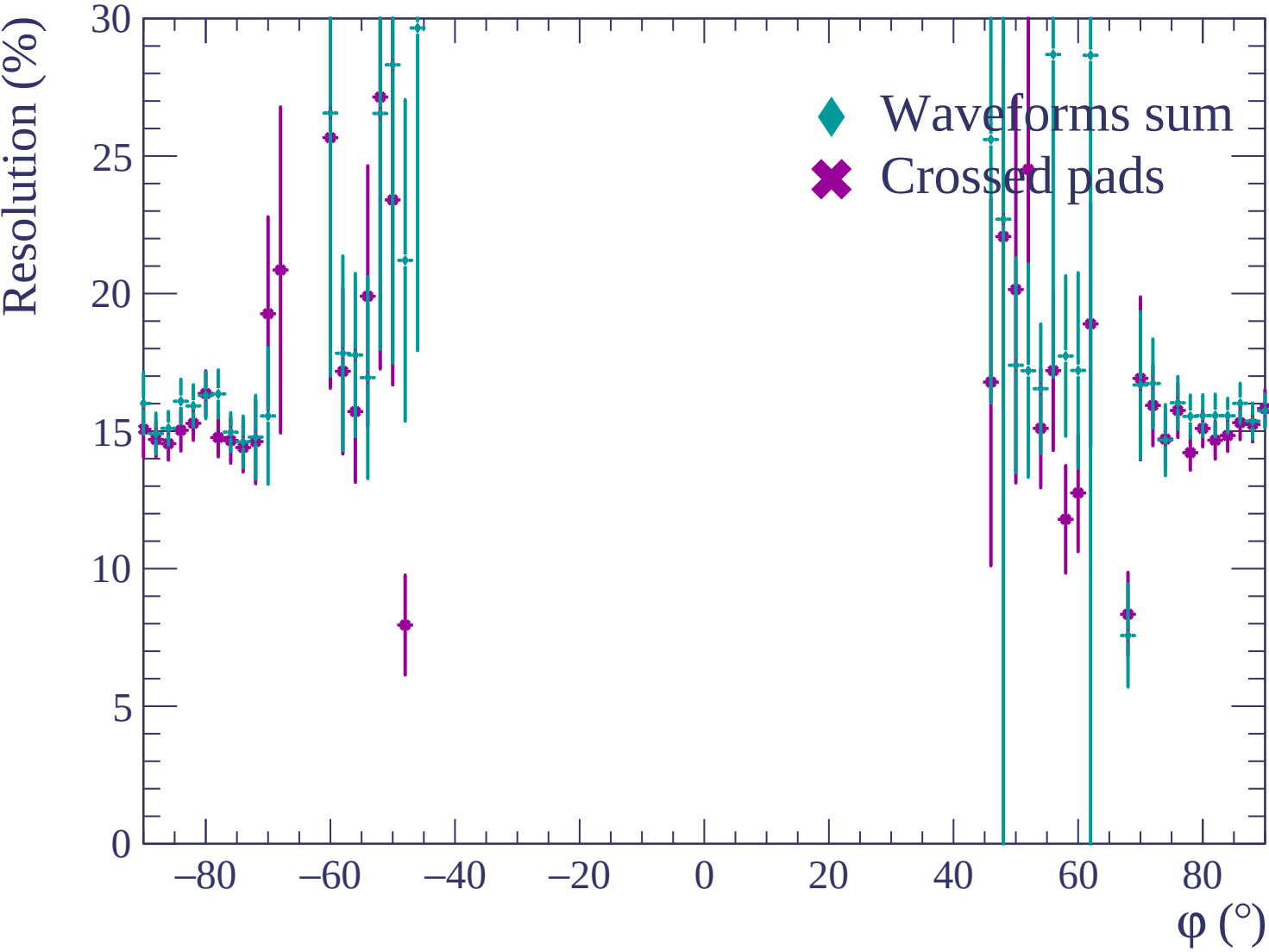


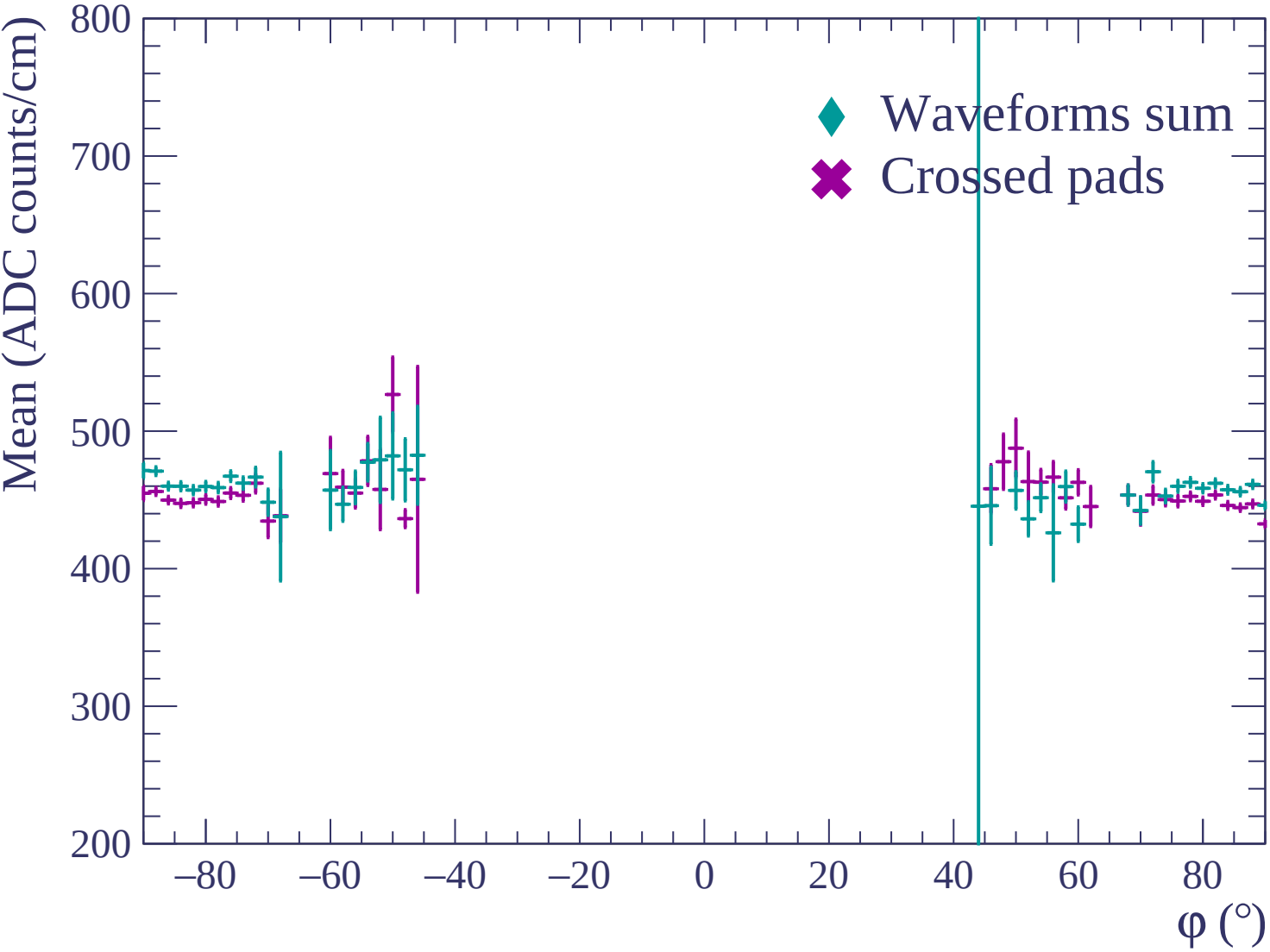


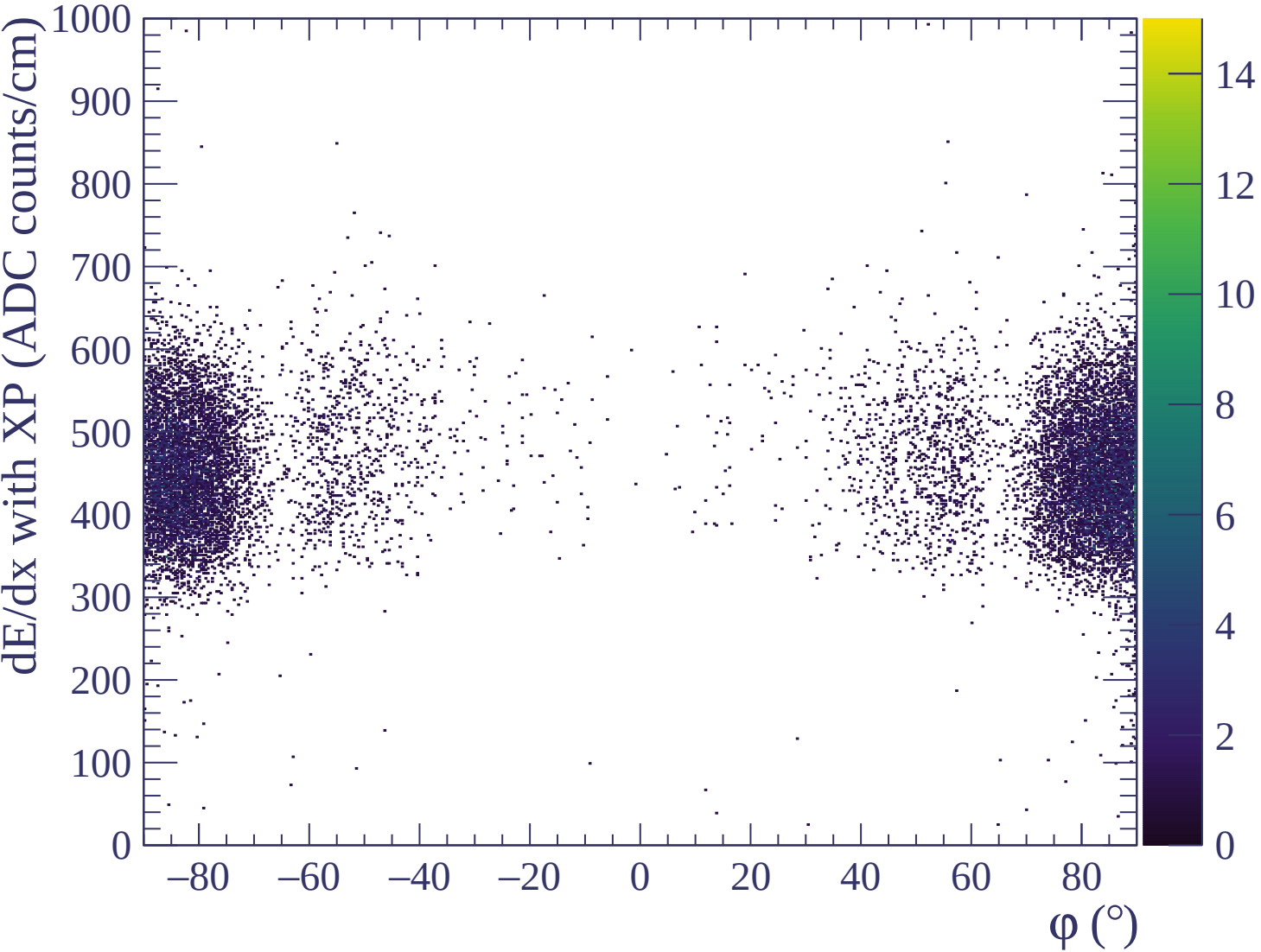


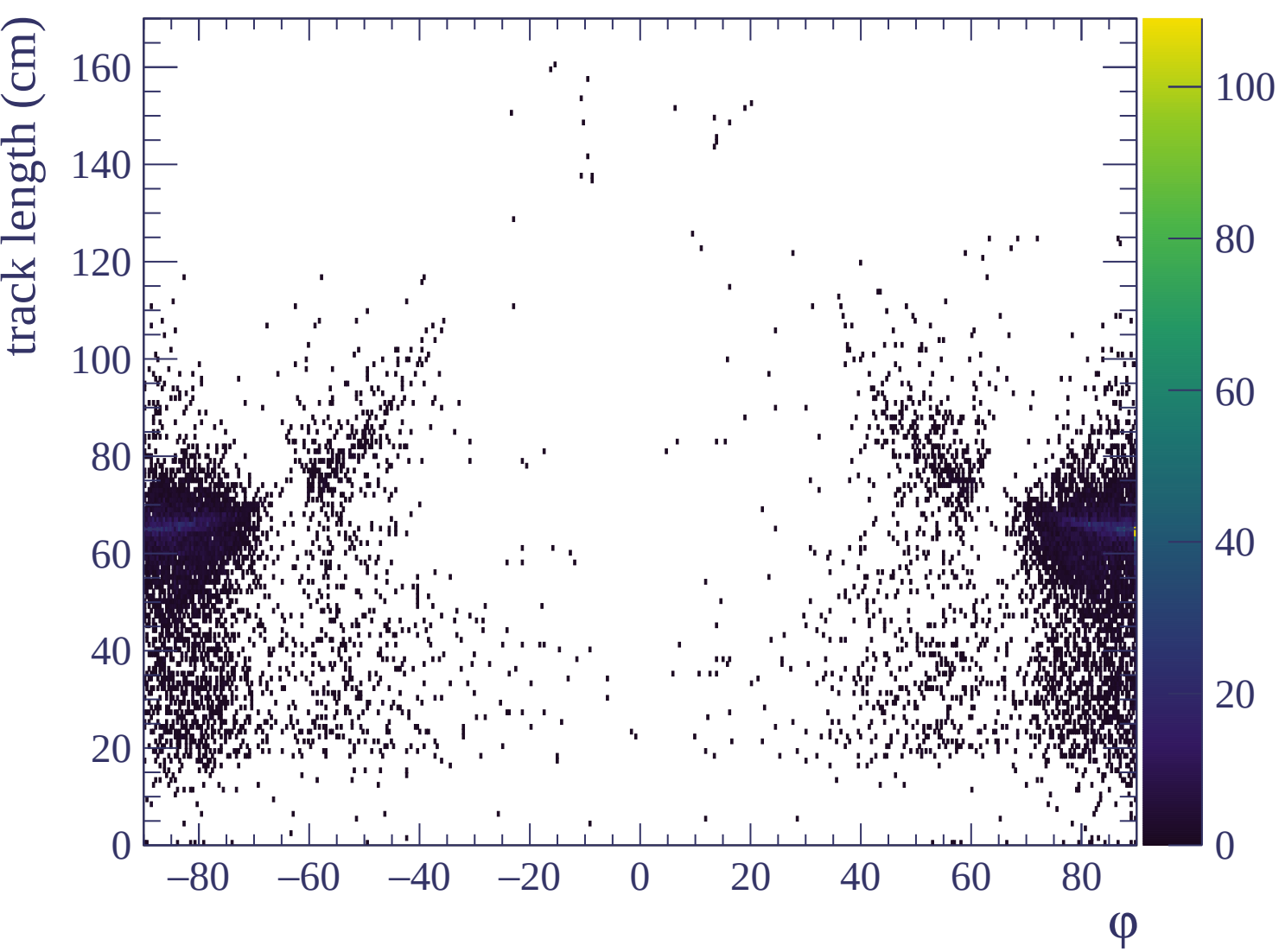


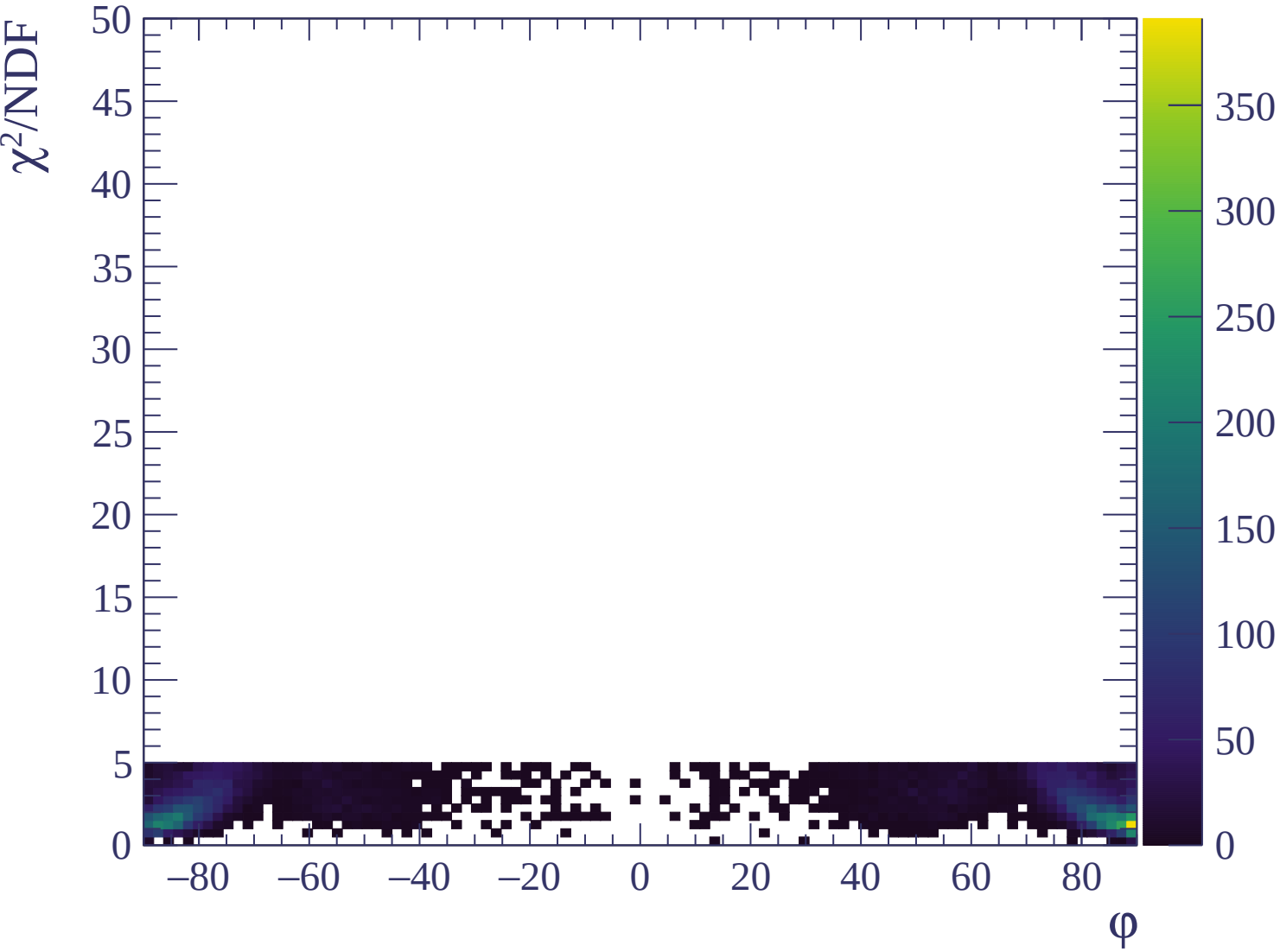


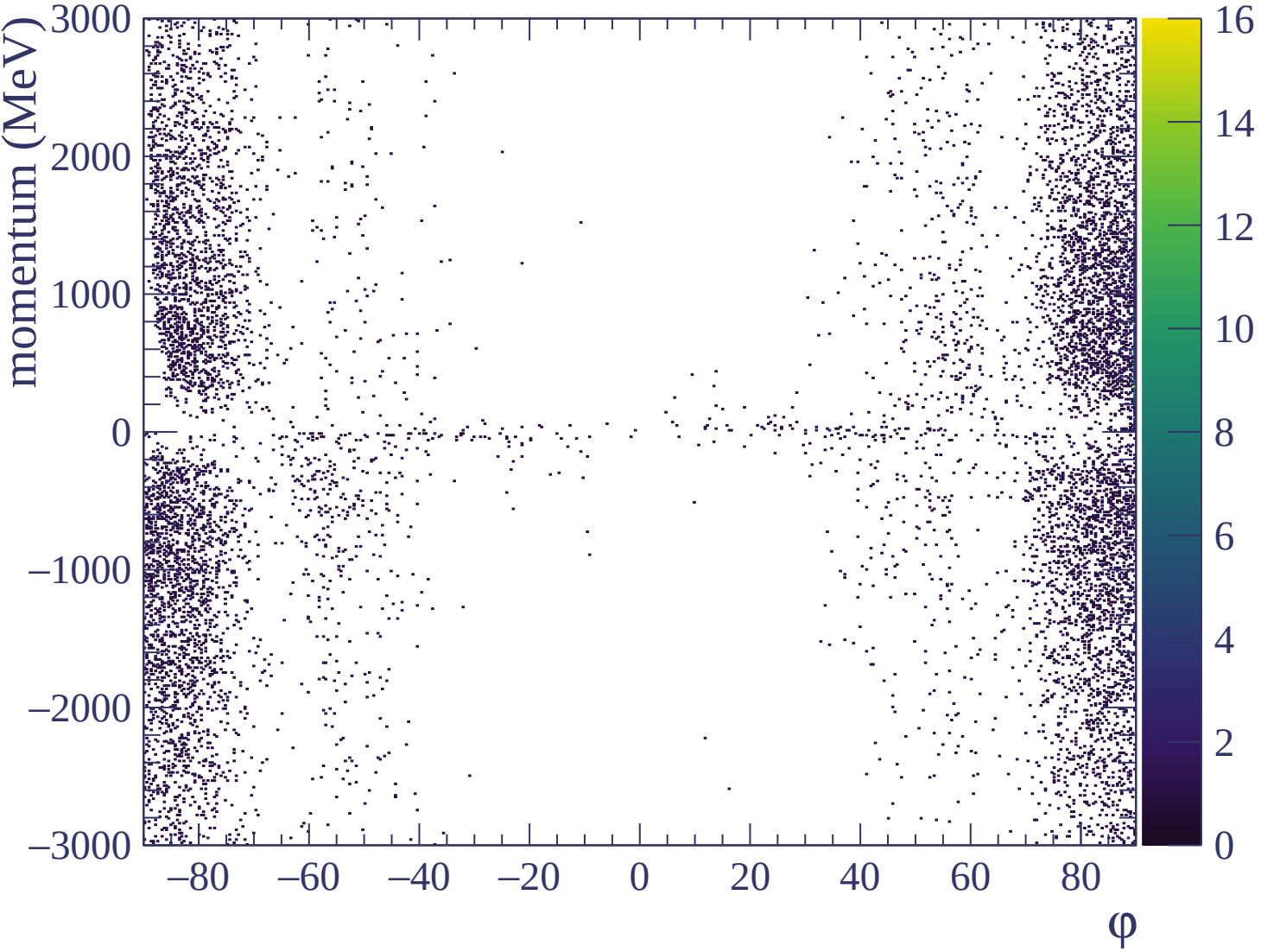


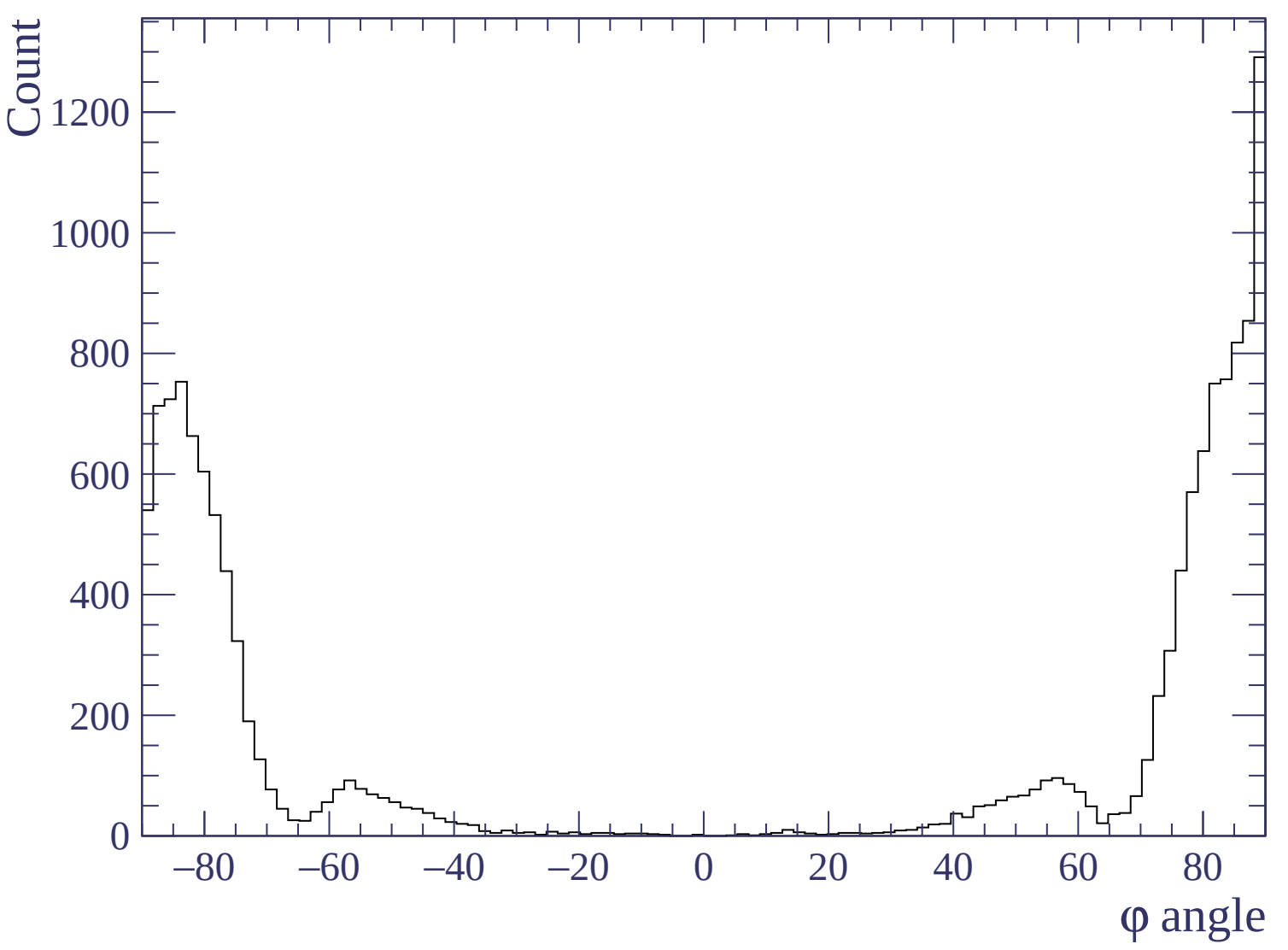




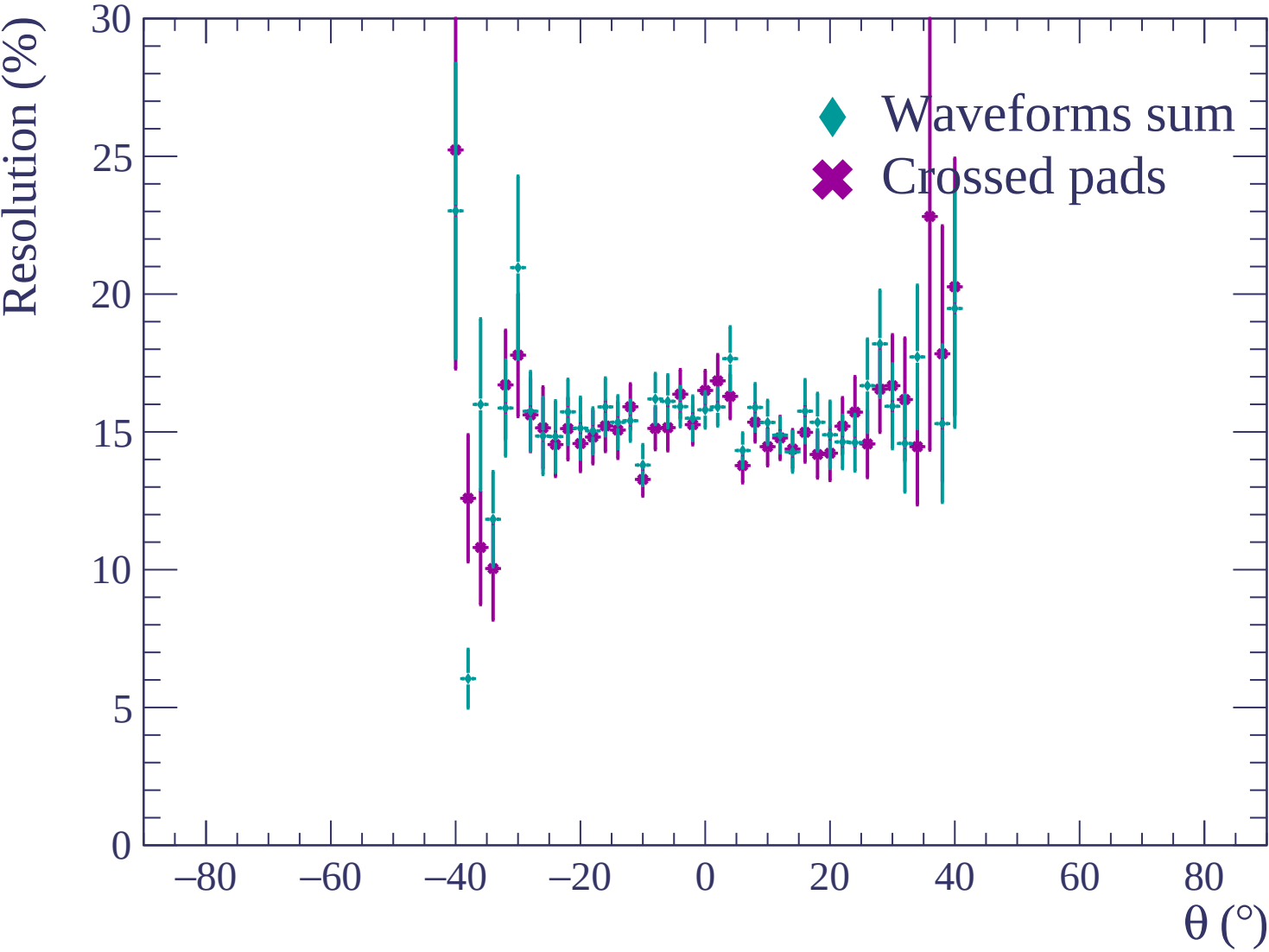


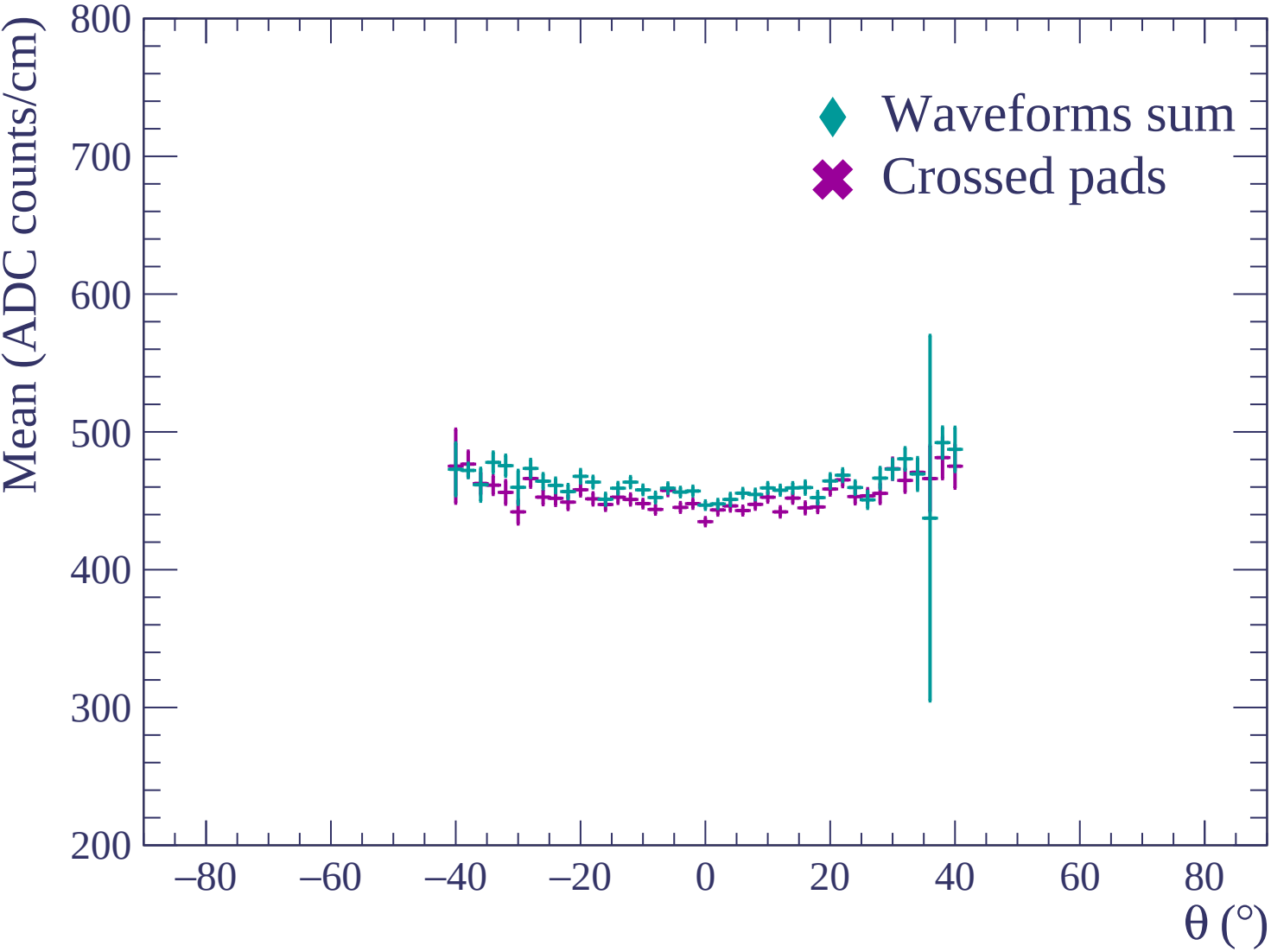


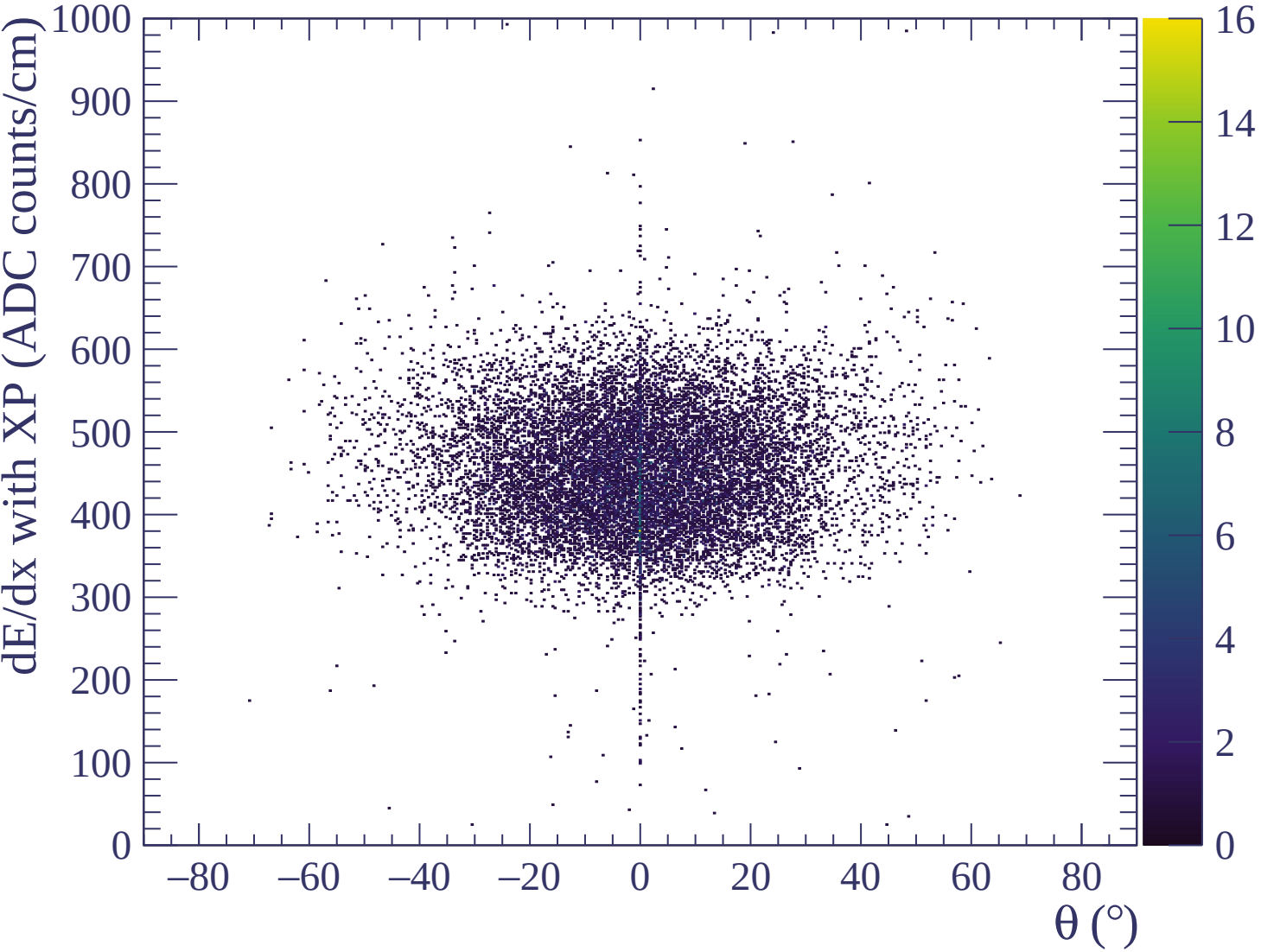


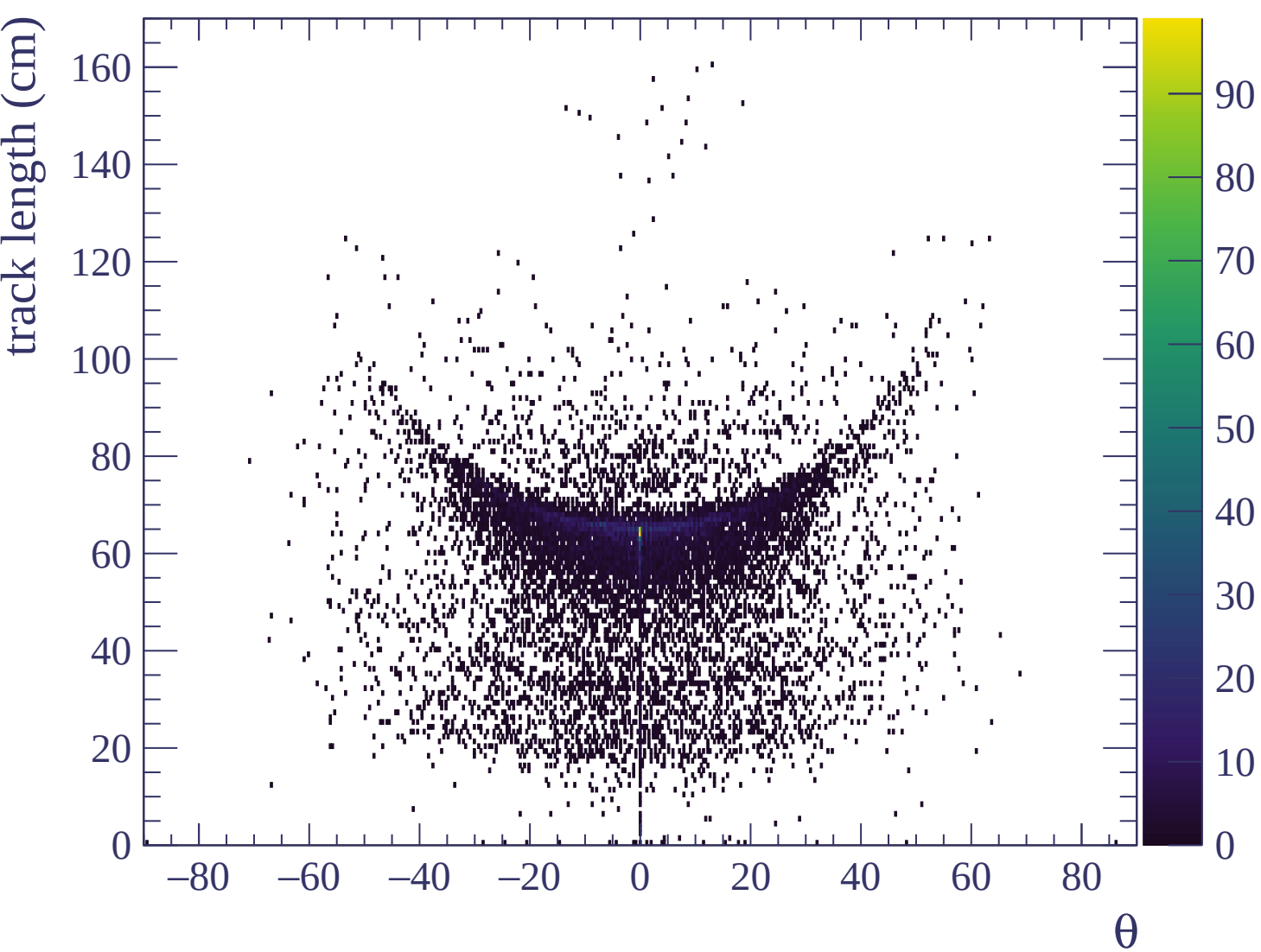


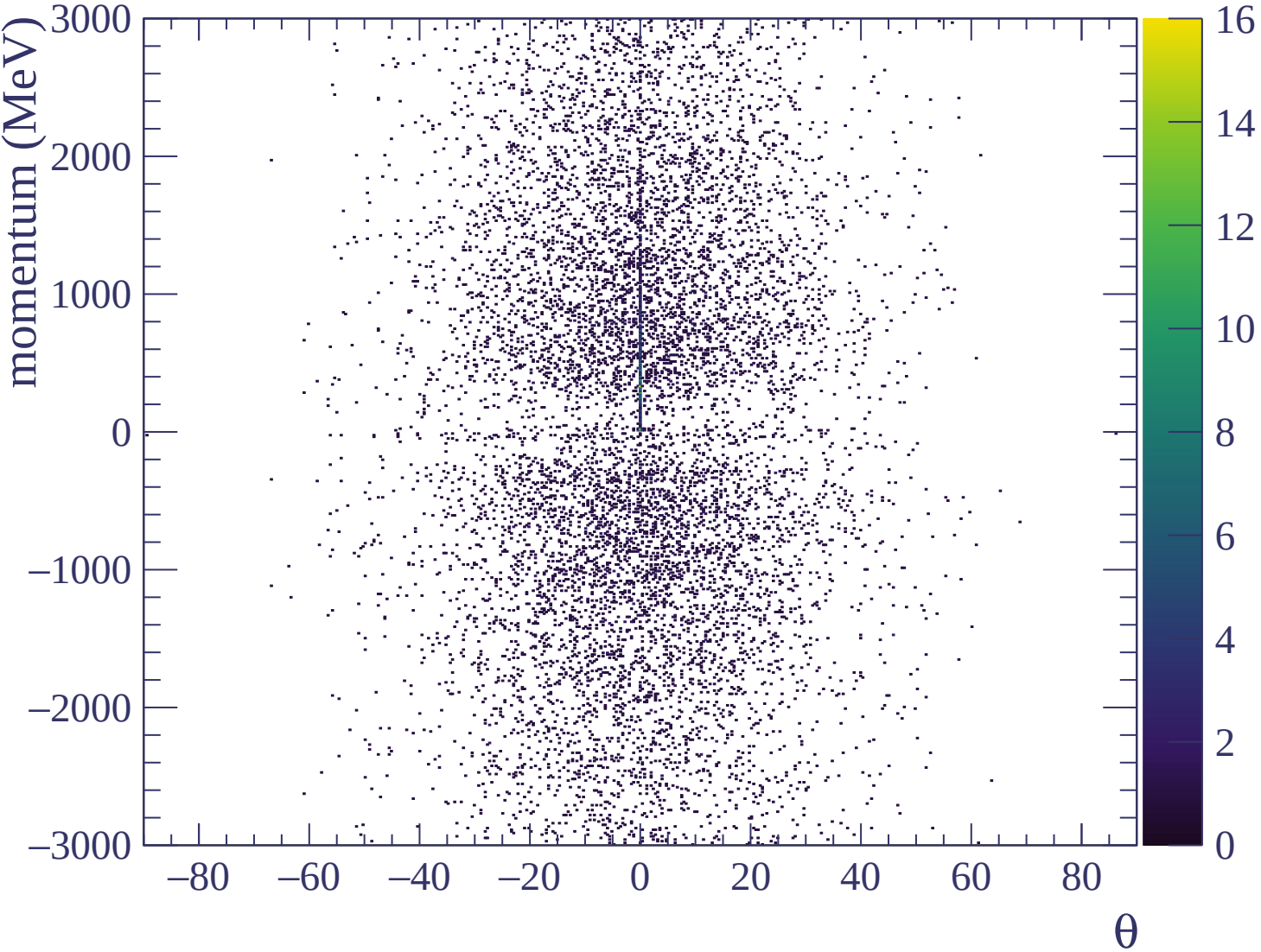


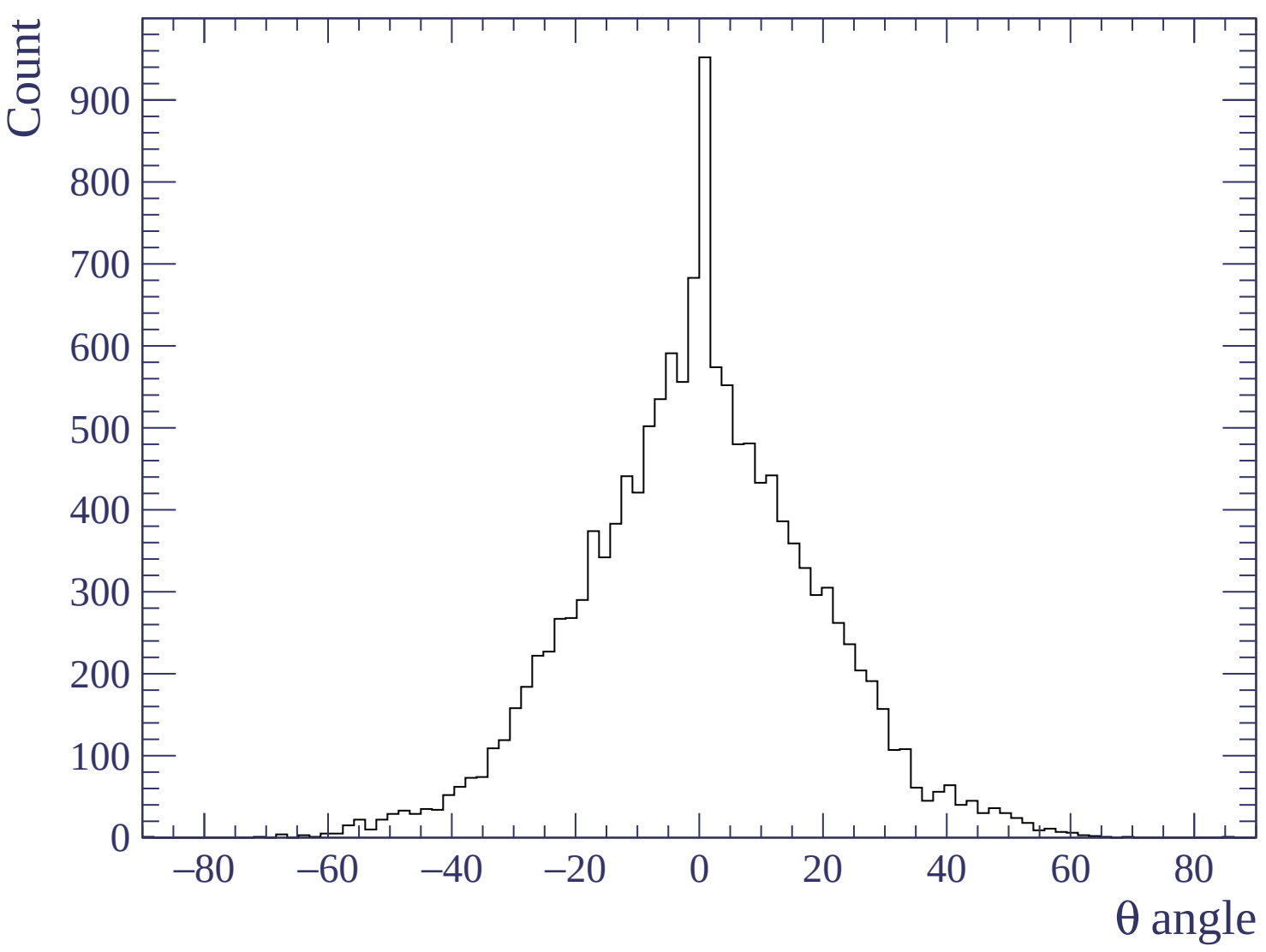


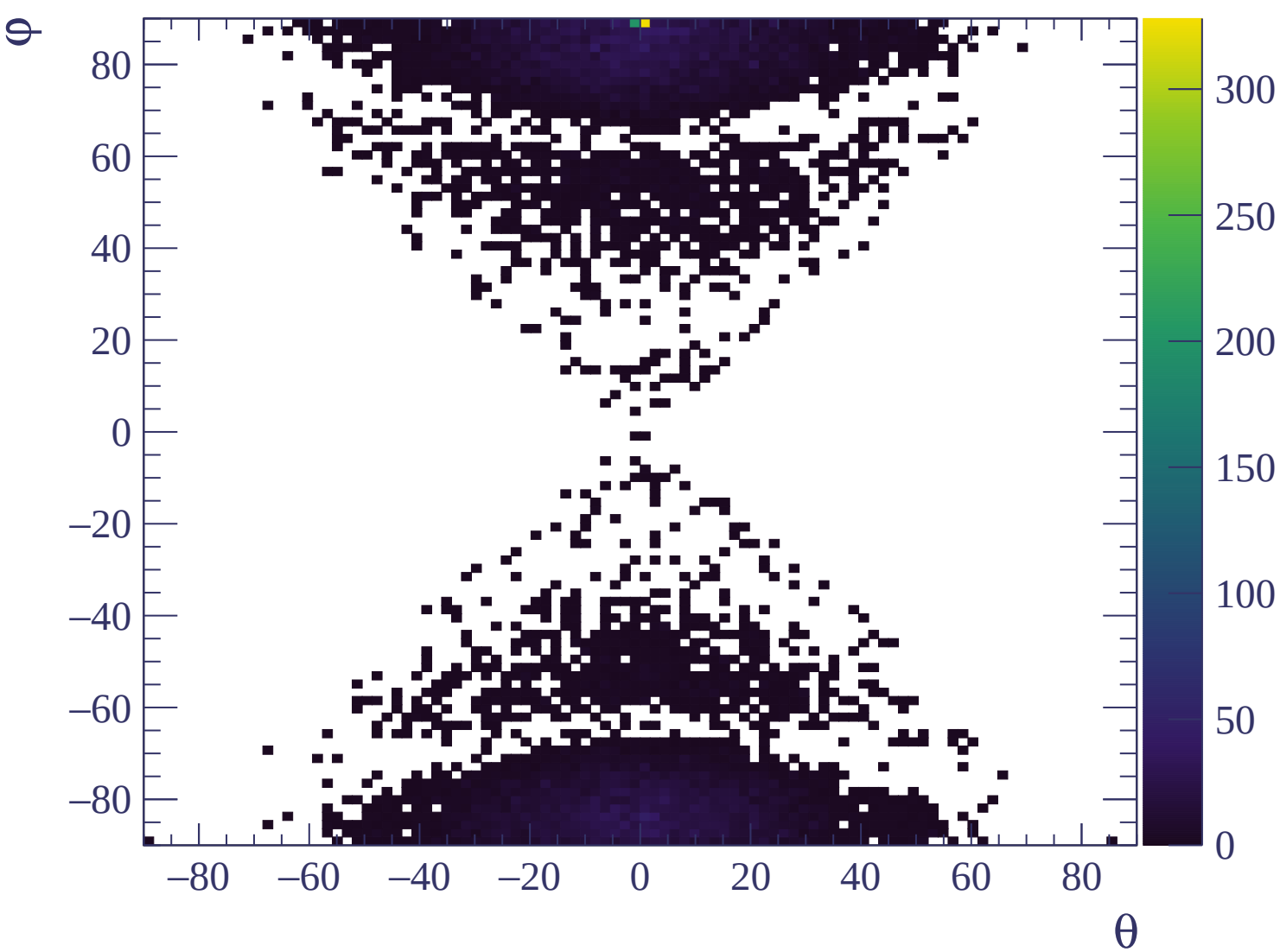


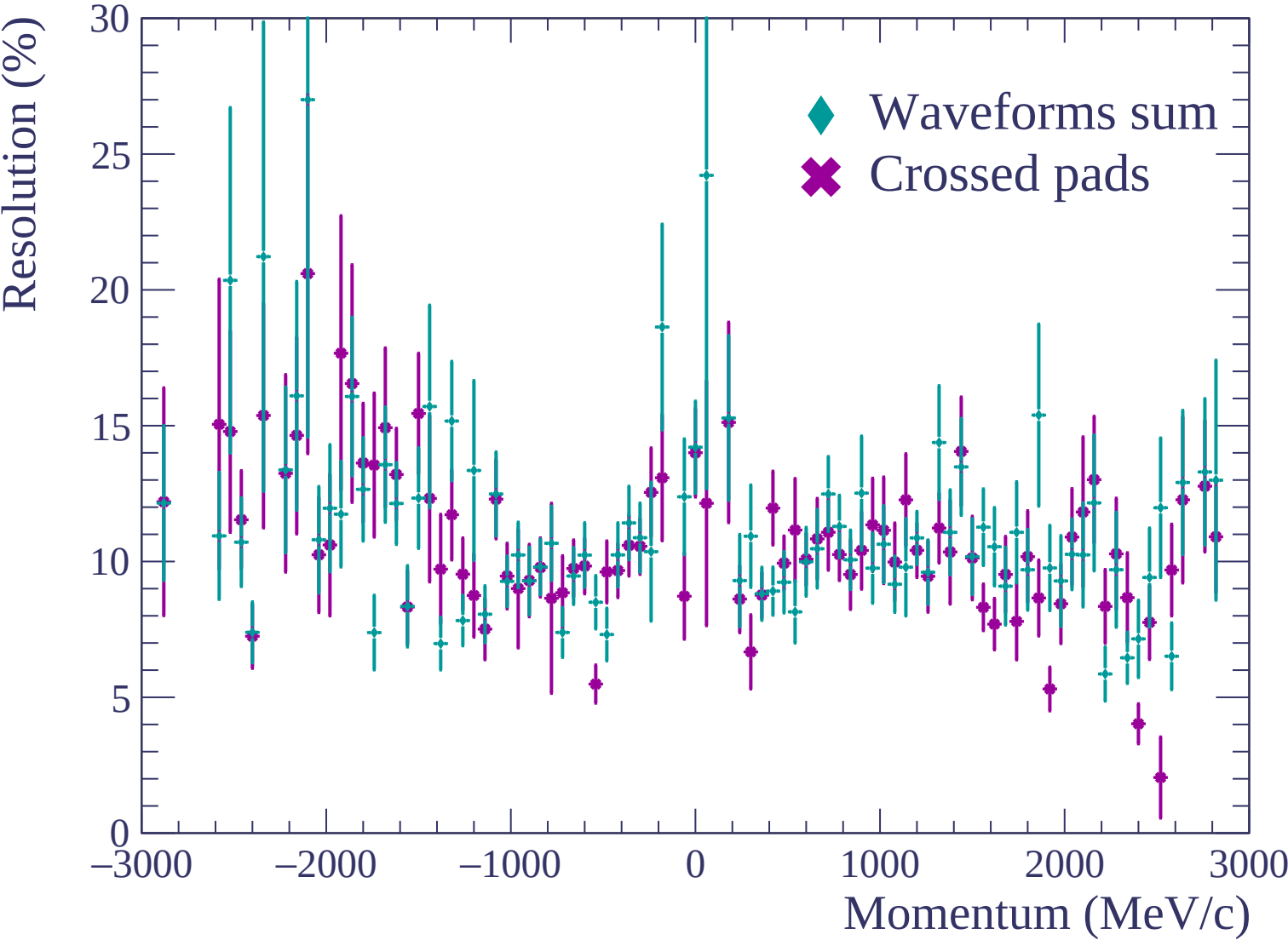




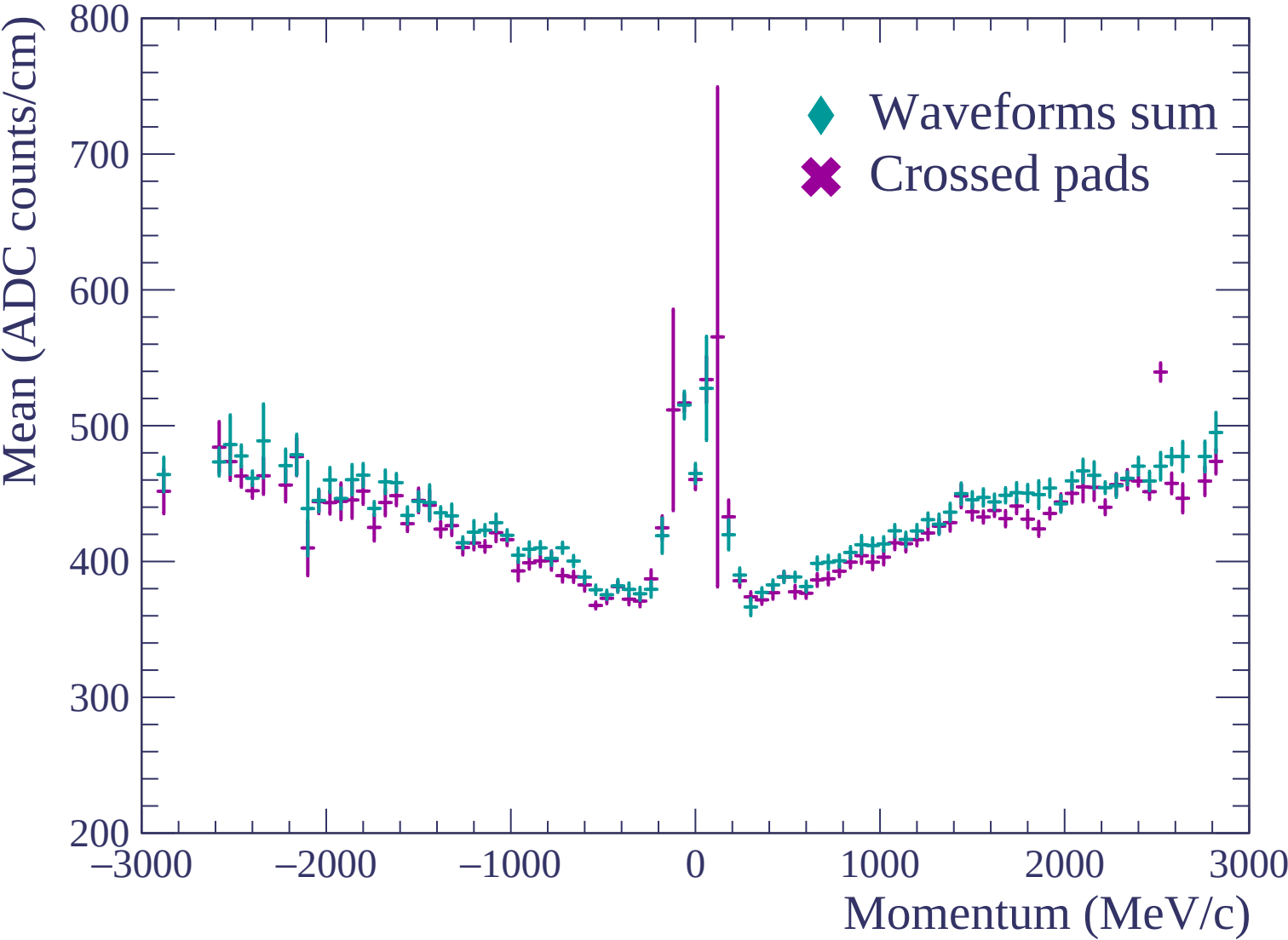


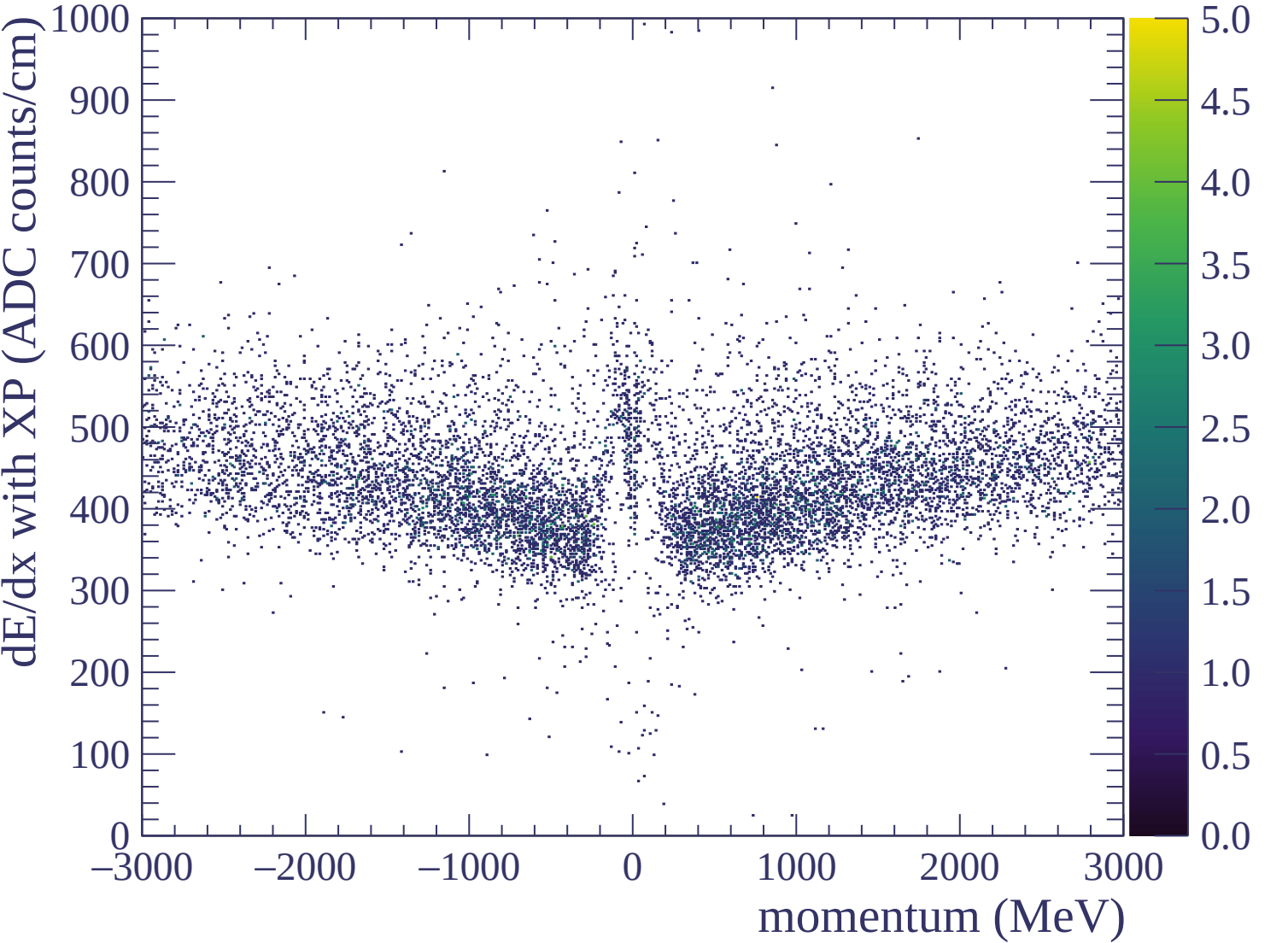


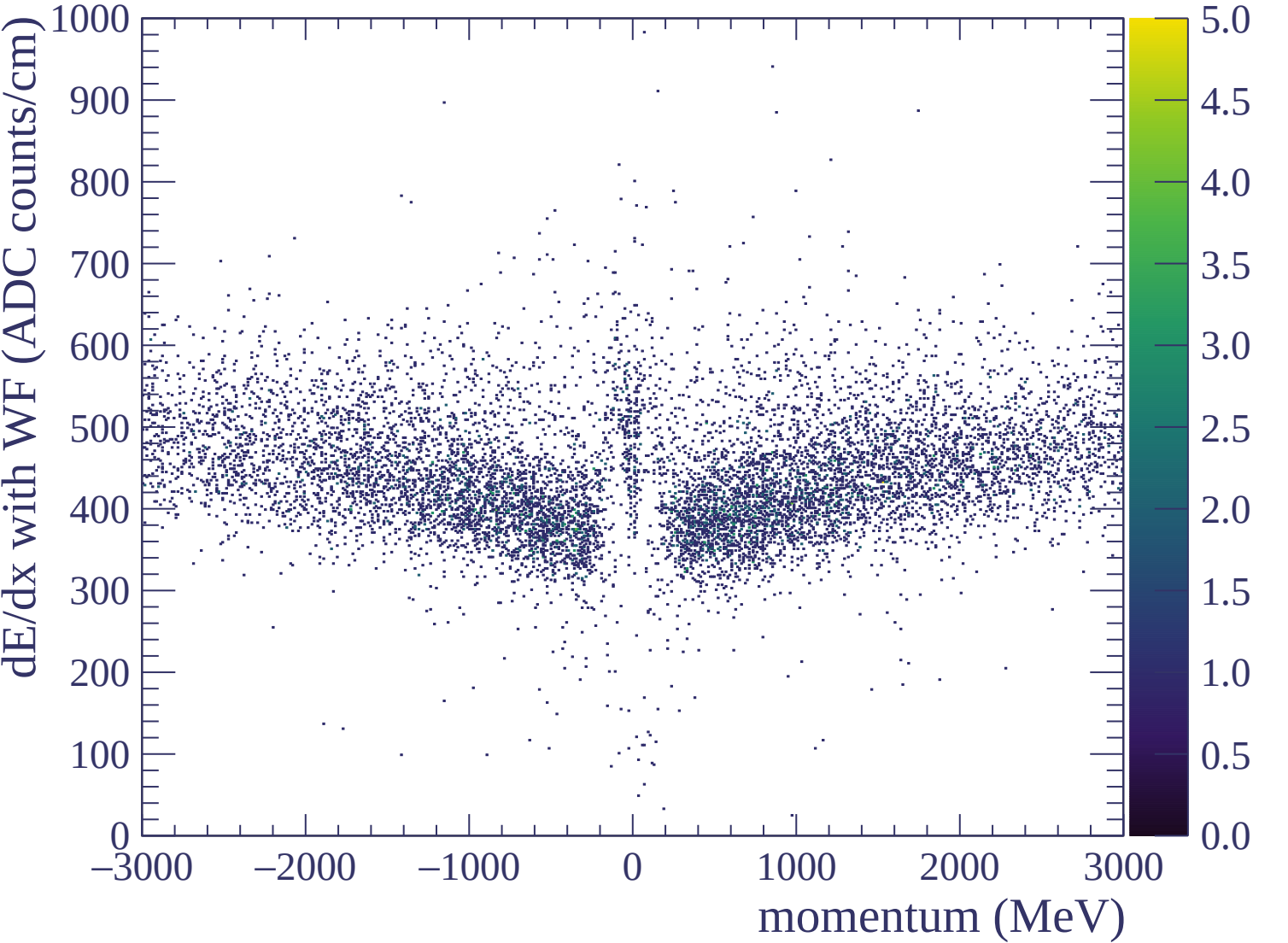


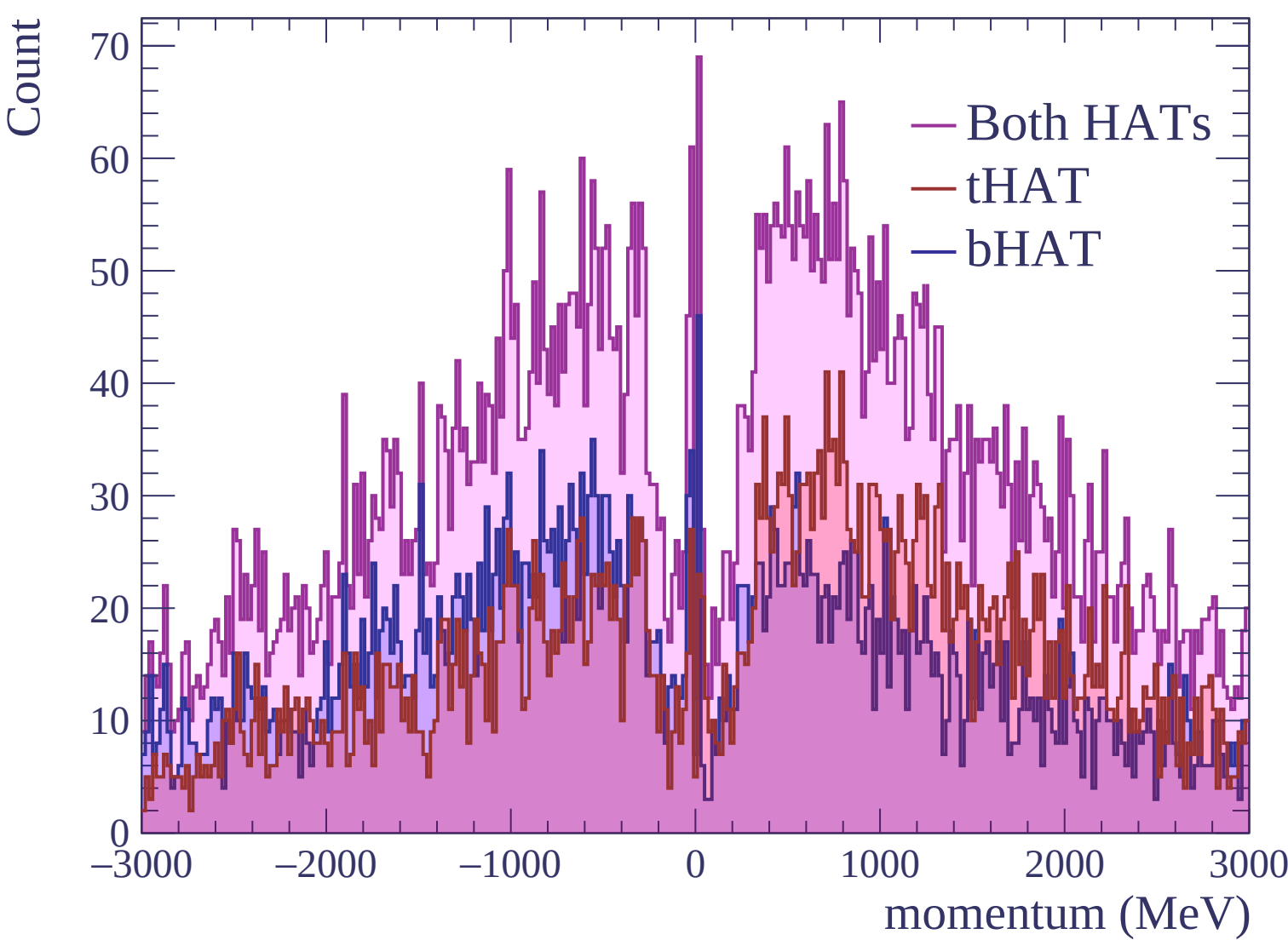


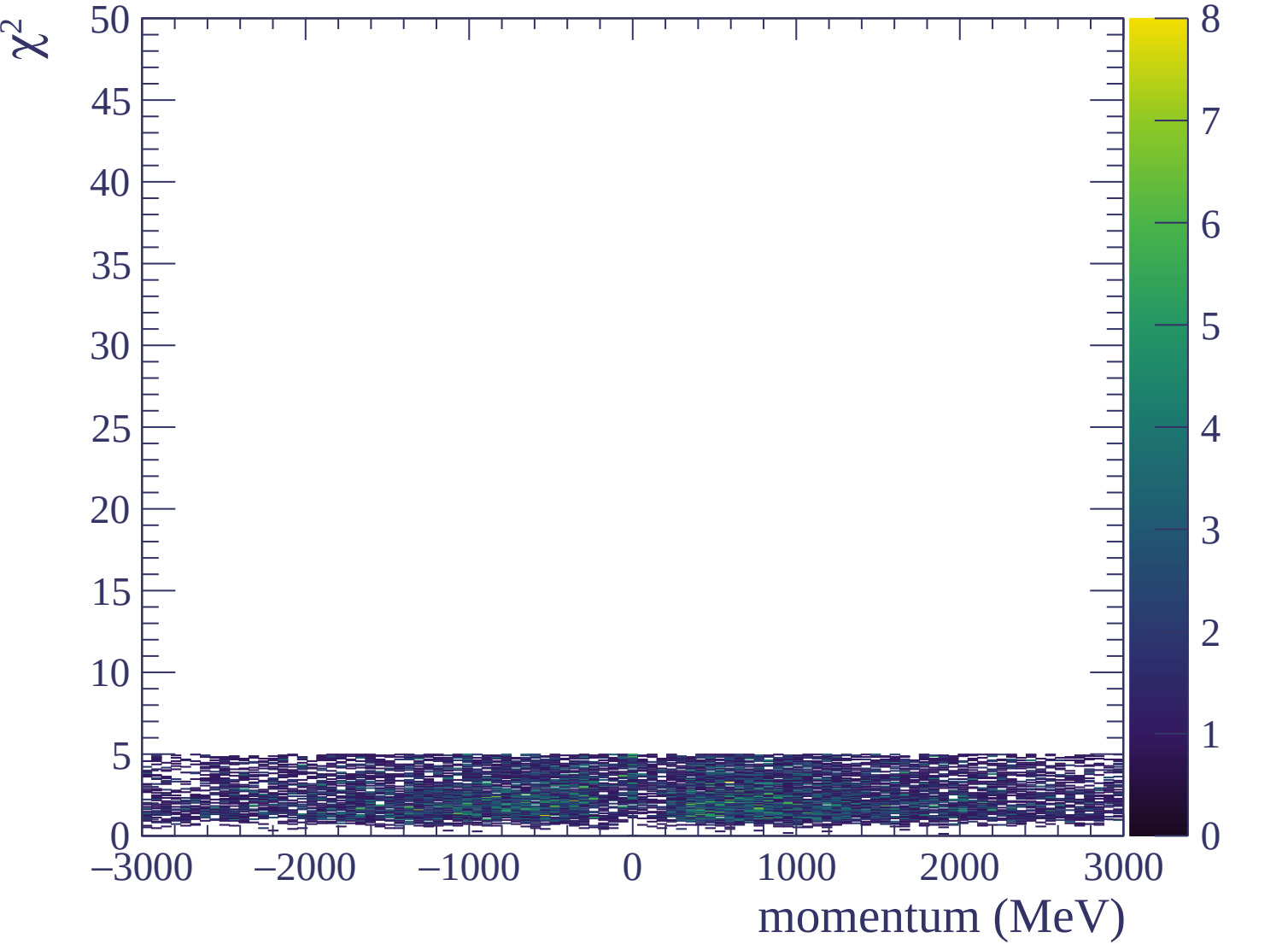


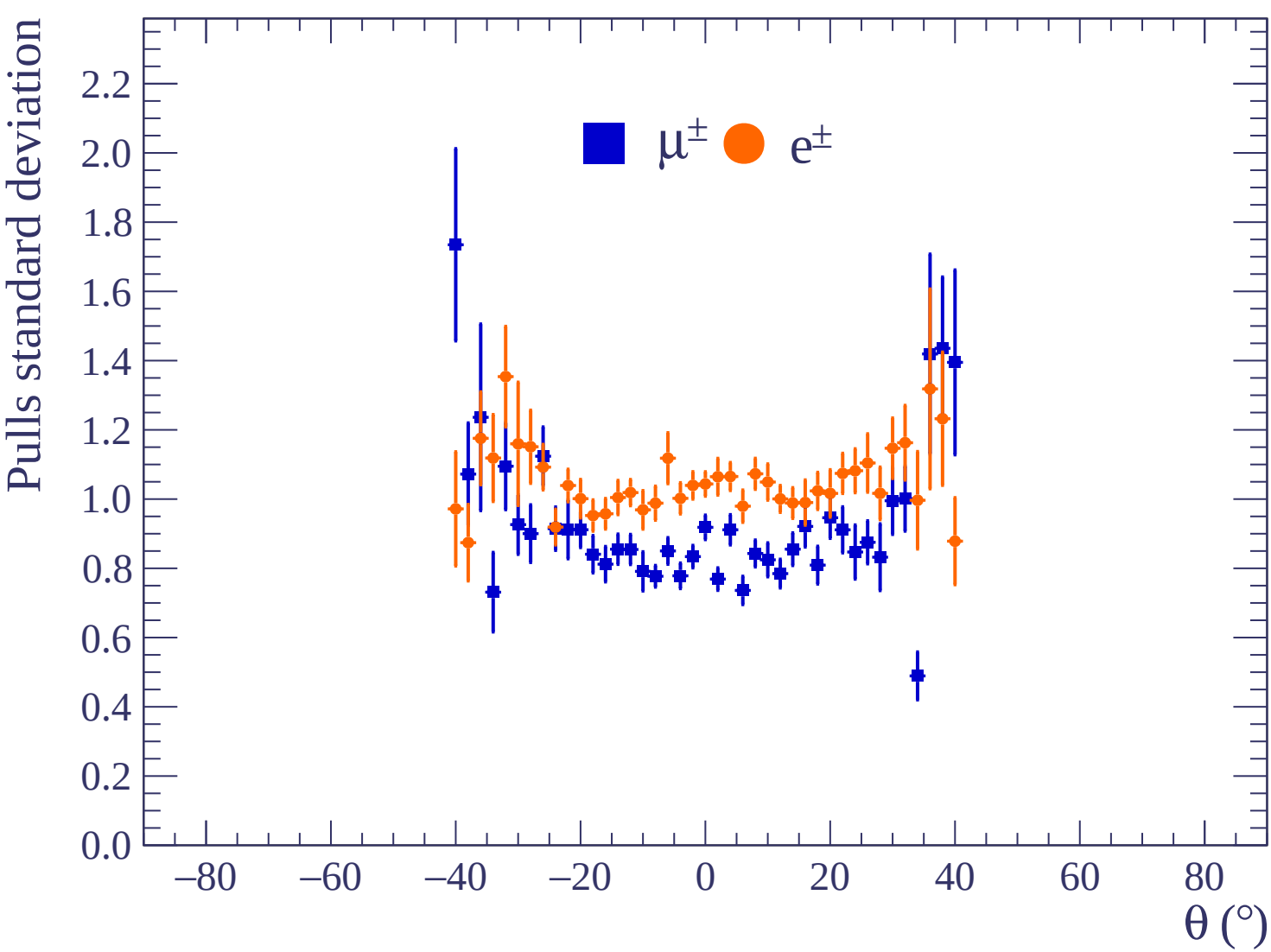


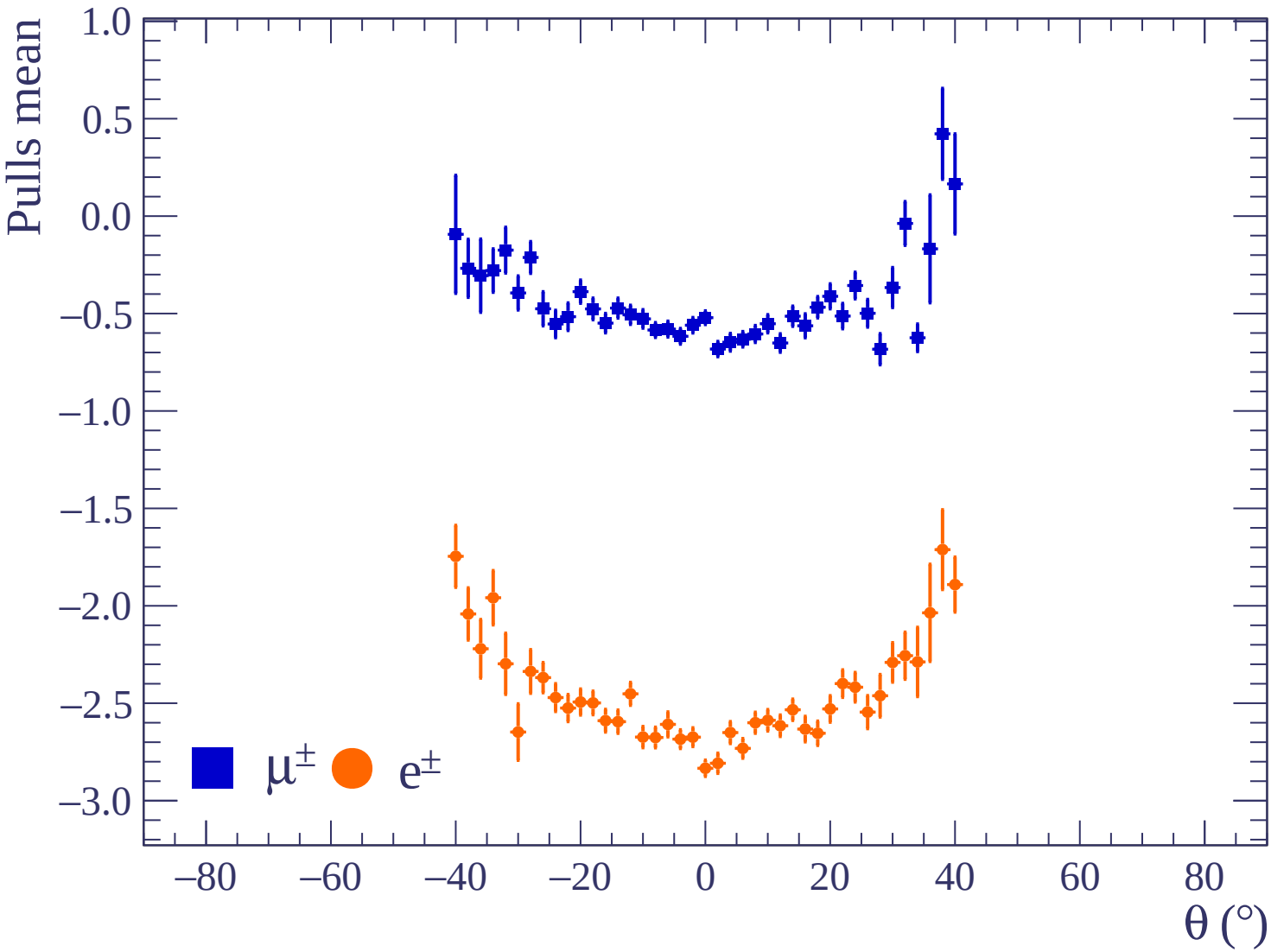


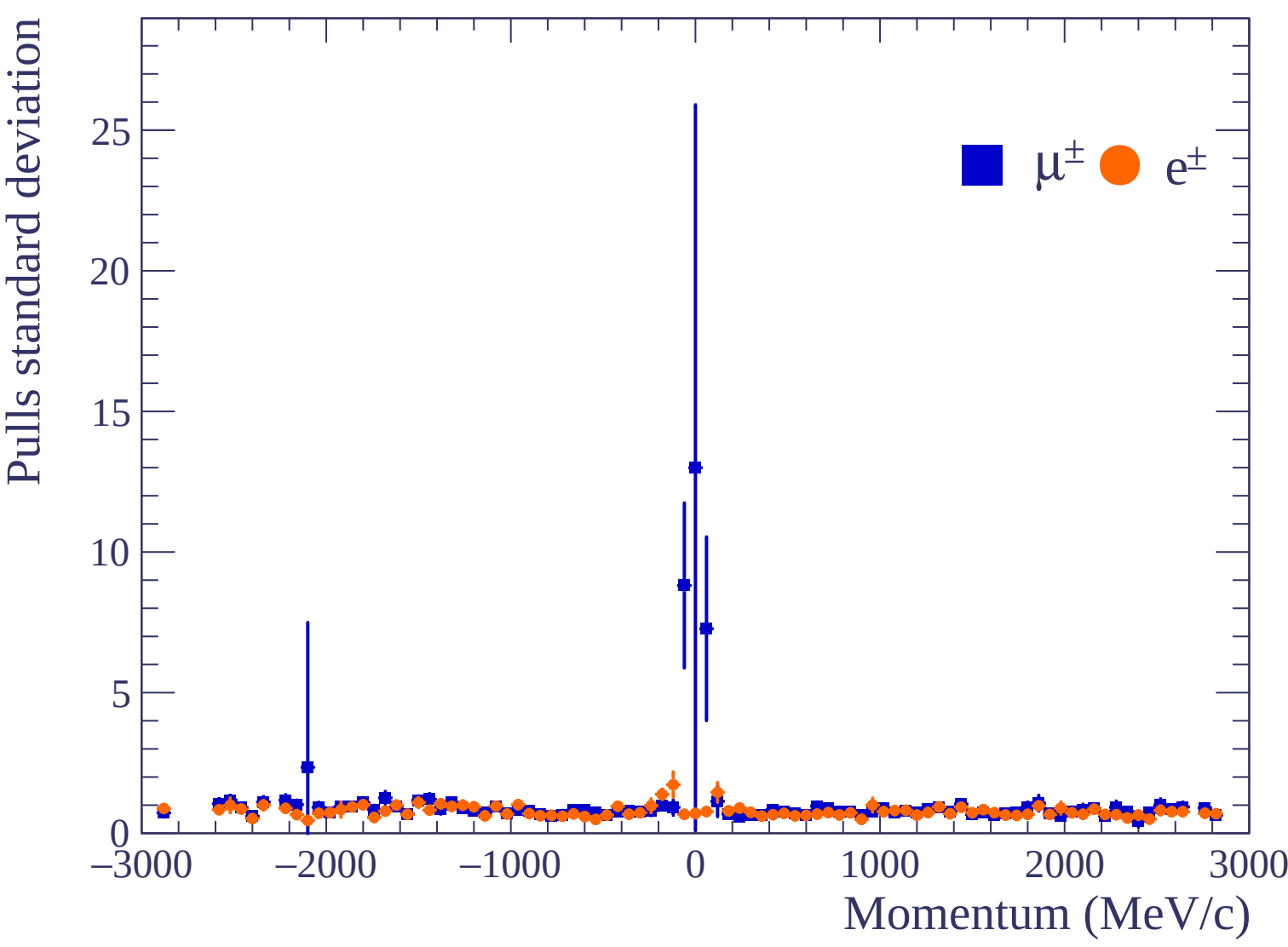




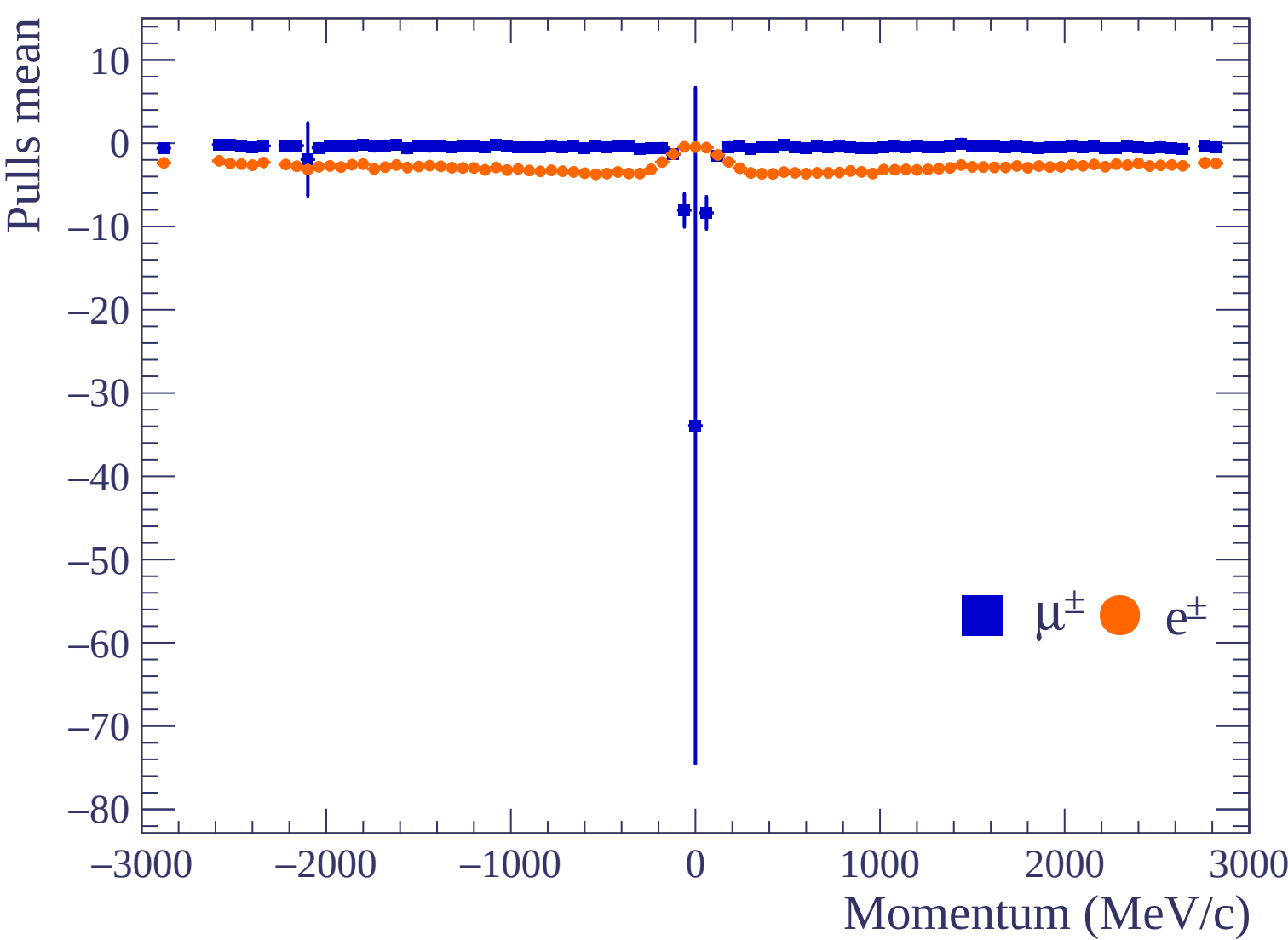


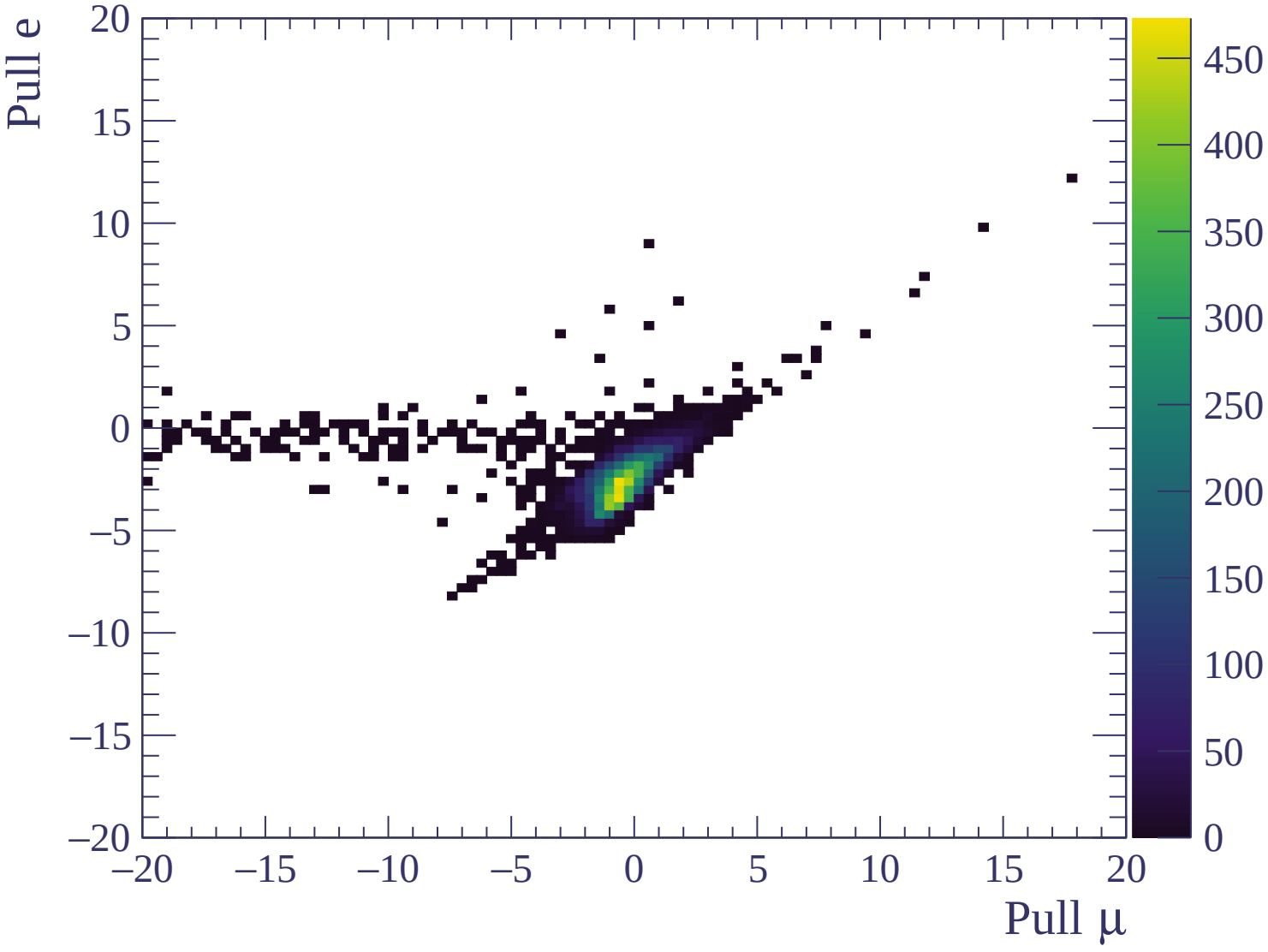


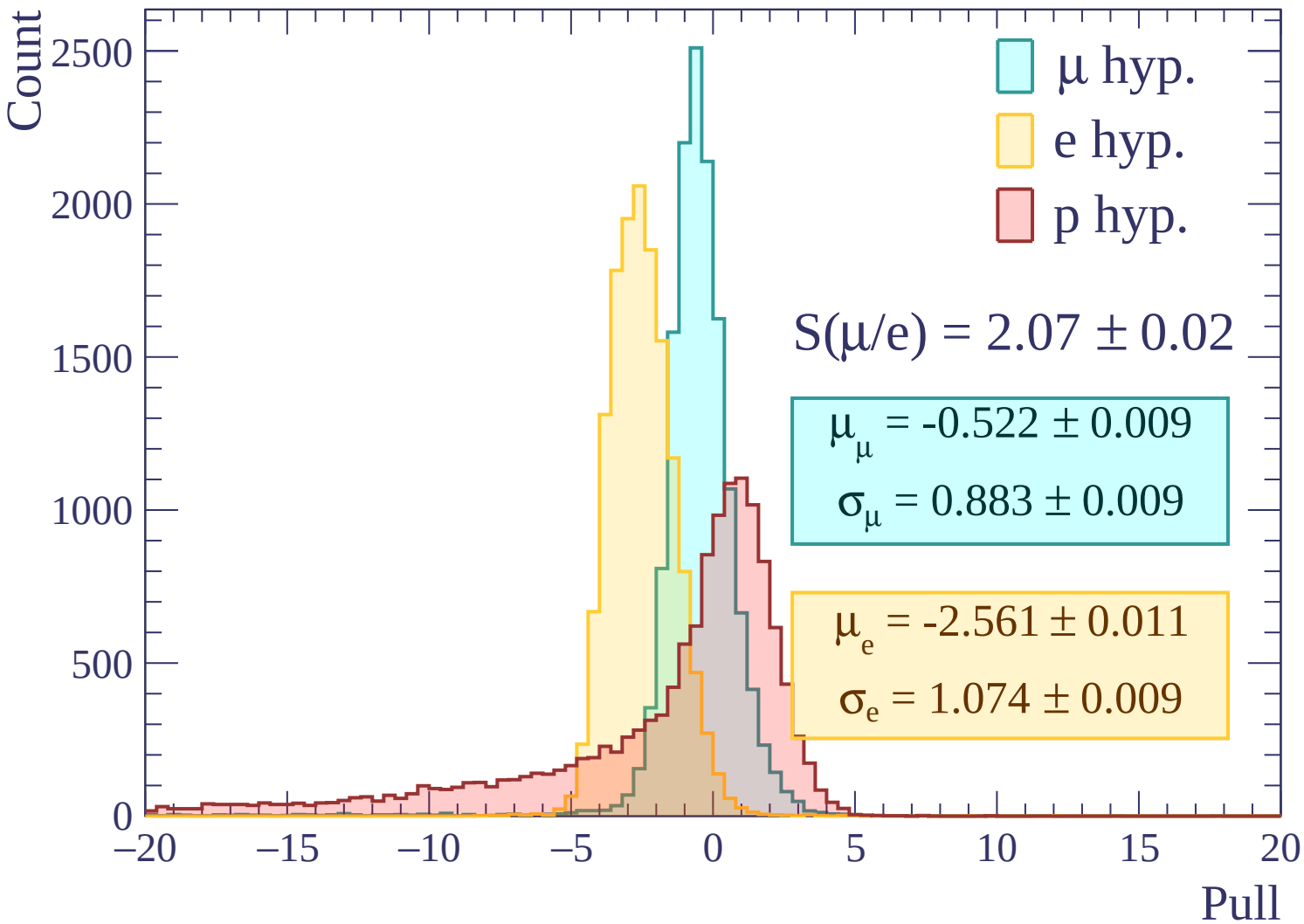


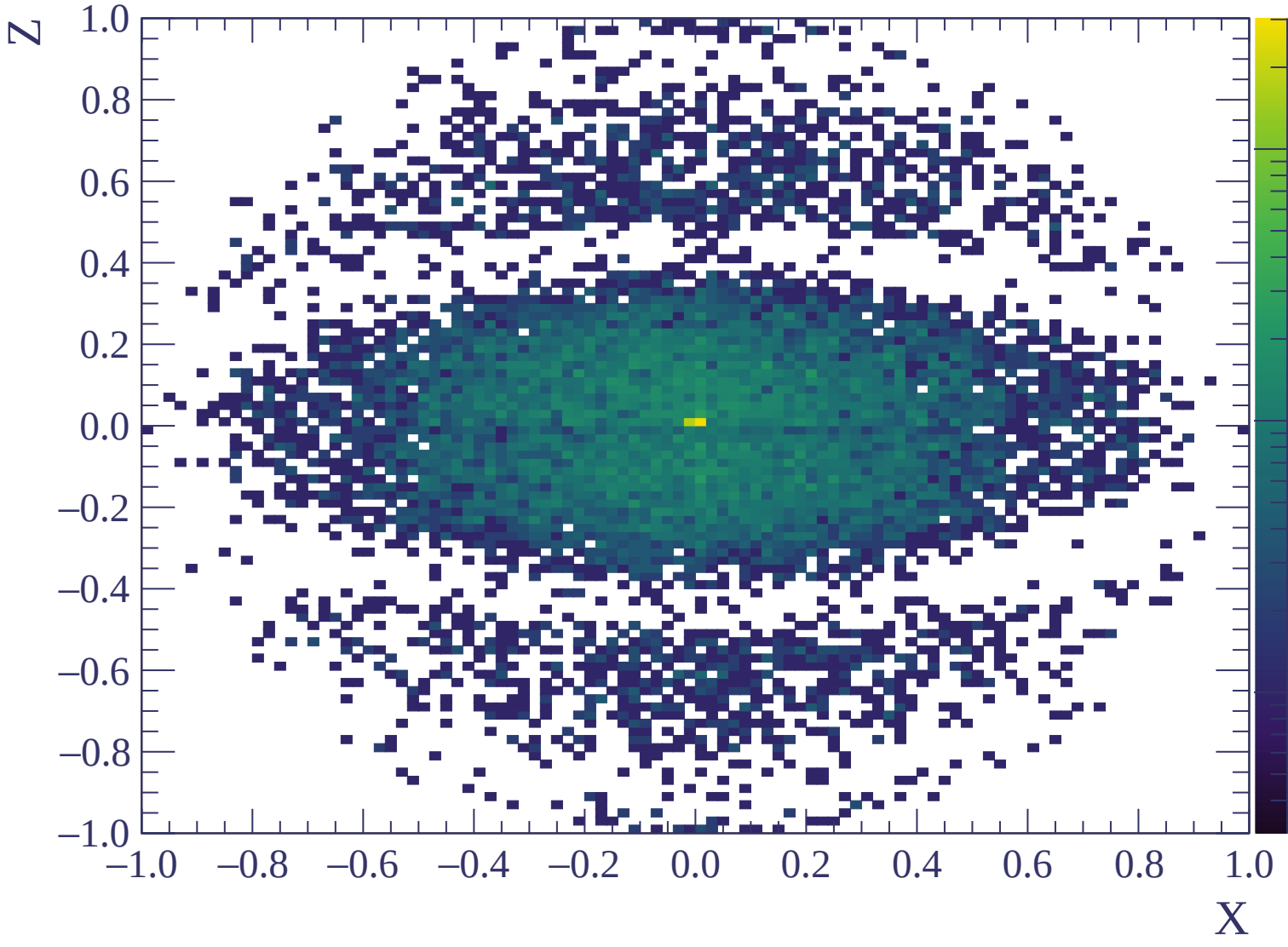


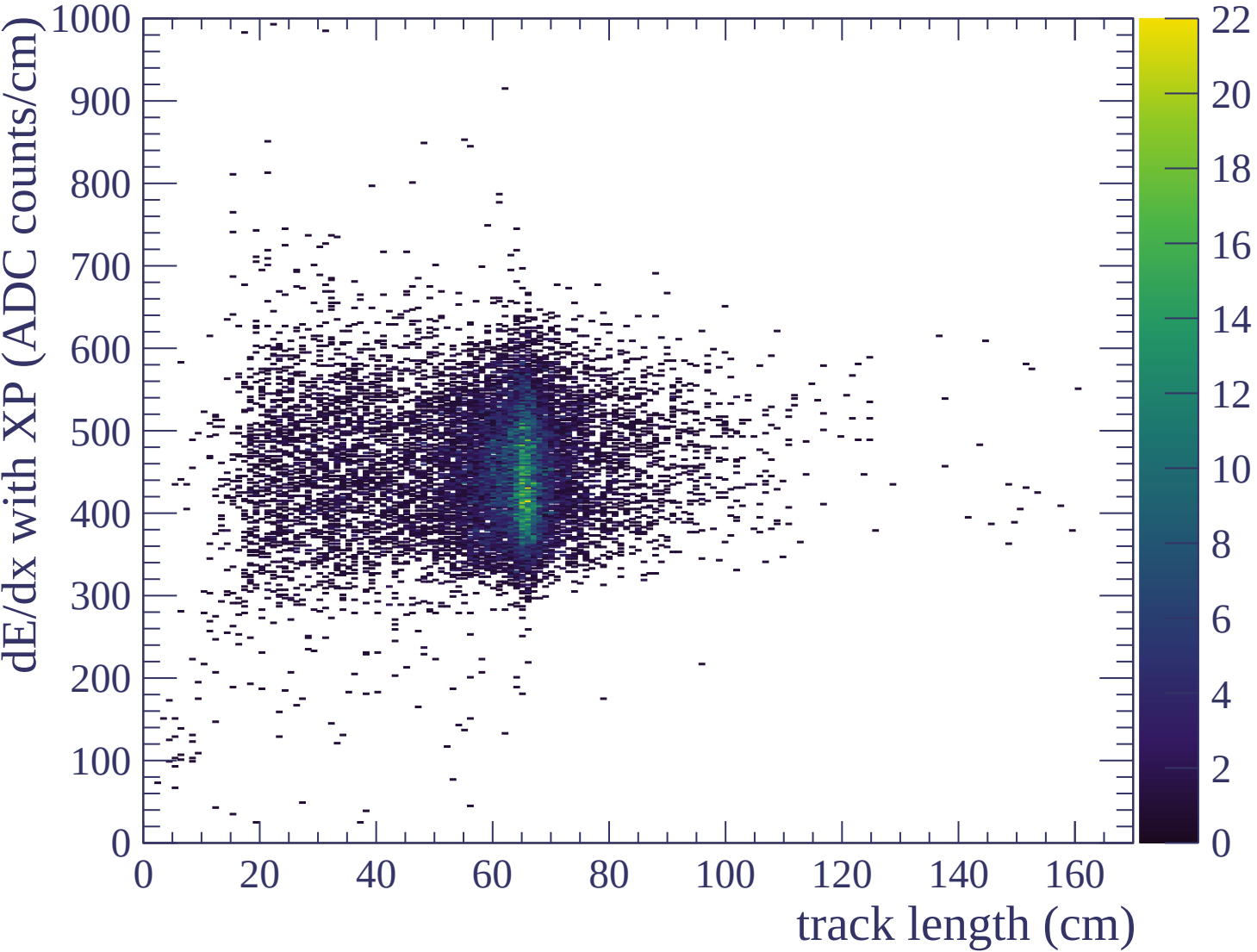


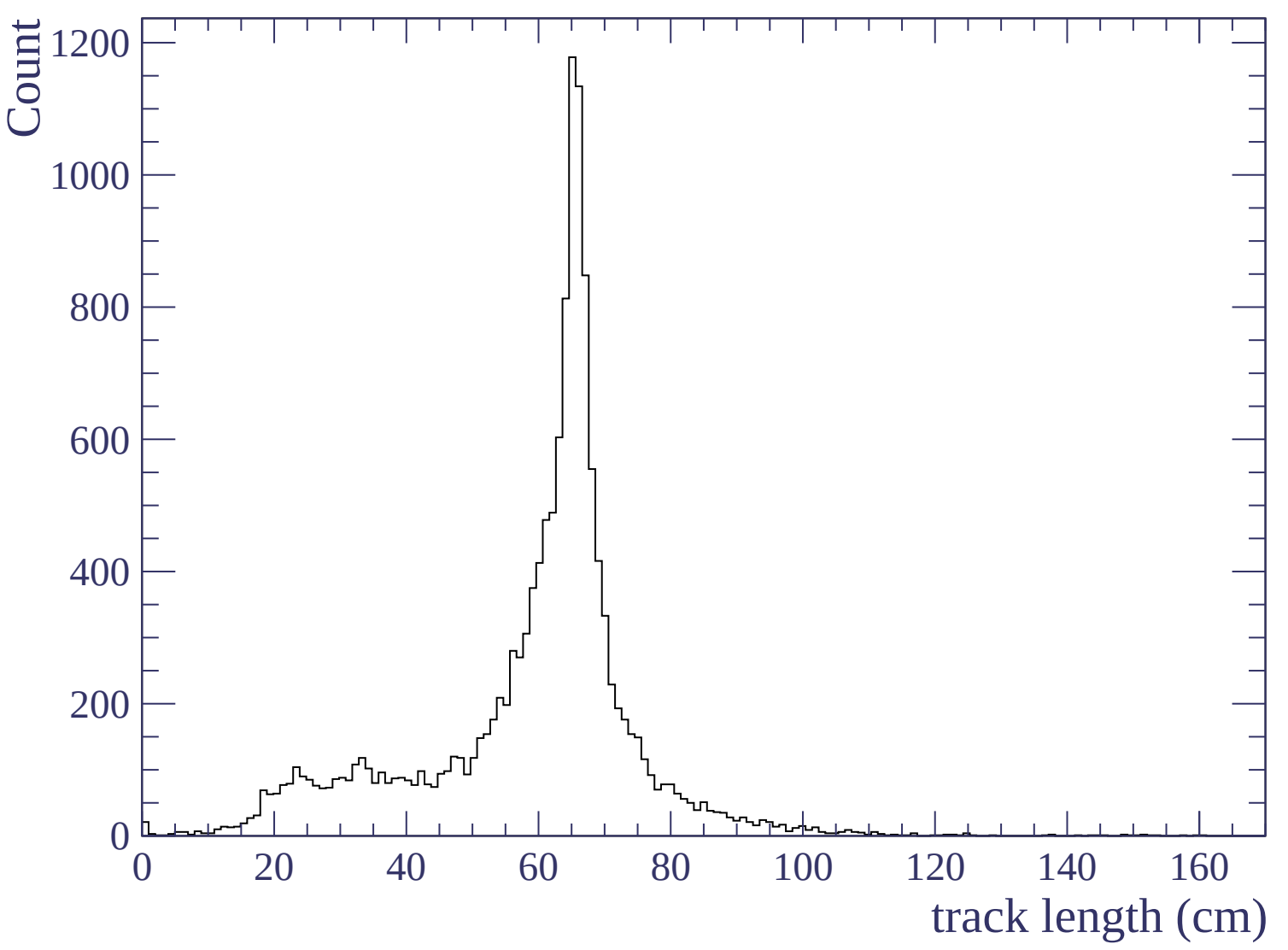


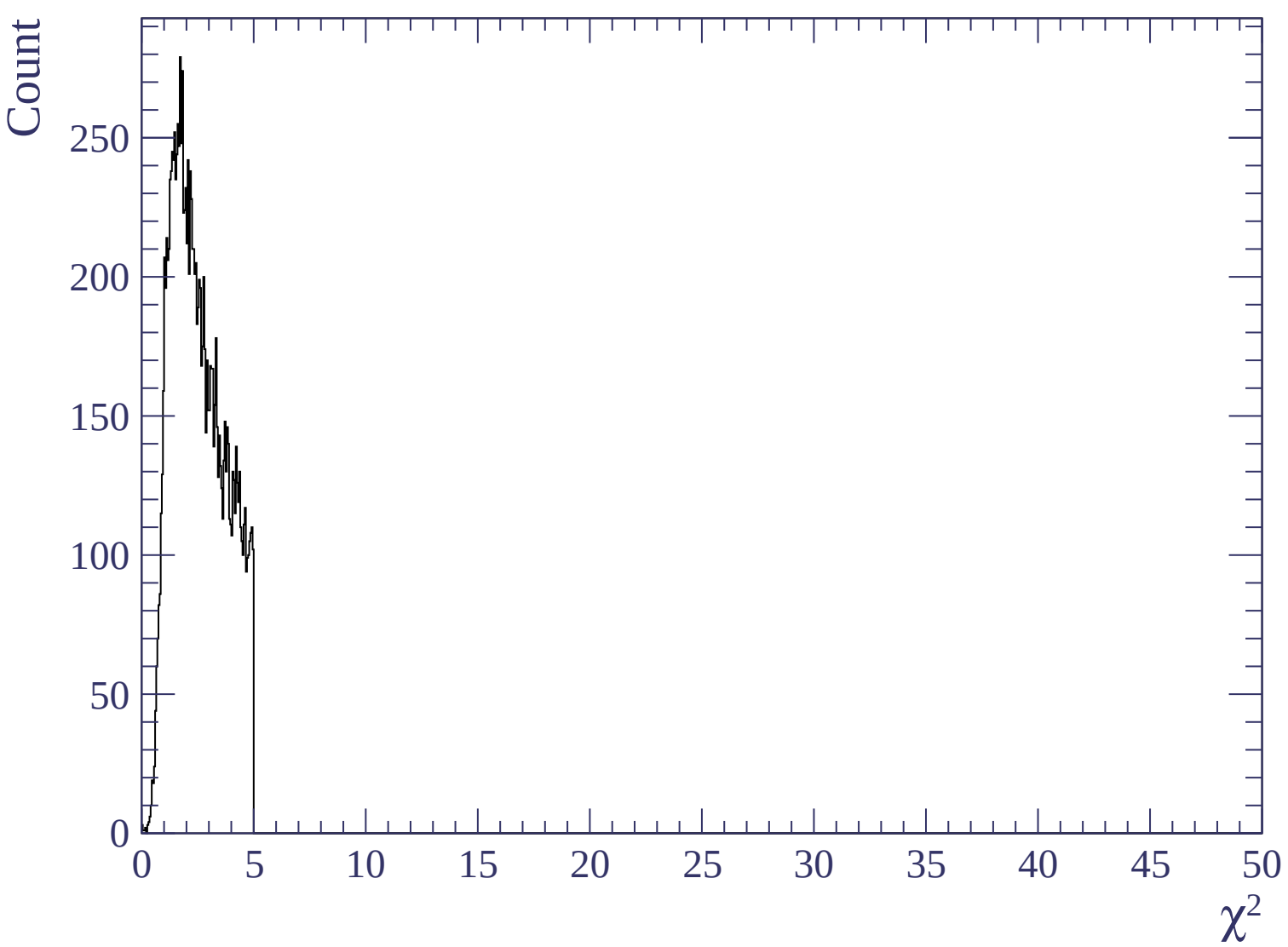


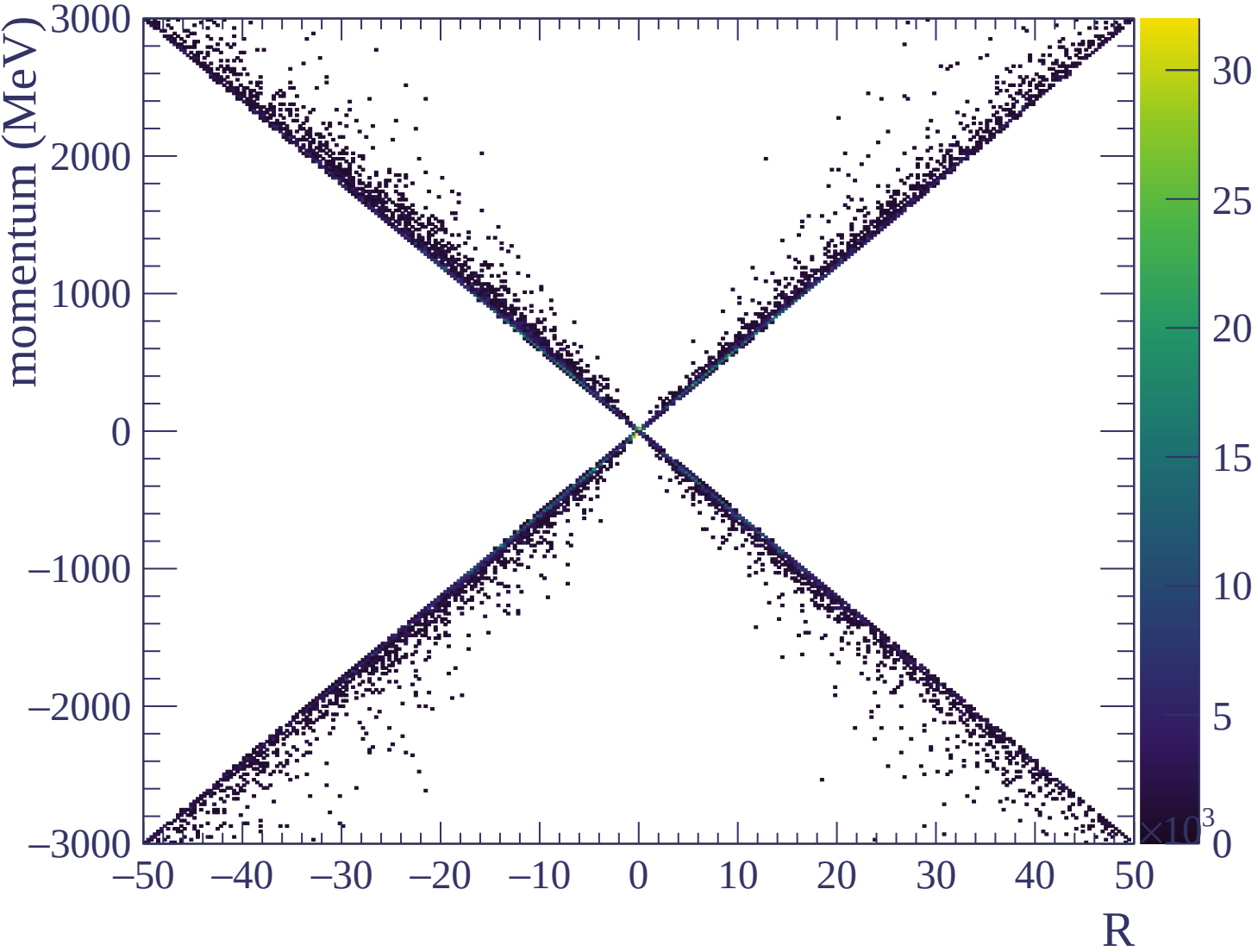




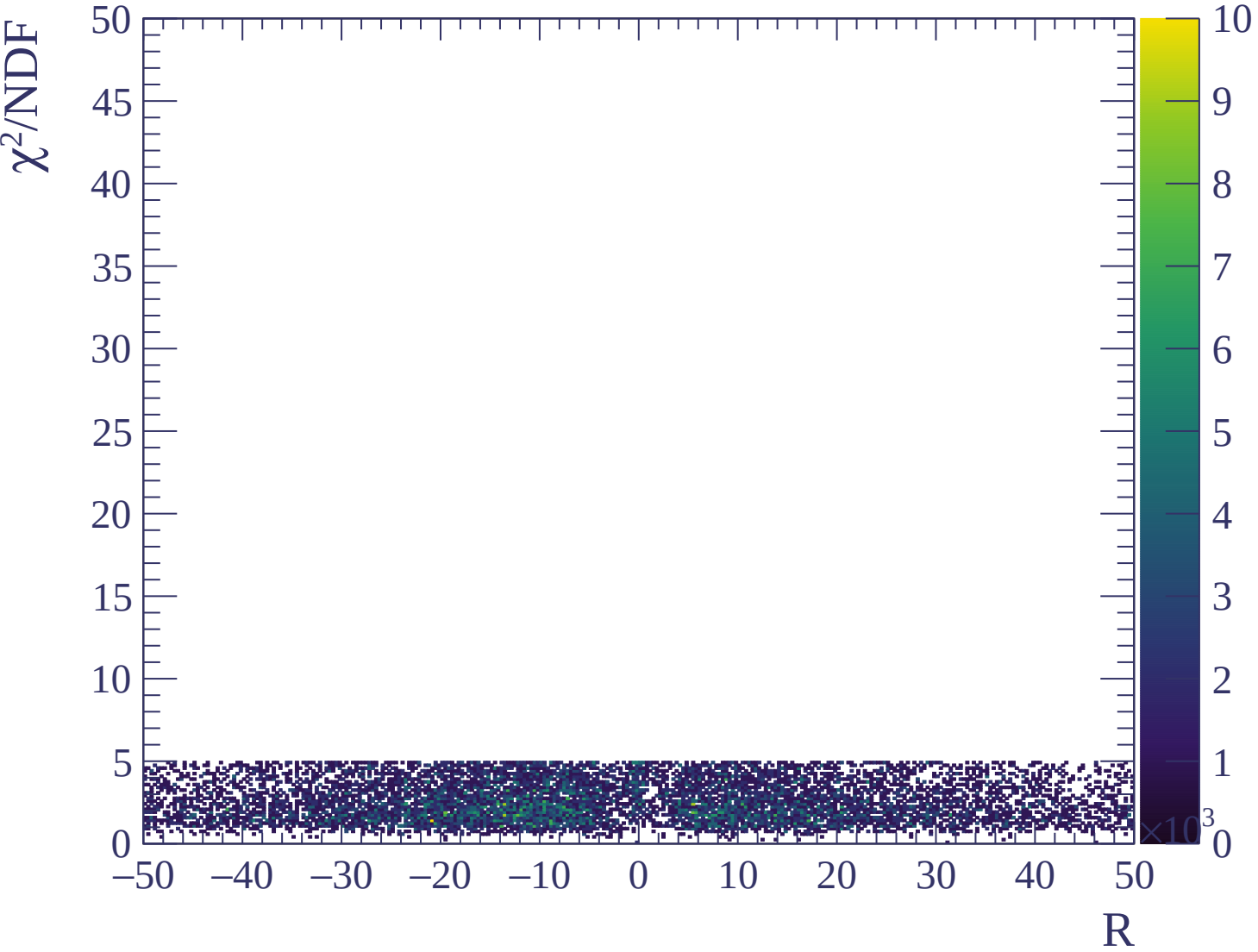






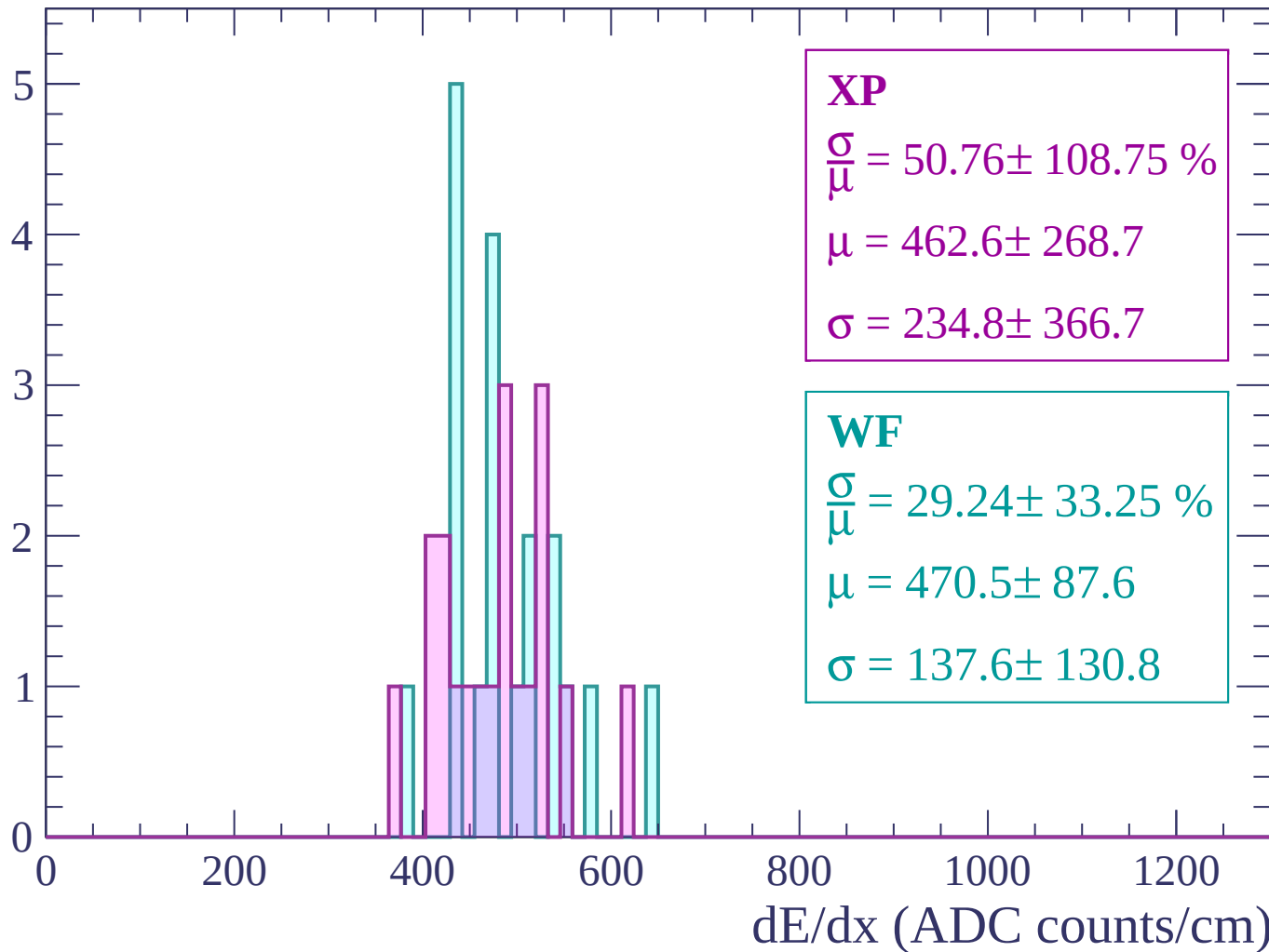






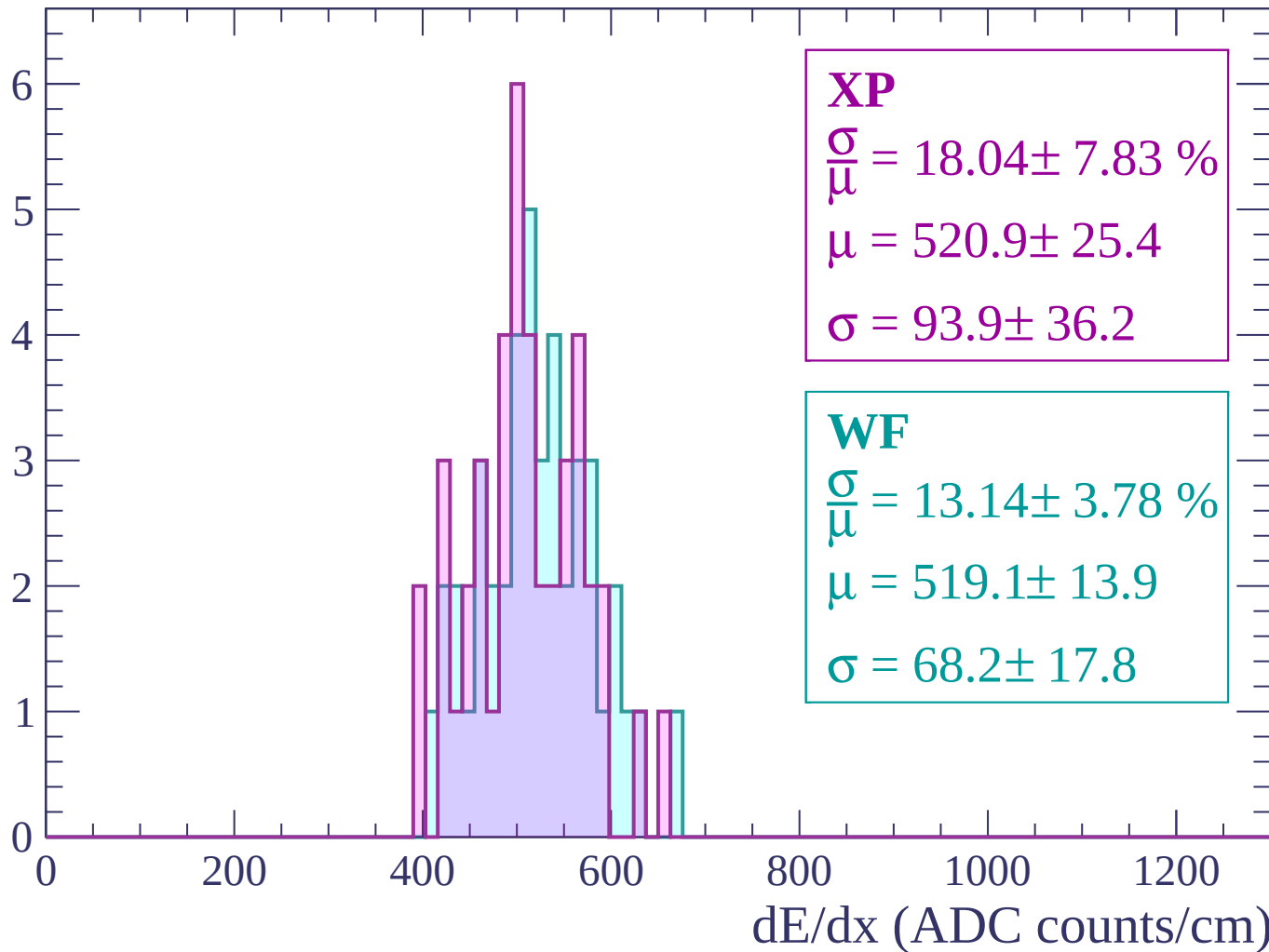
# Energy loss | -3000 < p < -2940

Count



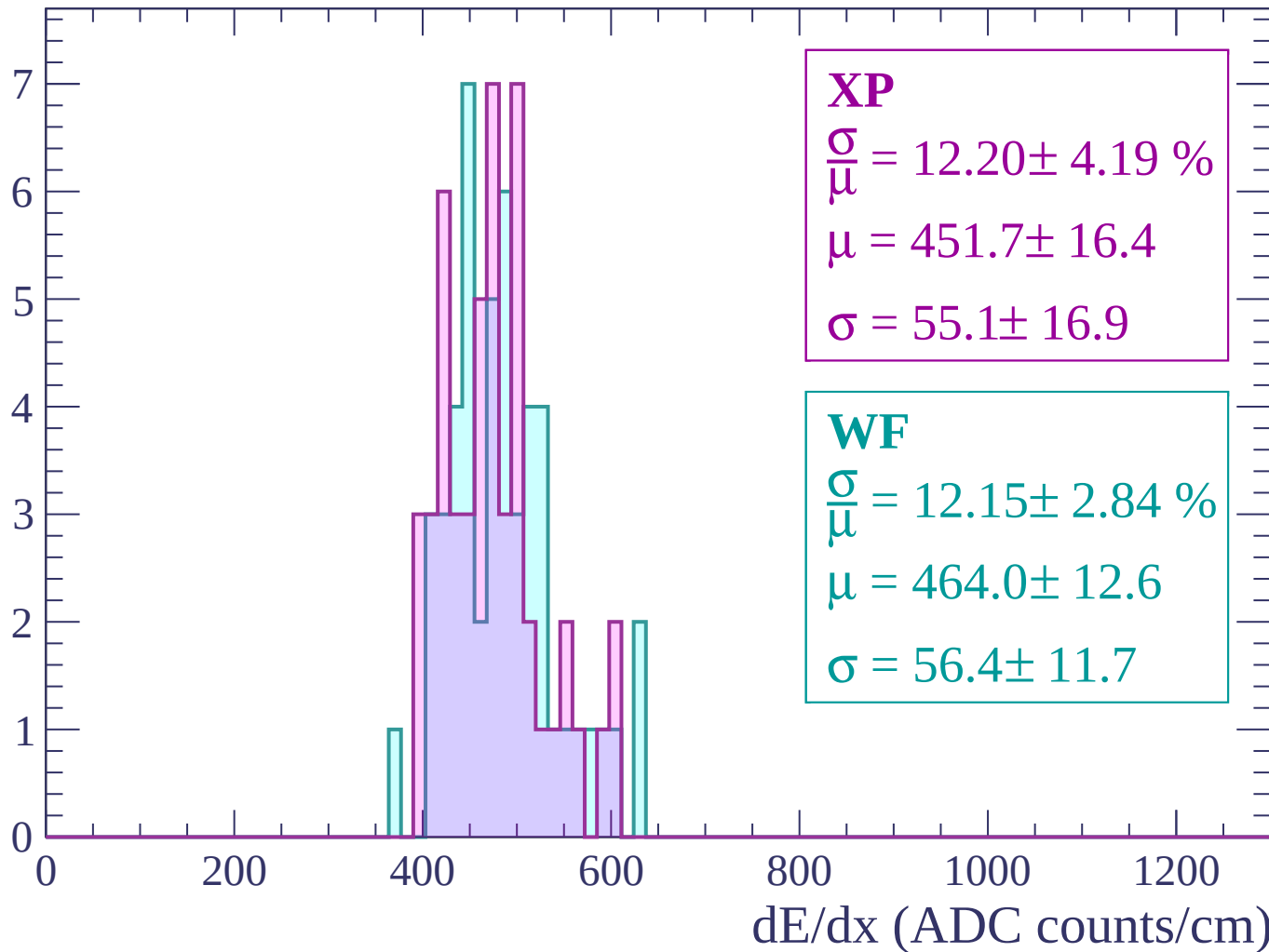
Energy loss |  $-2940 < p < -2880$

Count



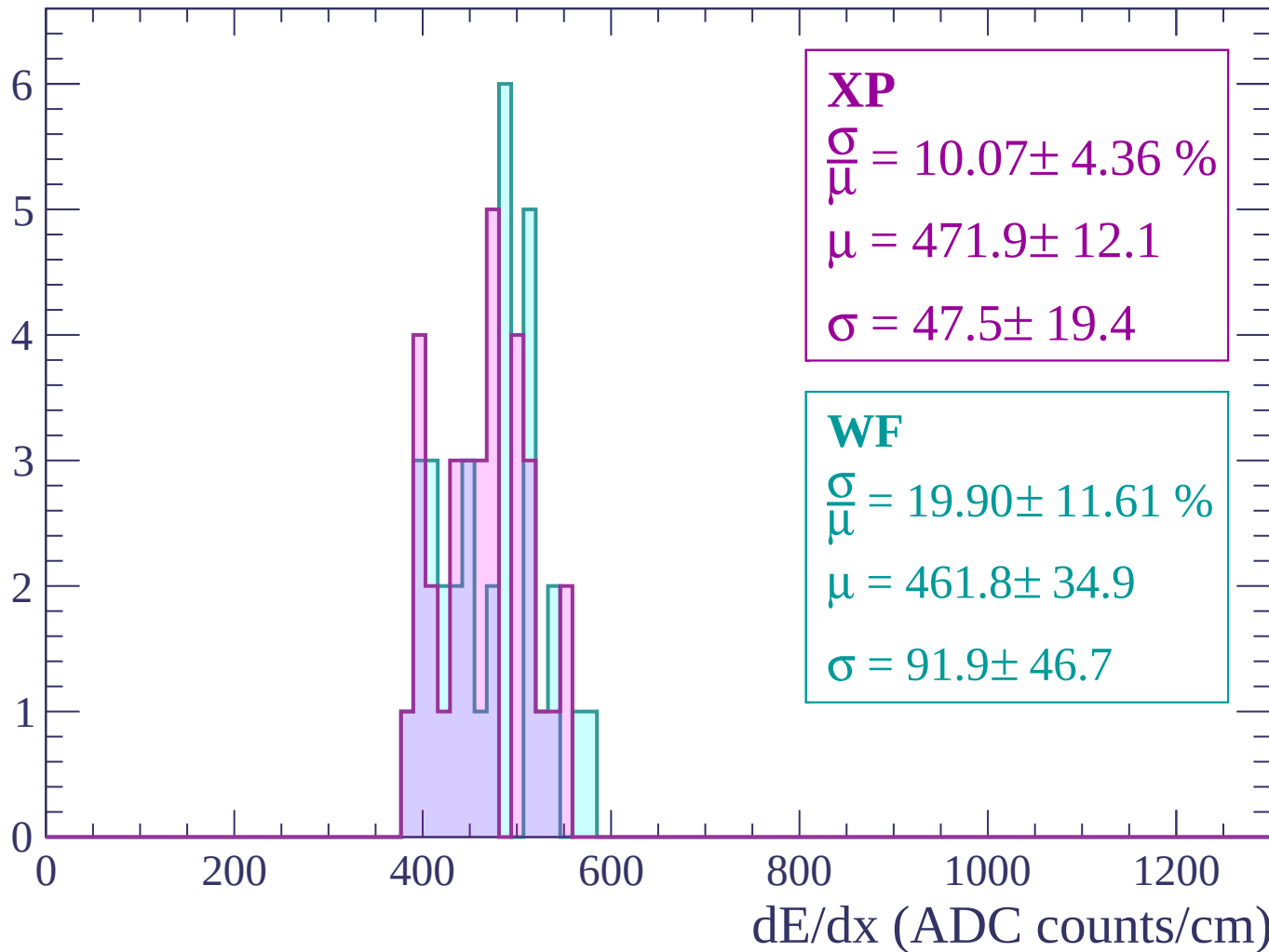
# Energy loss | -2880 < p < -2820

Count



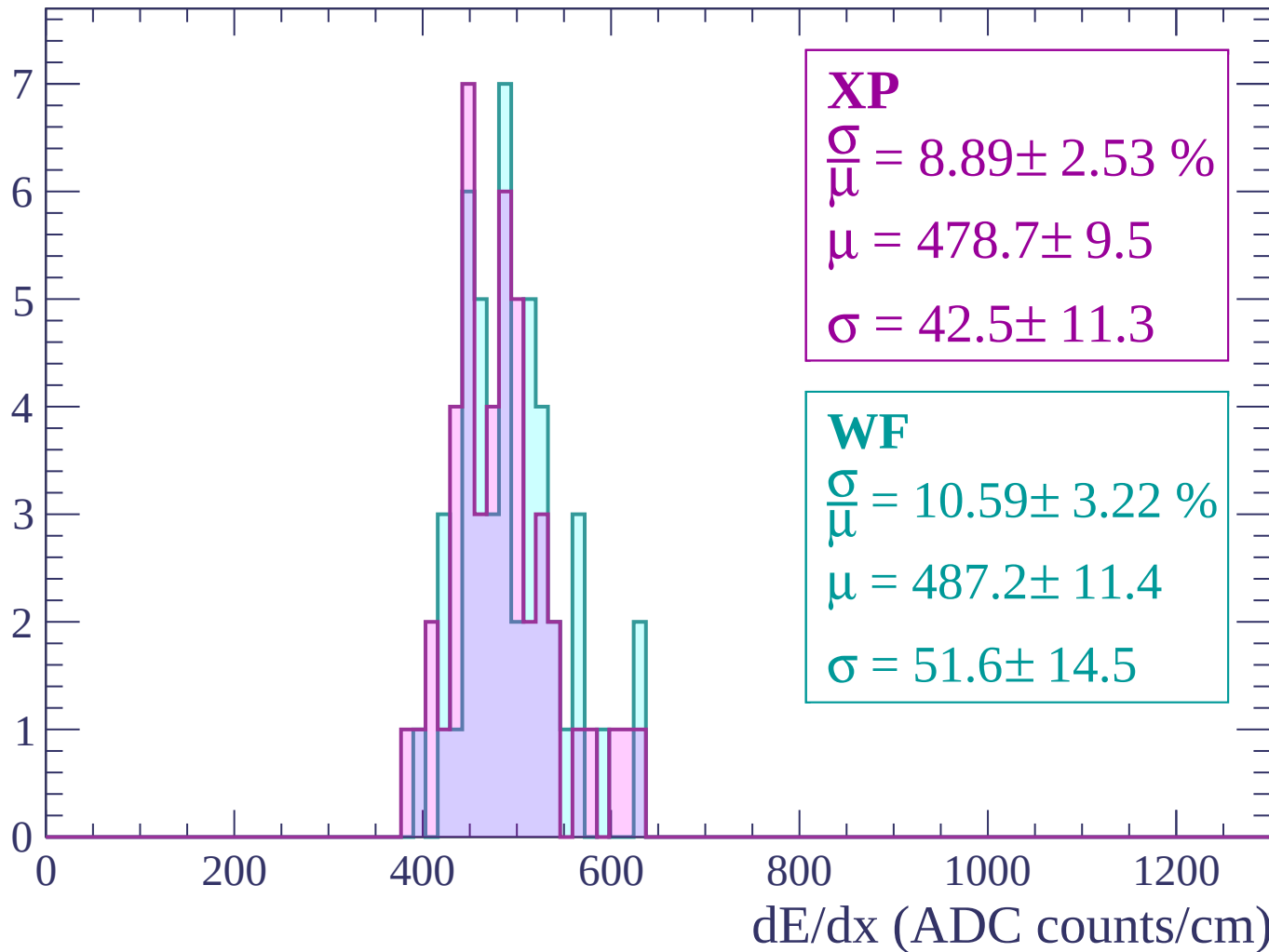
# Energy loss | -2820 < p < -2760

Count



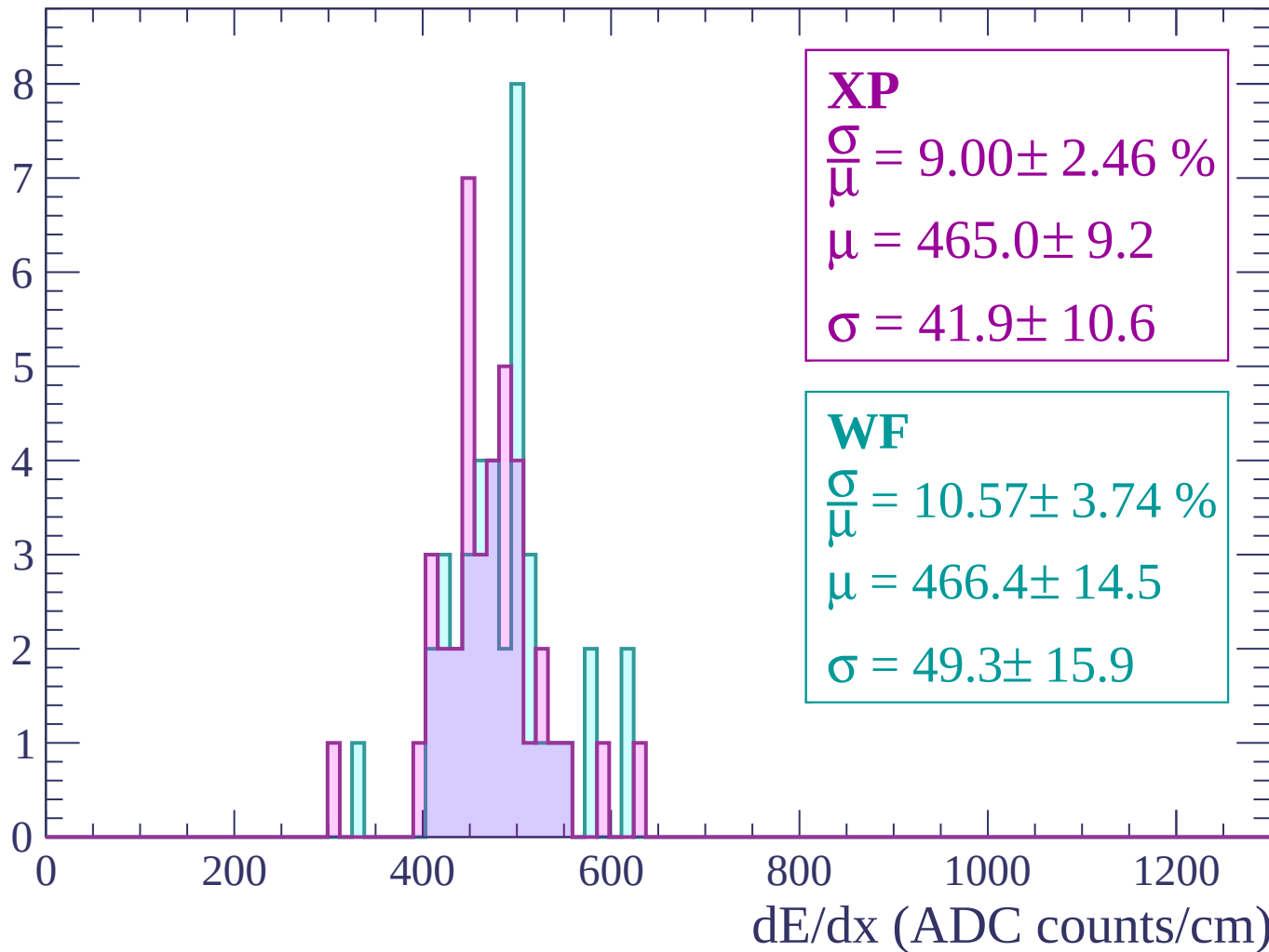
# Energy loss | -2760 < p < -2700

Count



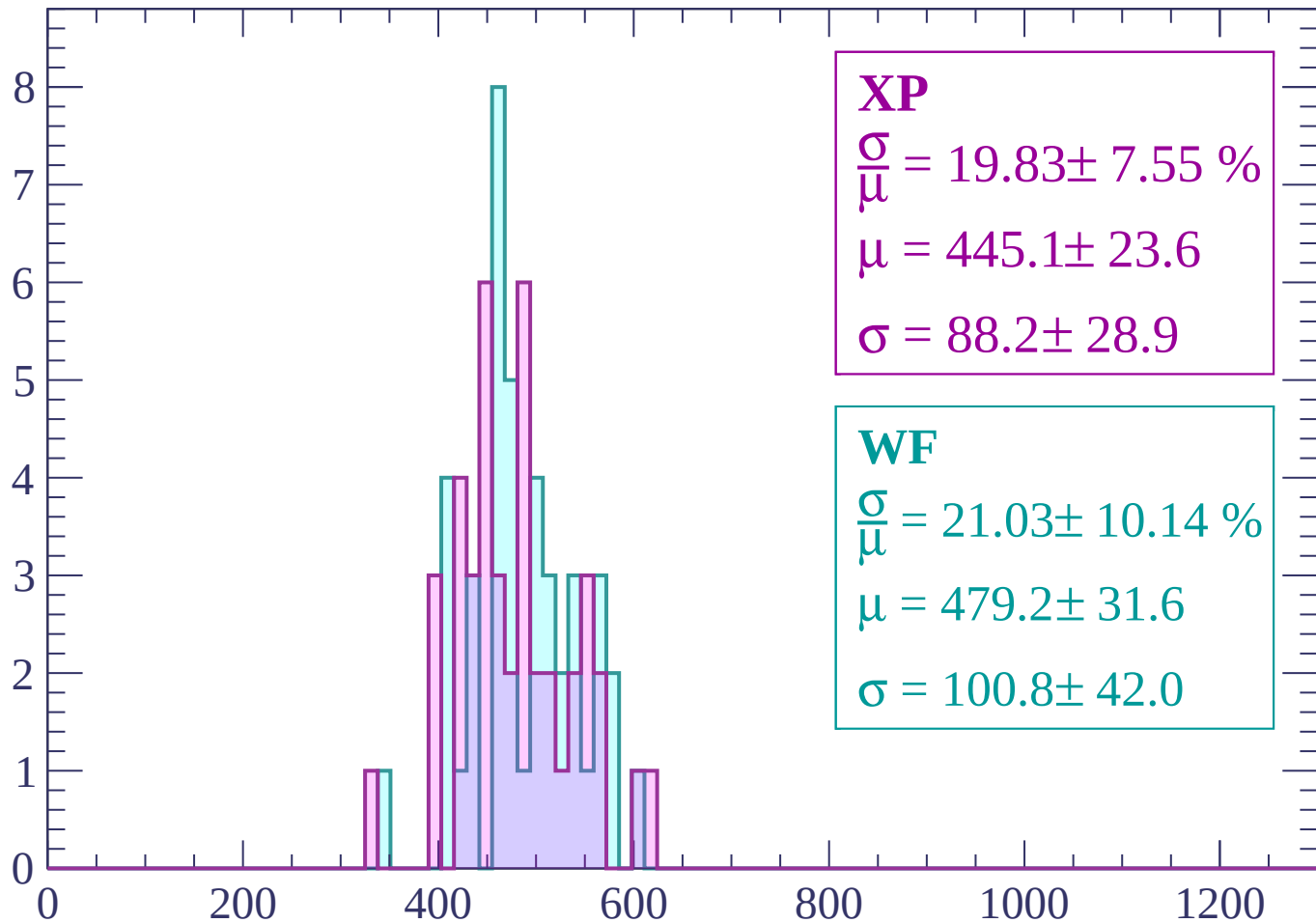
# Energy loss | -2700 < p < -2640

Count



# Energy loss | -2640 < p < -2580

Count

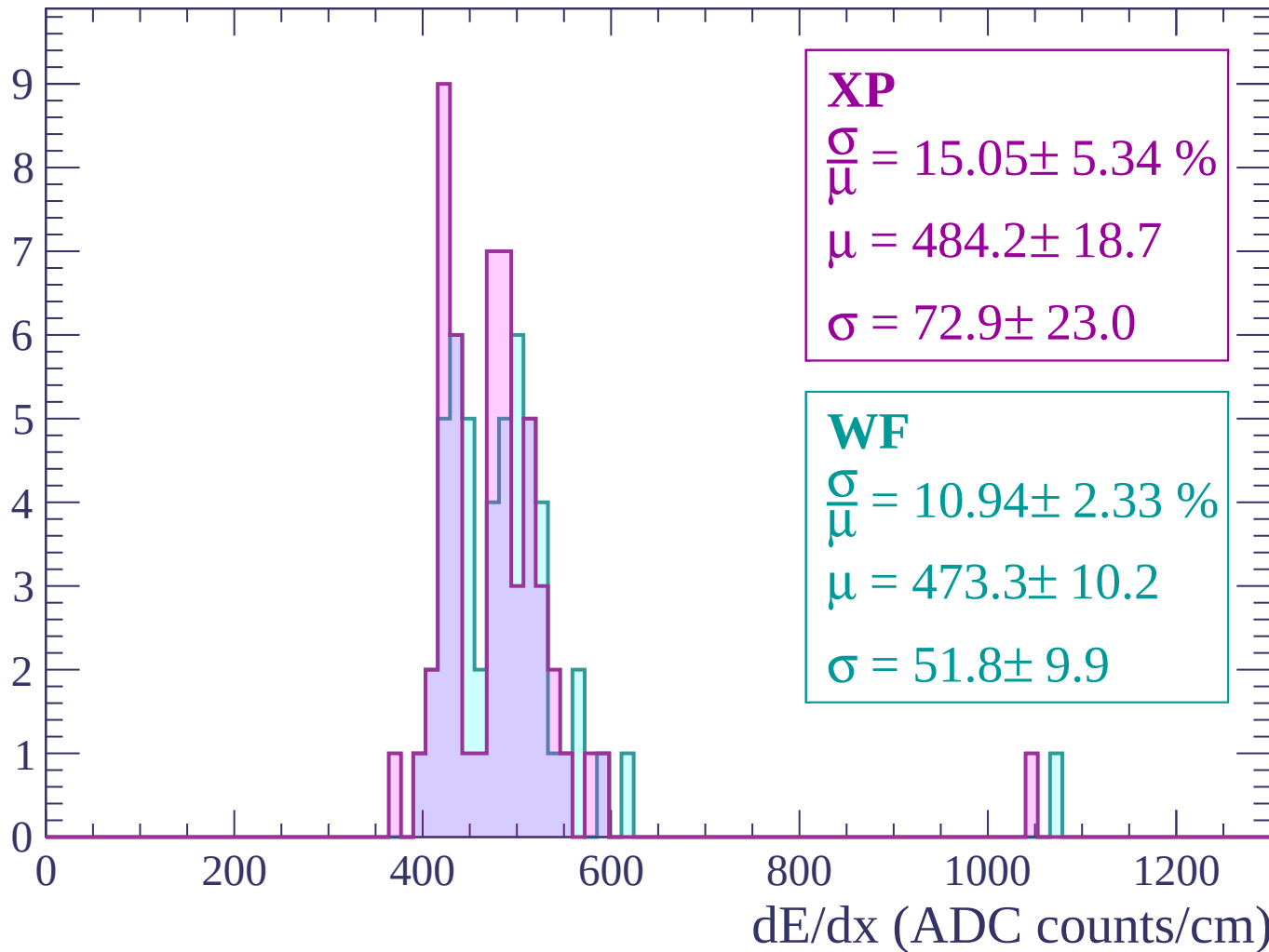


dE/dx (ADC counts/cm)



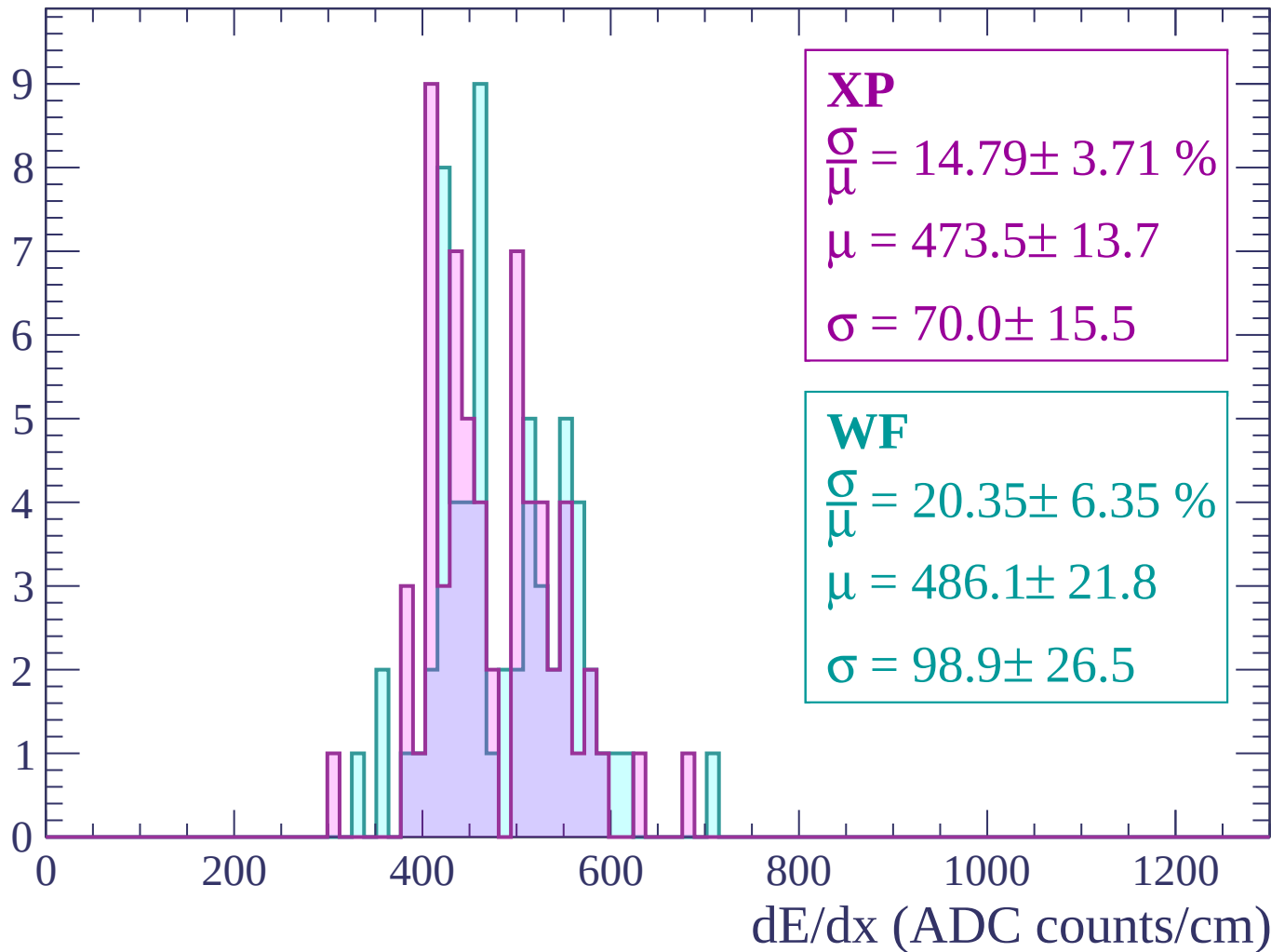
Energy loss |  $-2580 < p < -2520$

Count



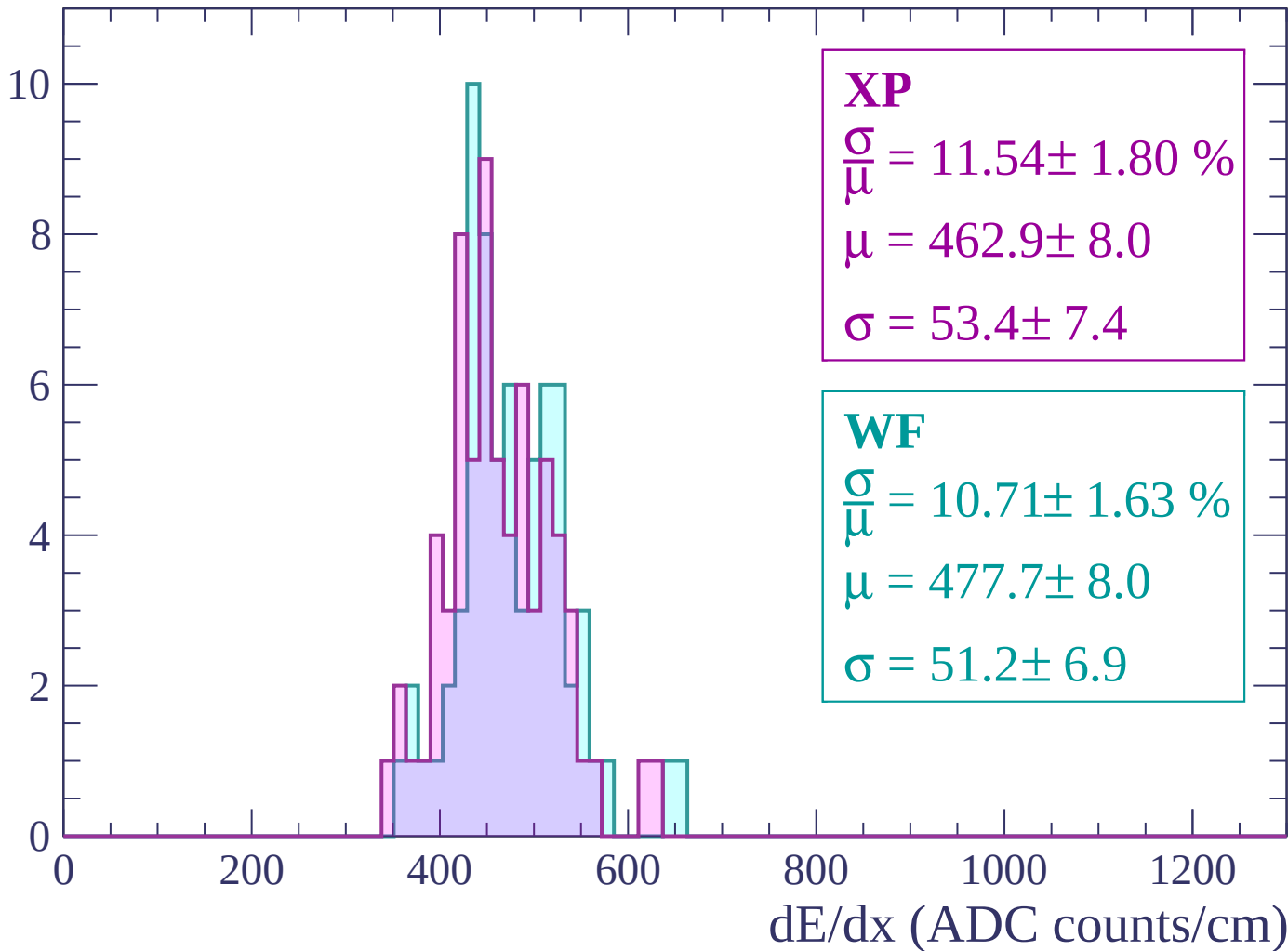
# Energy loss | -2520 < p < -2460

Count



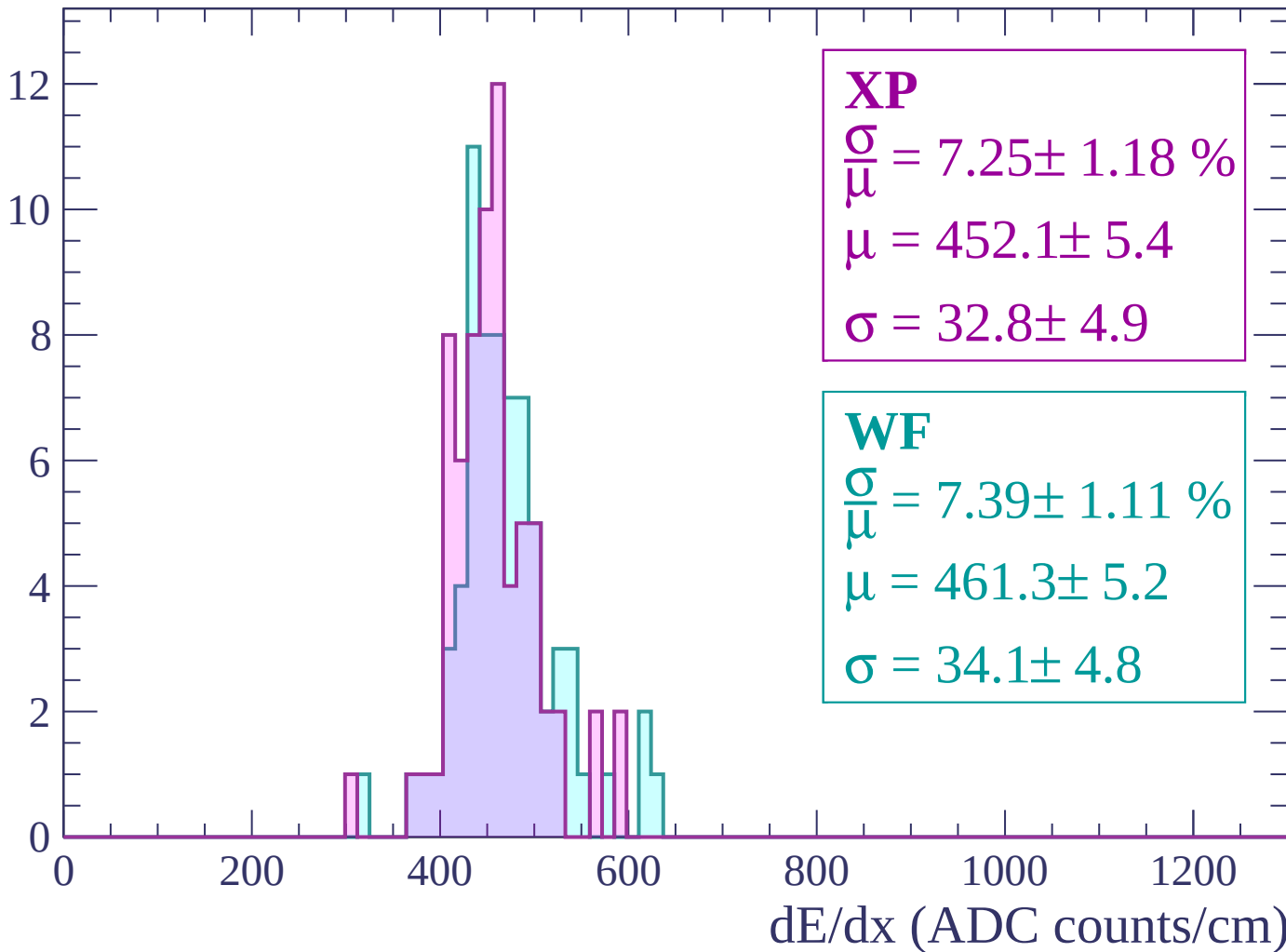
# Energy loss | -2460 < p < -2400

Count



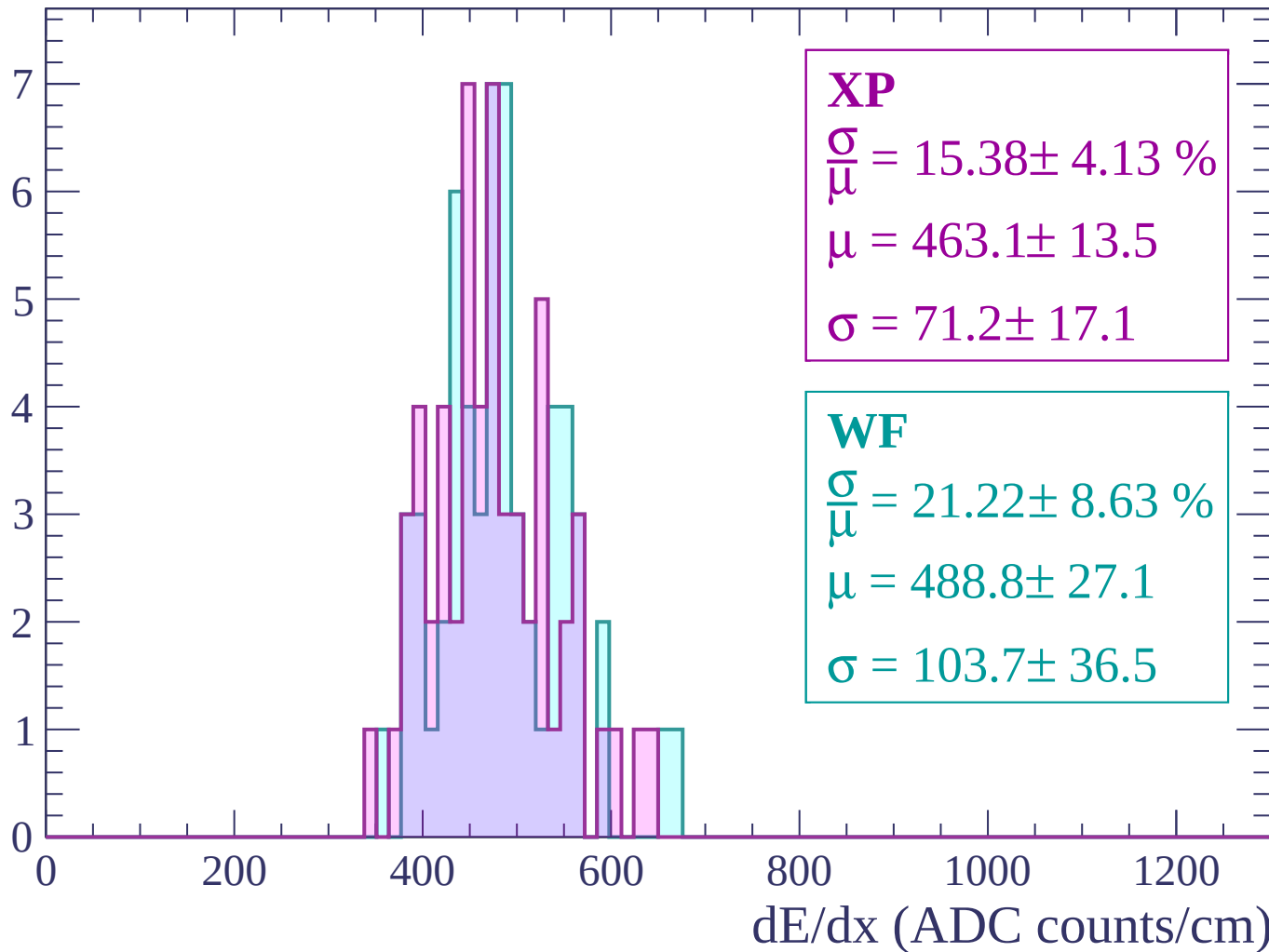
# Energy loss | -2400 < p < -2340

Count



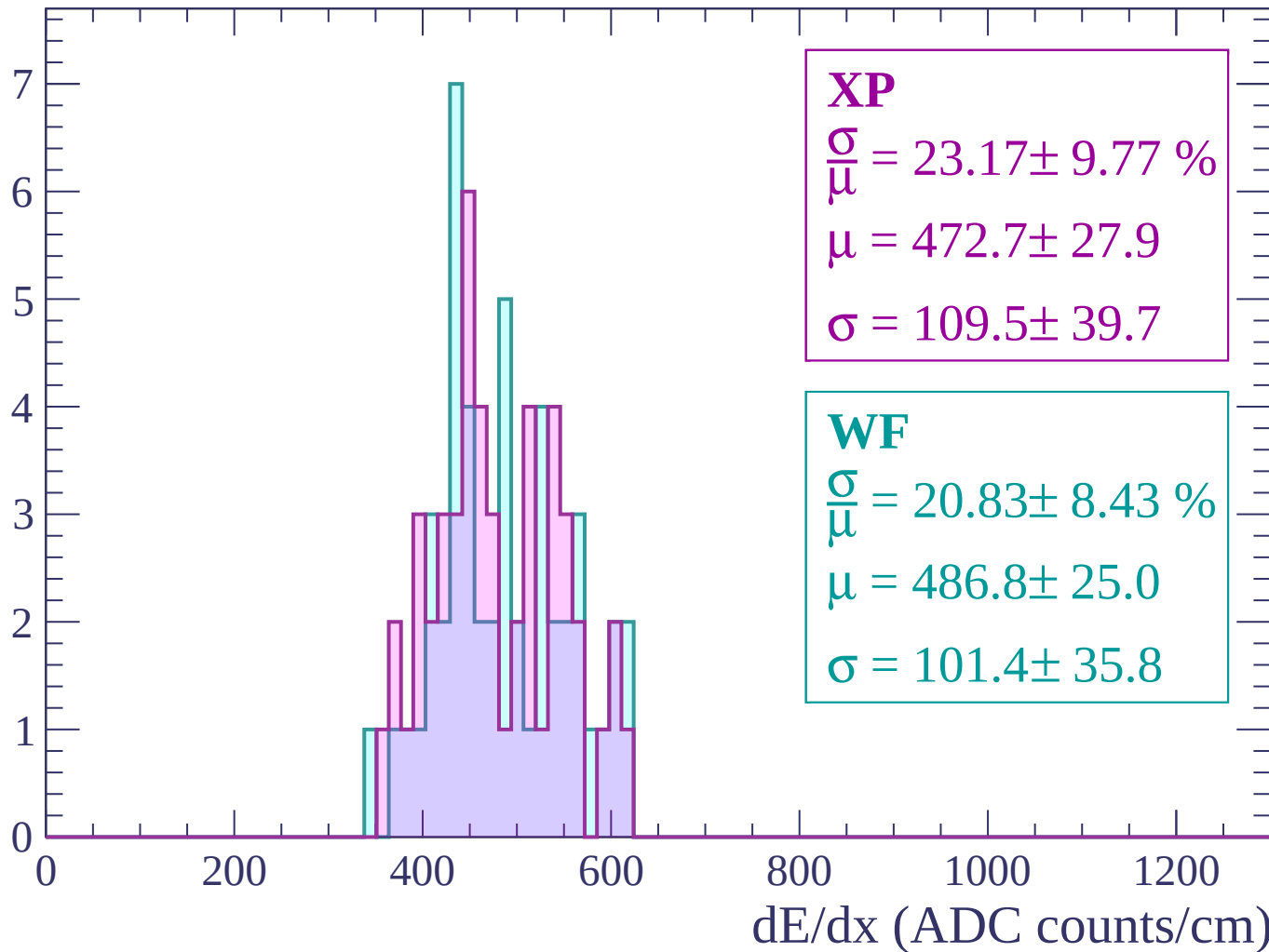
Energy loss |  $-2340 < p < -2280$

Count



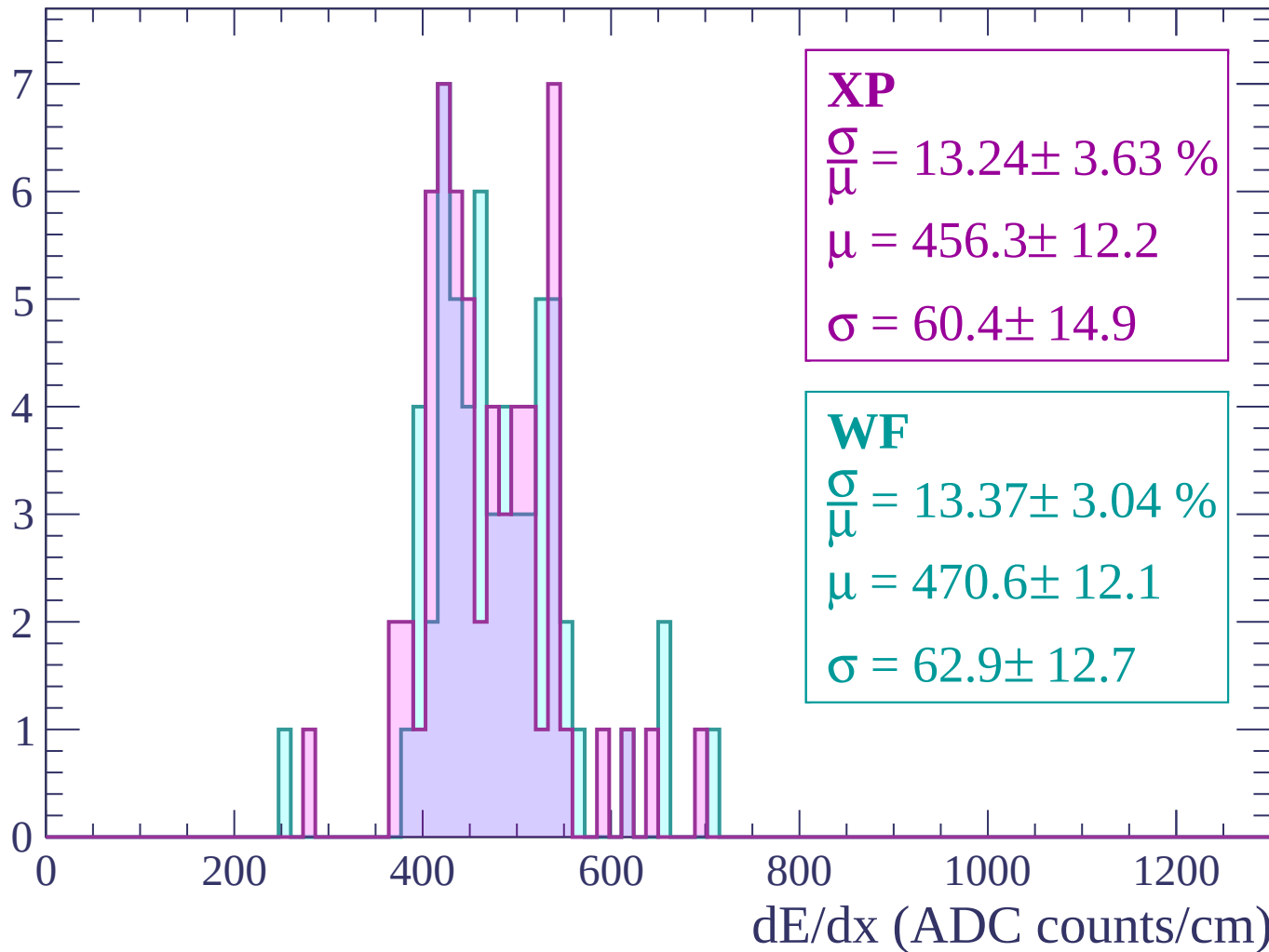
# Energy loss | -2280 < p < -2220

Count



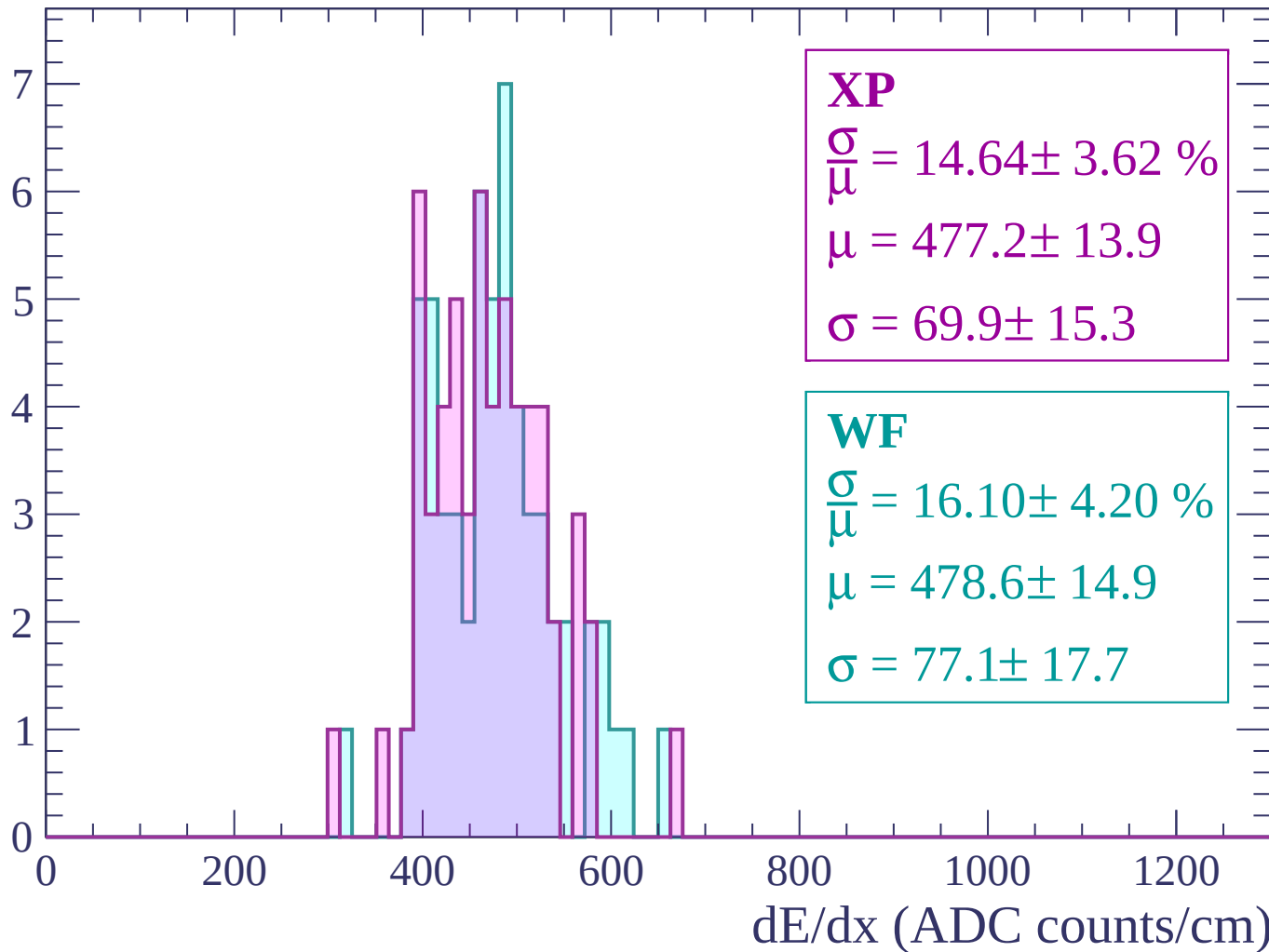
# Energy loss | -2220 < p < -2160

Count



# Energy loss | -2160 < p < -2100

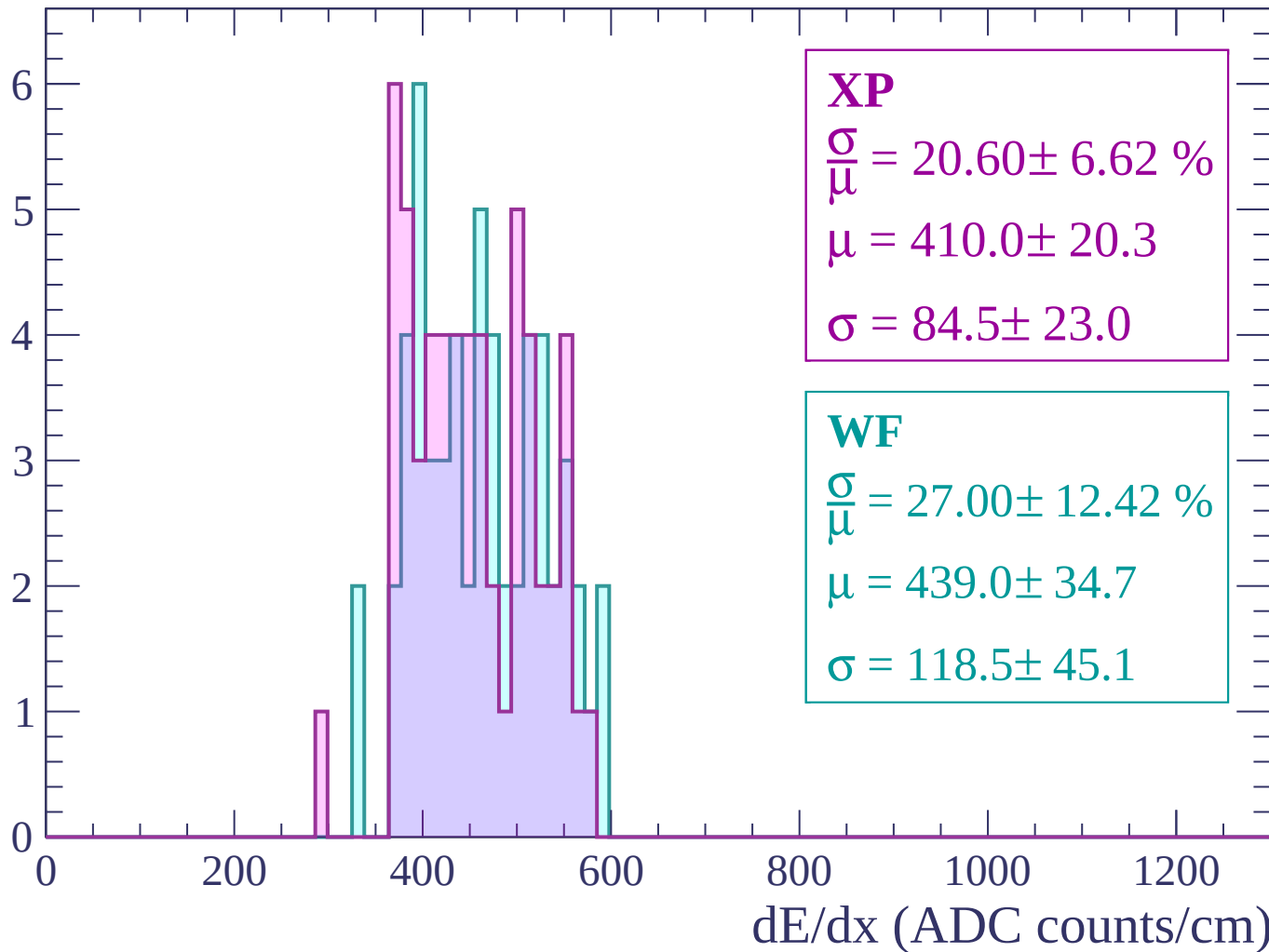
Count





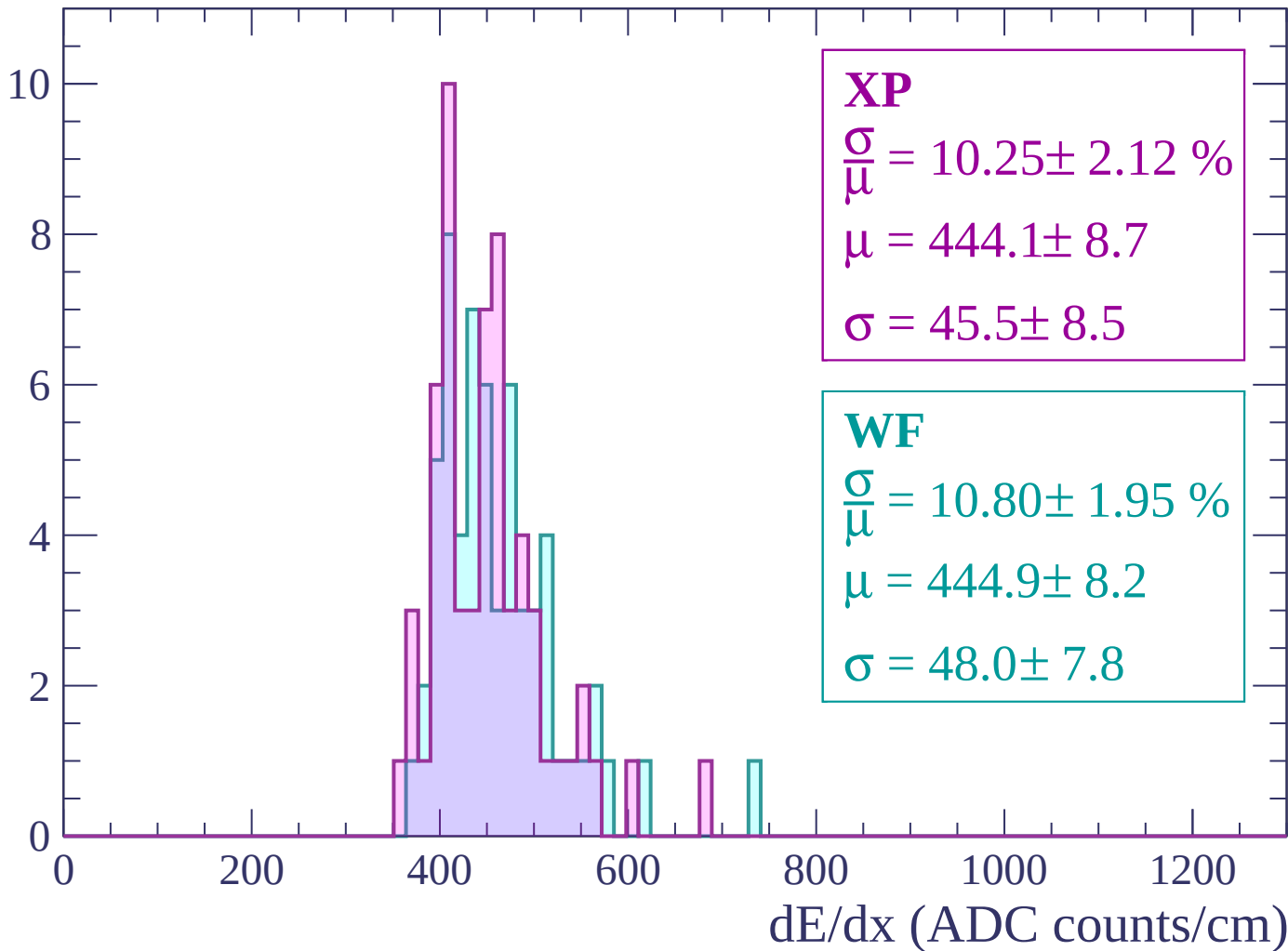
# Energy loss | -2100 < p < -2040

Count



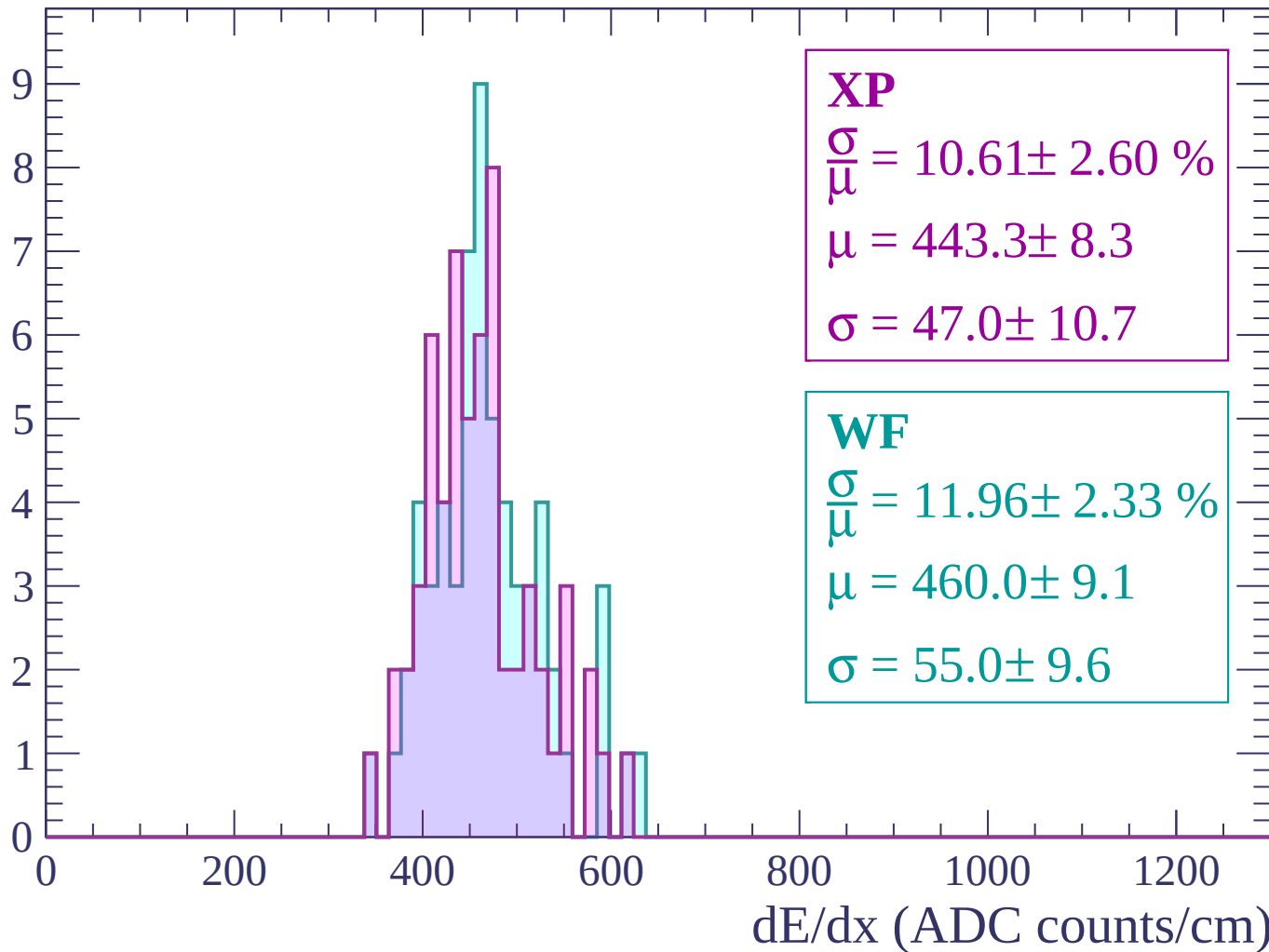
# Energy loss | -2040 < p < -1980

Count



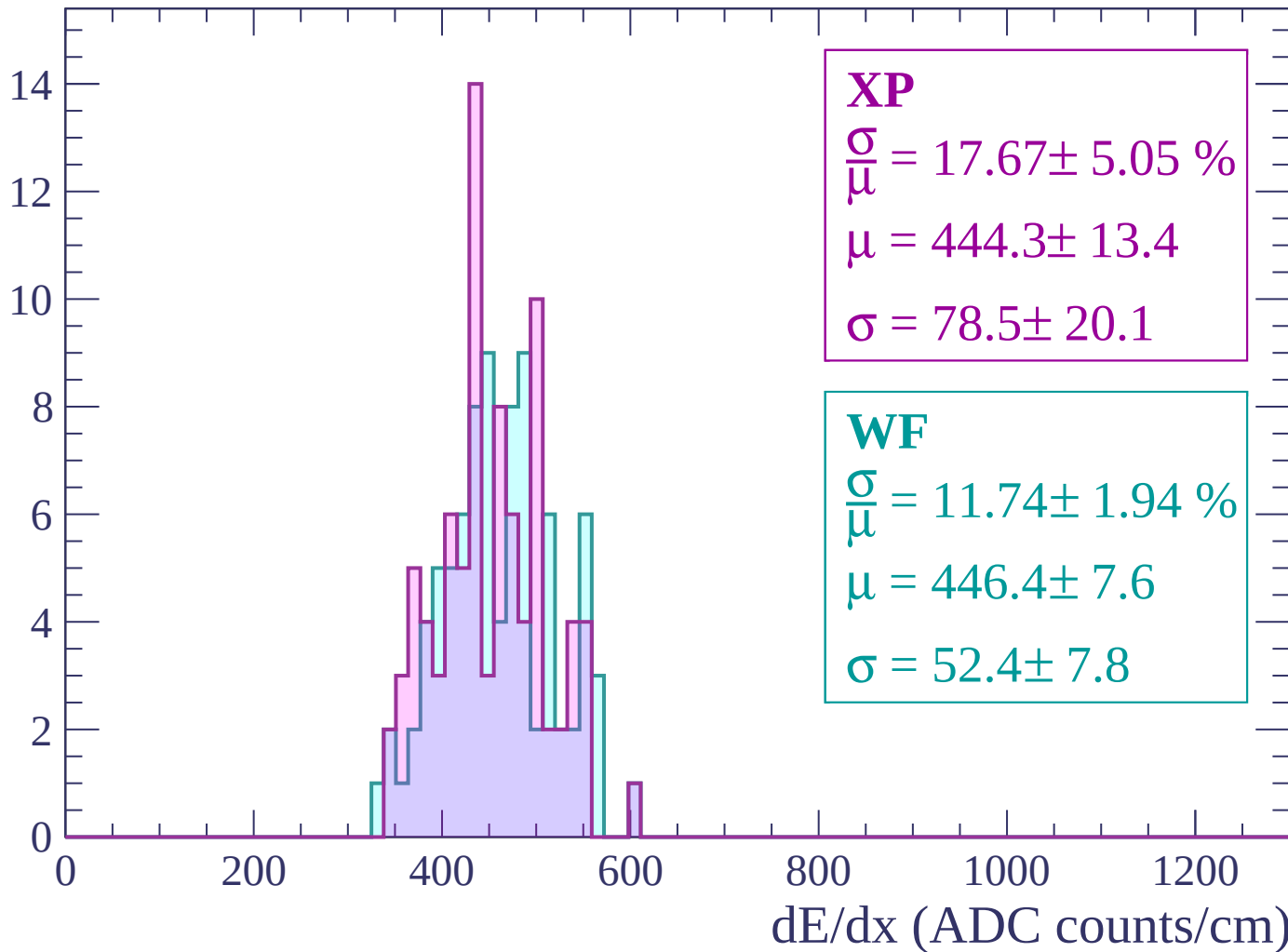
# Energy loss | $-1980 < p < -1920$

Count



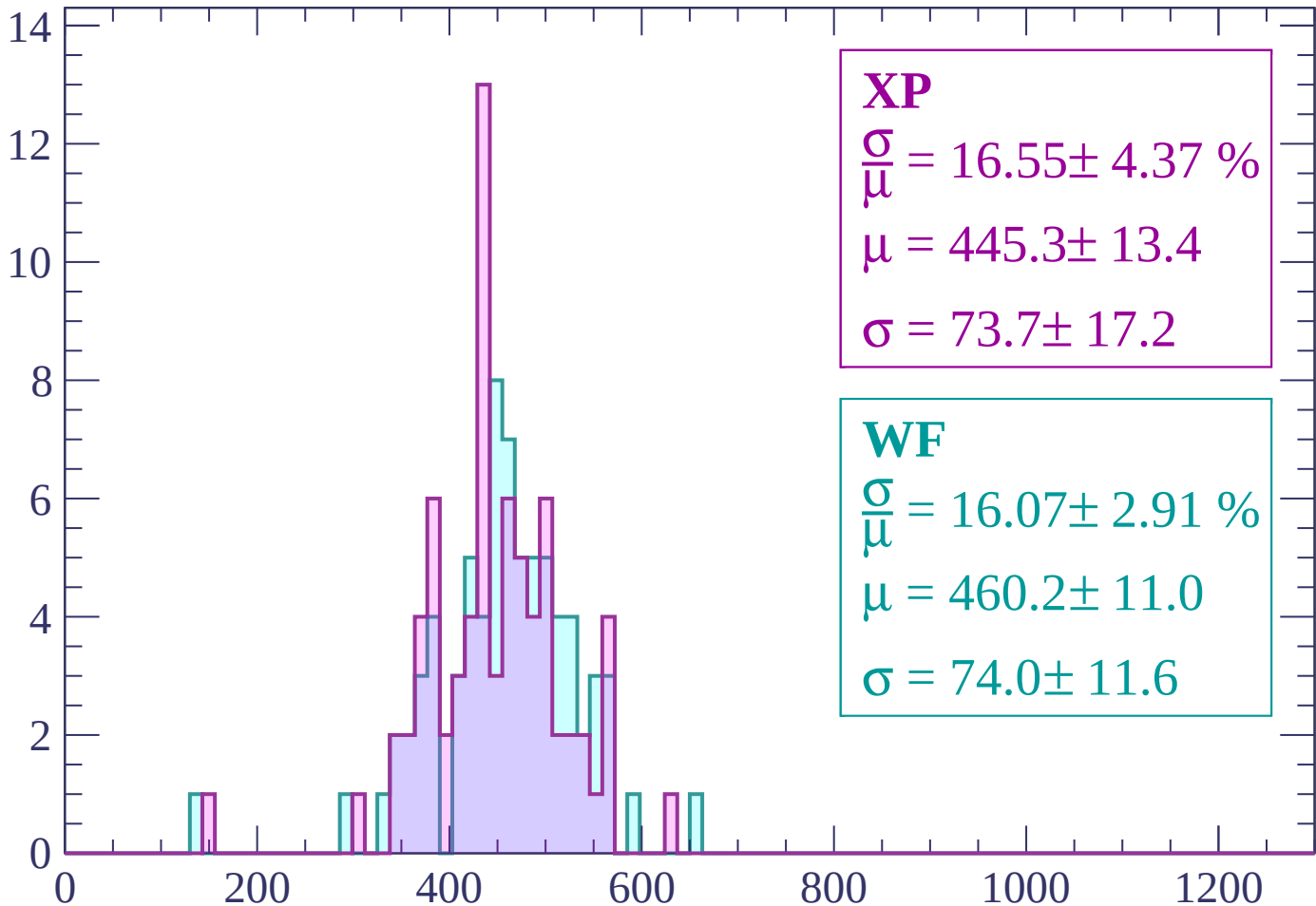
# Energy loss | -1920 < p < -1860

Count



# Energy loss | -1860 < p < -1800

Count



**XP**

$$\frac{\sigma}{\mu} = 16.55 \pm 4.37 \%$$

$$\mu = 445.3 \pm 13.4$$

$$\sigma = 73.7 \pm 17.2$$

**WF**

$$\frac{\sigma}{\mu} = 16.07 \pm 2.91 \%$$

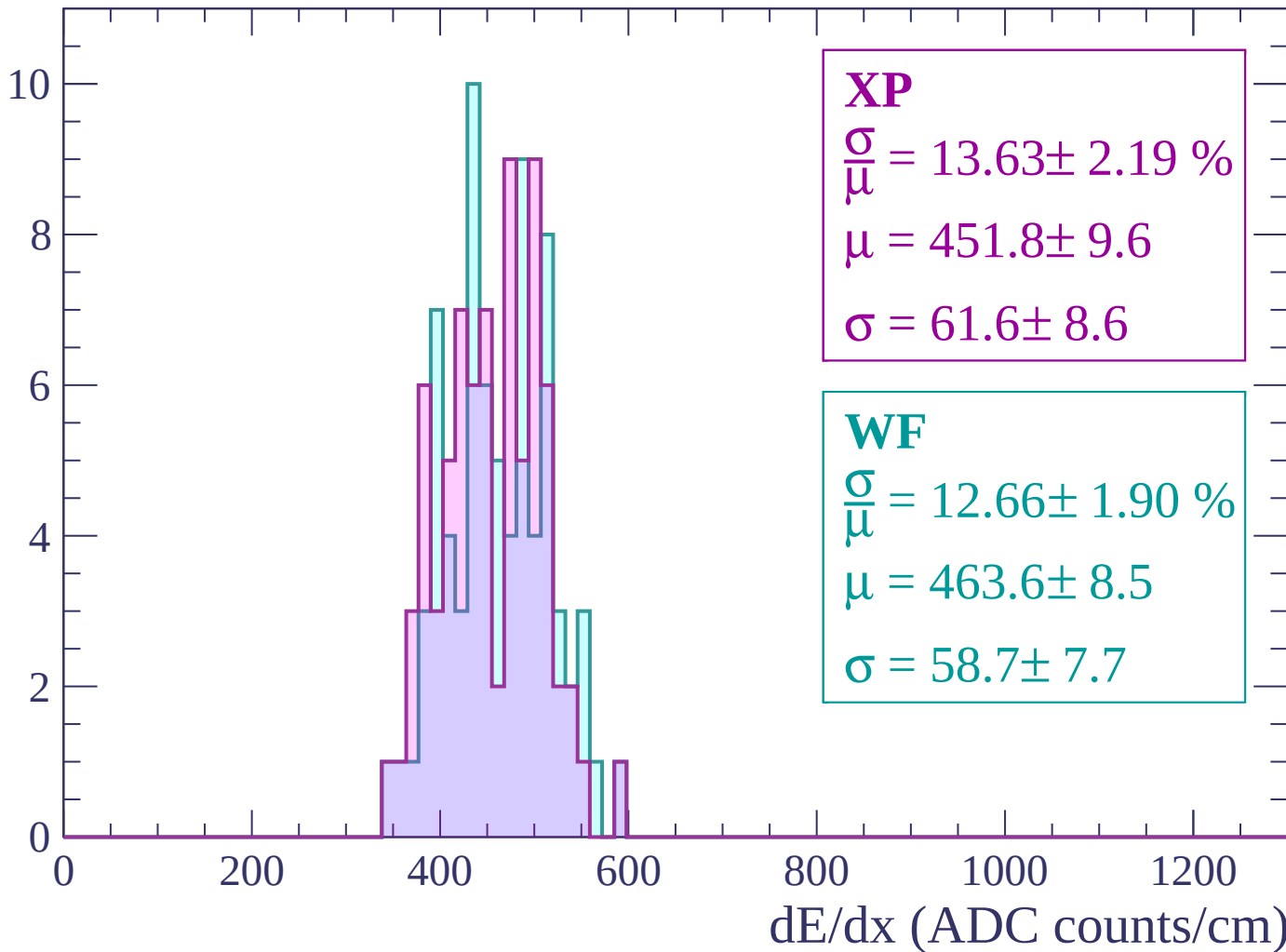
$$\mu = 460.2 \pm 11.0$$

$$\sigma = 74.0 \pm 11.6$$

$dE/dx$  (ADC counts/cm)

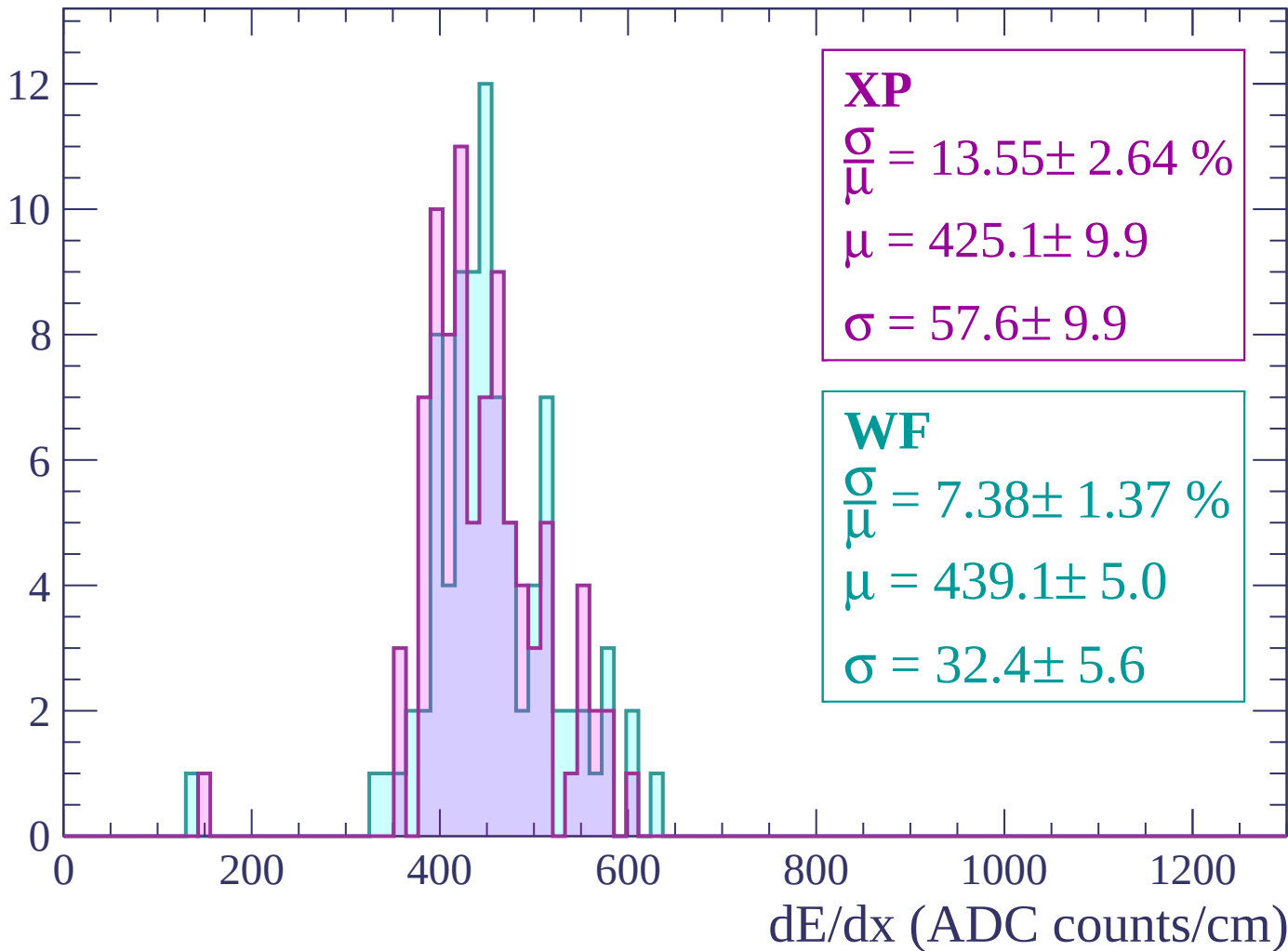
# Energy loss | -1800 < p < -1740

Count



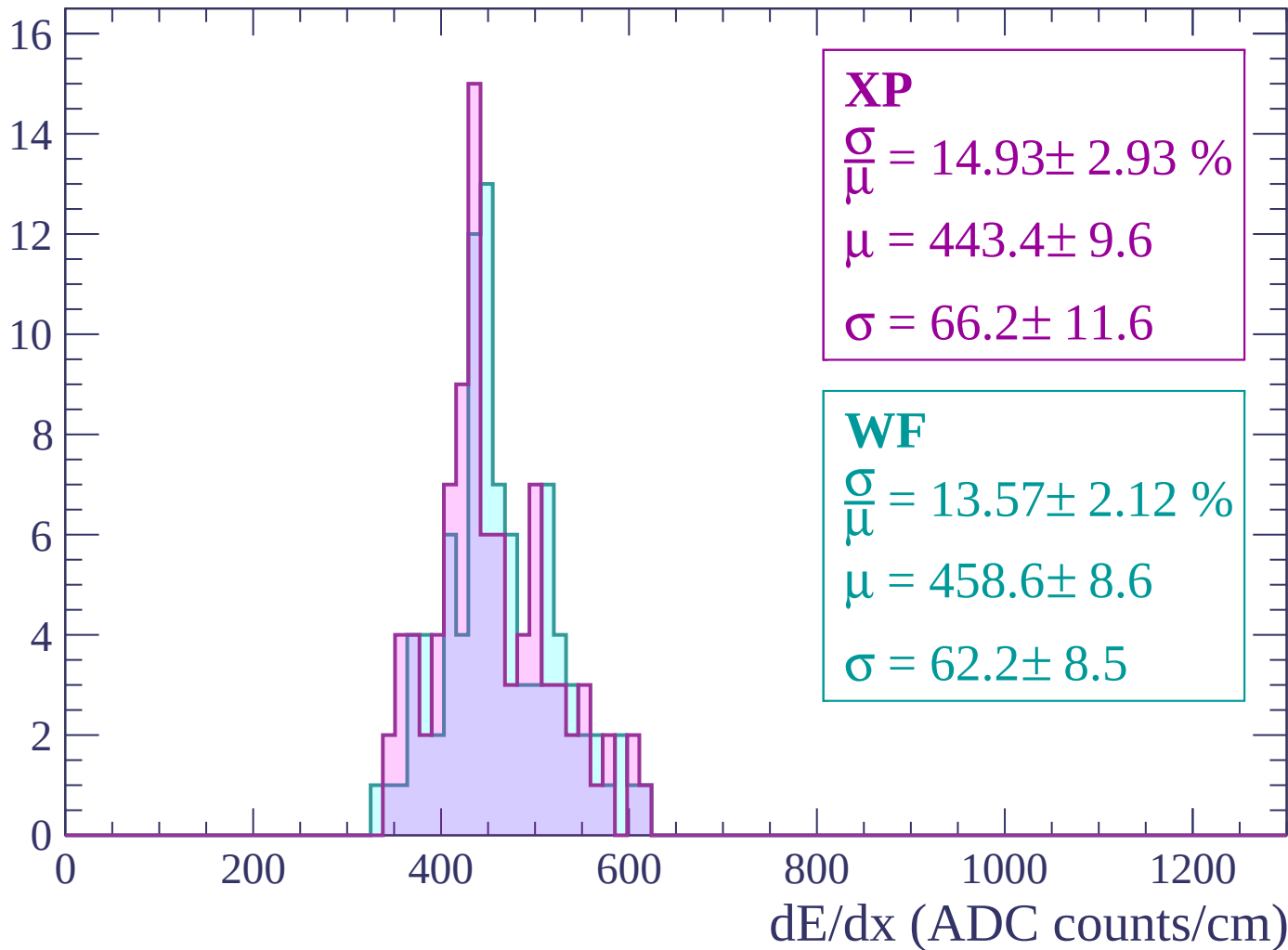
# Energy loss | -1740 < p < -1680

Count



# Energy loss | -1680 < p < -1620

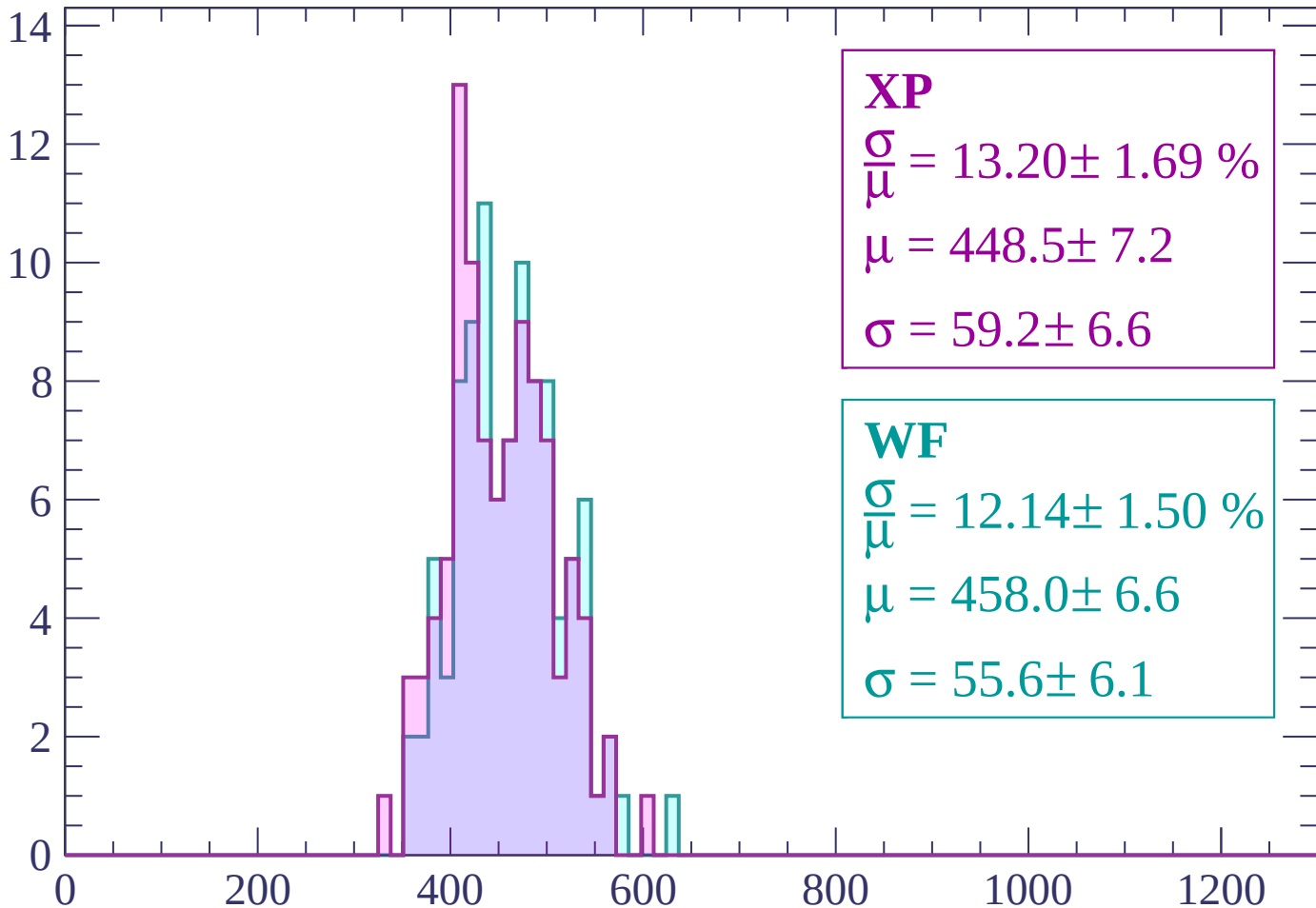
Count





# Energy loss | -1620 < p < -1560

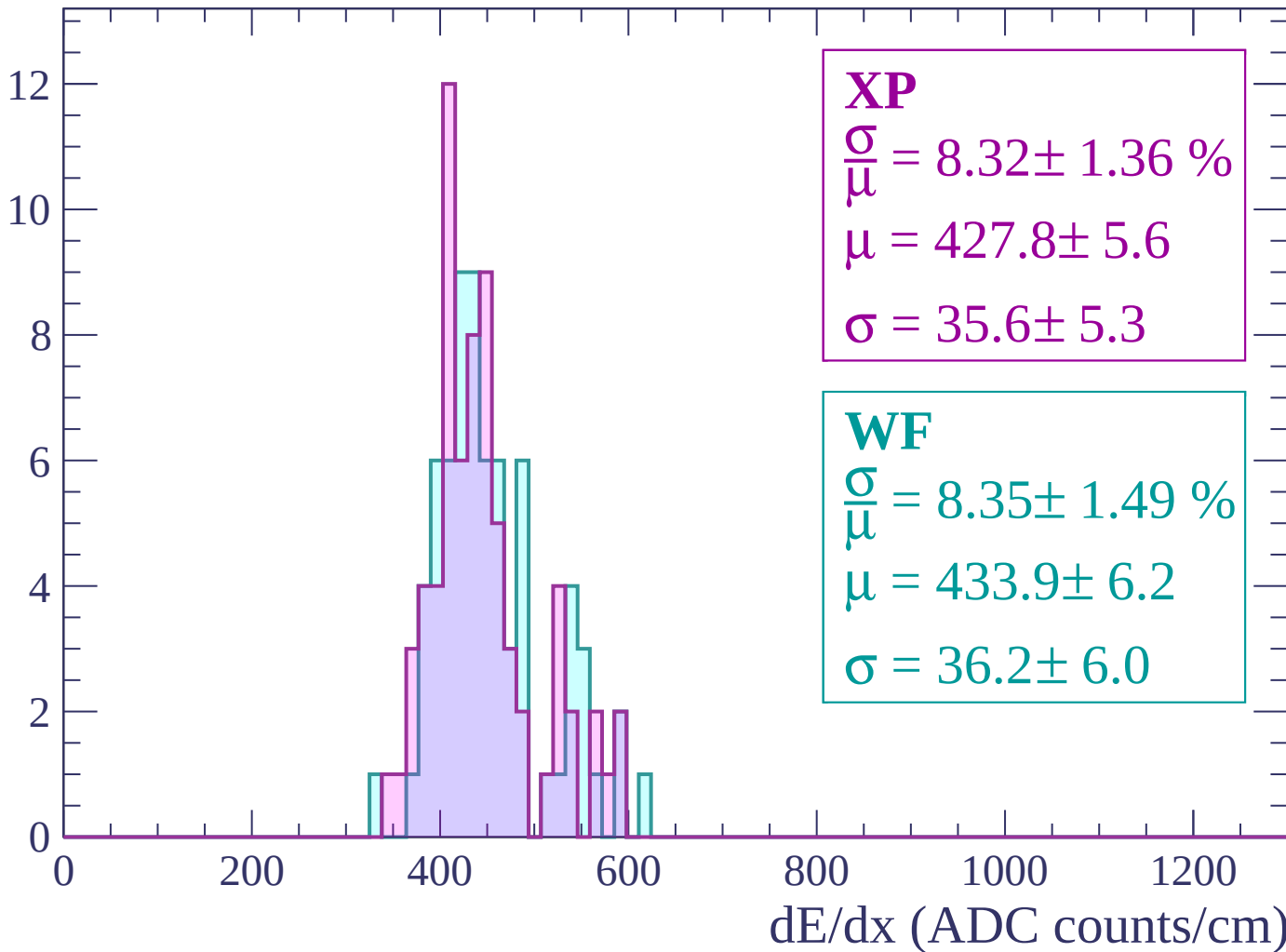
Count



dE/dx (ADC counts/cm)

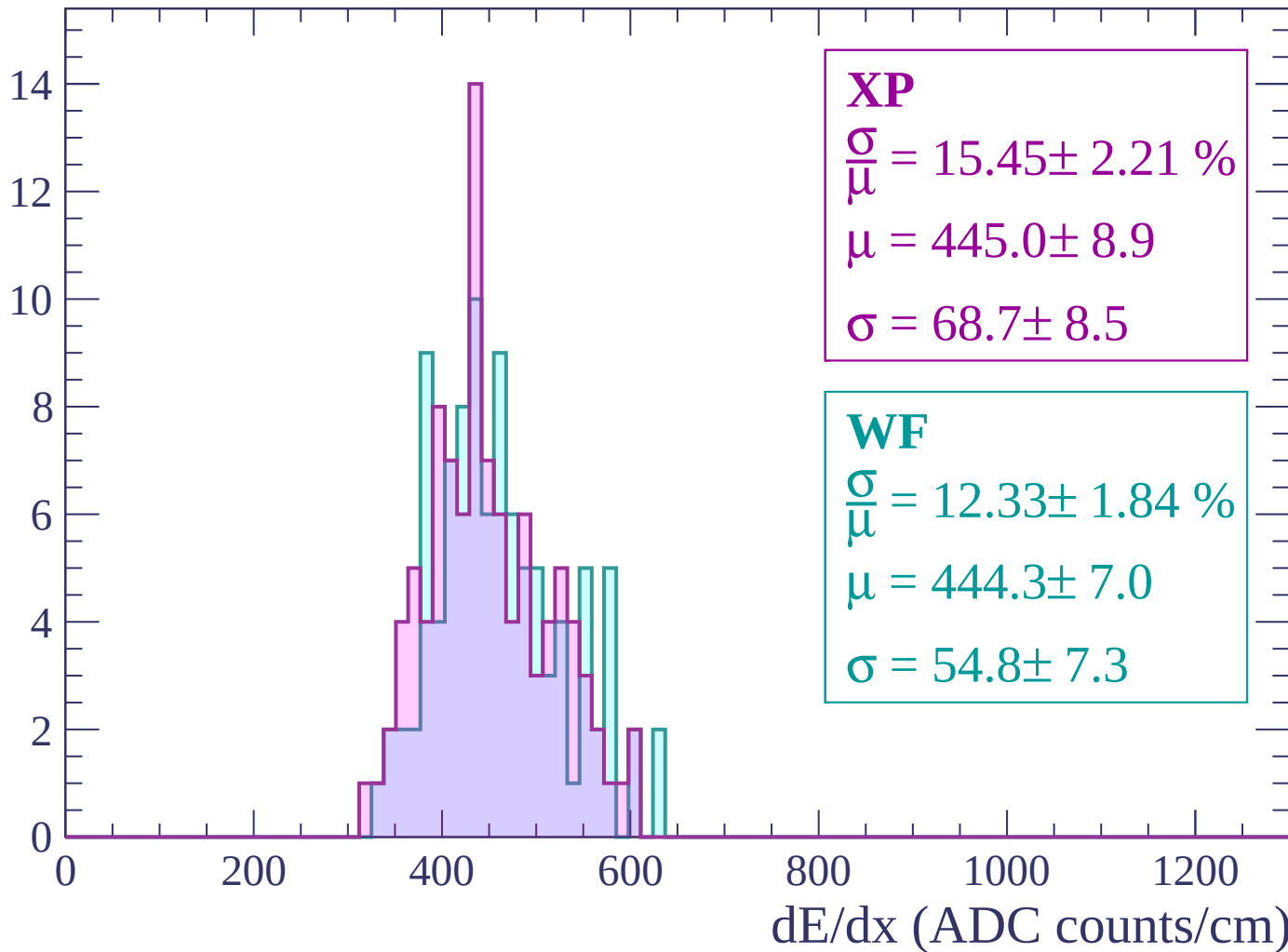
# Energy loss | -1560 < p < -1500

Count



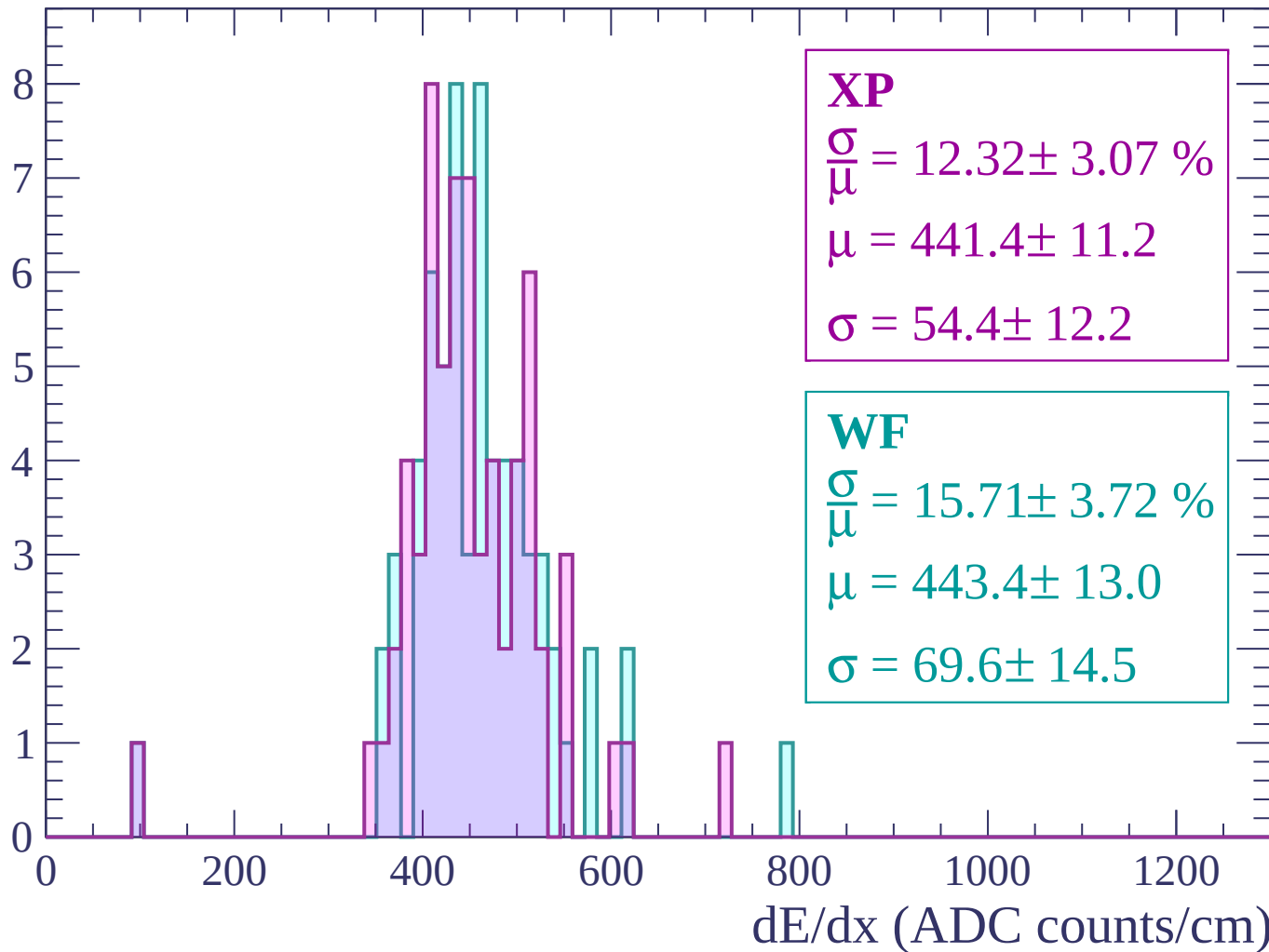
# Energy loss | -1500 < p < -1440

Count



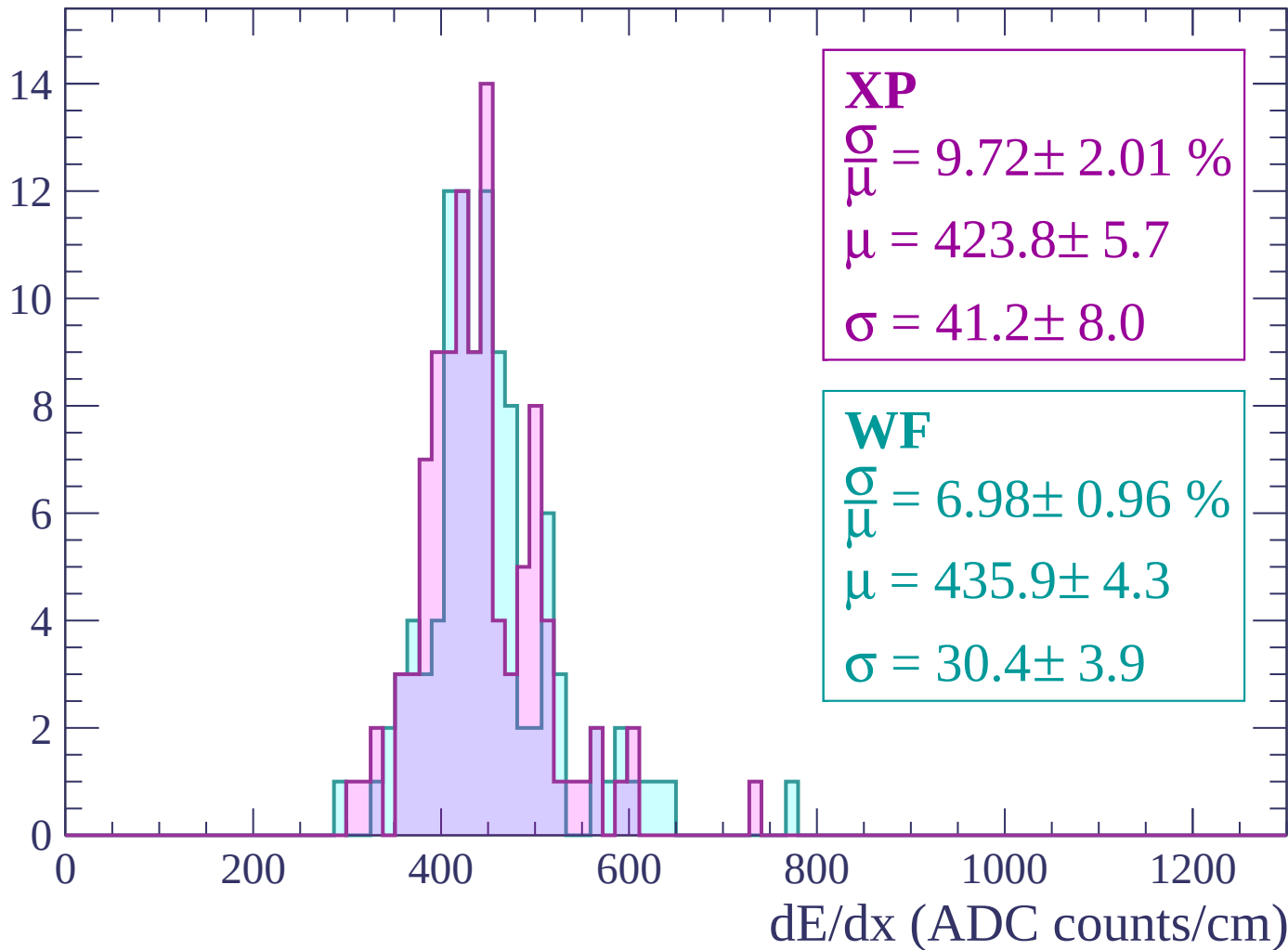
Energy loss |  $-1440 < p < -1380$

Count



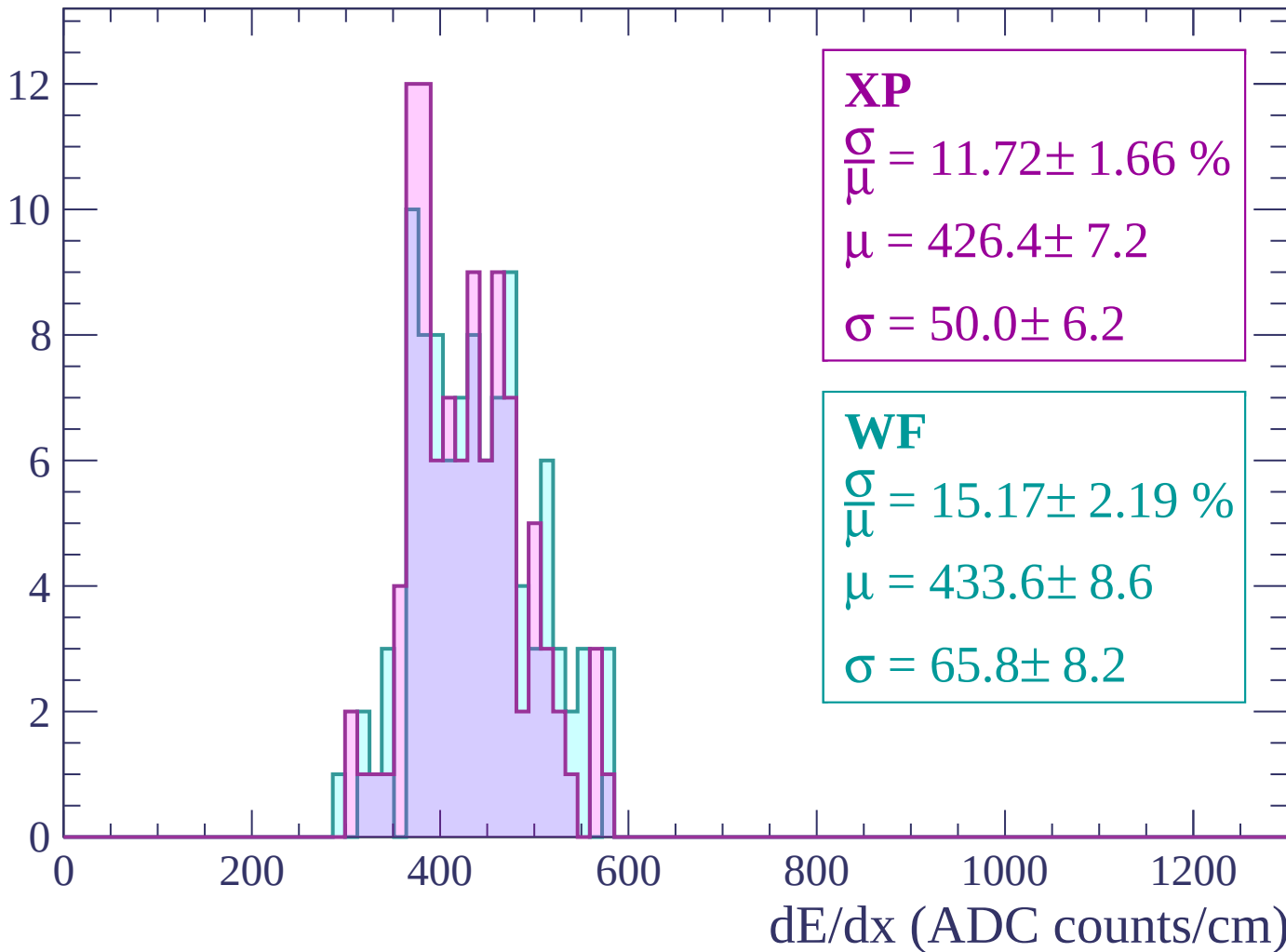
# Energy loss | -1380 < p < -1320

Count



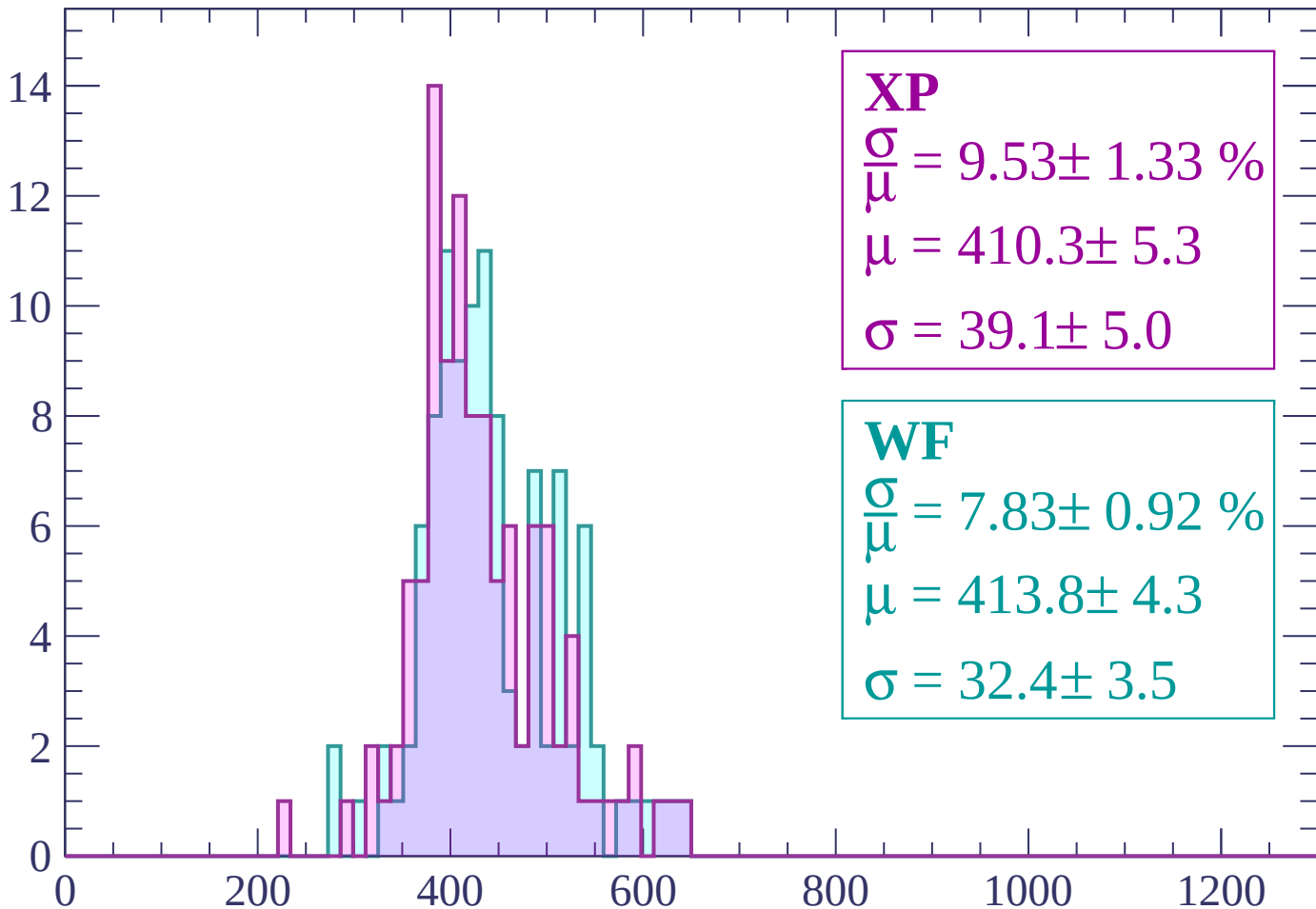
# Energy loss | -1320 < p < -1260

Count



Energy loss |  $-1260 < p < -1200$

Count



**XP**

$$\frac{\sigma}{\bar{\mu}} = 9.53 \pm 1.33 \%$$

$$\mu = 410.3 \pm 5.3$$

$$\sigma = 39.1 \pm 5.0$$

**WF**

$$\frac{\sigma}{\bar{\mu}} = 7.83 \pm 0.92 \%$$

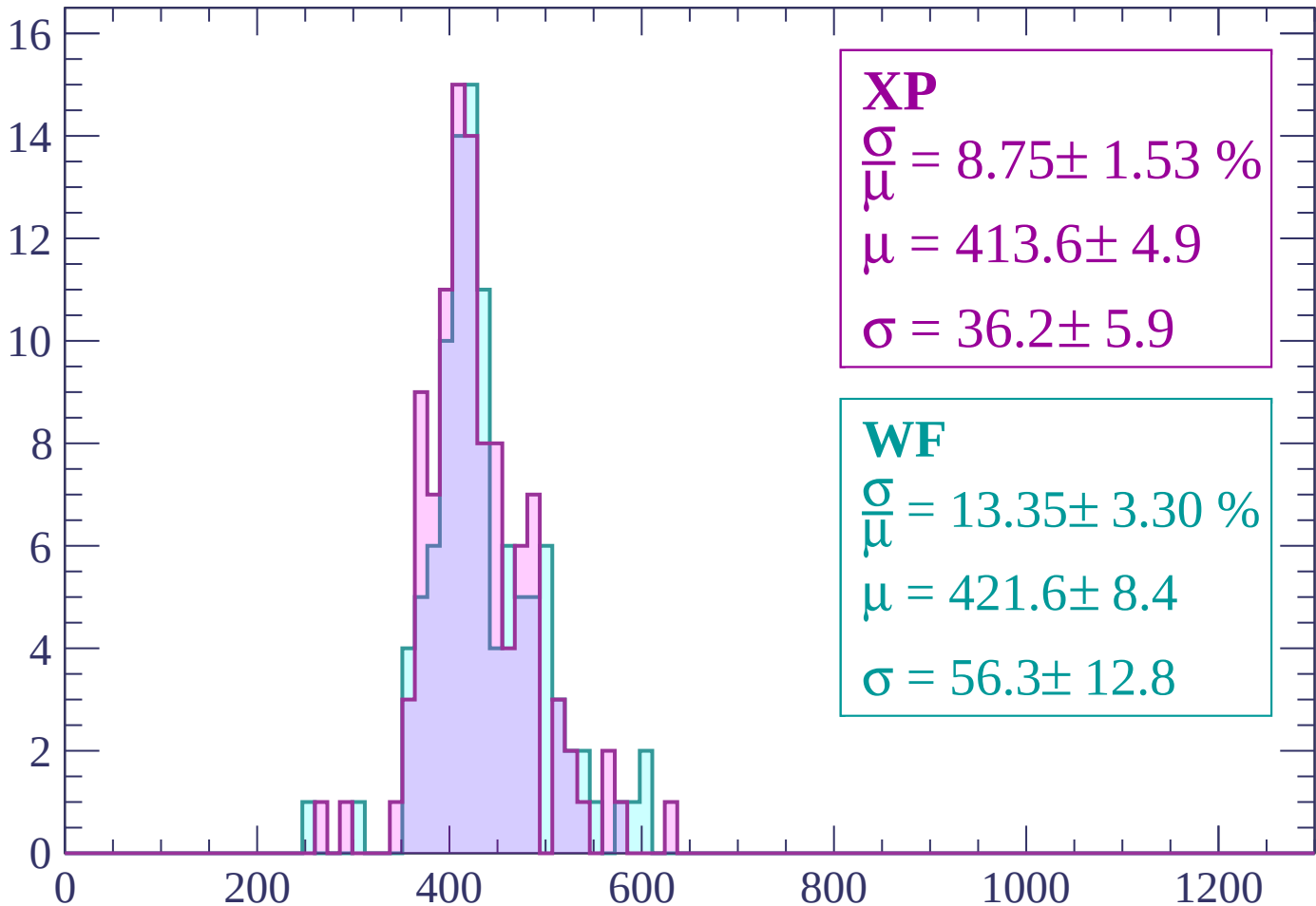
$$\mu = 413.8 \pm 4.3$$

$$\sigma = 32.4 \pm 3.5$$

$dE/dx$  (ADC counts/cm)

# Energy loss | -1200 < p < -1140

Count



**XP**

$$\frac{\sigma}{\mu} = 8.75 \pm 1.53 \%$$

$$\mu = 413.6 \pm 4.9$$

$$\sigma = 36.2 \pm 5.9$$

**WF**

$$\frac{\sigma}{\mu} = 13.35 \pm 3.30 \%$$

$$\mu = 421.6 \pm 8.4$$

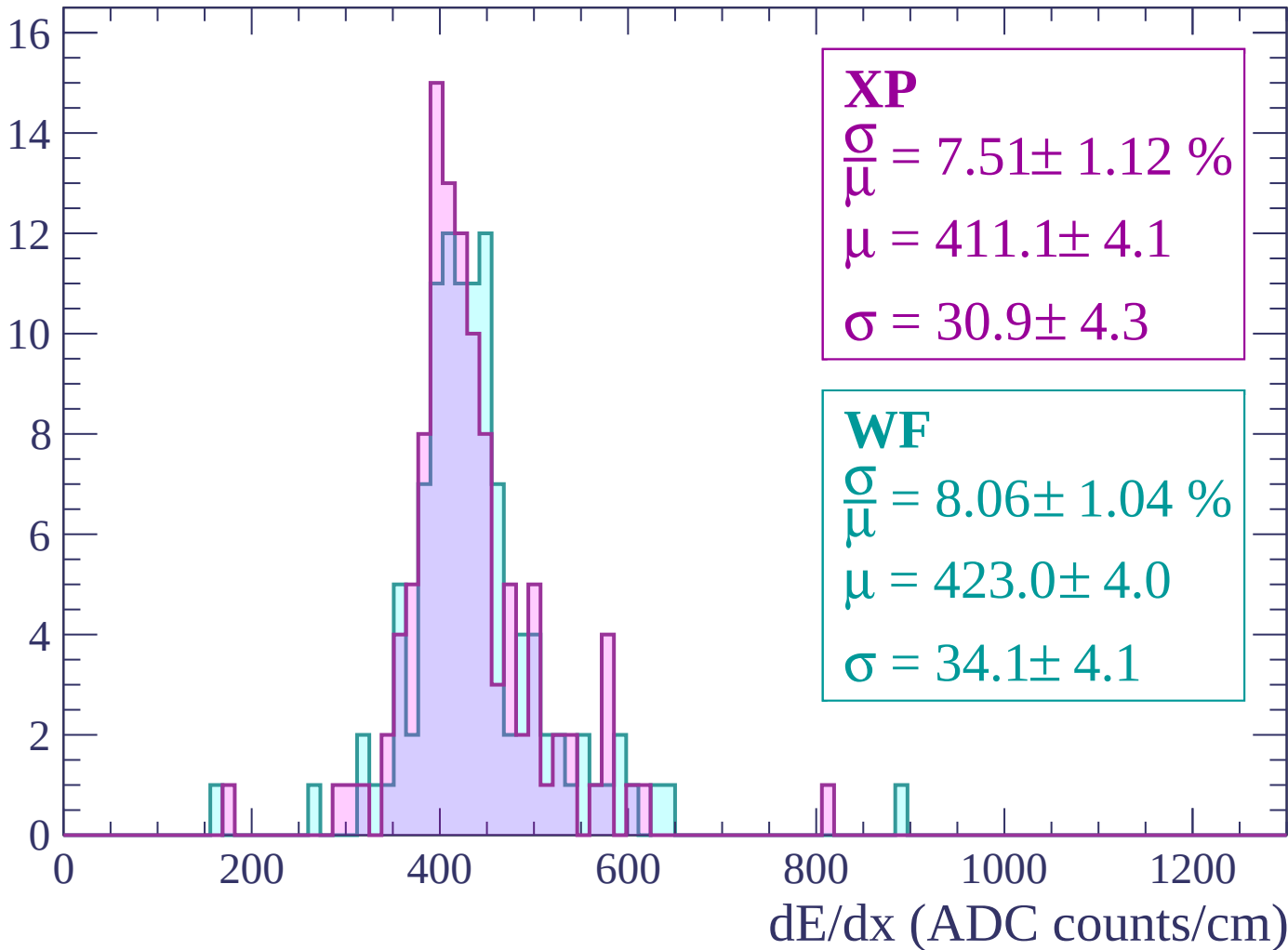
$$\sigma = 56.3 \pm 12.8$$

$dE/dx$  (ADC counts/cm)



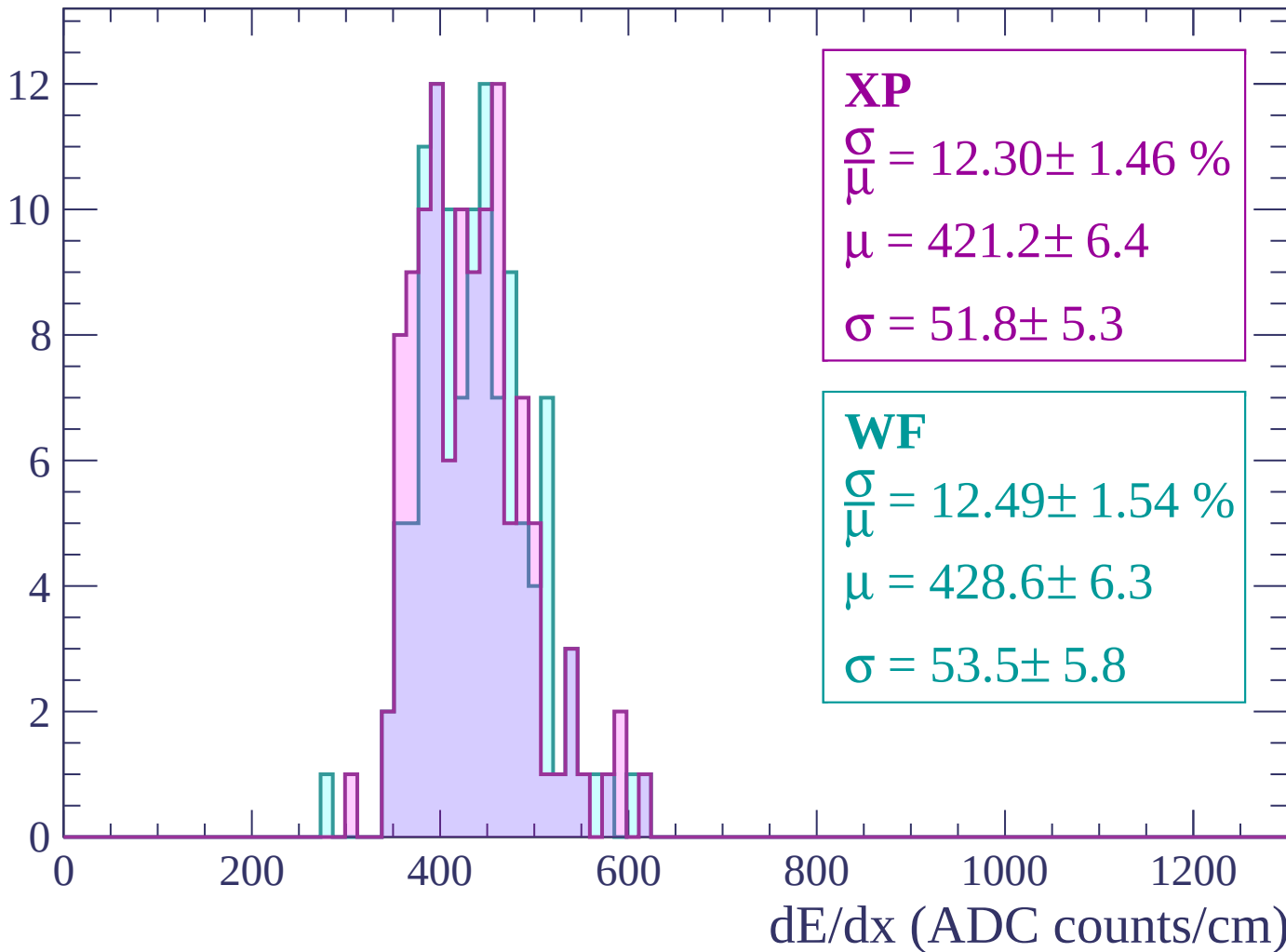
# Energy loss | -1140 < p < -1080

Count



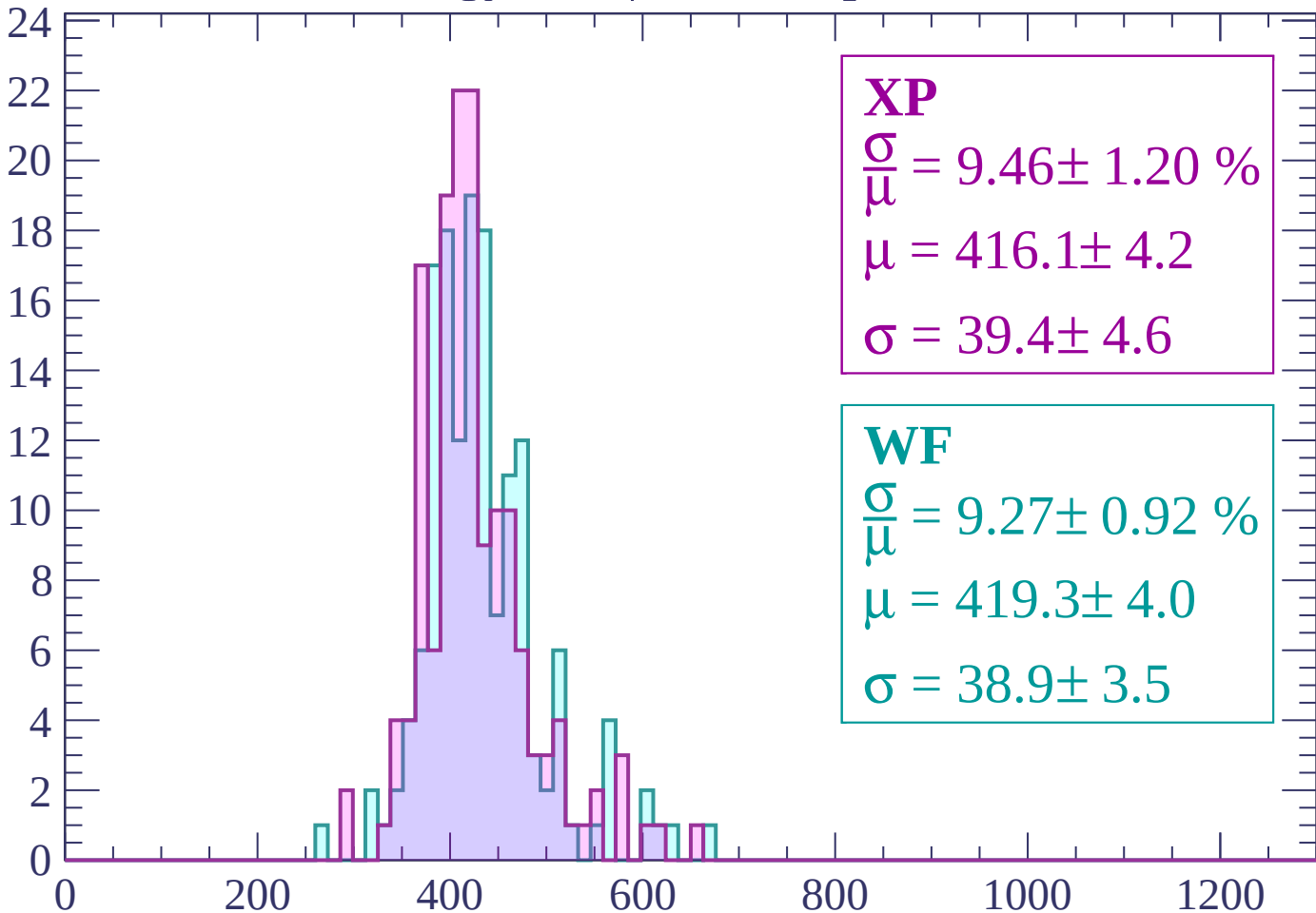
# Energy loss | -1080 < p < -1020

Count



Energy loss |  $-1020 < p < -960$

Count



**XP**

$$\frac{\sigma}{\bar{\mu}} = 9.46 \pm 1.20 \%$$

$$\mu = 416.1 \pm 4.2$$

$$\sigma = 39.4 \pm 4.6$$

**WF**

$$\frac{\sigma}{\bar{\mu}} = 9.27 \pm 0.92 \%$$

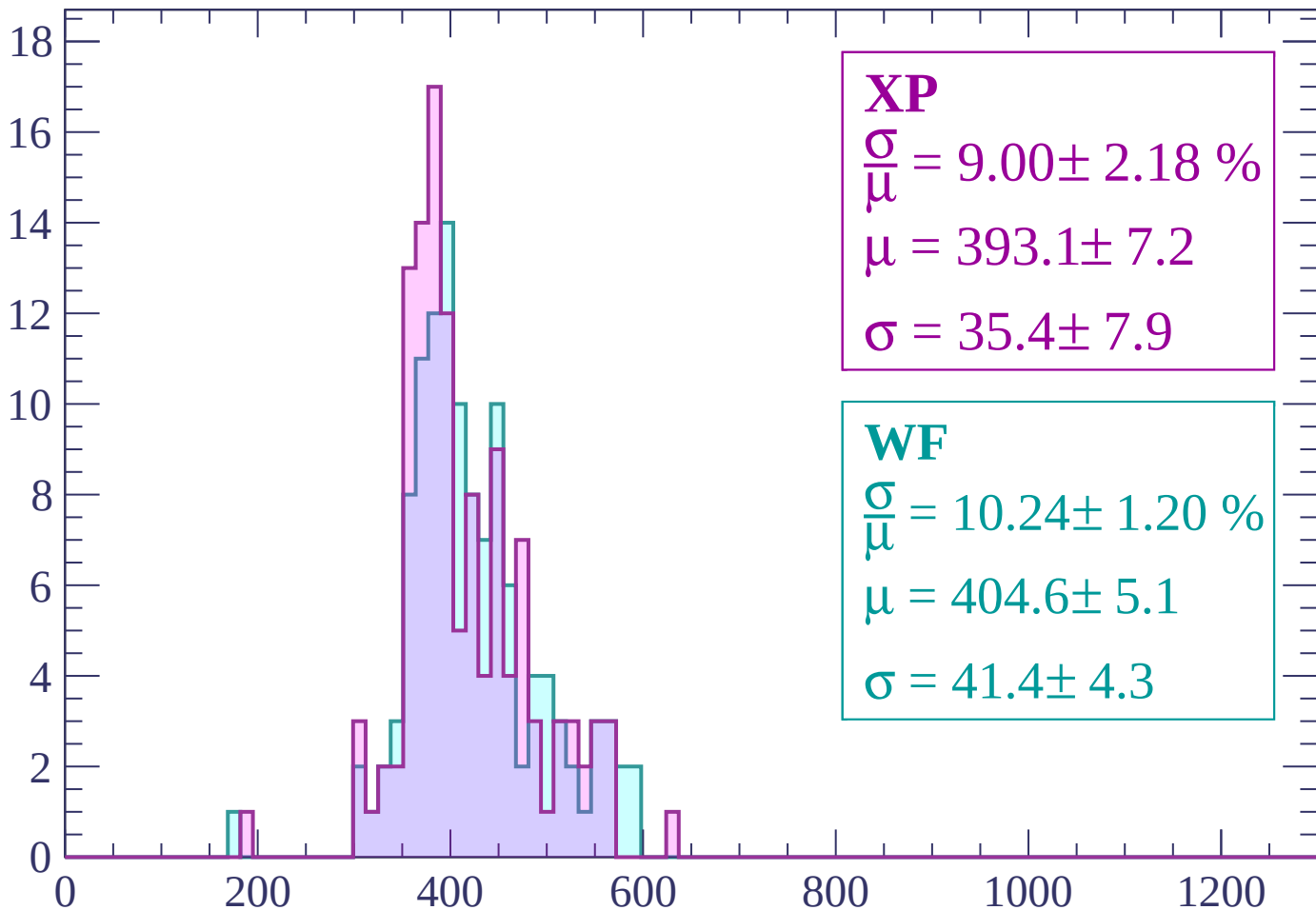
$$\mu = 419.3 \pm 4.0$$

$$\sigma = 38.9 \pm 3.5$$

dE/dx (ADC counts/cm)

Energy loss |  $-960 < p < -900$

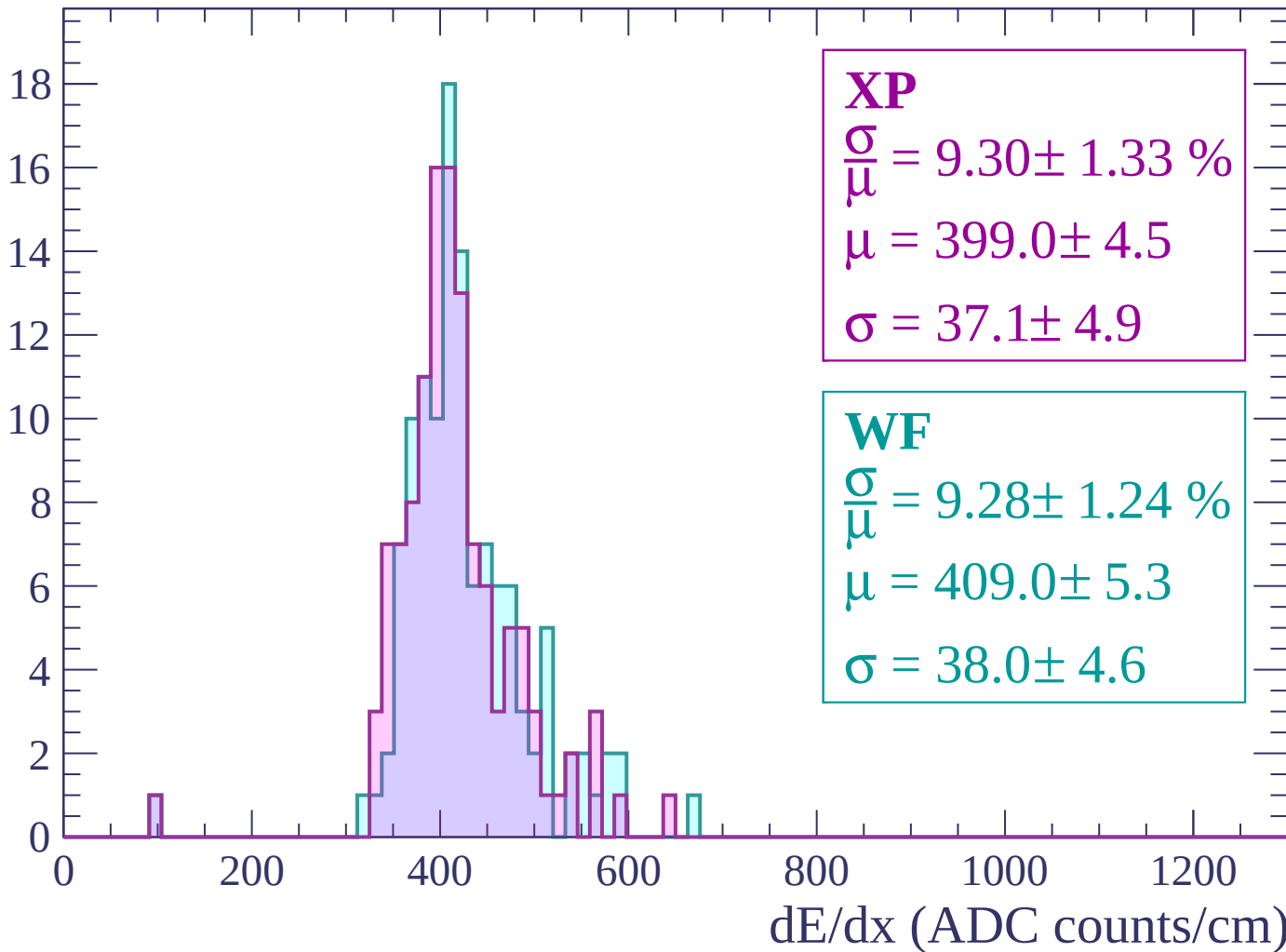
Count



dE/dx (ADC counts/cm)

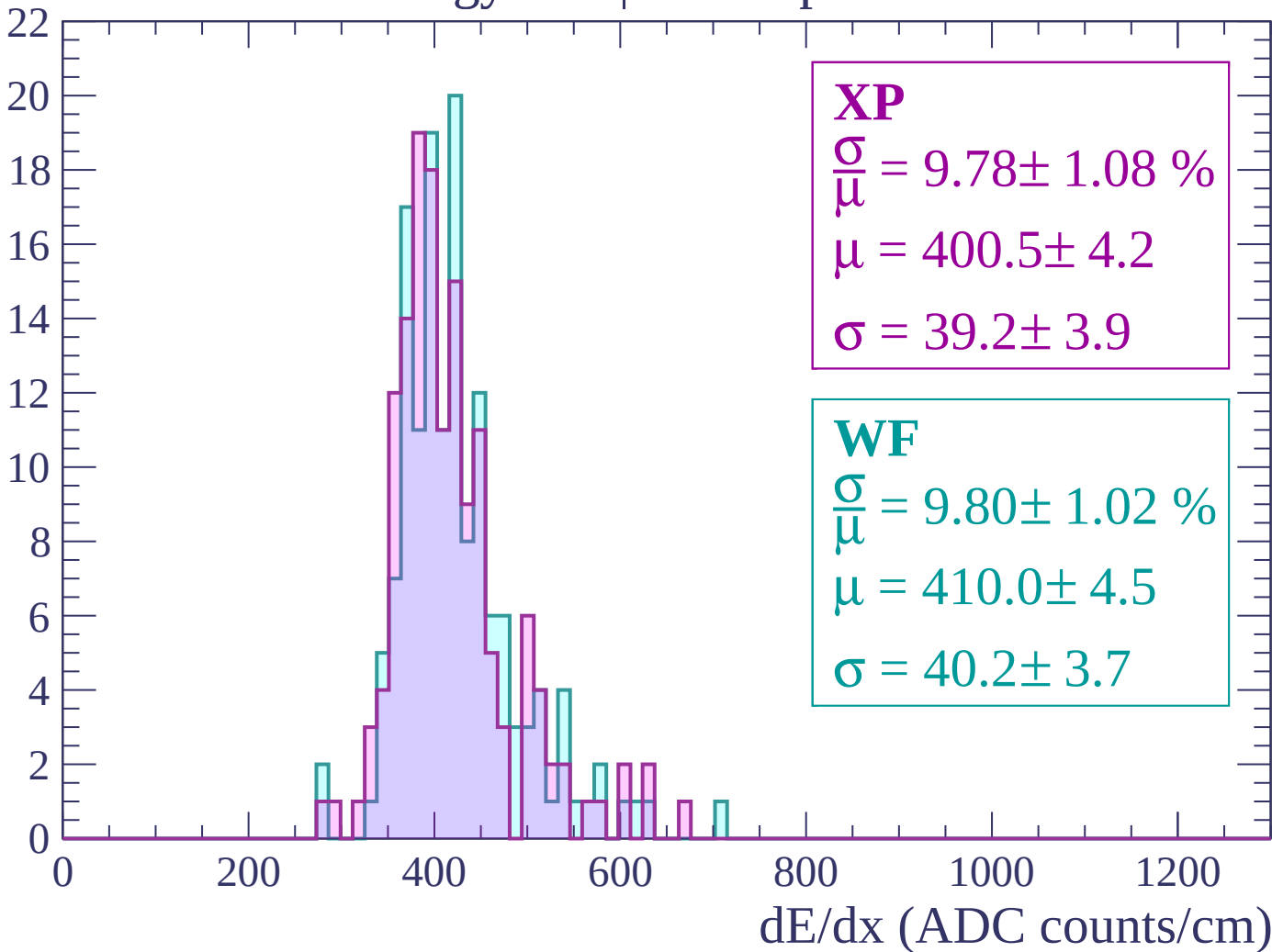
Energy loss |  $-900 < p < -840$

Count



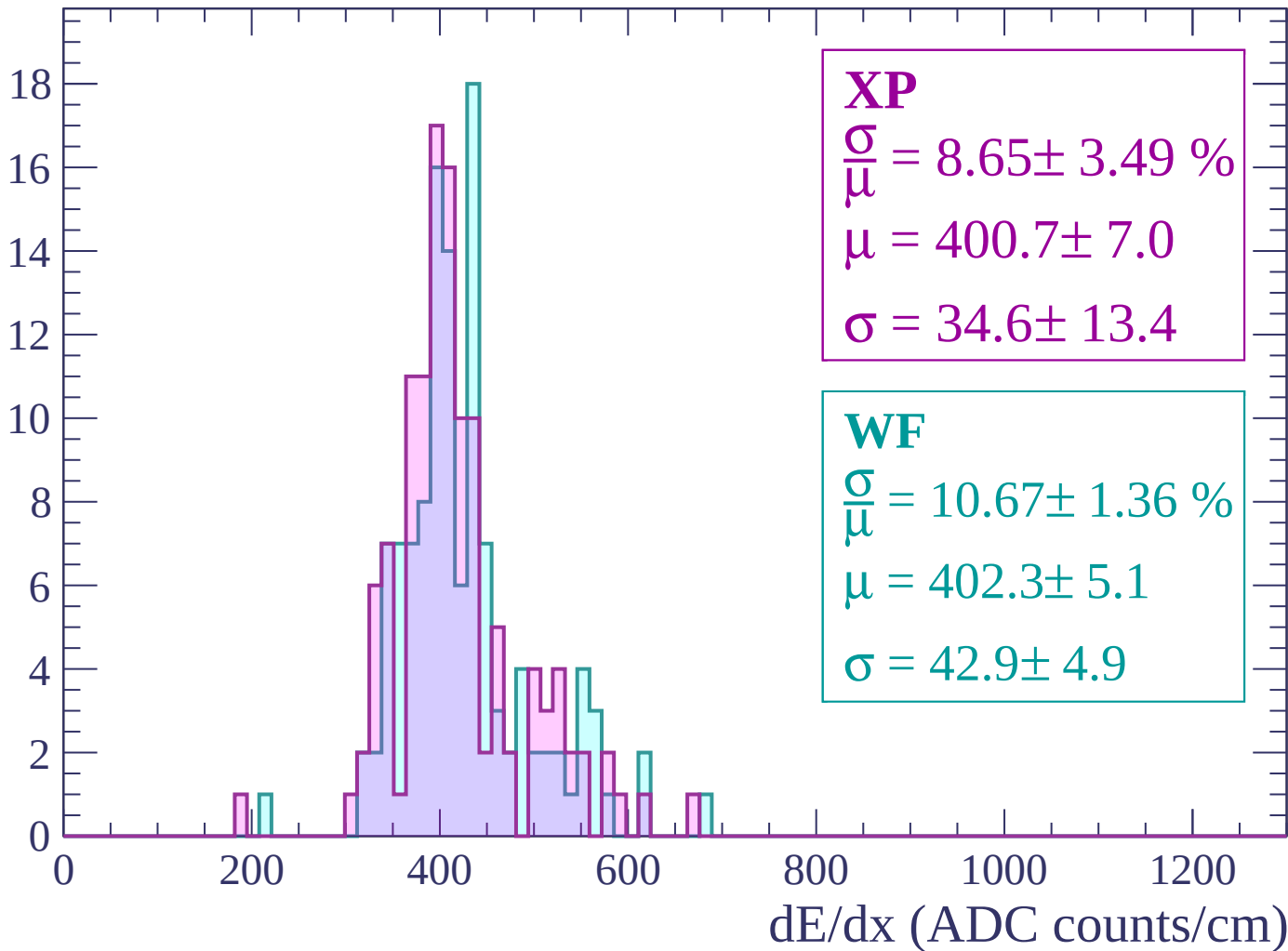
Energy loss |  $-840 < p < -780$

Count



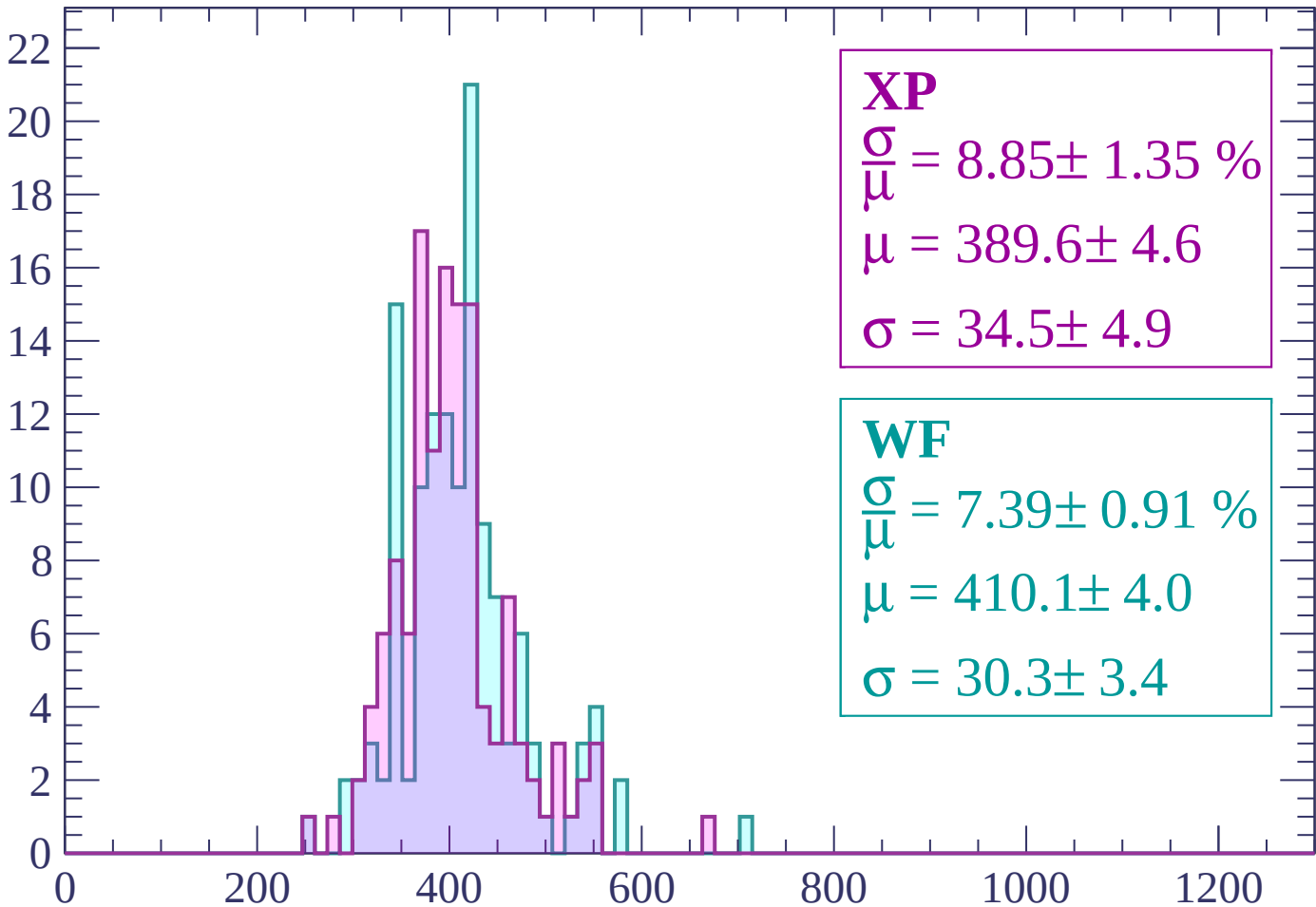
Energy loss |  $-780 < p < -720$

Count



Energy loss |  $-720 < p < -660$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.85 \pm 1.35 \%$$

$$\mu = 389.6 \pm 4.6$$

$$\sigma = 34.5 \pm 4.9$$

**WF**

$$\frac{\sigma}{\mu} = 7.39 \pm 0.91 \%$$

$$\mu = 410.1 \pm 4.0$$

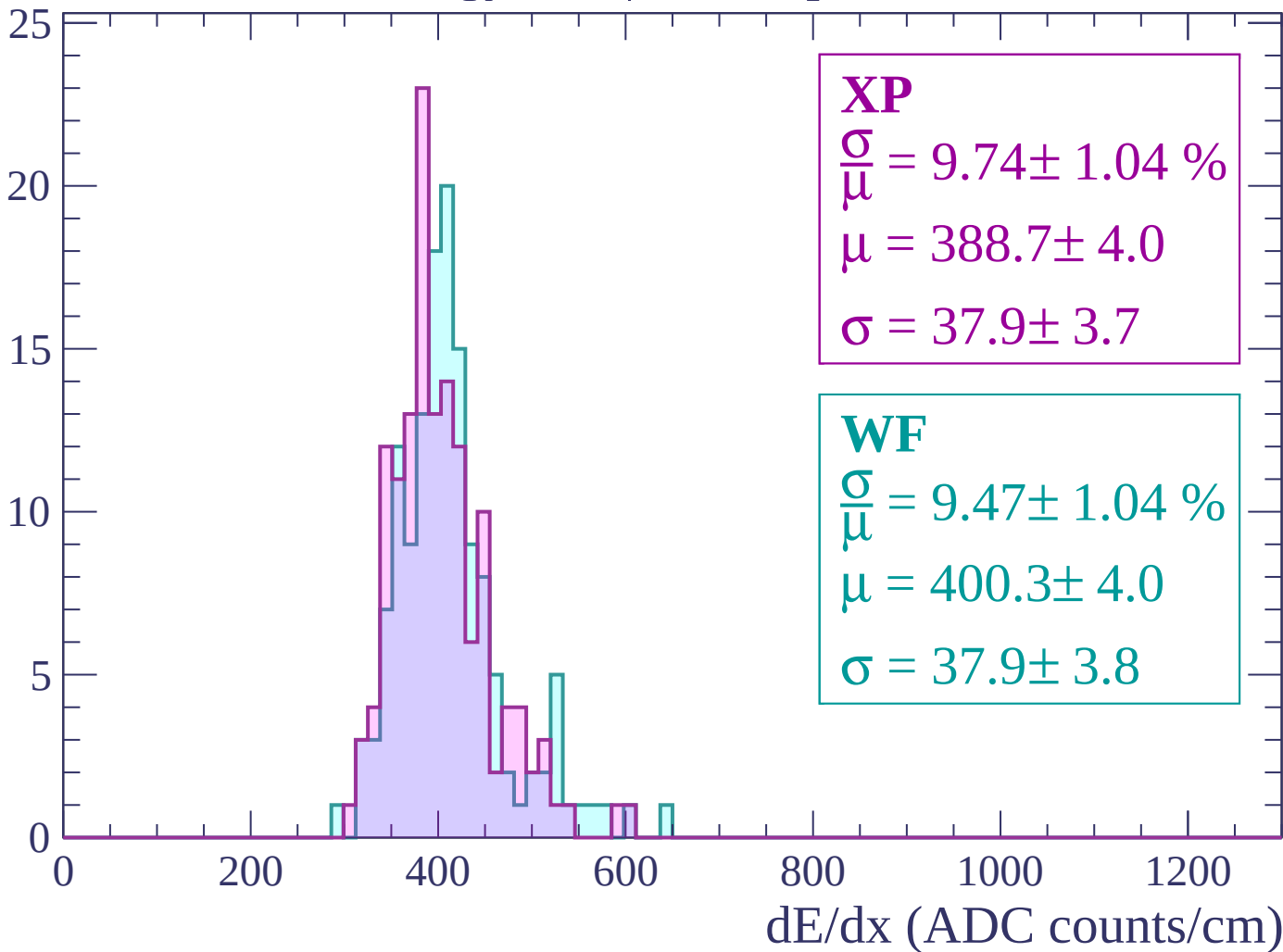
$$\sigma = 30.3 \pm 3.4$$

$dE/dx$  (ADC counts/cm)



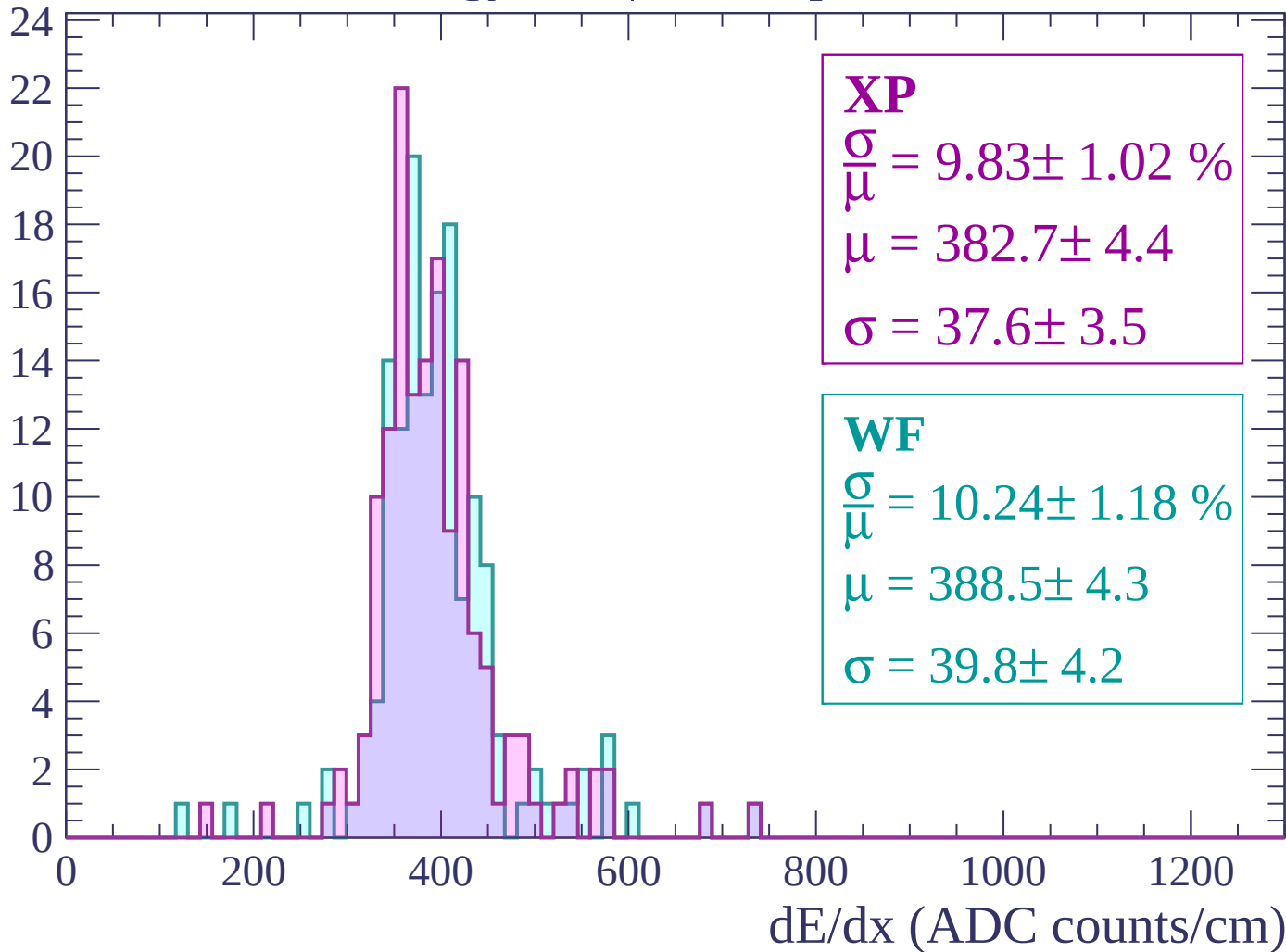
Energy loss |  $-660 < p < -600$

Count



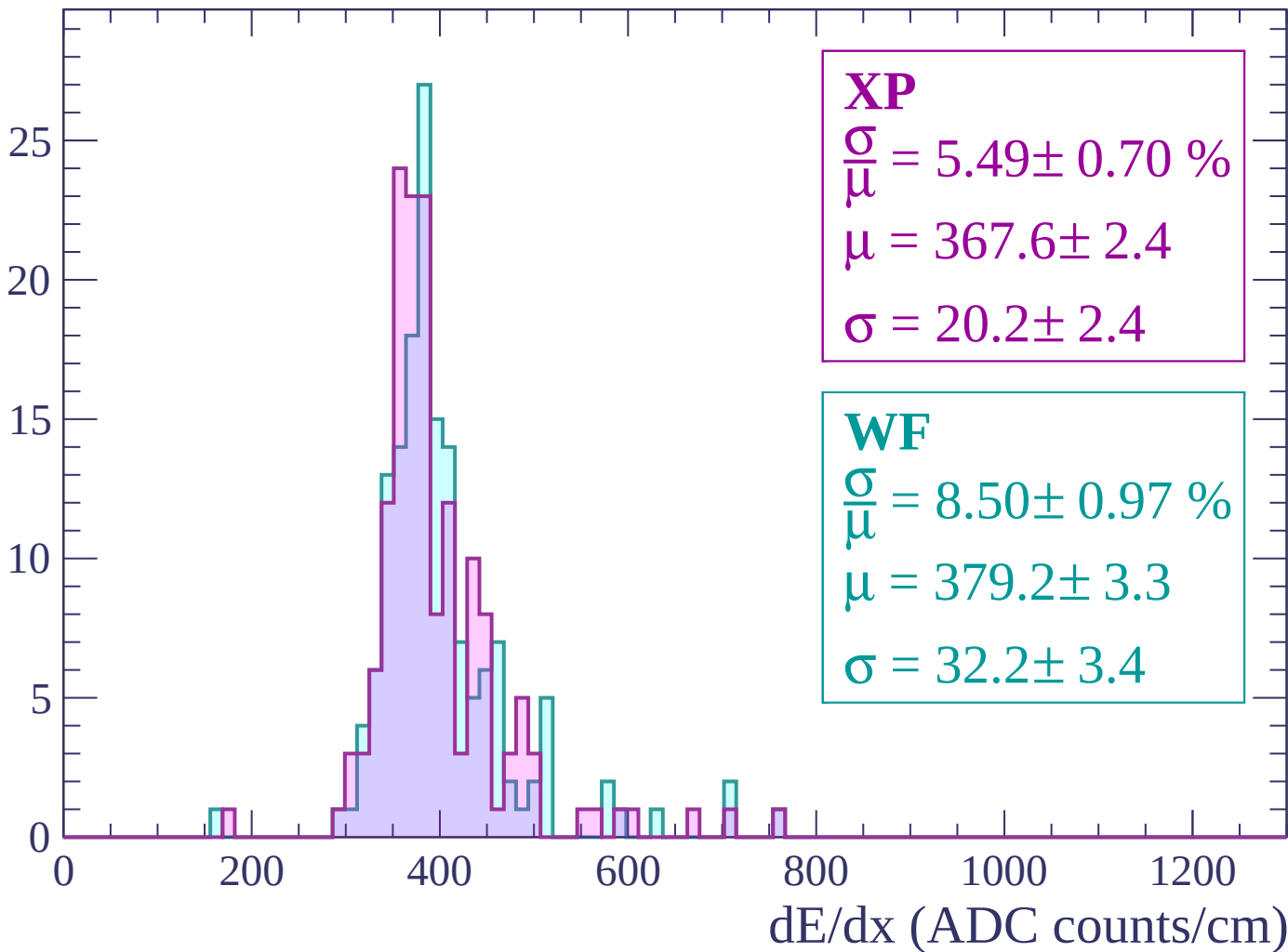
# Energy loss | $-600 < p < -540$

Count



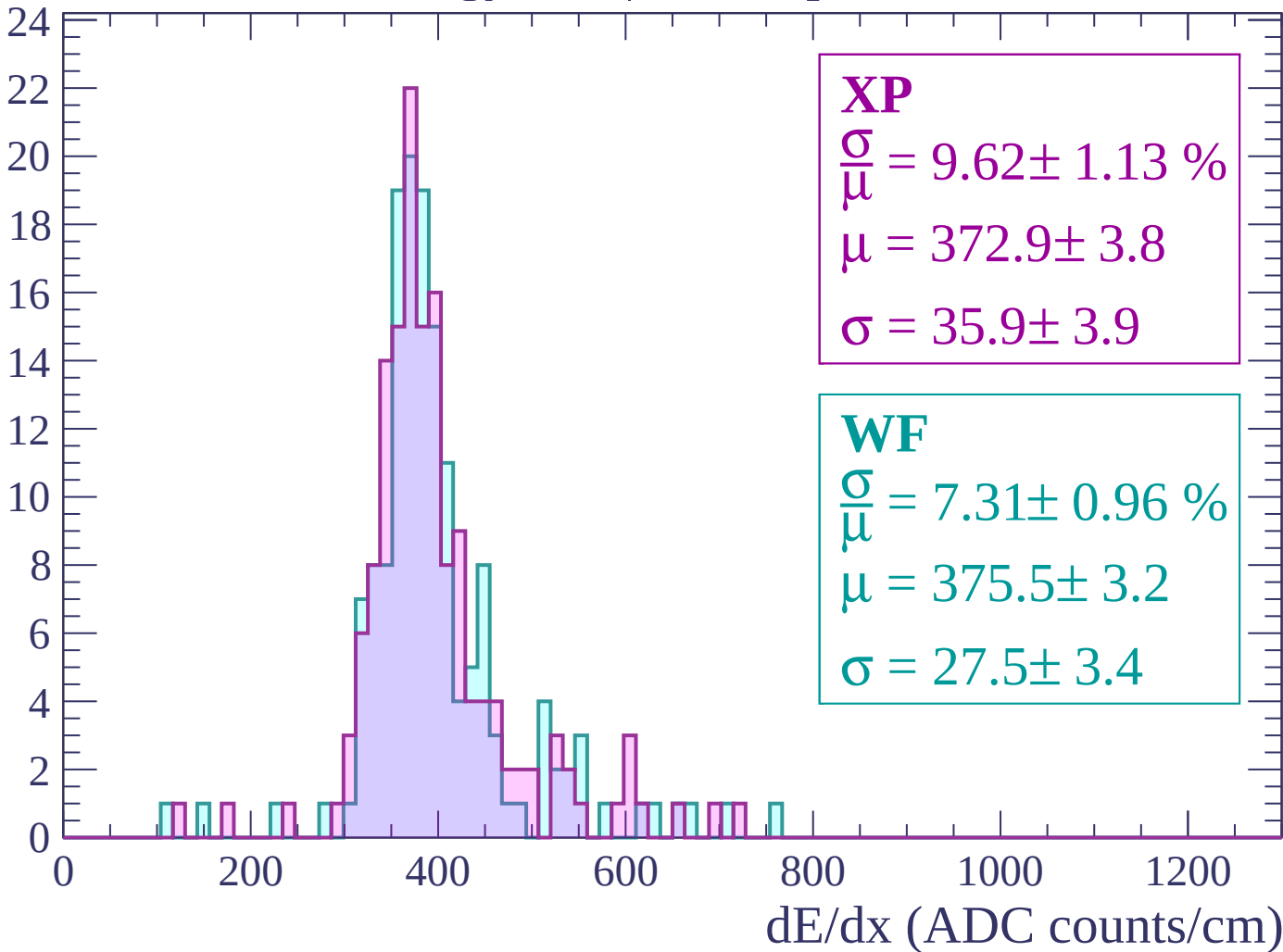
Energy loss |  $-540 < p < -480$

Count



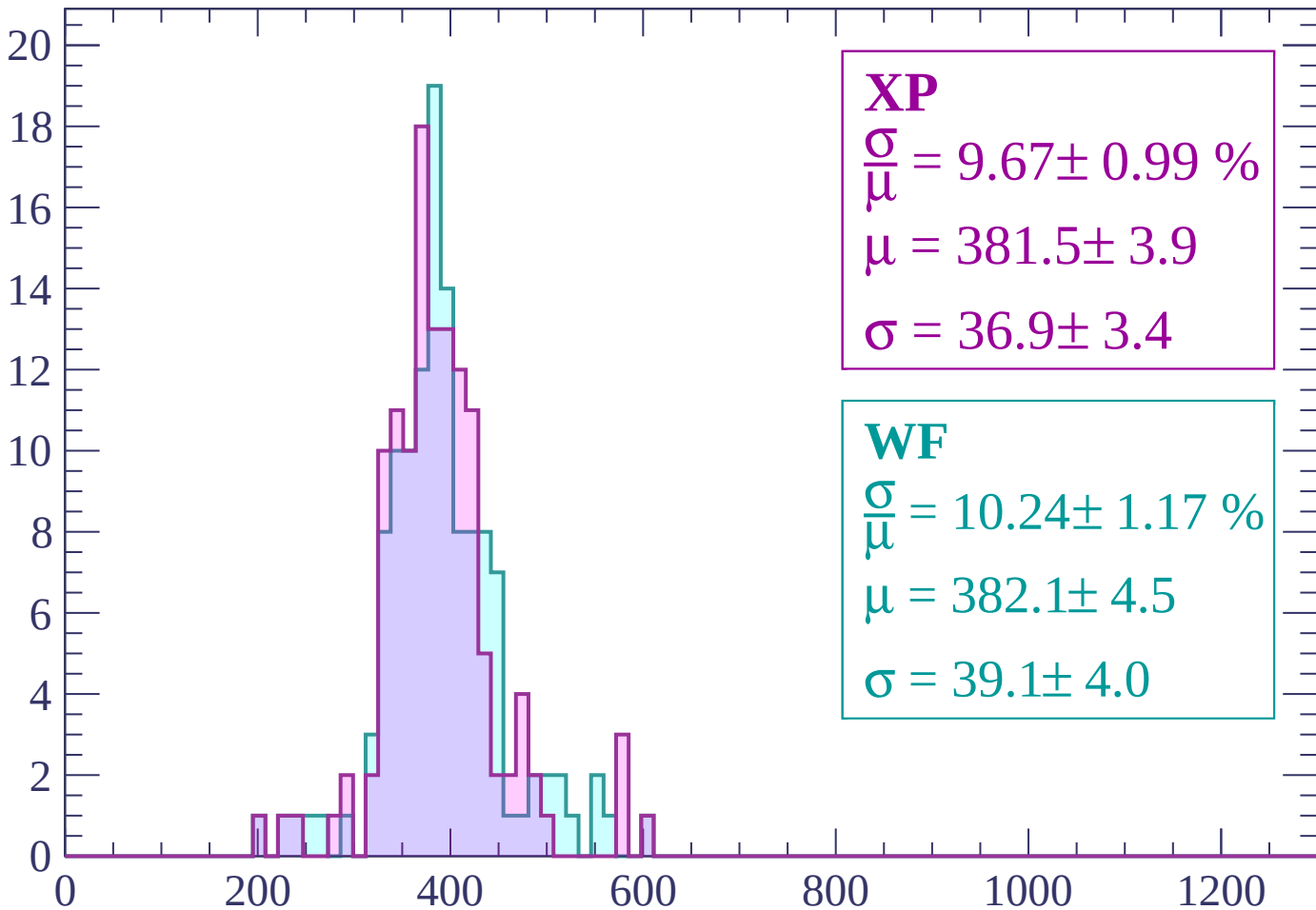
Energy loss |  $-480 < p < -420$

Count



Energy loss |  $-420 < p < -360$

Count



**XP**

$$\frac{\sigma}{\mu} = 9.67 \pm 0.99 \%$$

$$\mu = 381.5 \pm 3.9$$

$$\sigma = 36.9 \pm 3.4$$

**WF**

$$\frac{\sigma}{\mu} = 10.24 \pm 1.17 \%$$

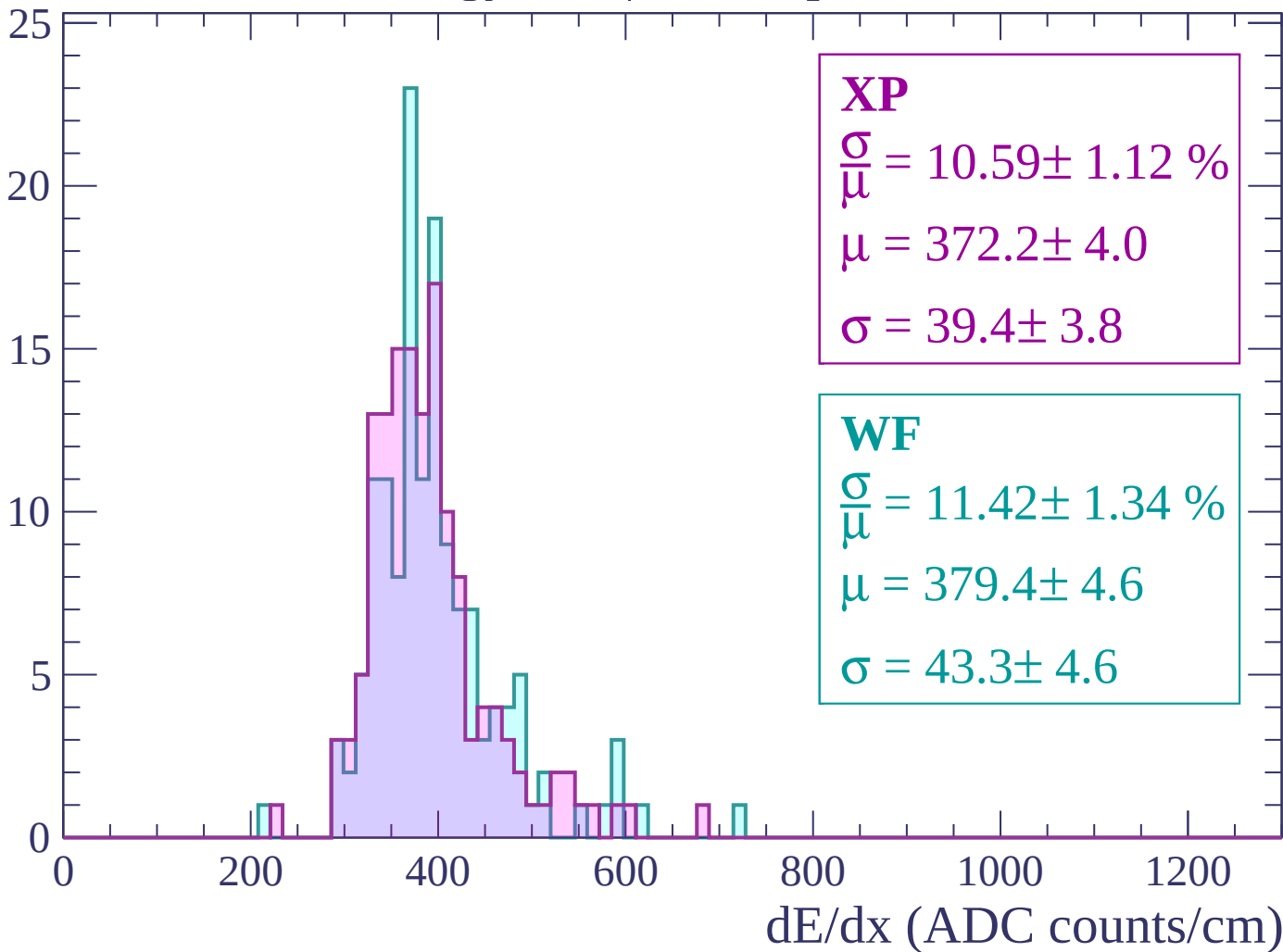
$$\mu = 382.1 \pm 4.5$$

$$\sigma = 39.1 \pm 4.0$$

$dE/dx$  (ADC counts/cm)

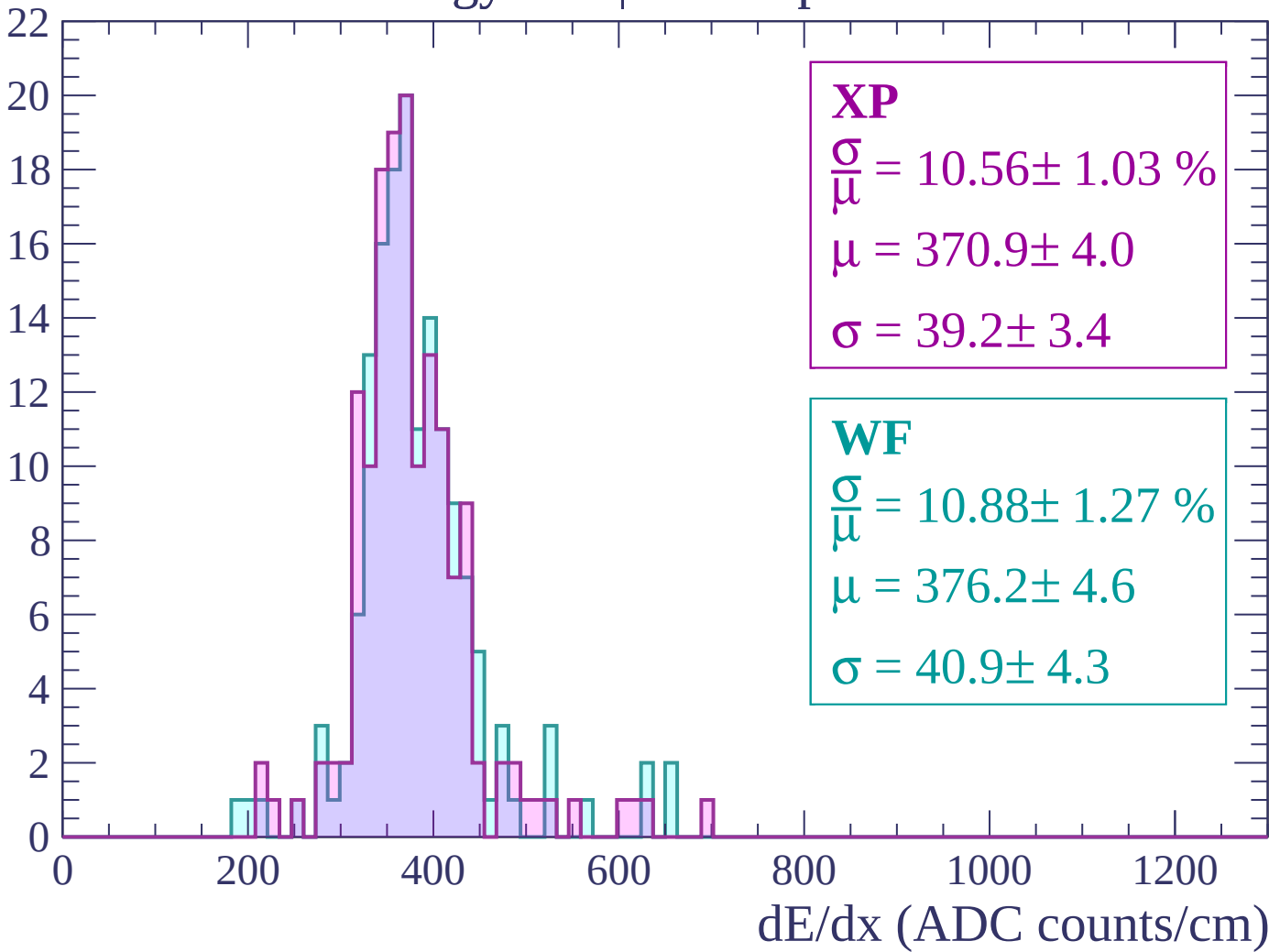
Energy loss |  $-360 < p < -300$

Count



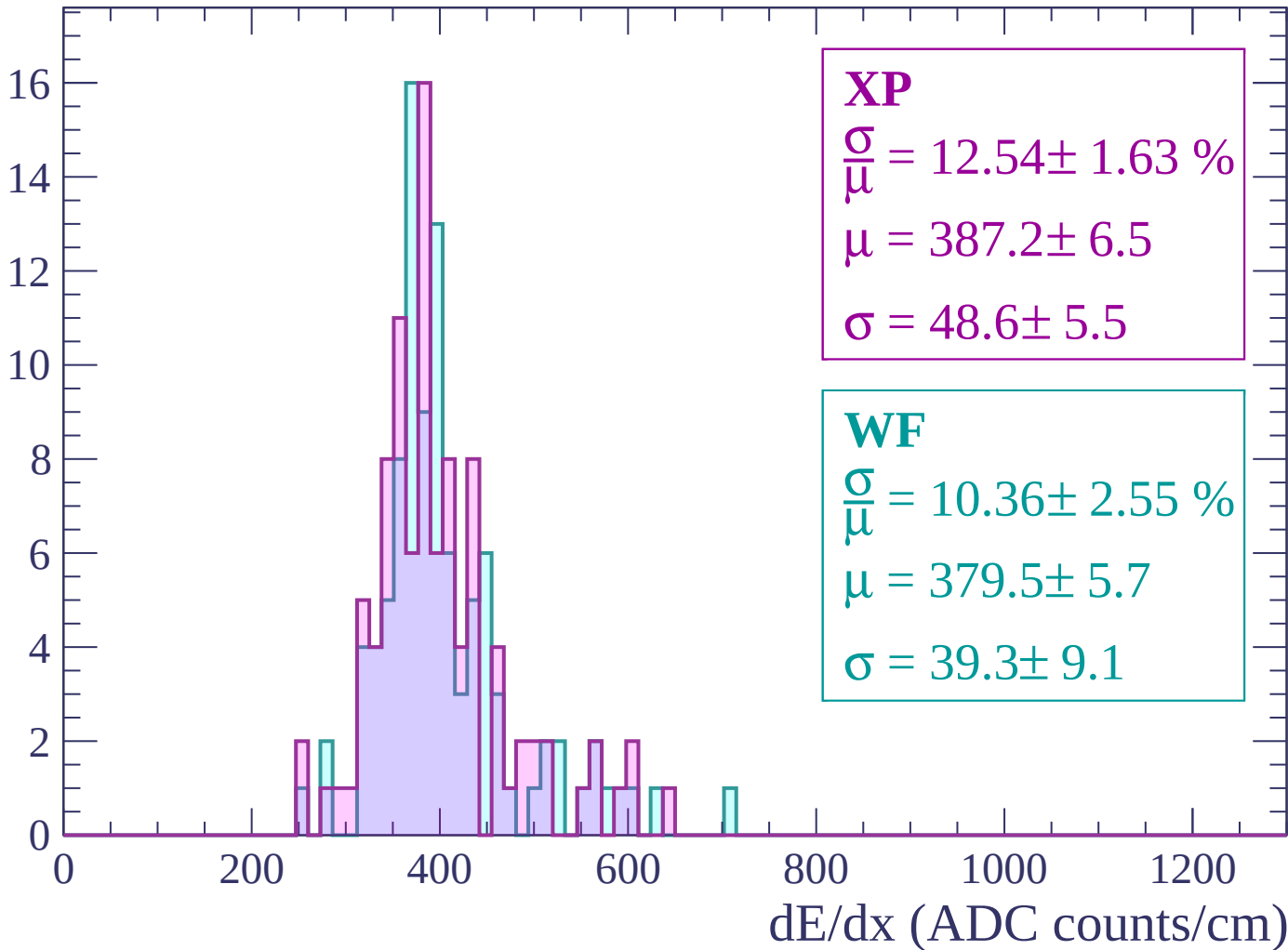
# Energy loss | -300 < p < -240

Count



# Energy loss | -240 < p < -180

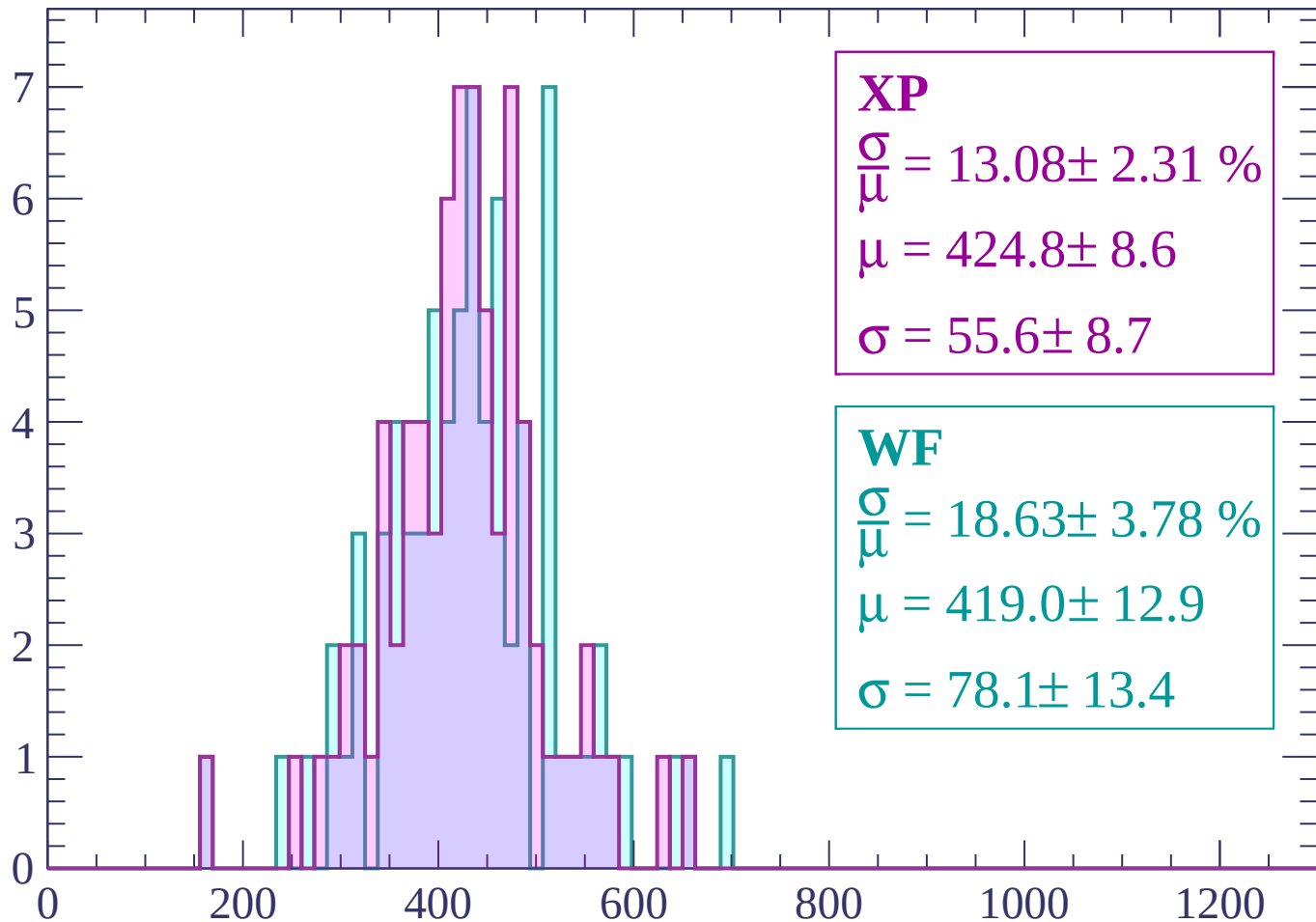
Count





Energy loss |  $-180 < p < -120$

Count



**XP**

$$\frac{\sigma}{\mu} = 13.08 \pm 2.31 \%$$

$$\mu = 424.8 \pm 8.6$$

$$\sigma = 55.6 \pm 8.7$$

**WF**

$$\frac{\sigma}{\mu} = 18.63 \pm 3.78 \%$$

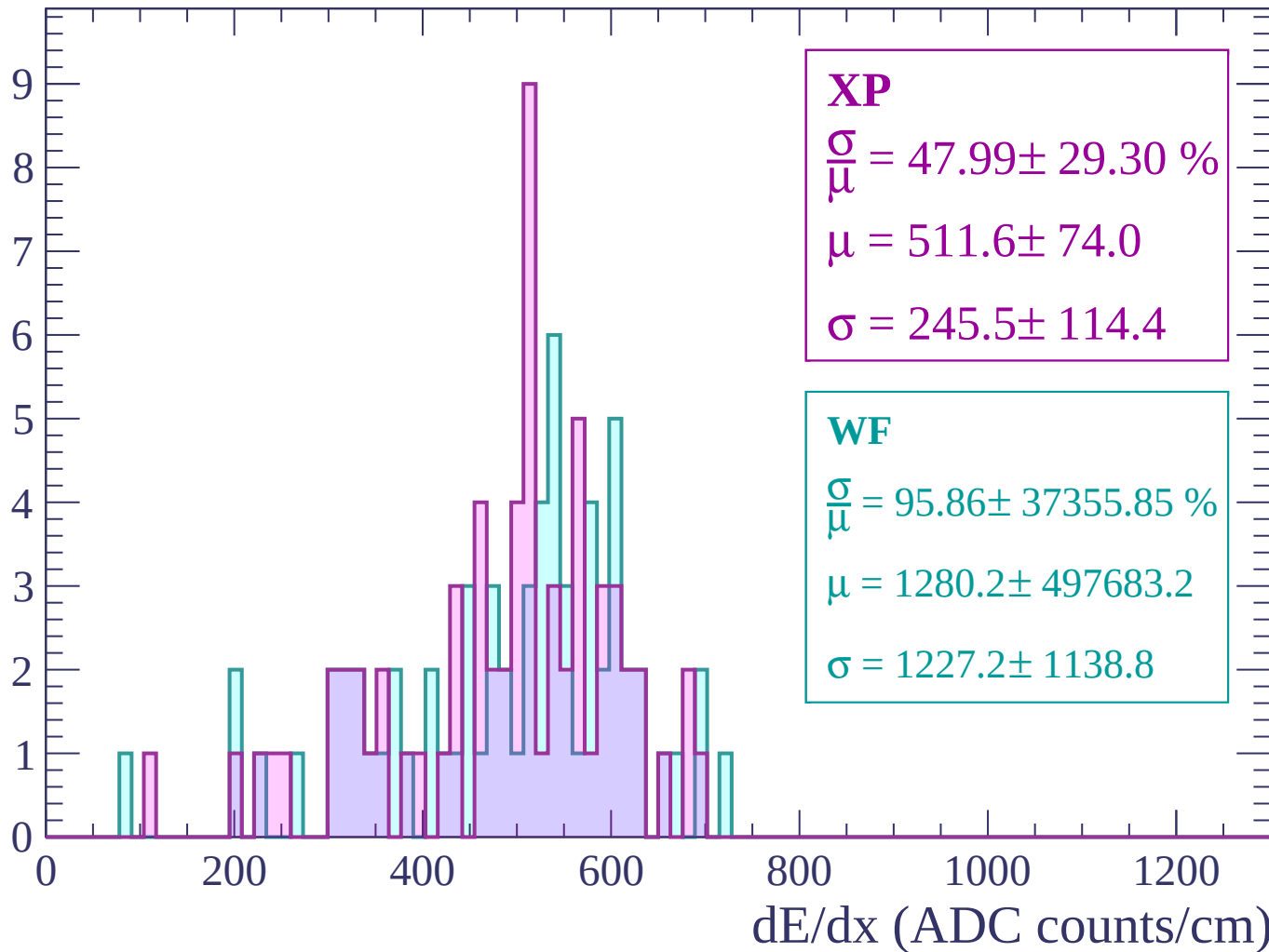
$$\mu = 419.0 \pm 12.9$$

$$\sigma = 78.1 \pm 13.4$$

dE/dx (ADC counts/cm)

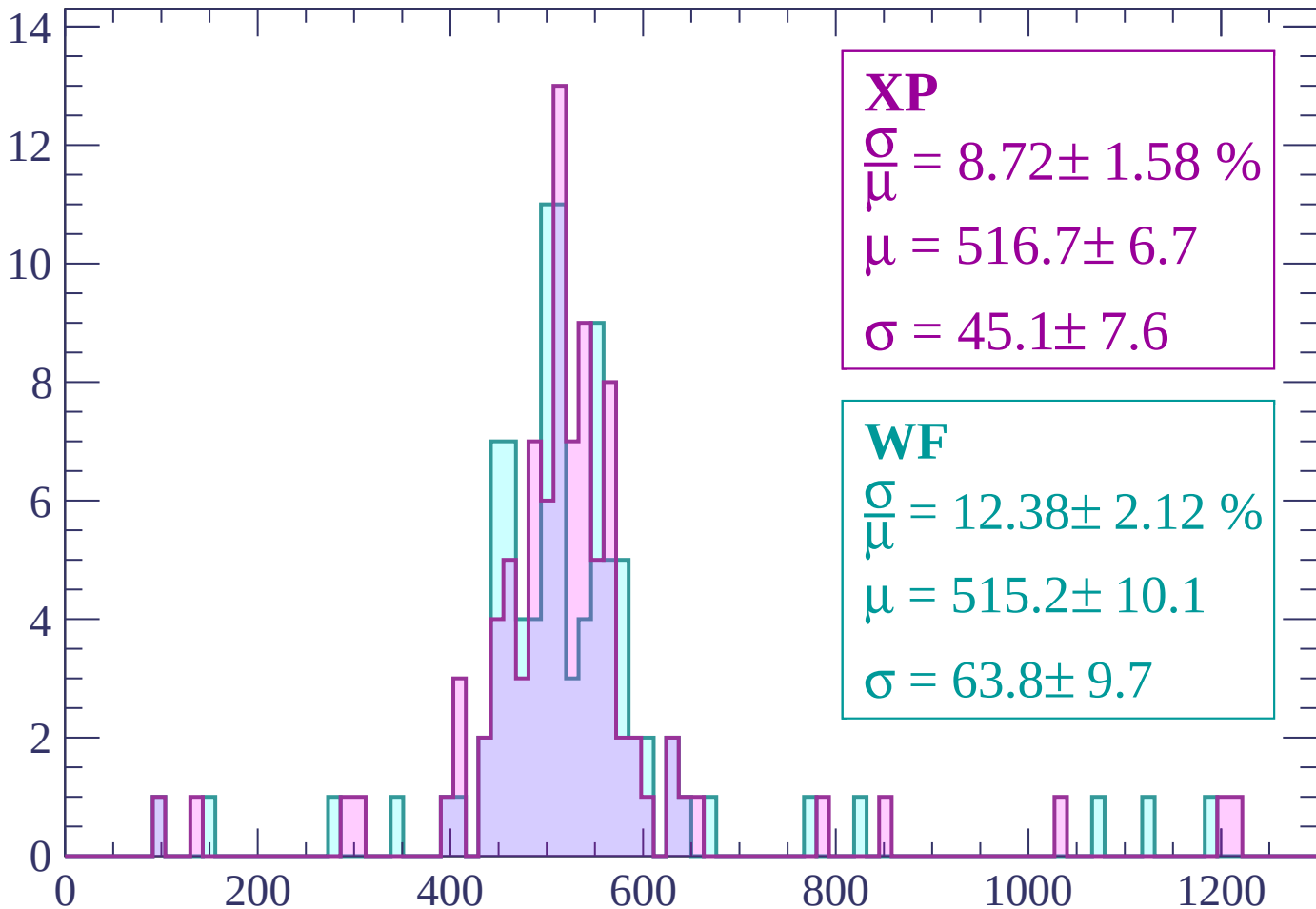
Energy loss |  $-120 < p < -60$

Count



Energy loss |  $-60 < p < 0$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.72 \pm 1.58 \%$$

$$\mu = 516.7 \pm 6.7$$

$$\sigma = 45.1 \pm 7.6$$

**WF**

$$\frac{\sigma}{\mu} = 12.38 \pm 2.12 \%$$

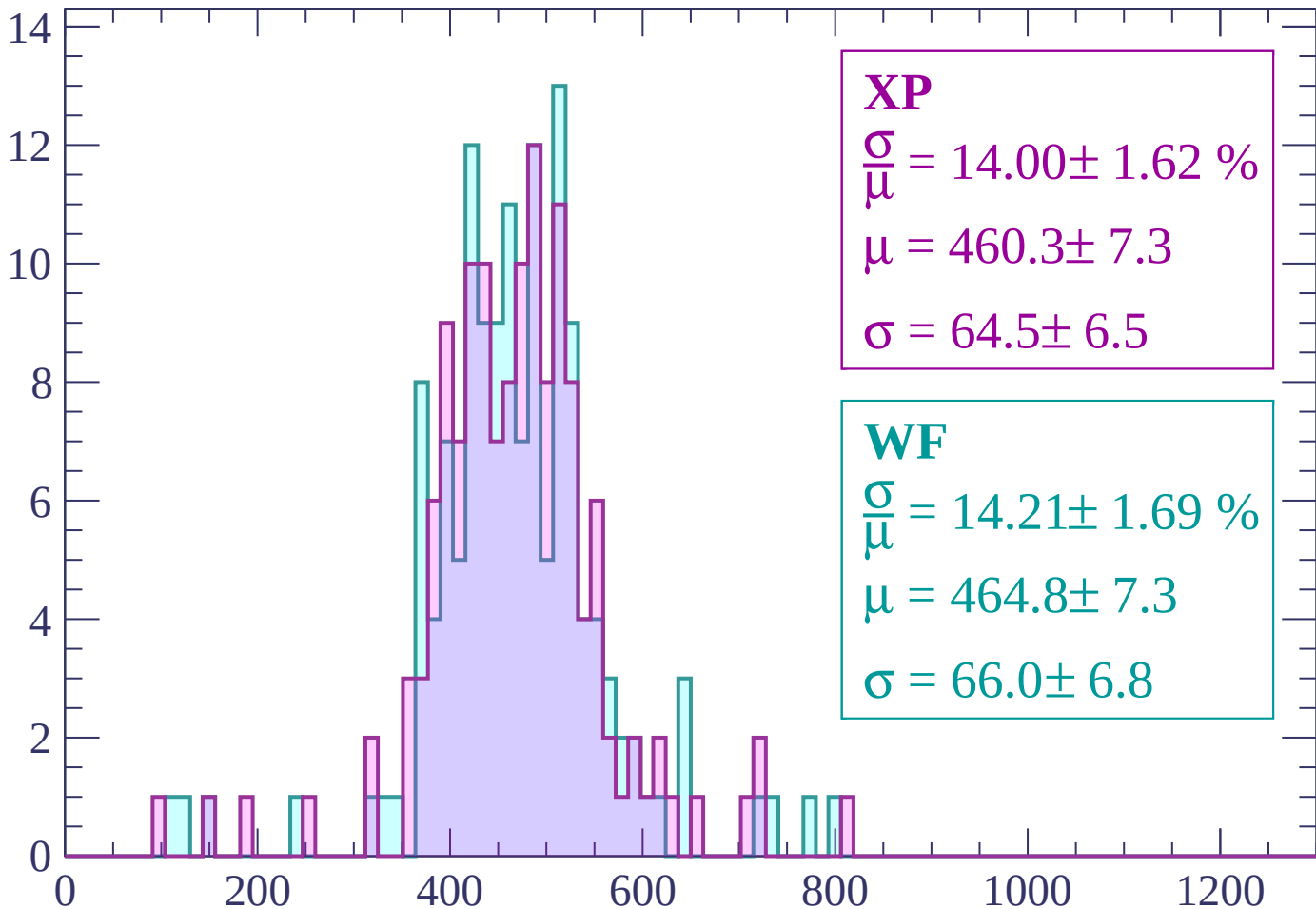
$$\mu = 515.2 \pm 10.1$$

$$\sigma = 63.8 \pm 9.7$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $0 < p < 60$

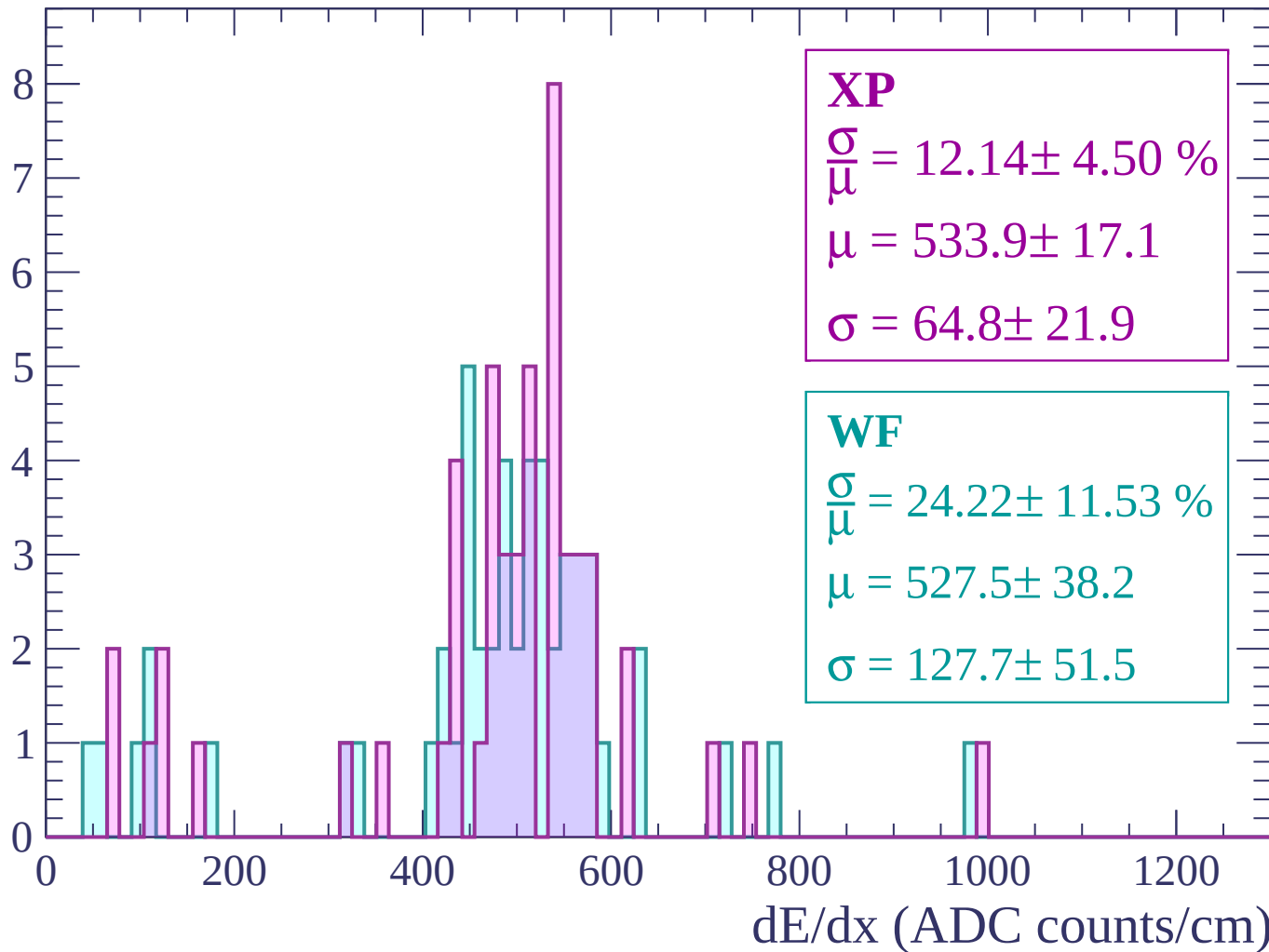
Count



dE/dx (ADC counts/cm)

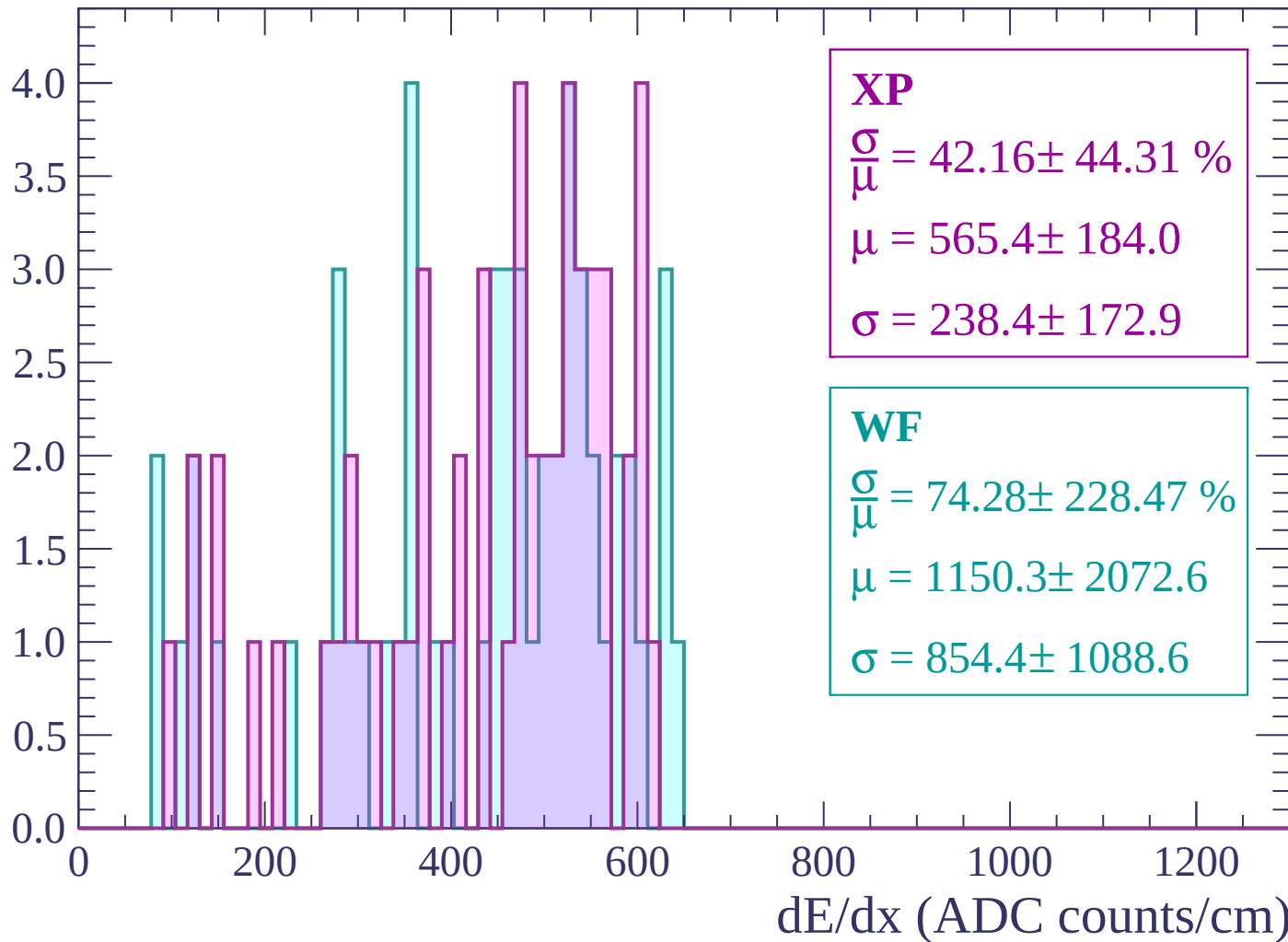
Energy loss |  $60 < p < 120$

Count



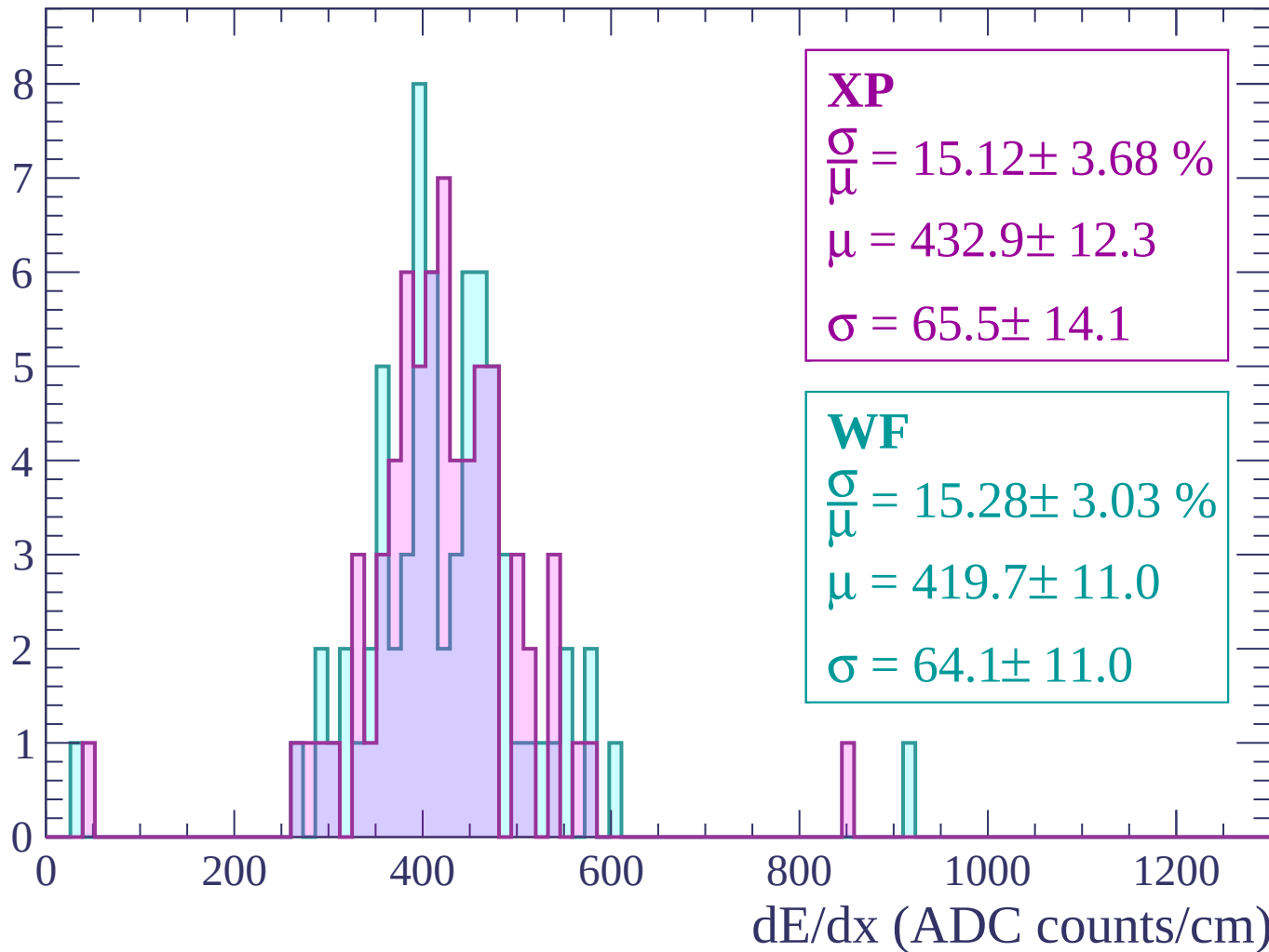
# Energy loss | $120 < p < 180$

Count



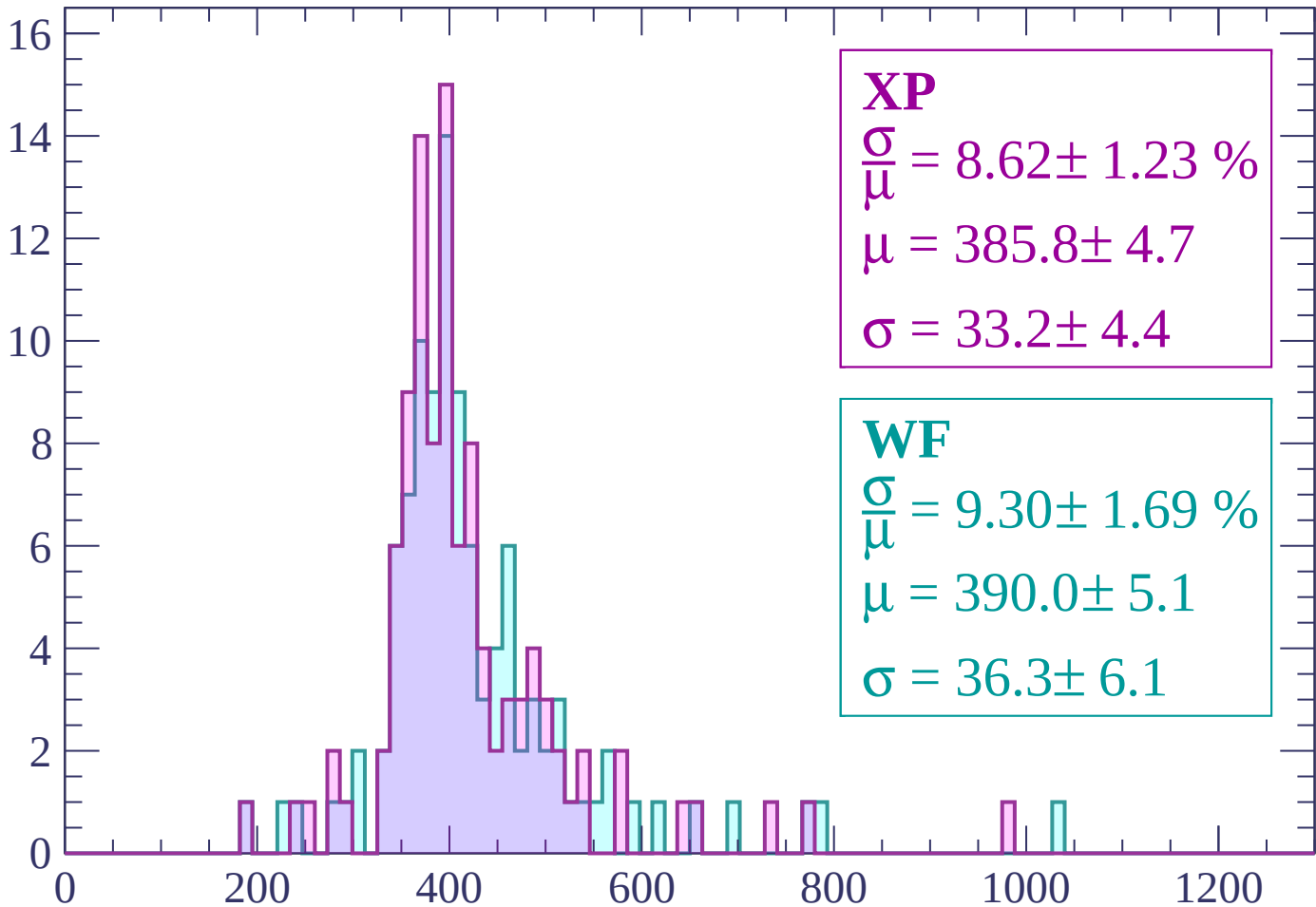
Energy loss |  $180 < p < 240$

Count



Energy loss |  $240 < p < 300$

Count

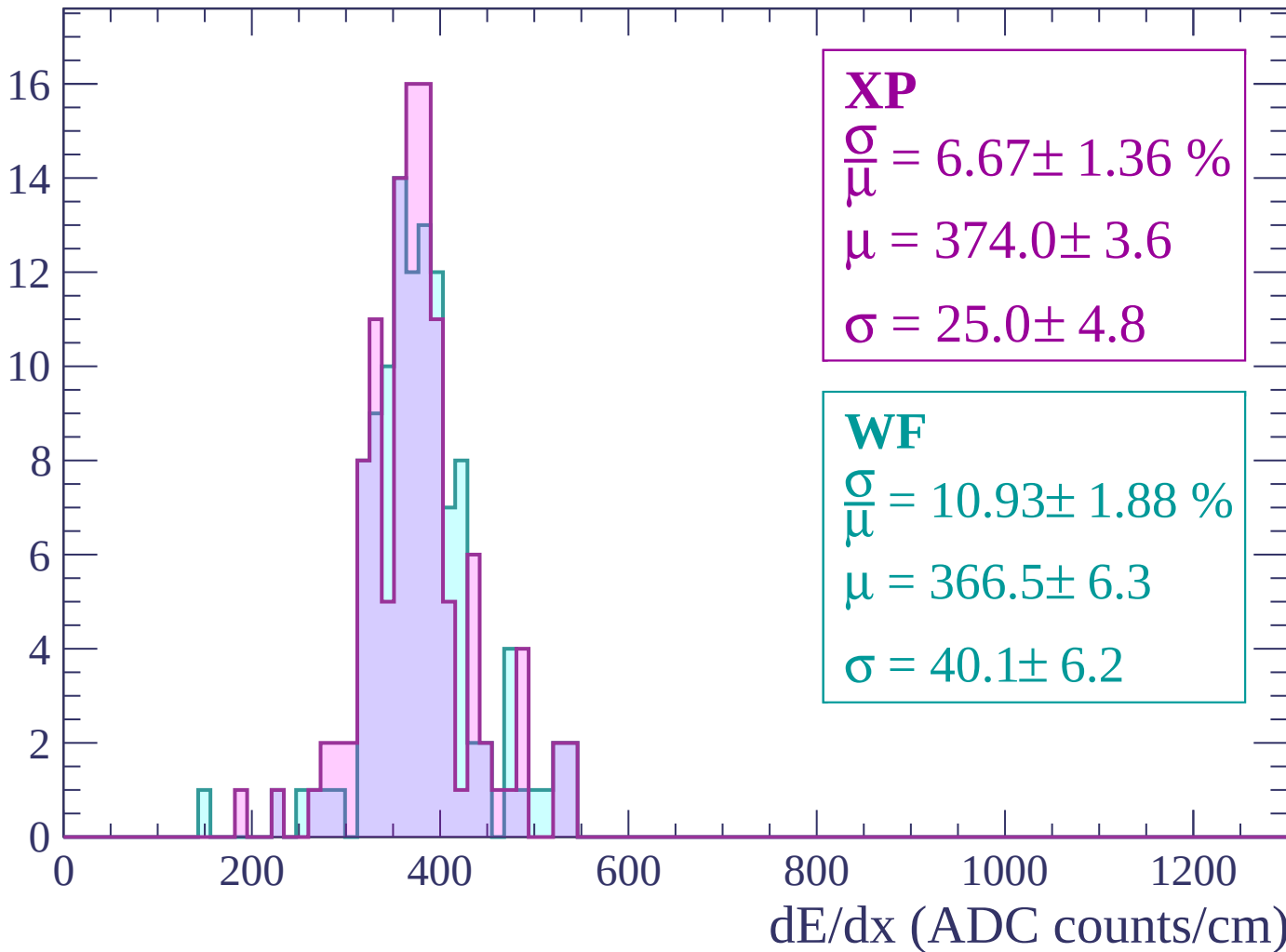


dE/dx (ADC counts/cm)



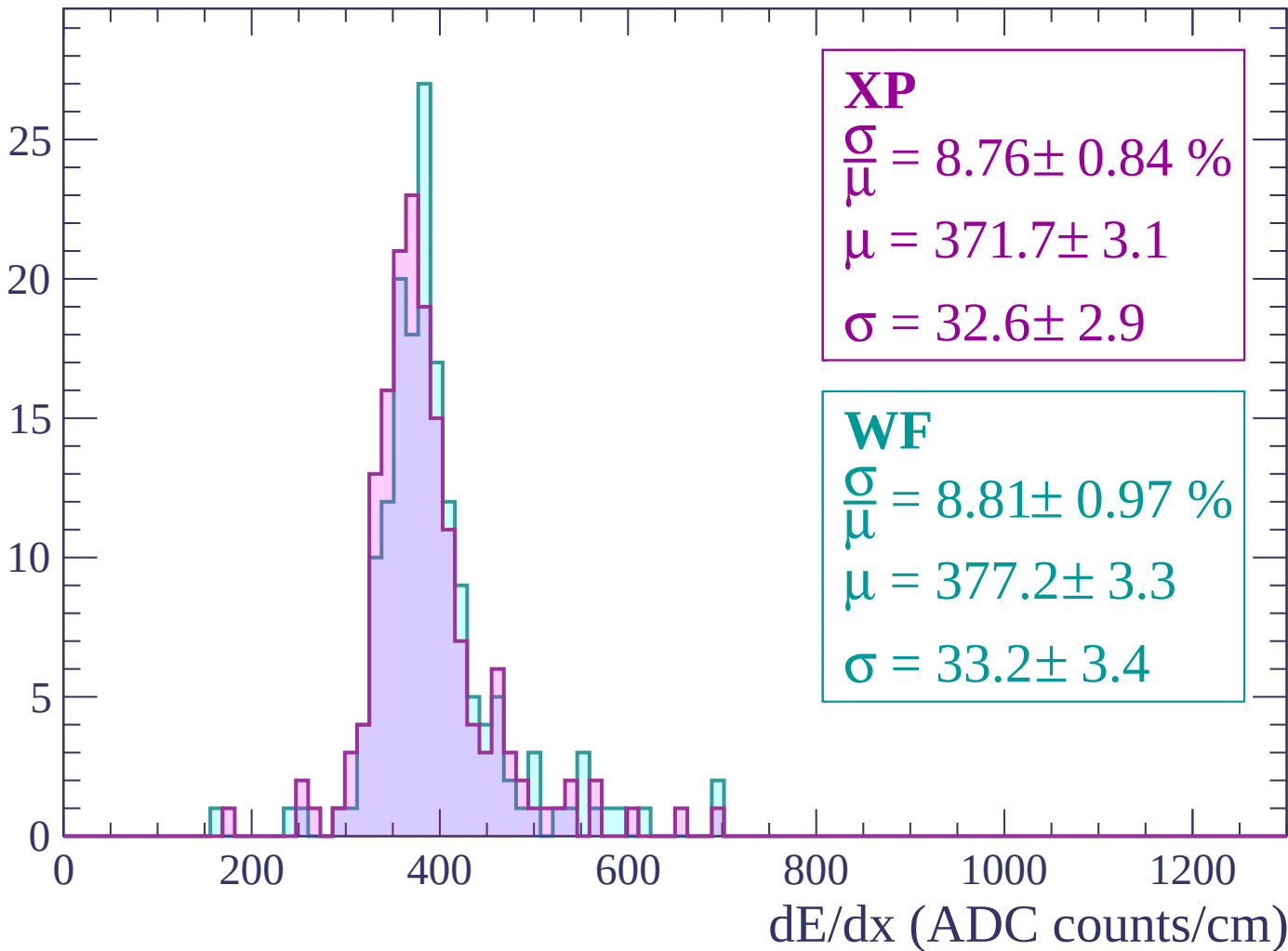
Energy loss |  $300 < p < 360$

Count



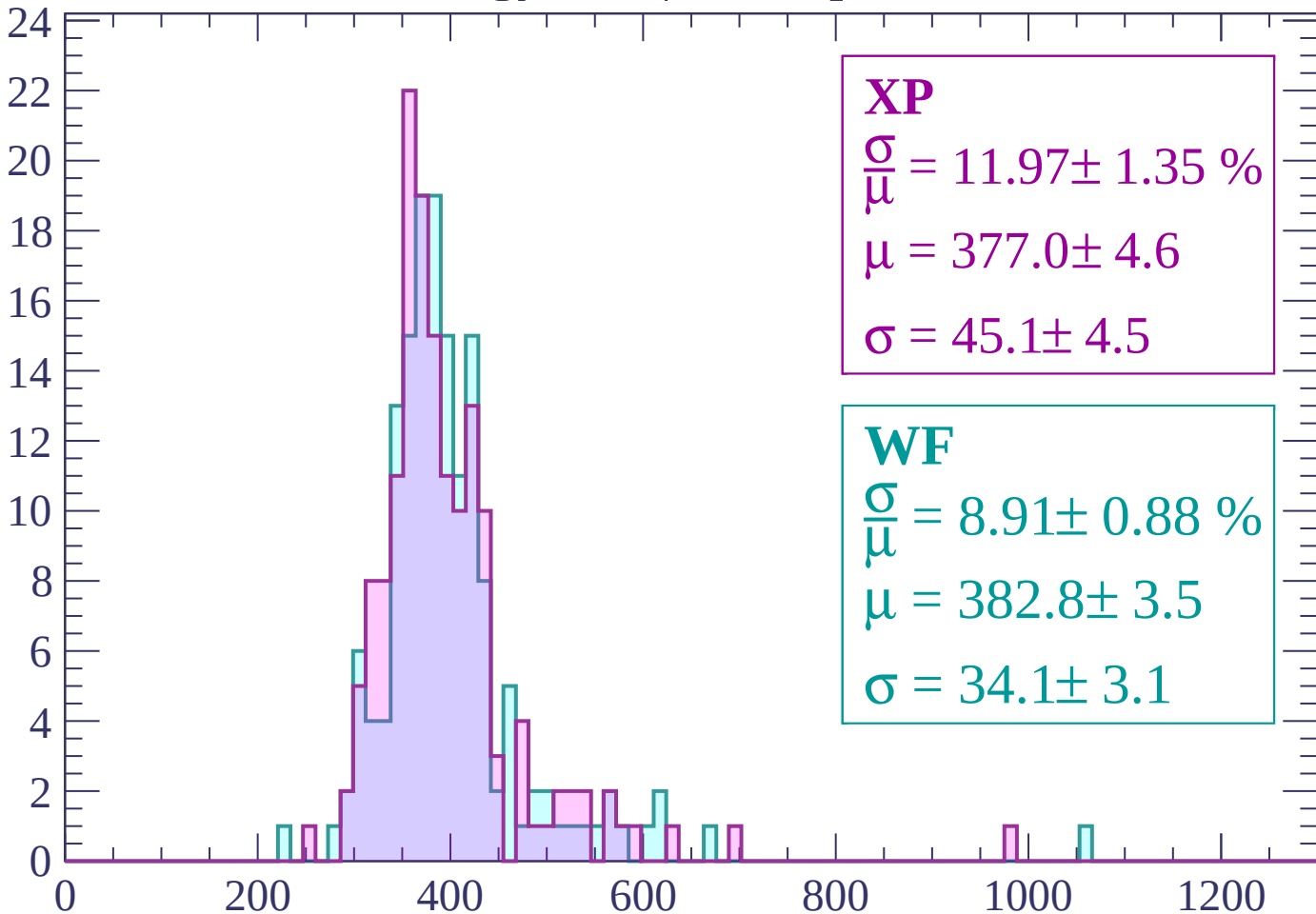
Energy loss |  $360 < p < 420$

Count



Energy loss |  $420 < p < 480$

Count



**XP**

$$\frac{\sigma}{\mu} = 11.97 \pm 1.35 \%$$

$$\mu = 377.0 \pm 4.6$$

$$\sigma = 45.1 \pm 4.5$$

**WF**

$$\frac{\sigma}{\mu} = 8.91 \pm 0.88 \%$$

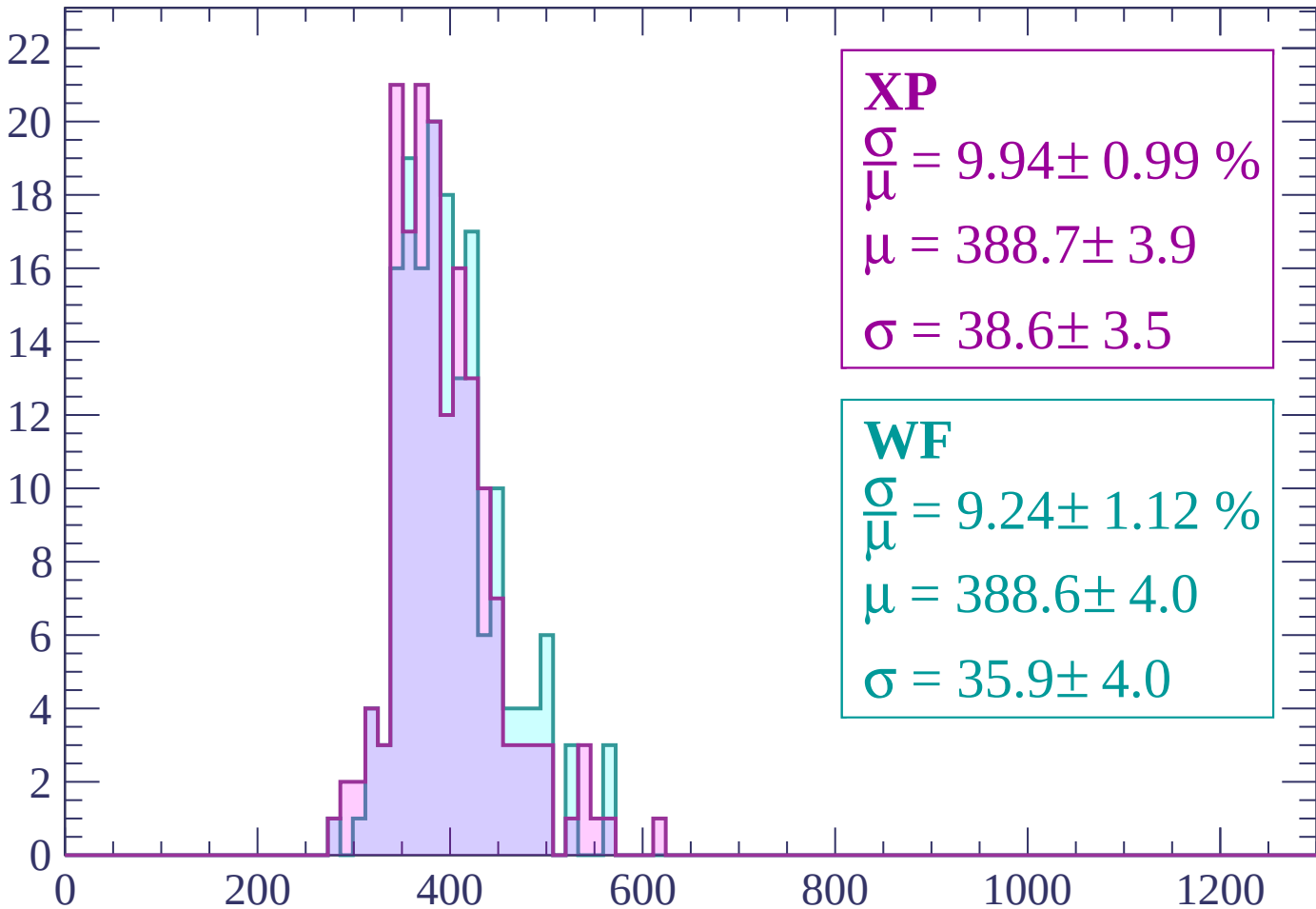
$$\mu = 382.8 \pm 3.5$$

$$\sigma = 34.1 \pm 3.1$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $480 < p < 540$

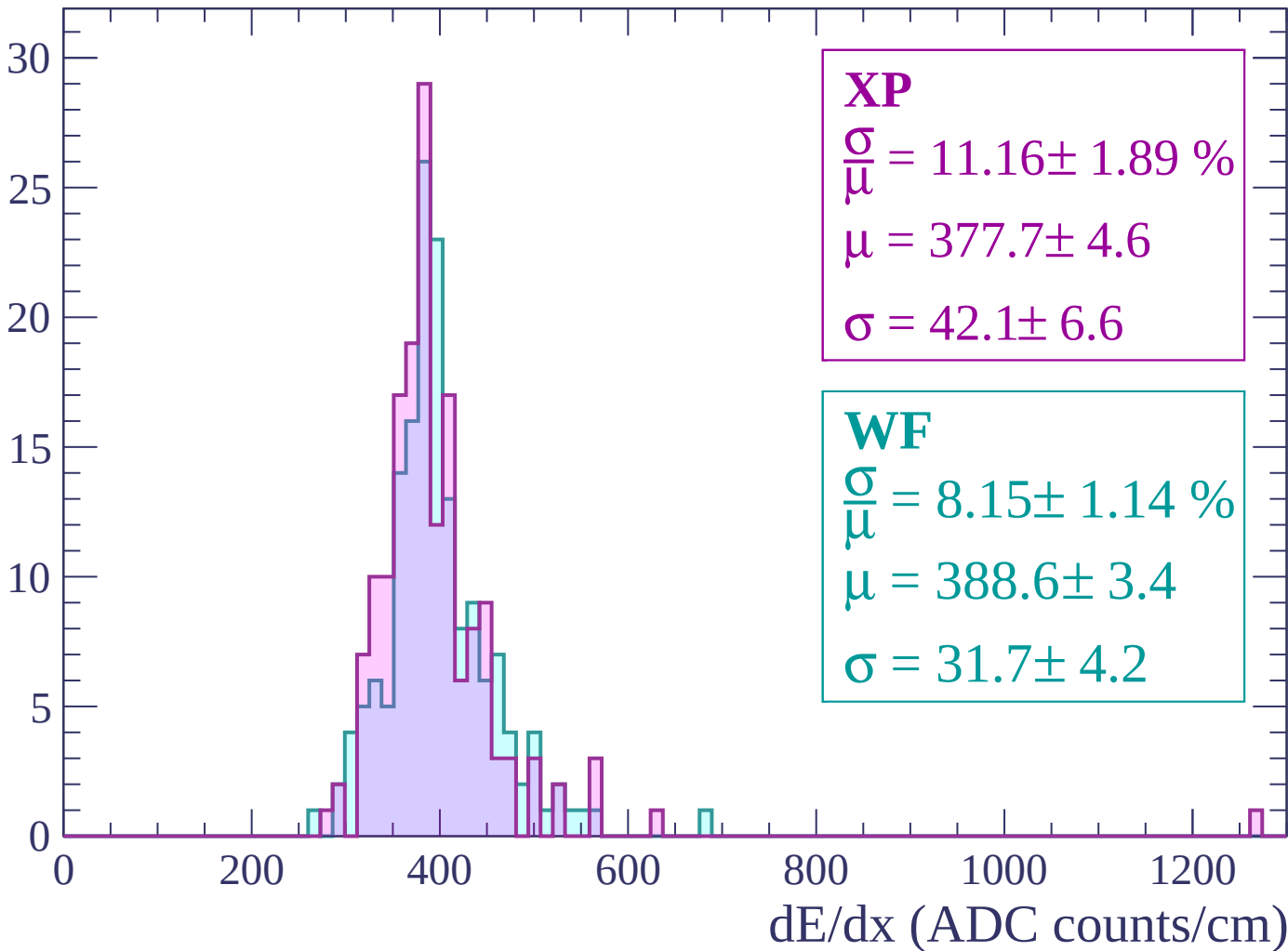
Count



dE/dx (ADC counts/cm)

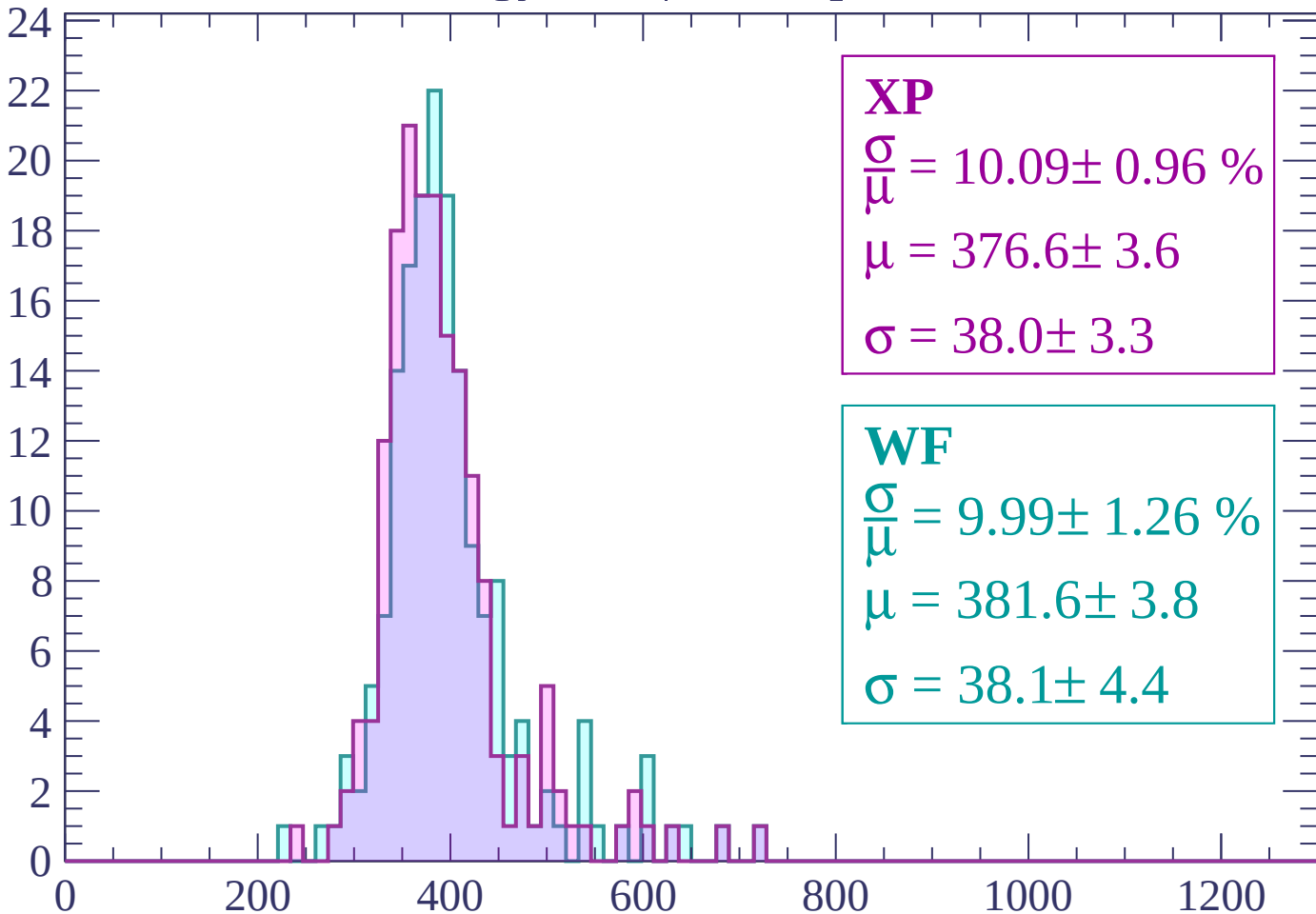
Energy loss |  $540 < p < 600$

Count



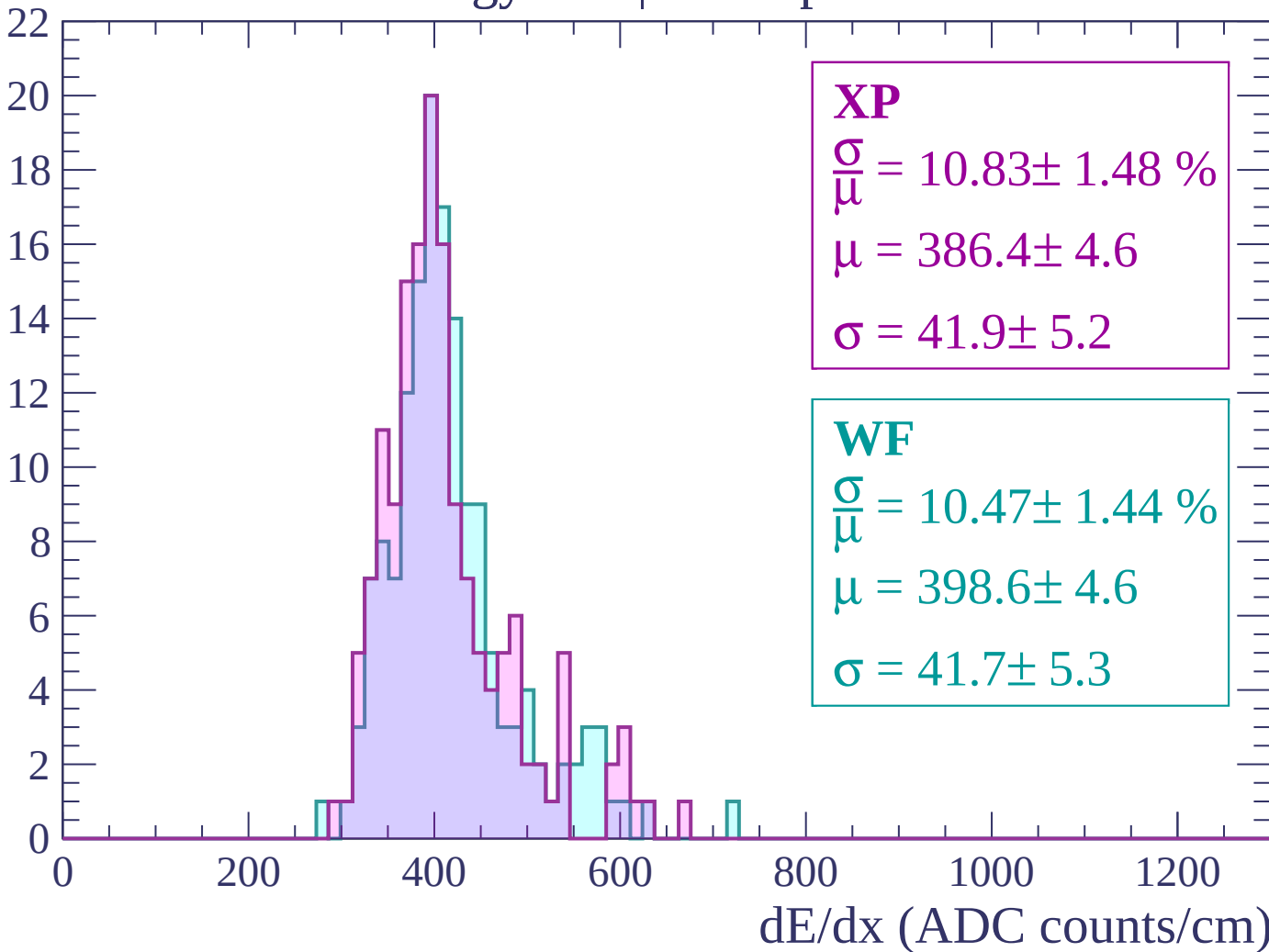
Energy loss |  $600 < p < 660$

Count



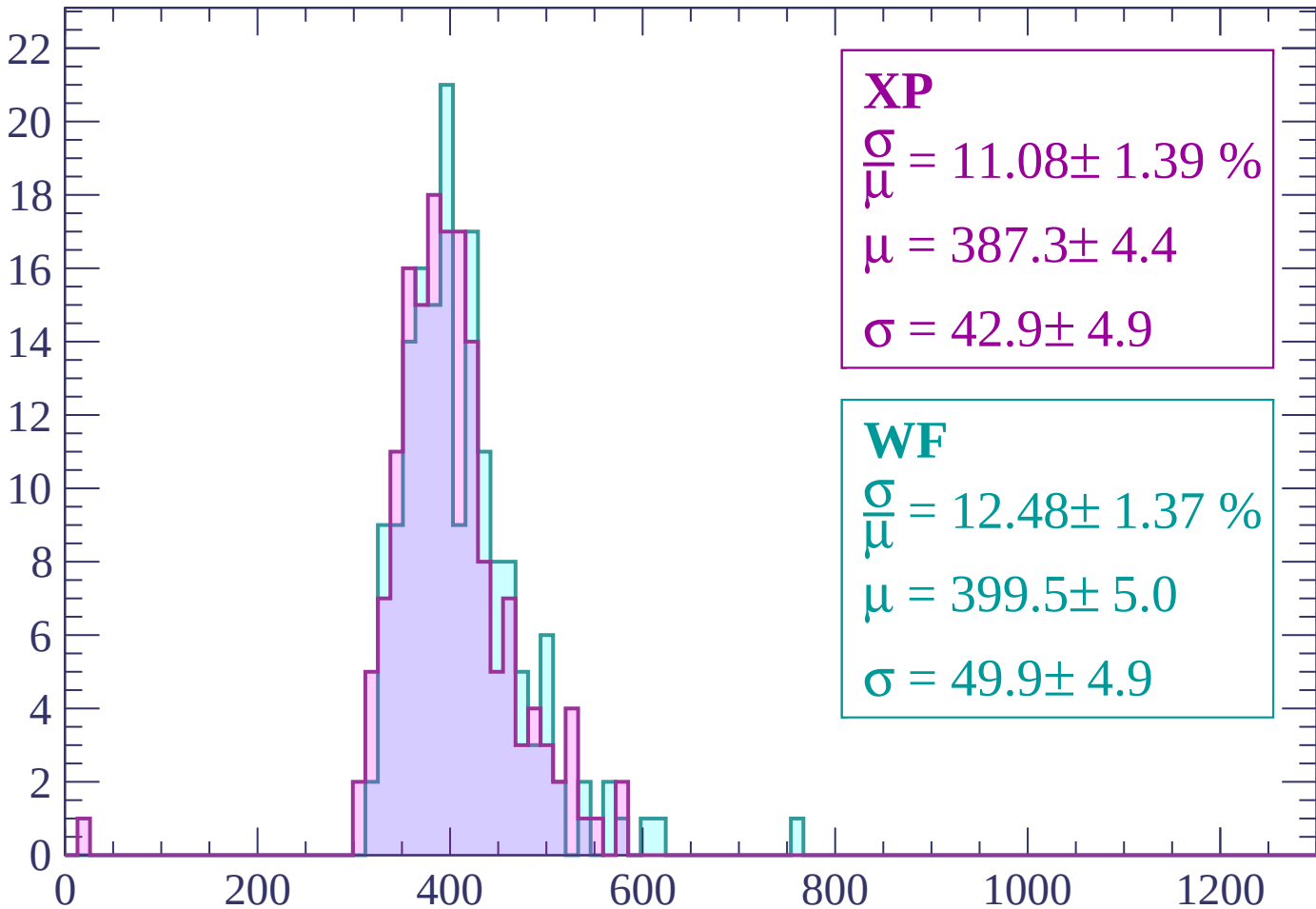
Energy loss |  $660 < p < 720$

Count



Energy loss |  $720 < p < 780$

Count



**XP**

$$\frac{\sigma}{\mu} = 11.08 \pm 1.39 \%$$

$$\mu = 387.3 \pm 4.4$$

$$\sigma = 42.9 \pm 4.9$$

**WF**

$$\frac{\sigma}{\mu} = 12.48 \pm 1.37 \%$$

$$\mu = 399.5 \pm 5.0$$

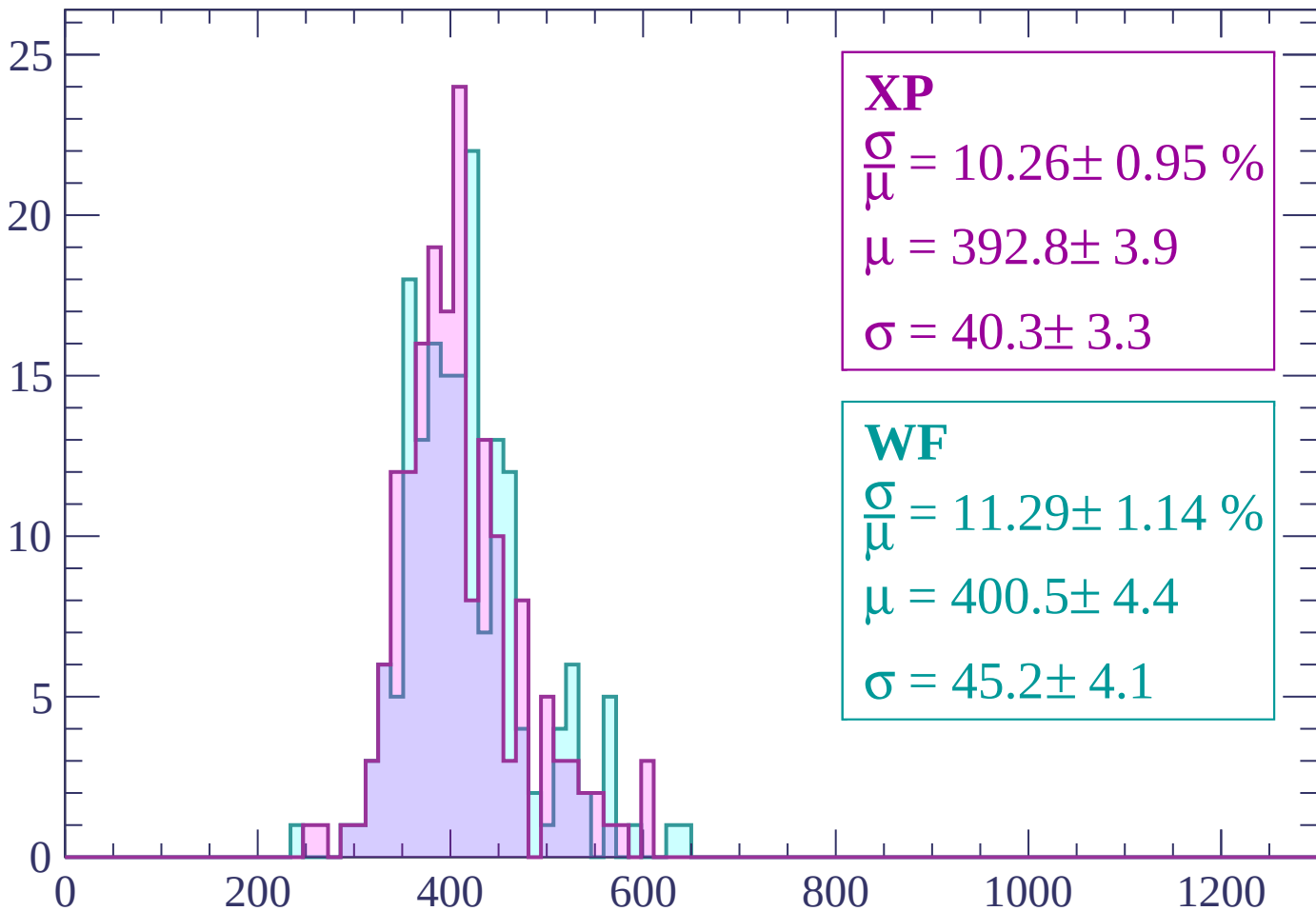
$$\sigma = 49.9 \pm 4.9$$

$dE/dx$  (ADC counts/cm)



Energy loss |  $780 < p < 840$

Count



**XP**

$$\frac{\sigma}{\bar{\mu}} = 10.26 \pm 0.95 \%$$

$$\mu = 392.8 \pm 3.9$$

$$\sigma = 40.3 \pm 3.3$$

**WF**

$$\frac{\sigma}{\bar{\mu}} = 11.29 \pm 1.14 \%$$

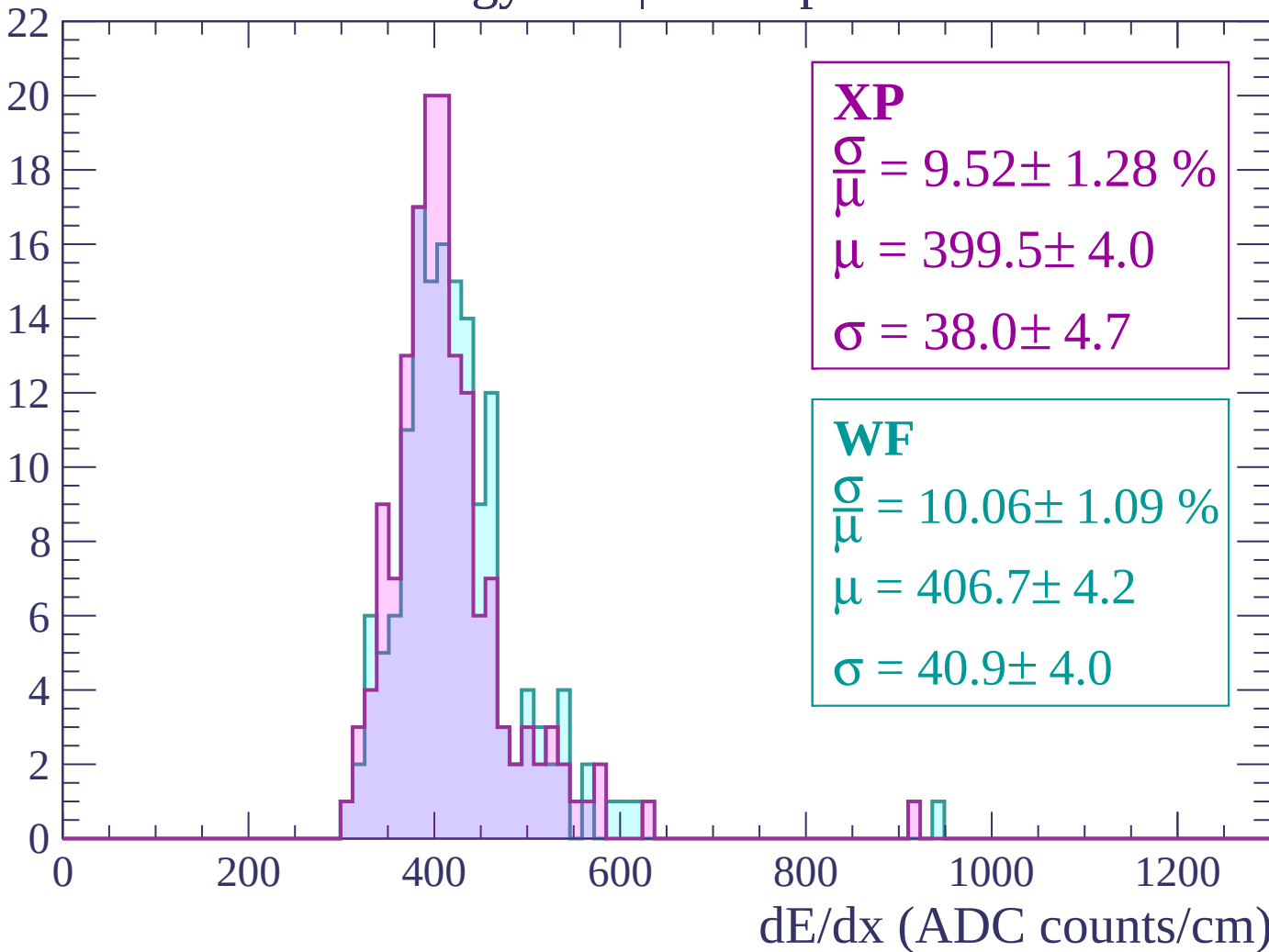
$$\mu = 400.5 \pm 4.4$$

$$\sigma = 45.2 \pm 4.1$$

$dE/dx$  (ADC counts/cm)

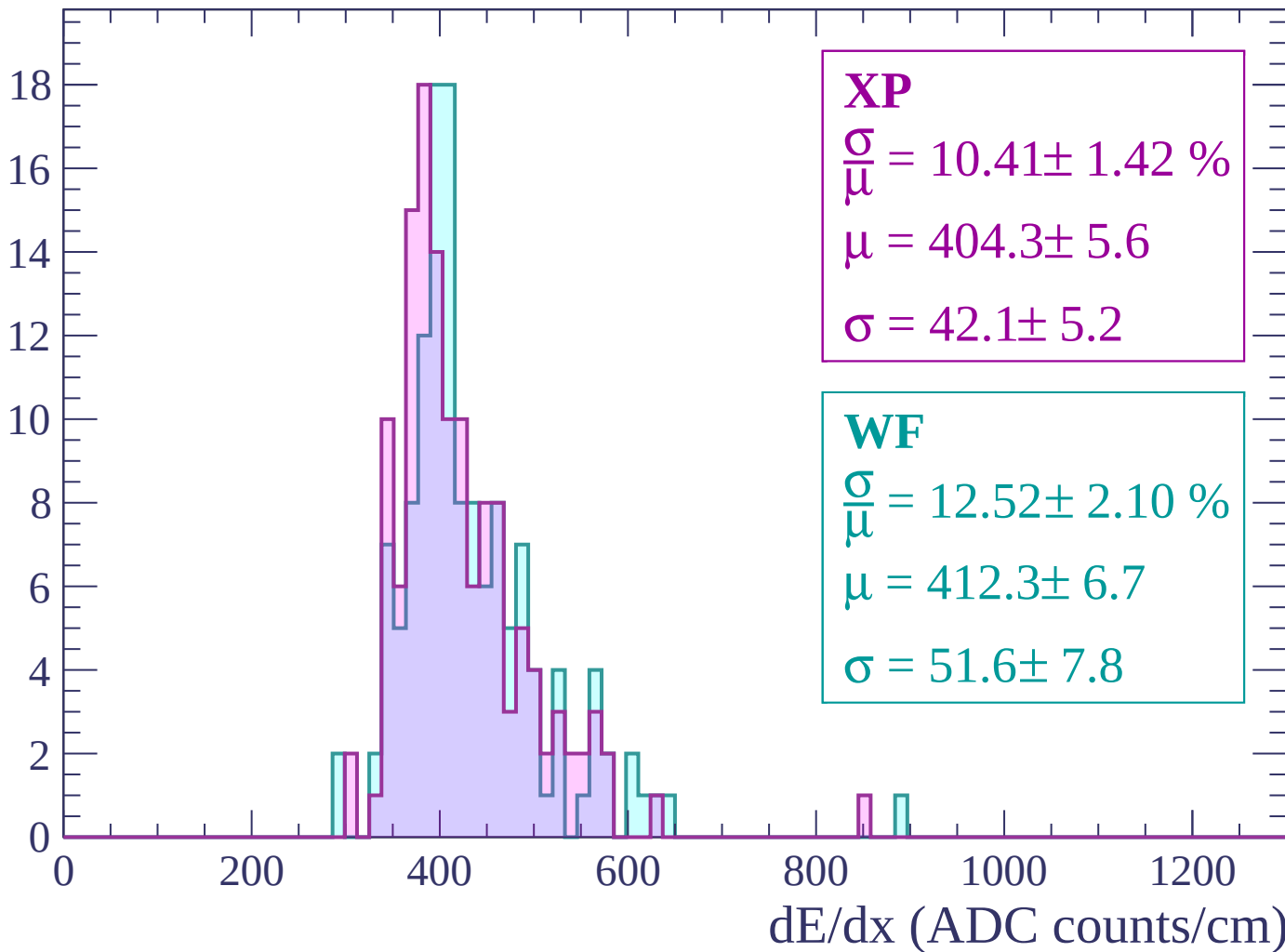
Energy loss |  $840 < p < 900$

Count



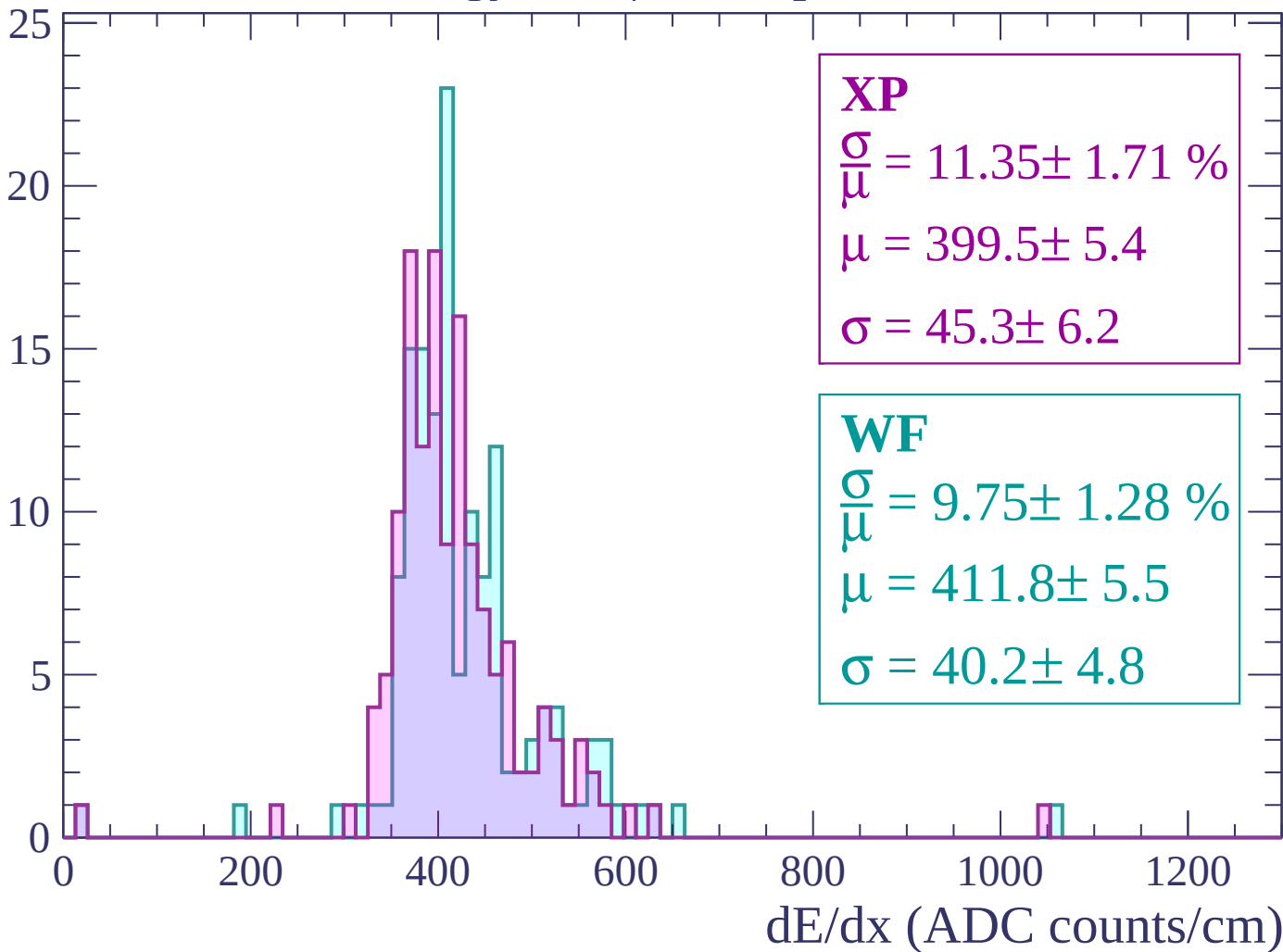
Energy loss |  $900 < p < 960$

Count



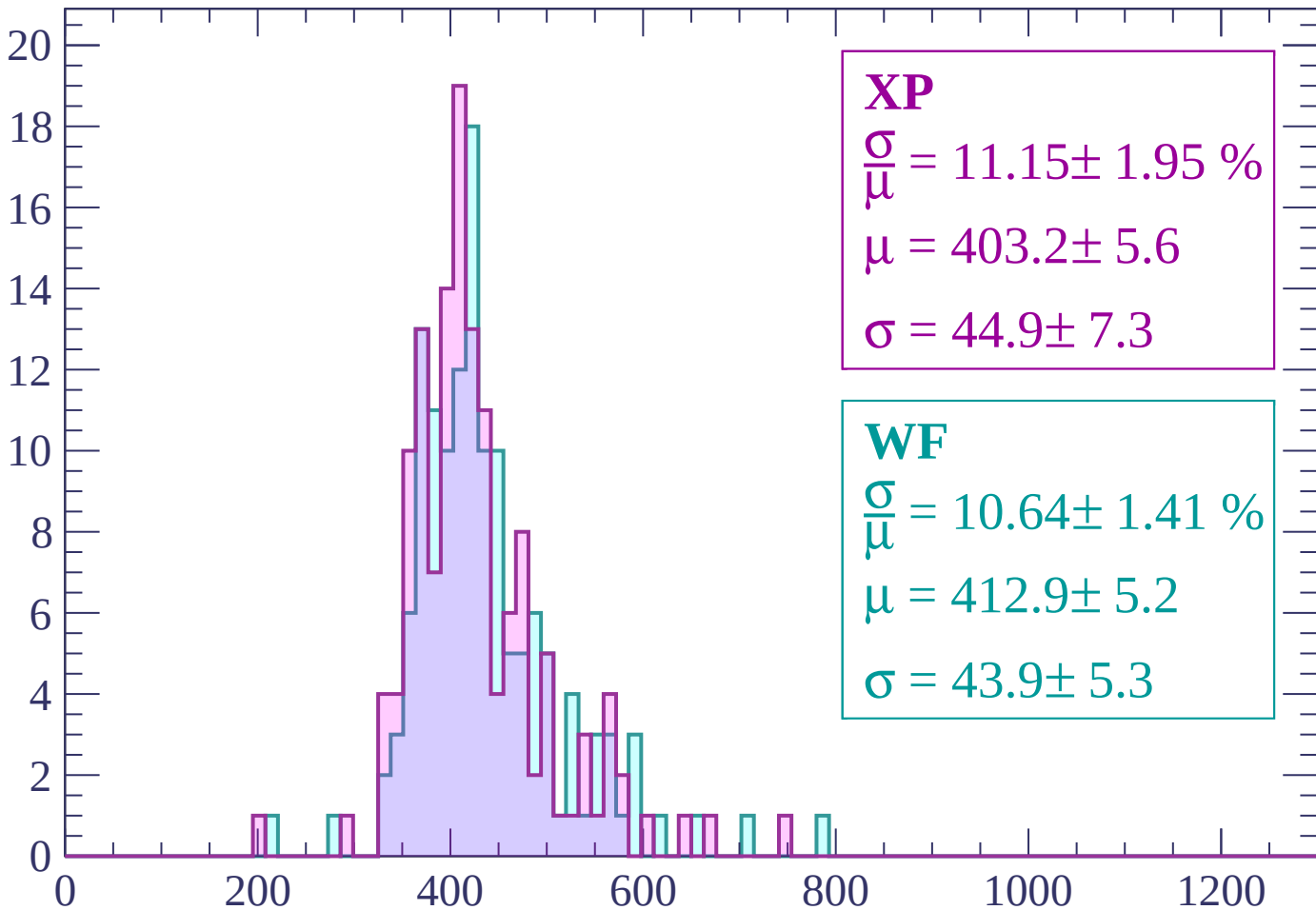
# Energy loss | $960 < p < 1020$

Count



Energy loss |  $1020 < p < 1080$

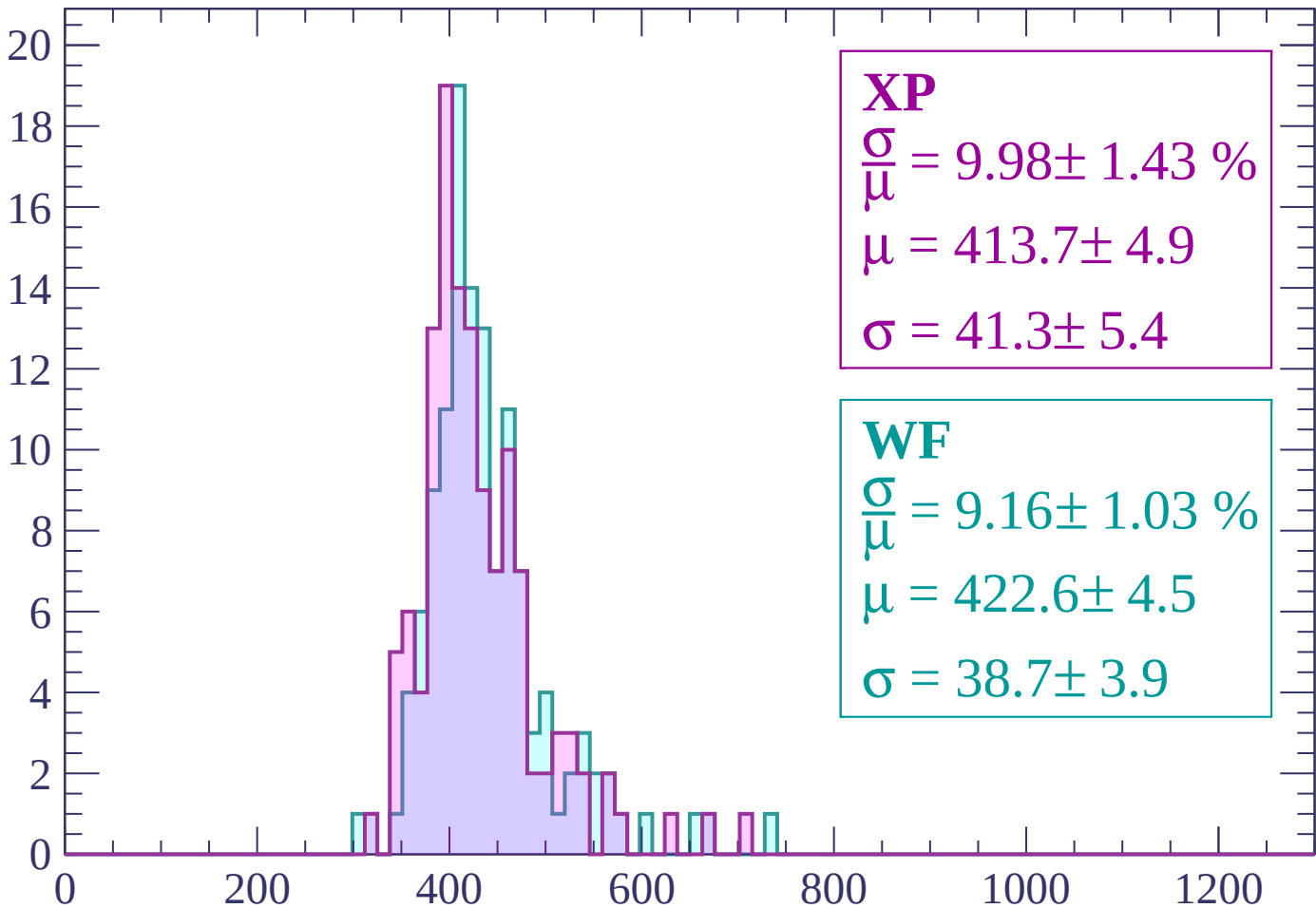
Count



dE/dx (ADC counts/cm)

Energy loss |  $1080 < p < 1140$

Count



**XP**

$$\frac{\sigma}{\mu} = 9.98 \pm 1.43 \%$$

$$\mu = 413.7 \pm 4.9$$

$$\sigma = 41.3 \pm 5.4$$

**WF**

$$\frac{\sigma}{\mu} = 9.16 \pm 1.03 \%$$

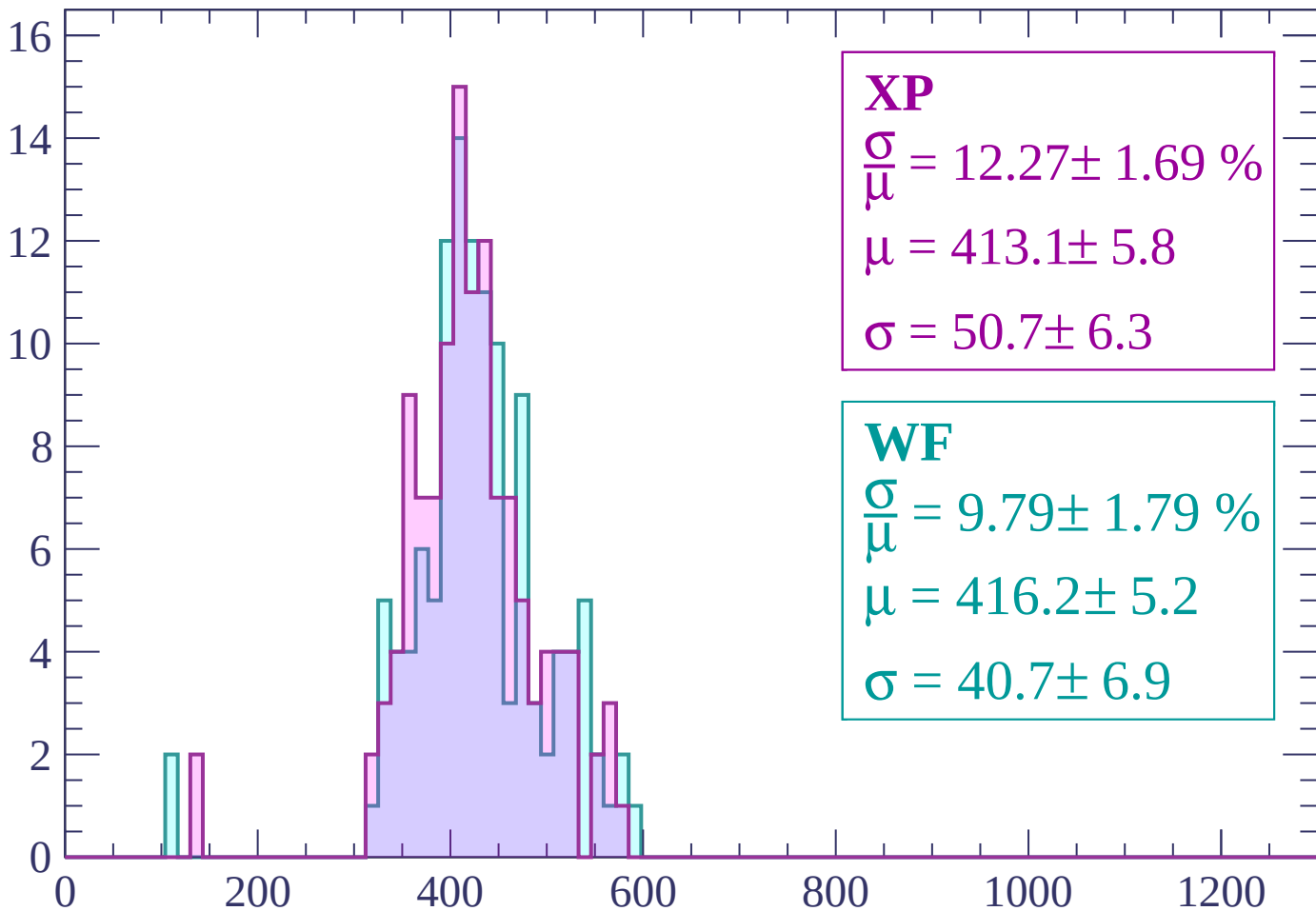
$$\mu = 422.6 \pm 4.5$$

$$\sigma = 38.7 \pm 3.9$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $1140 < p < 1200$

Count



**XP**

$$\frac{\sigma}{\bar{\mu}} = 12.27 \pm 1.69 \%$$

$$\mu = 413.1 \pm 5.8$$

$$\sigma = 50.7 \pm 6.3$$

**WF**

$$\frac{\sigma}{\bar{\mu}} = 9.79 \pm 1.79 \%$$

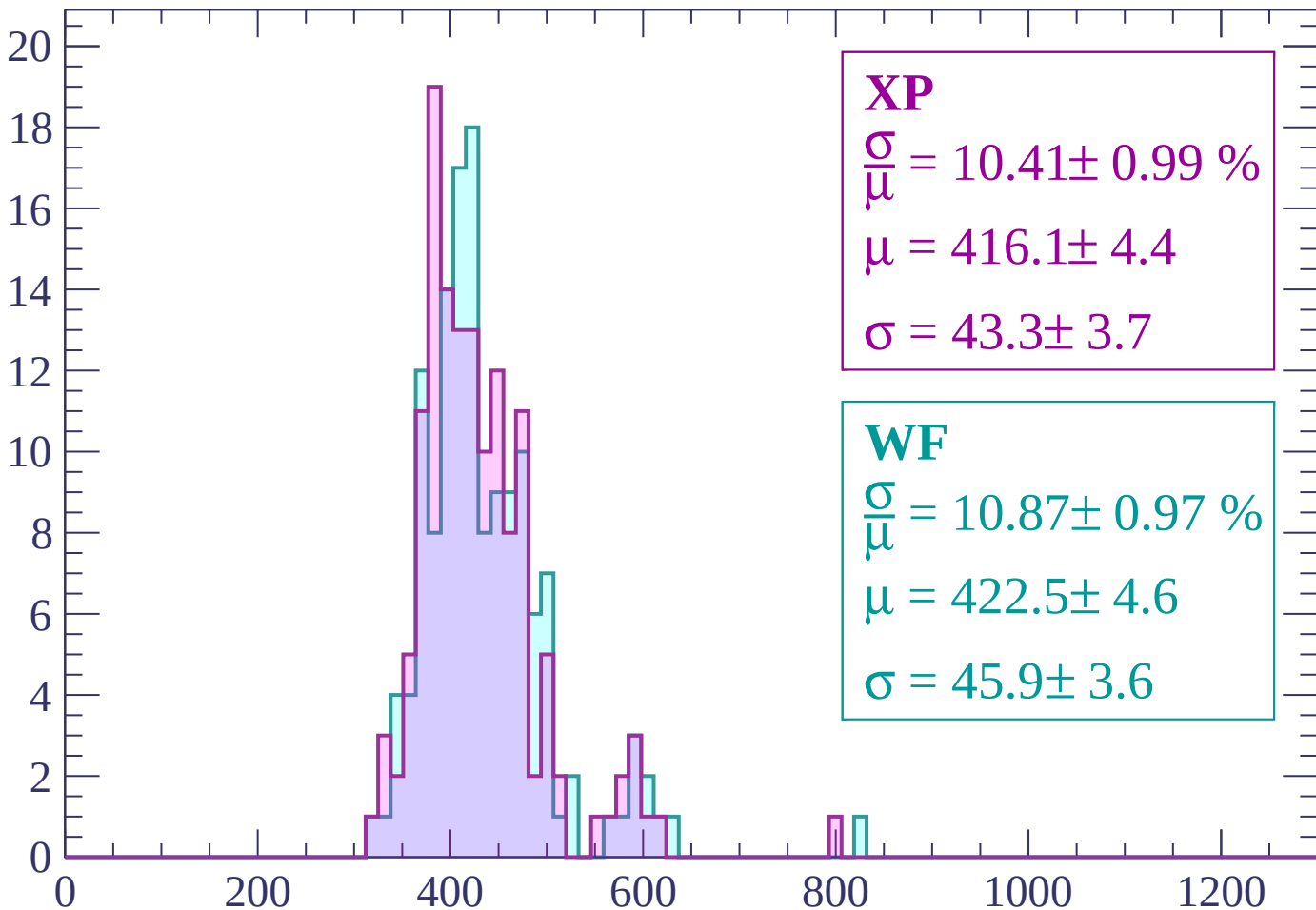
$$\mu = 416.2 \pm 5.2$$

$$\sigma = 40.7 \pm 6.9$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $1200 < p < 1260$

Count

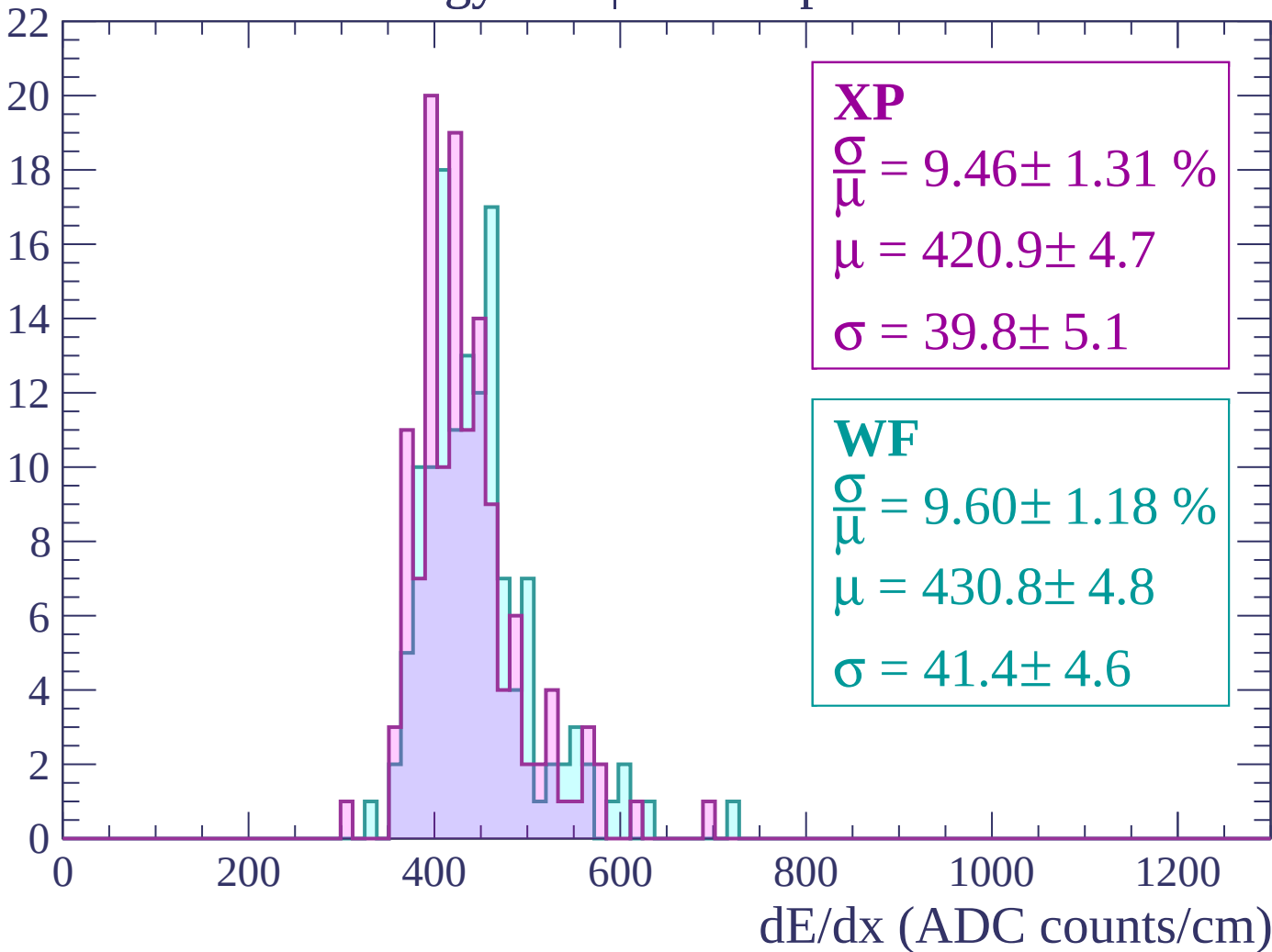


dE/dx (ADC counts/cm)



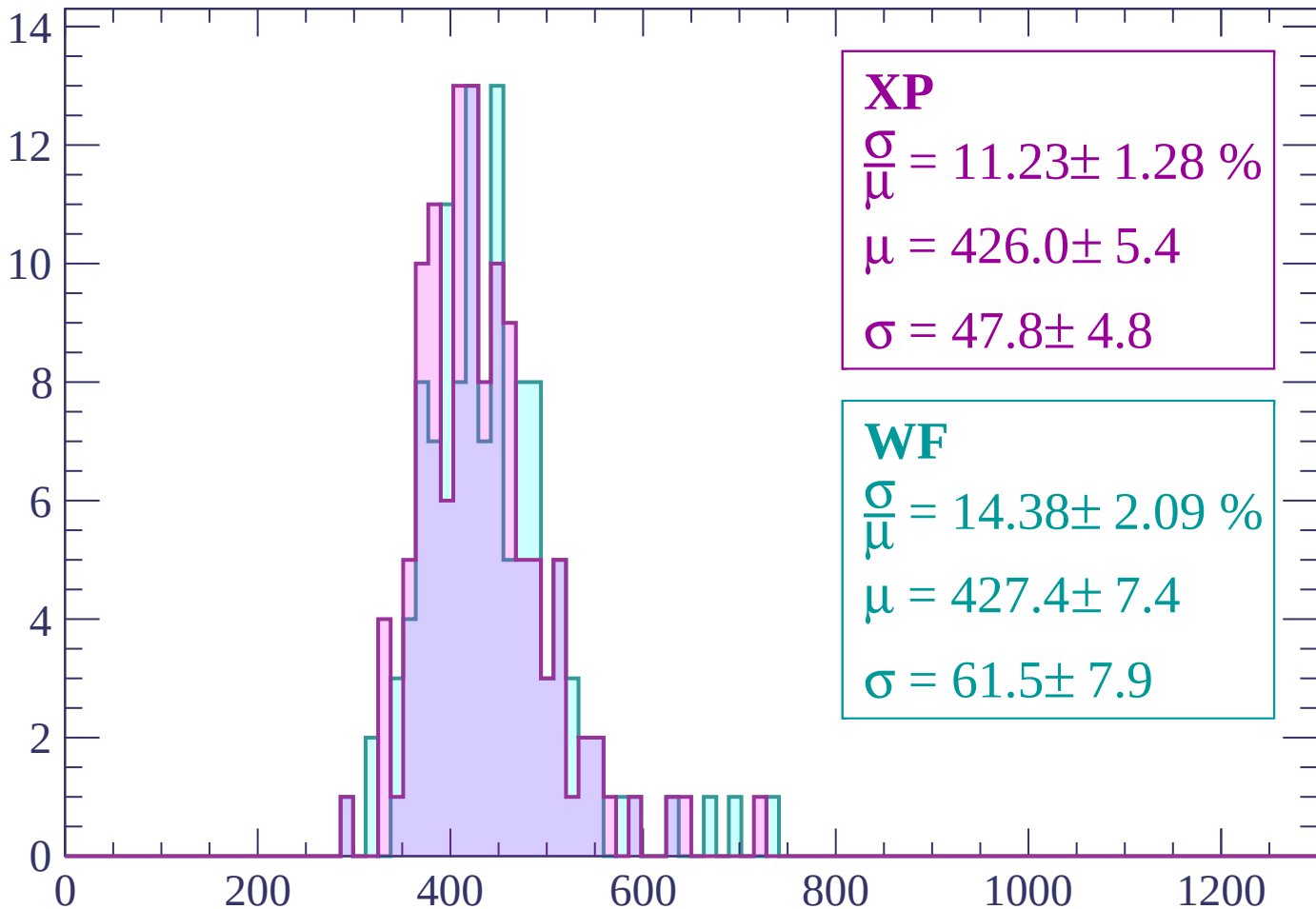
Energy loss |  $1260 < p < 1320$

Count



# Energy loss | $1320 < p < 1380$

Count



**XP**

$$\frac{\sigma}{\mu} = 11.23 \pm 1.28 \%$$

$$\mu = 426.0 \pm 5.4$$

$$\sigma = 47.8 \pm 4.8$$

**WF**

$$\frac{\sigma}{\mu} = 14.38 \pm 2.09 \%$$

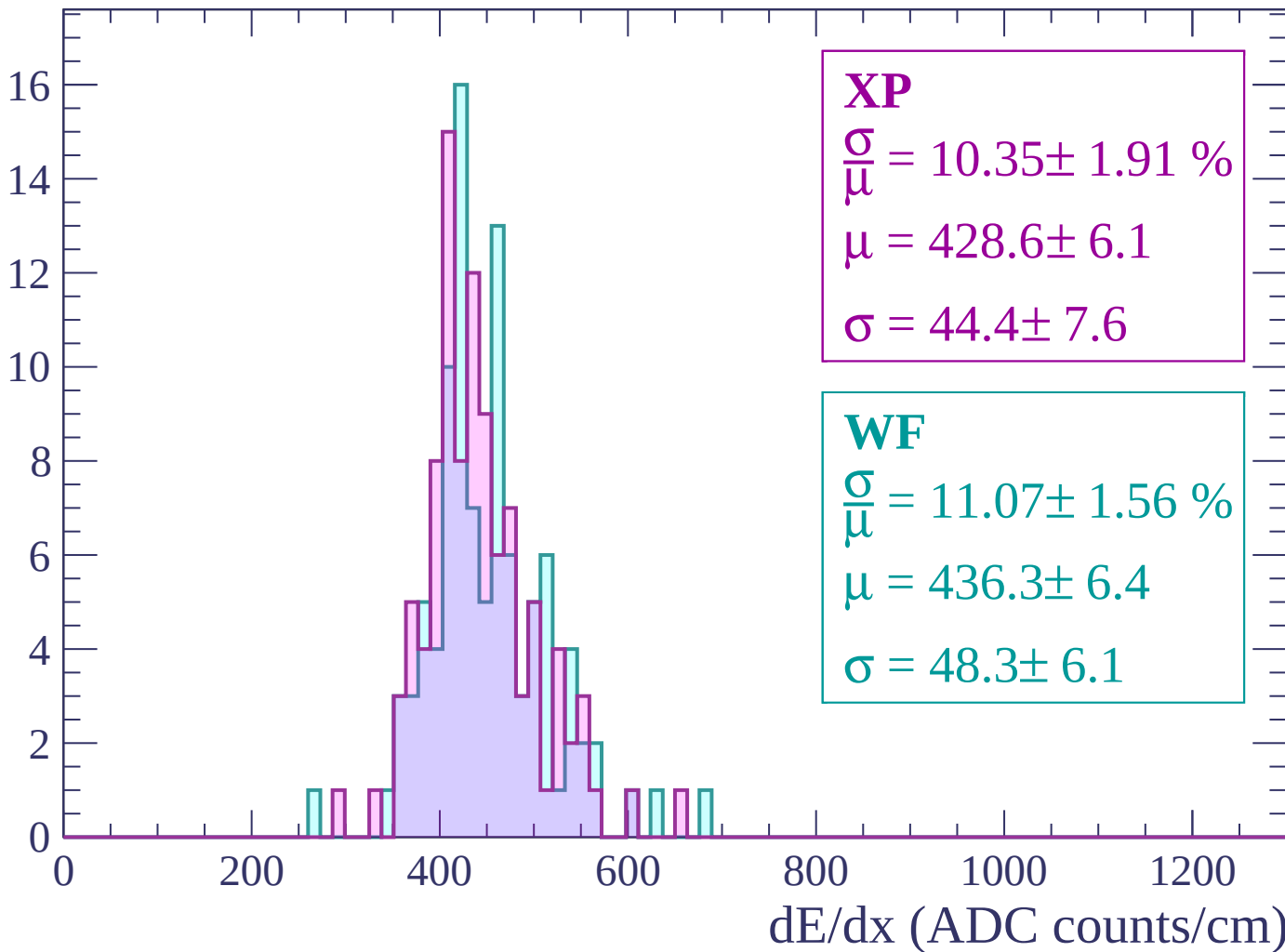
$$\mu = 427.4 \pm 7.4$$

$$\sigma = 61.5 \pm 7.9$$

$dE/dx$  (ADC counts/cm)

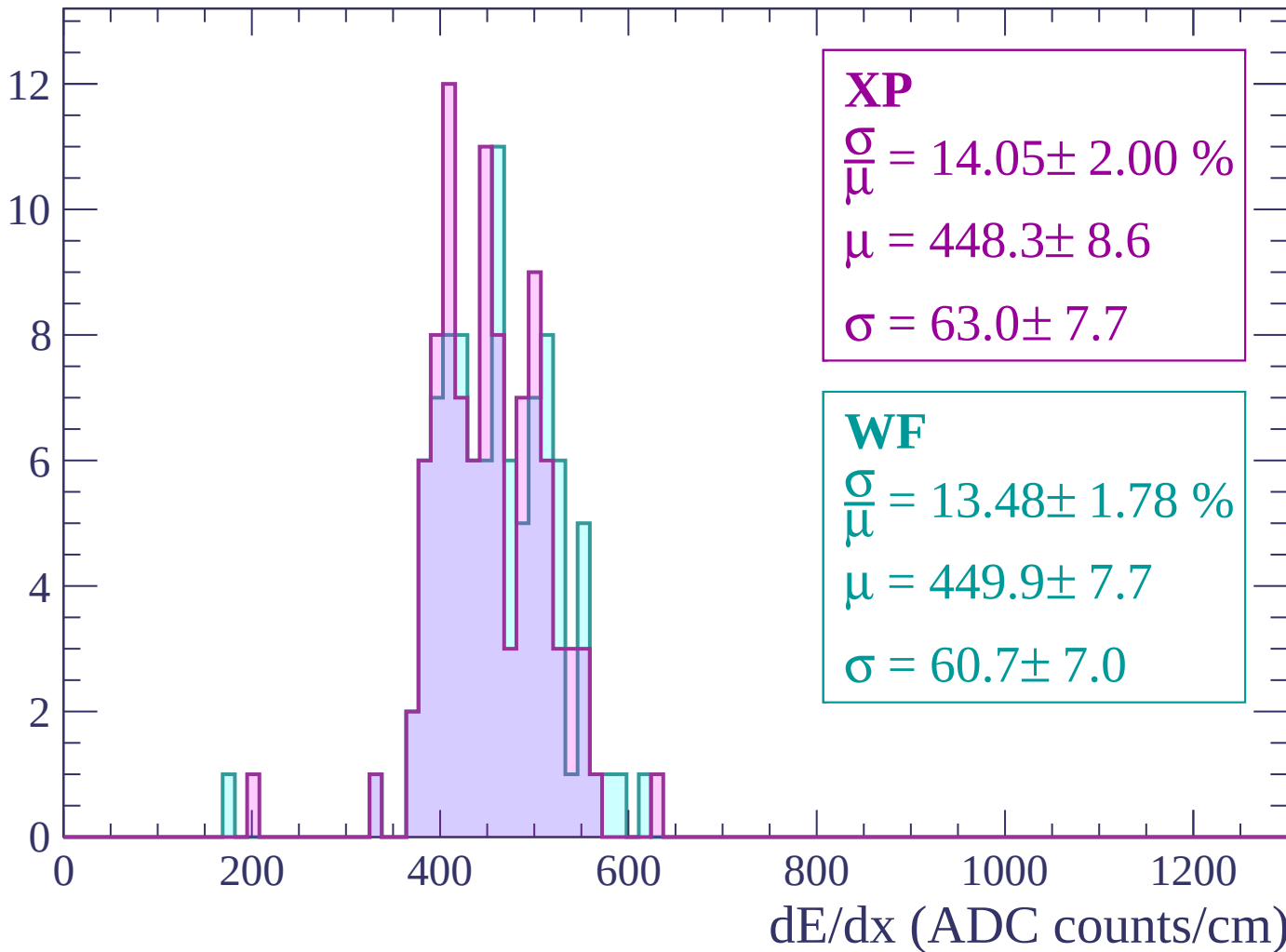
Energy loss |  $1380 < p < 1440$

Count



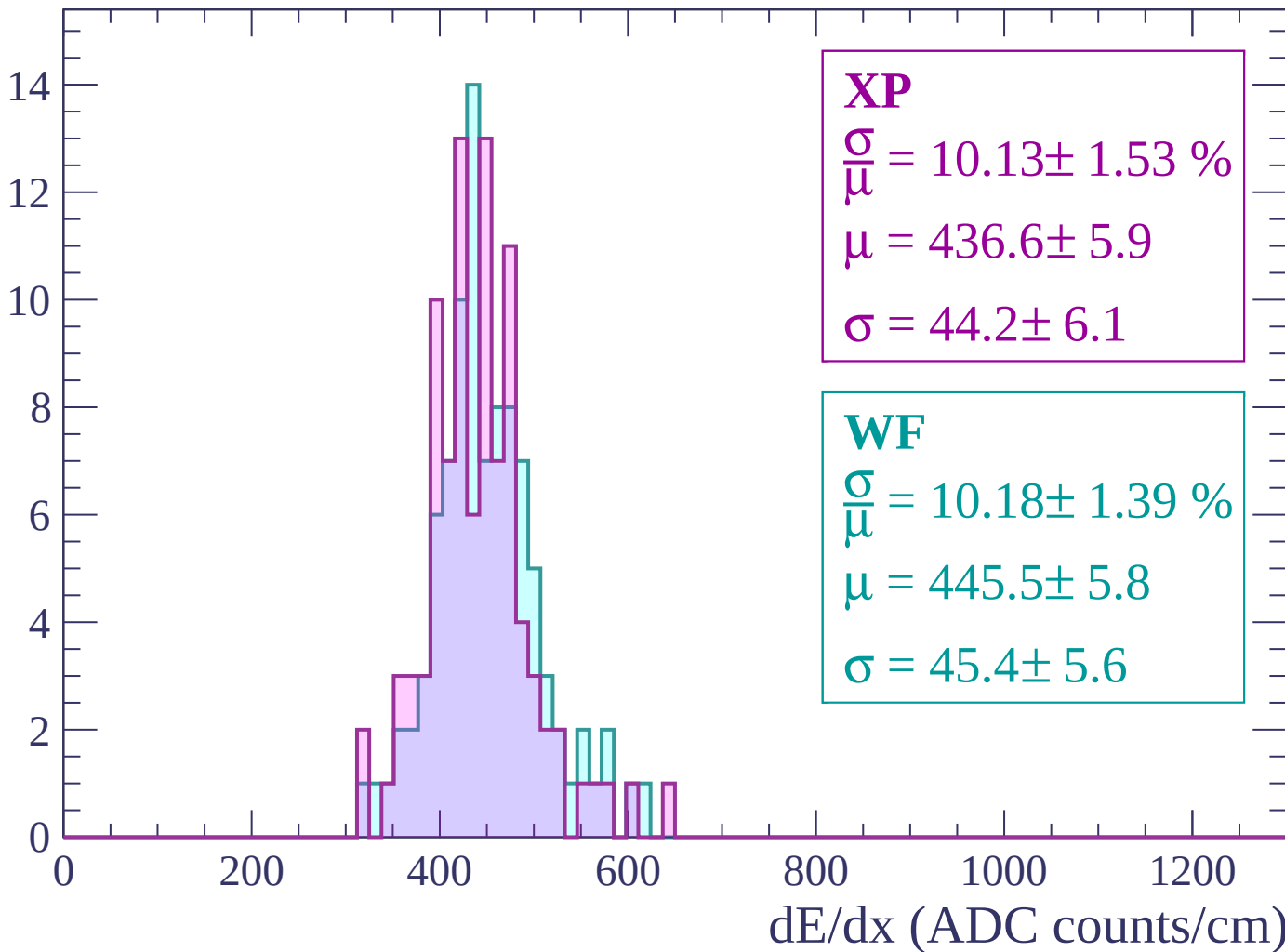
Energy loss |  $1440 < p < 1500$

Count



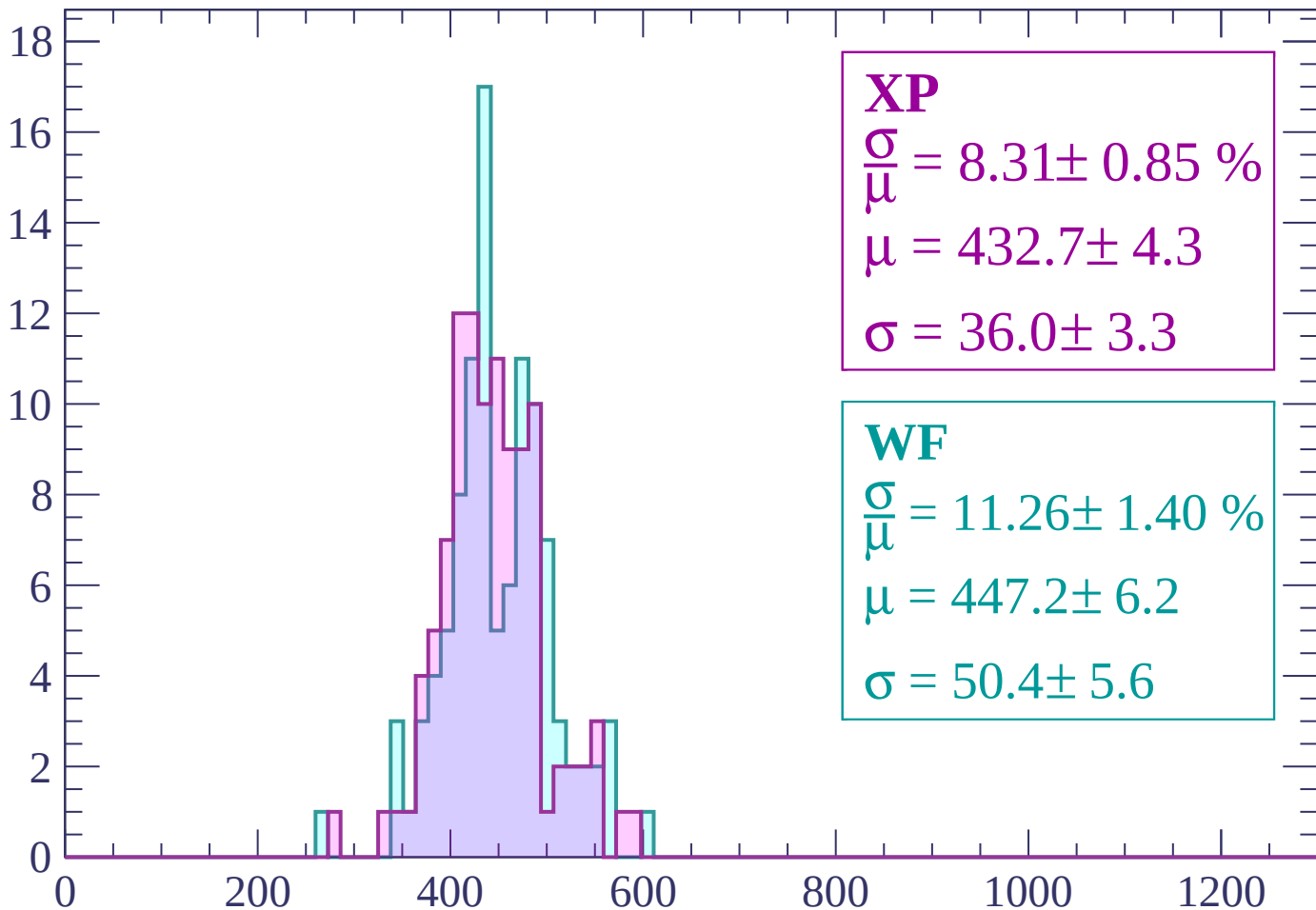
Energy loss |  $1500 < p < 1560$

Count



Energy loss |  $1560 < p < 1620$

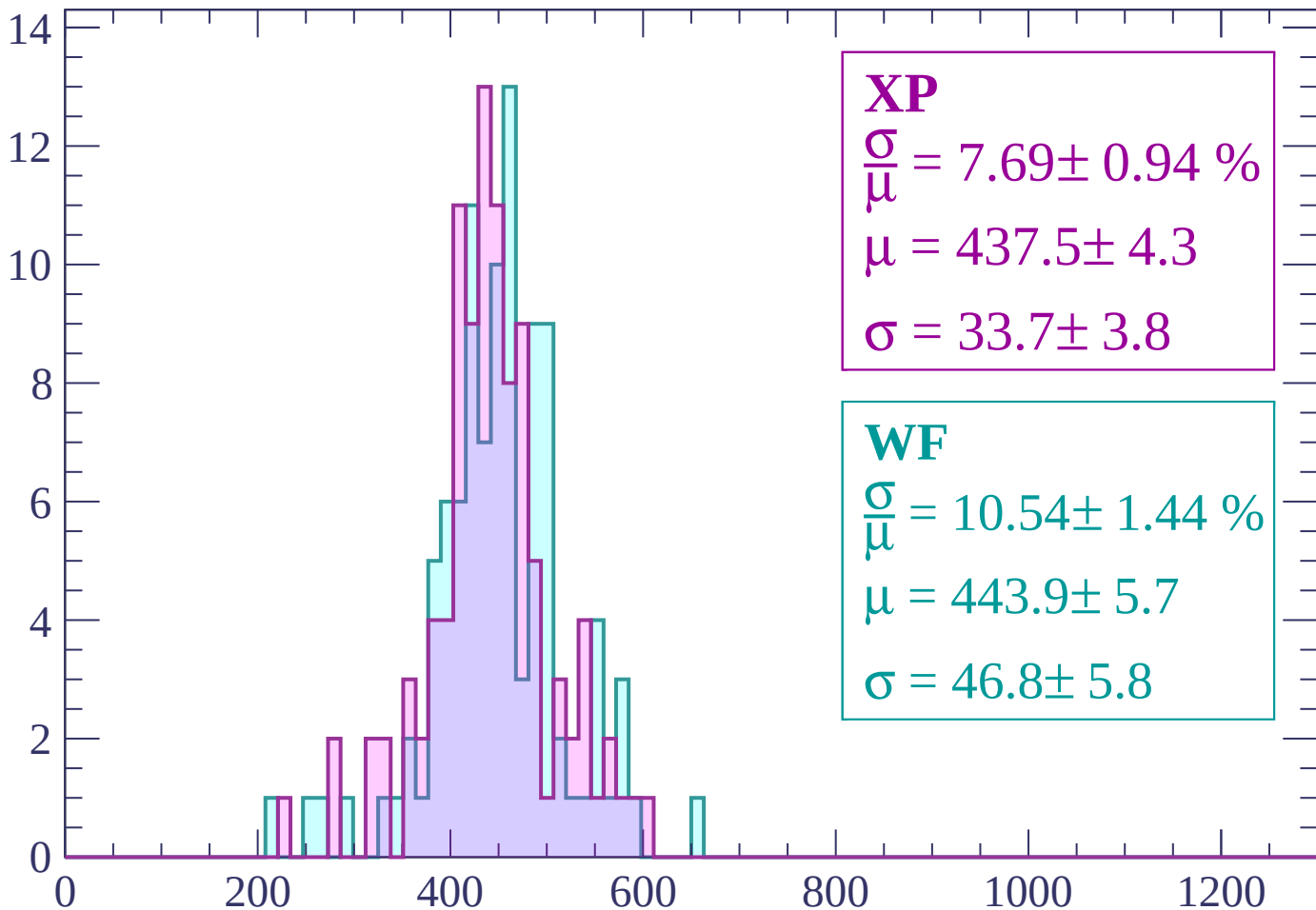
Count



dE/dx (ADC counts/cm)

Energy loss |  $1620 < p < 1680$

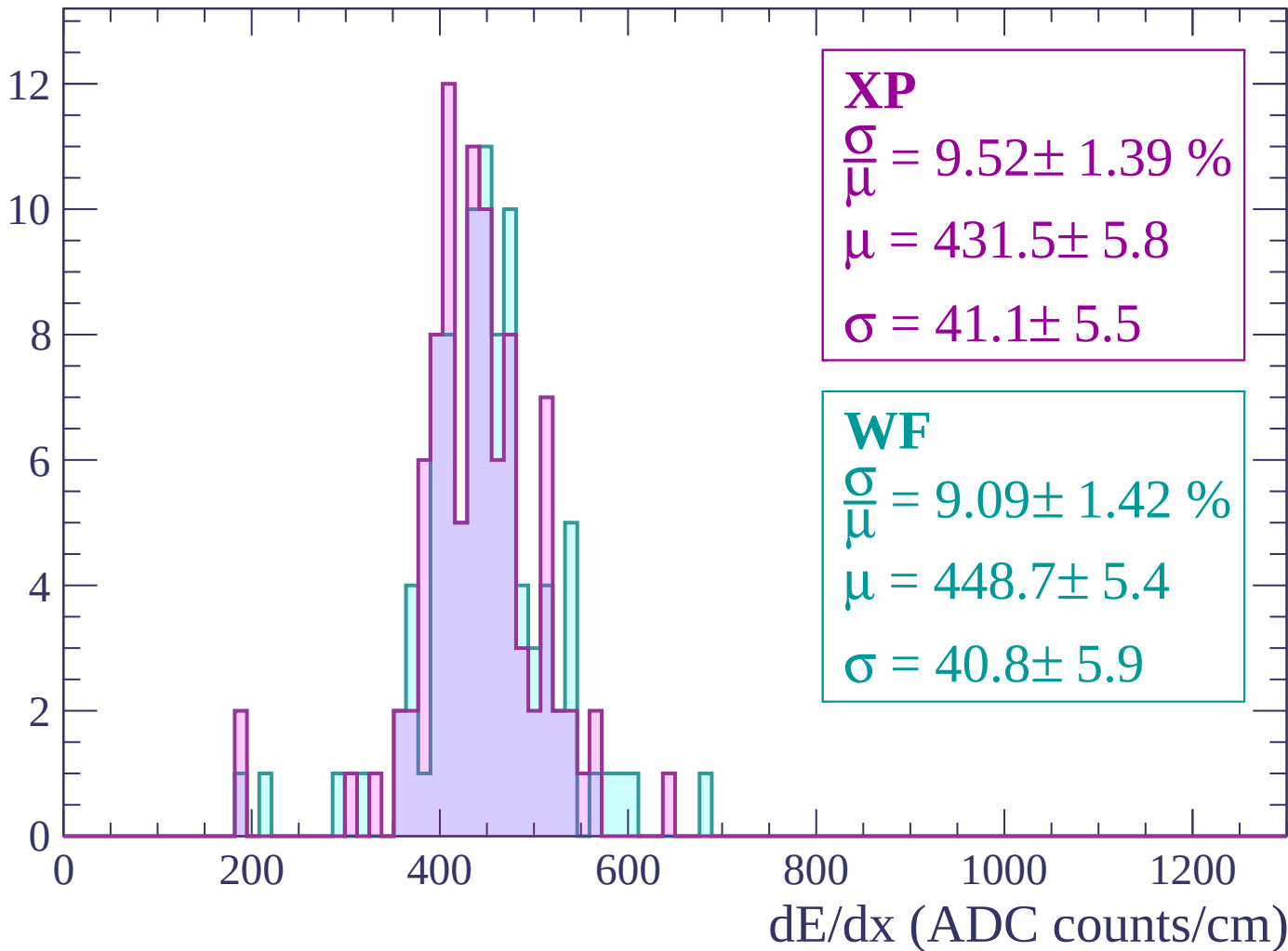
Count



dE/dx (ADC counts/cm)

Energy loss |  $1680 < p < 1740$

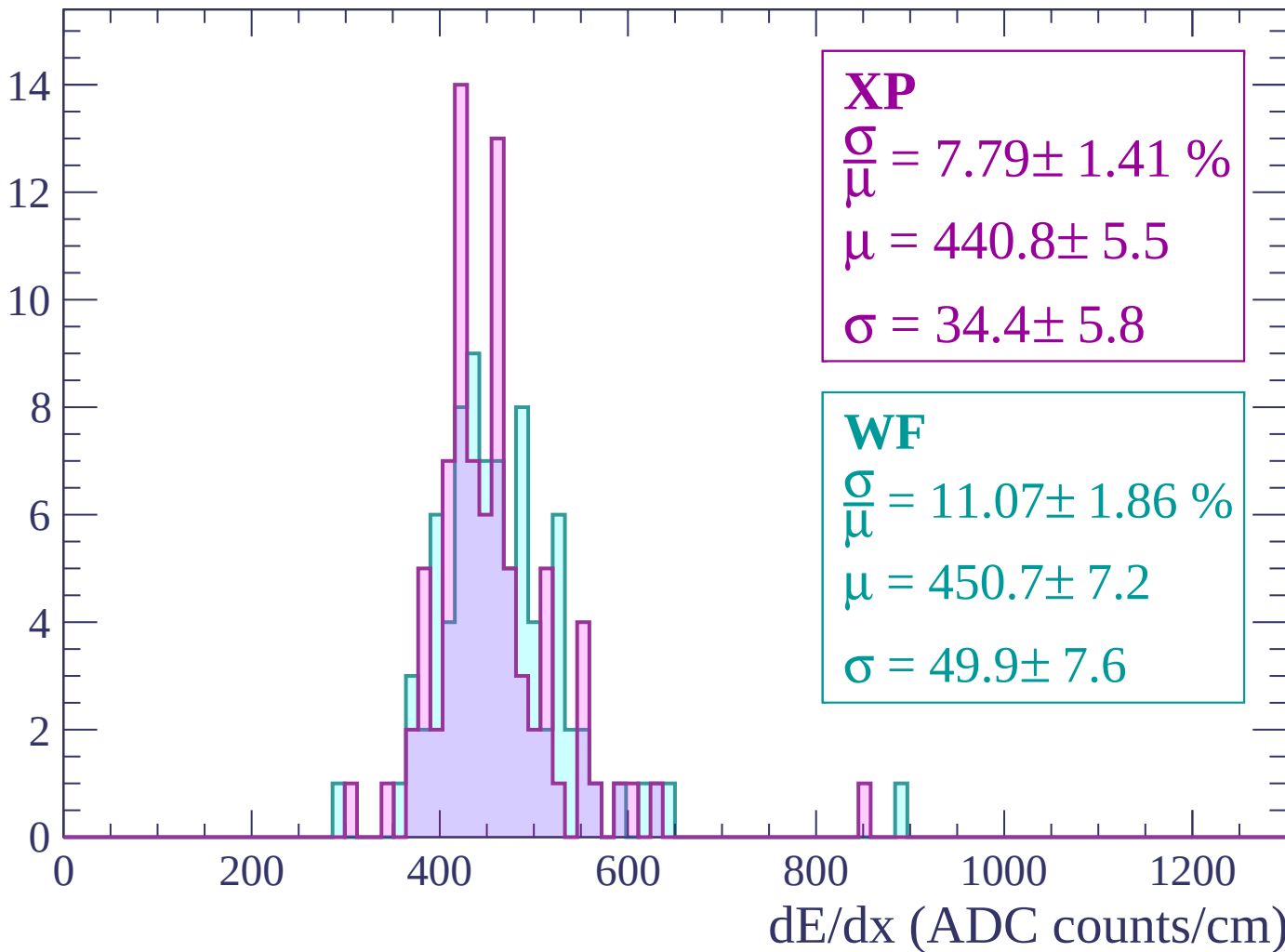
Count





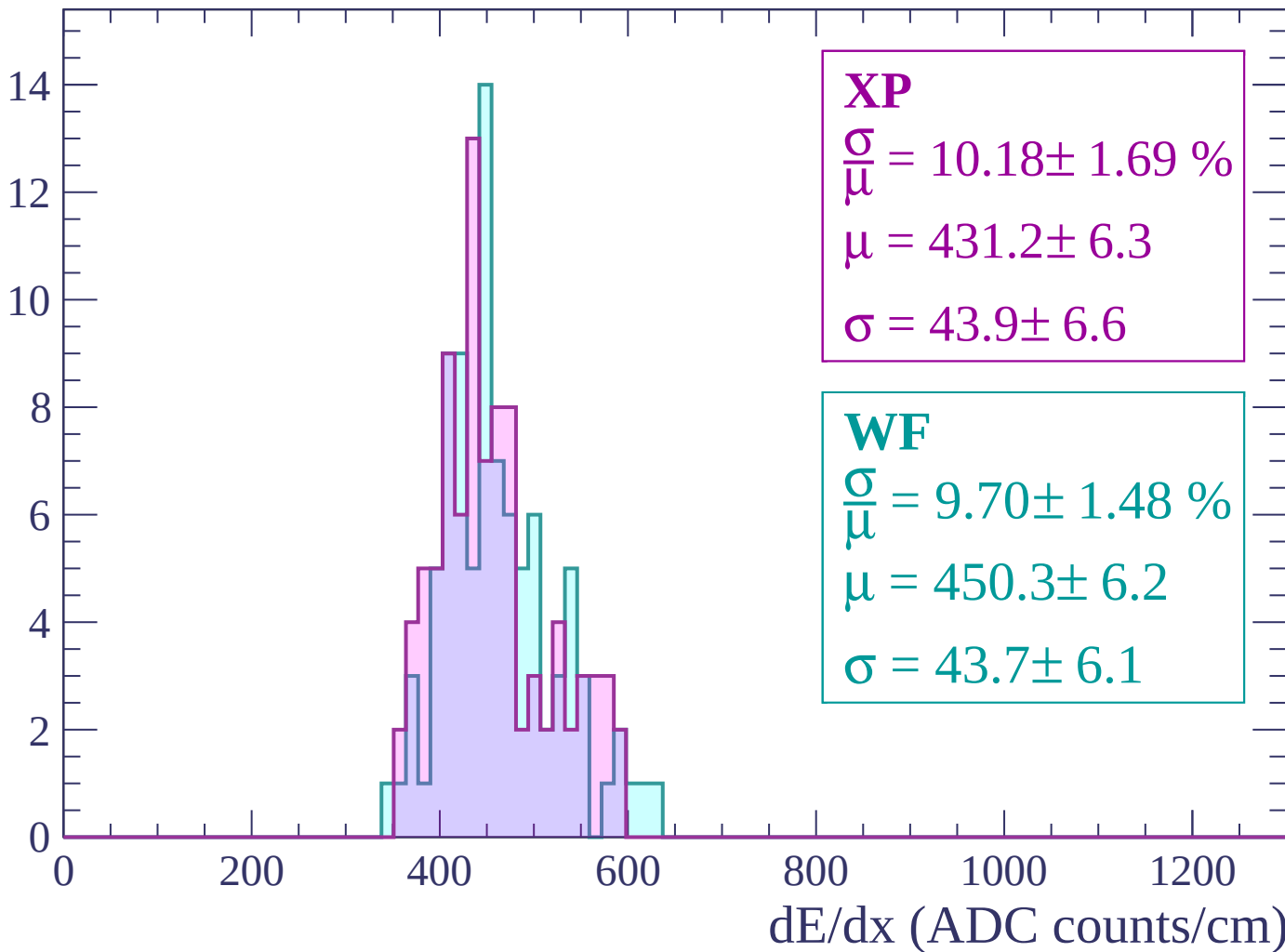
Energy loss |  $1740 < p < 1800$

Count



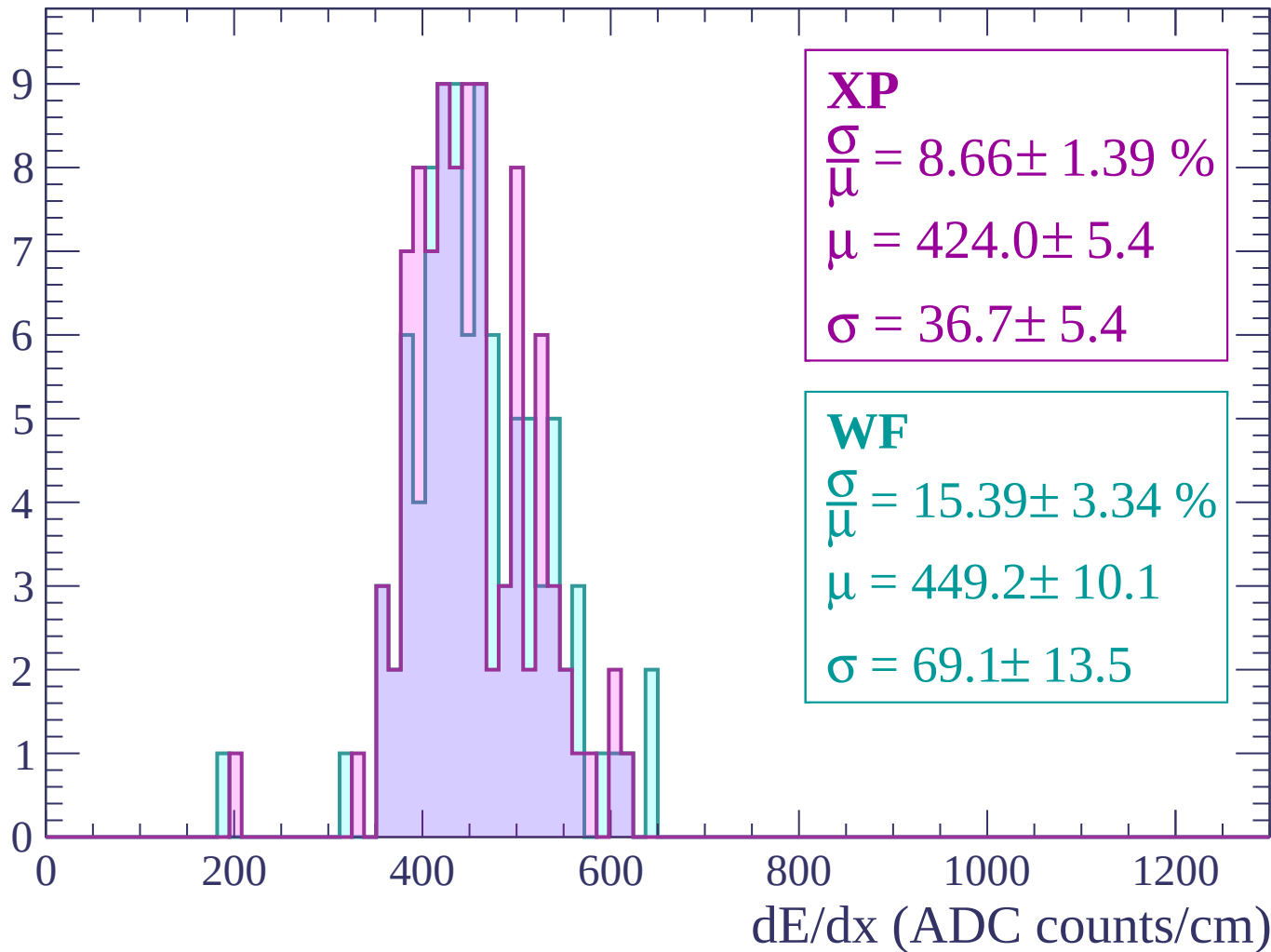
Energy loss |  $1800 < p < 1860$

Count



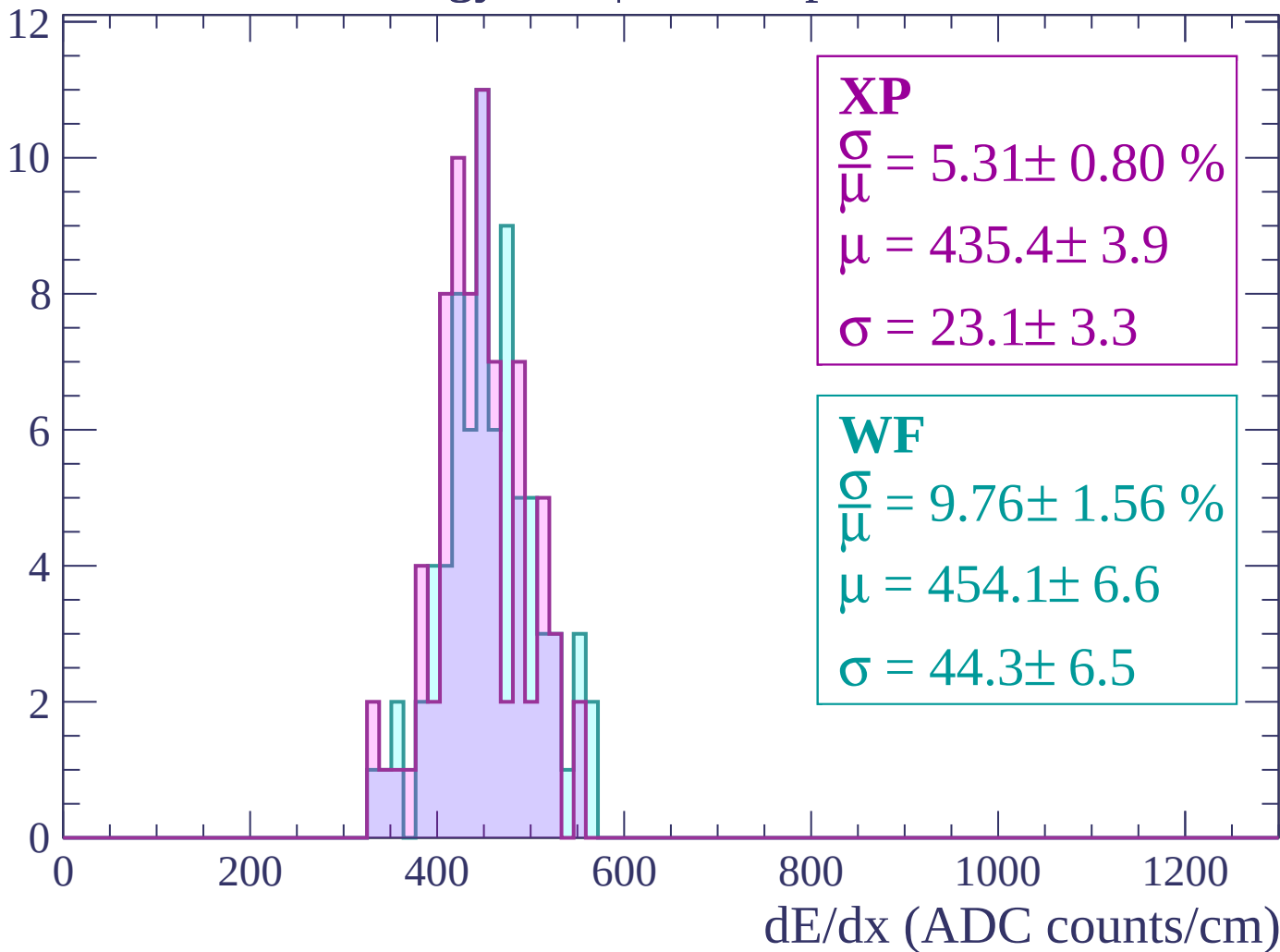
Energy loss |  $1860 < p < 1920$

Count



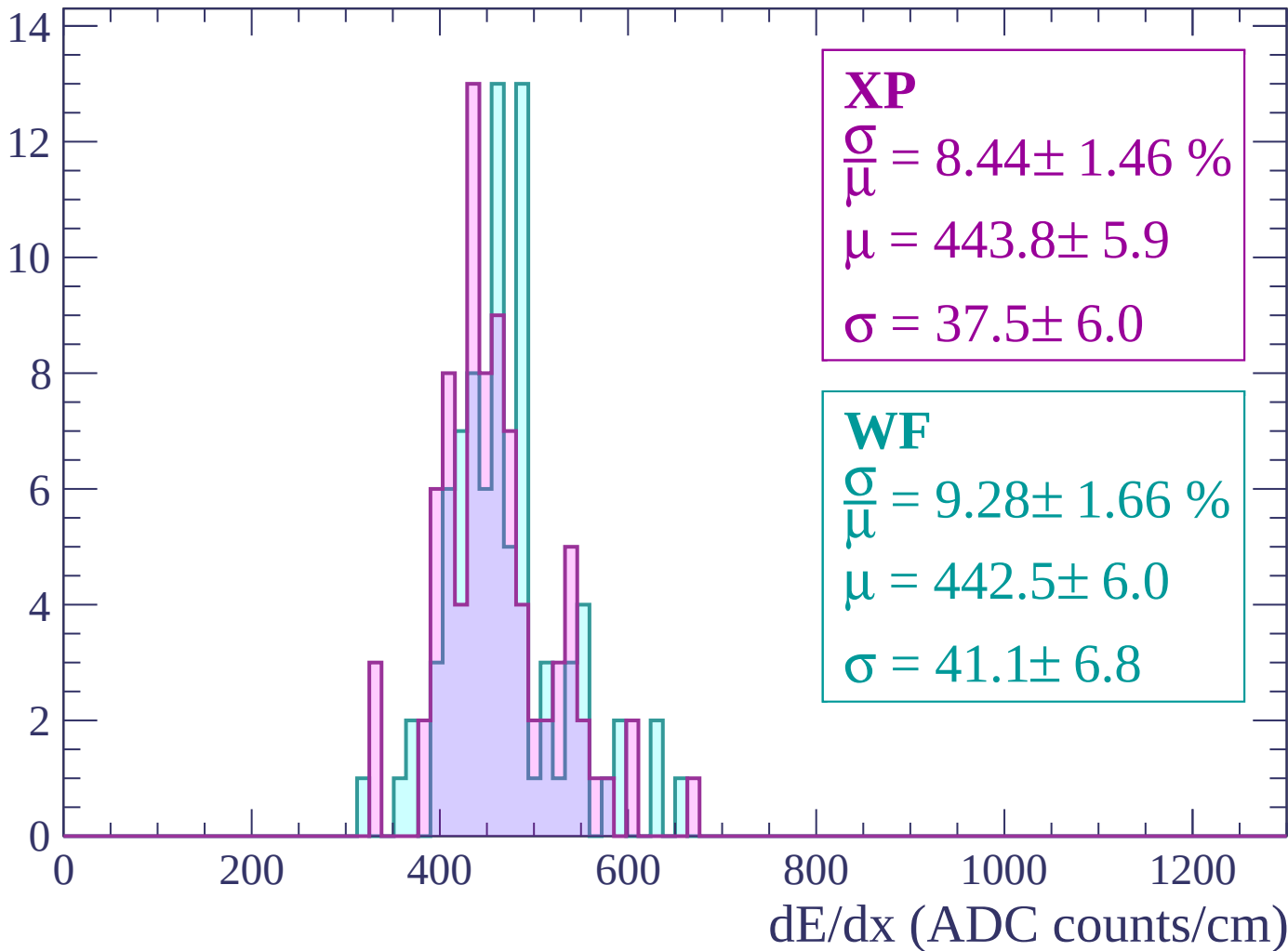
Energy loss |  $1920 < p < 1980$

Count



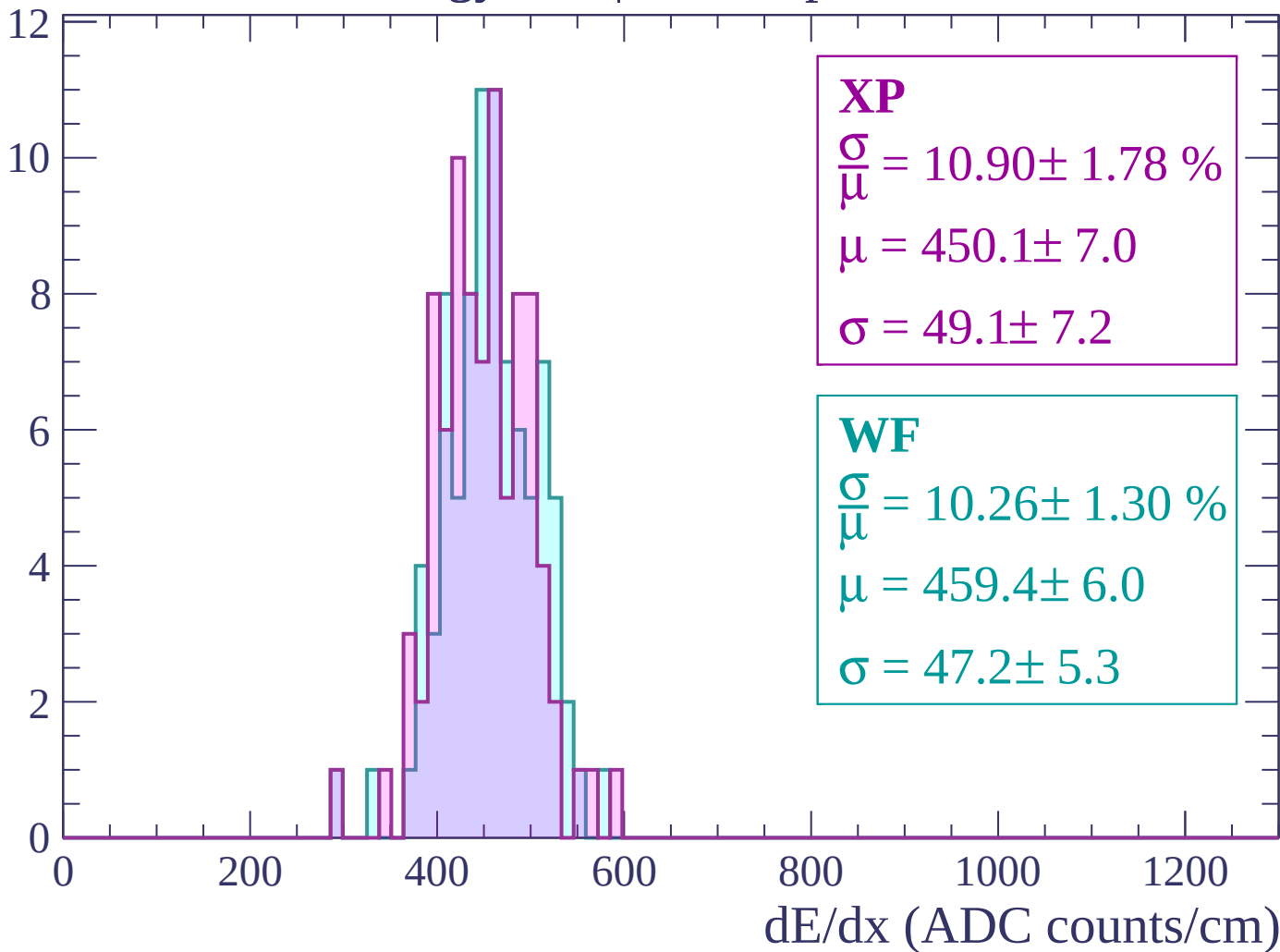
# Energy loss | $1980 < p < 2040$

Count



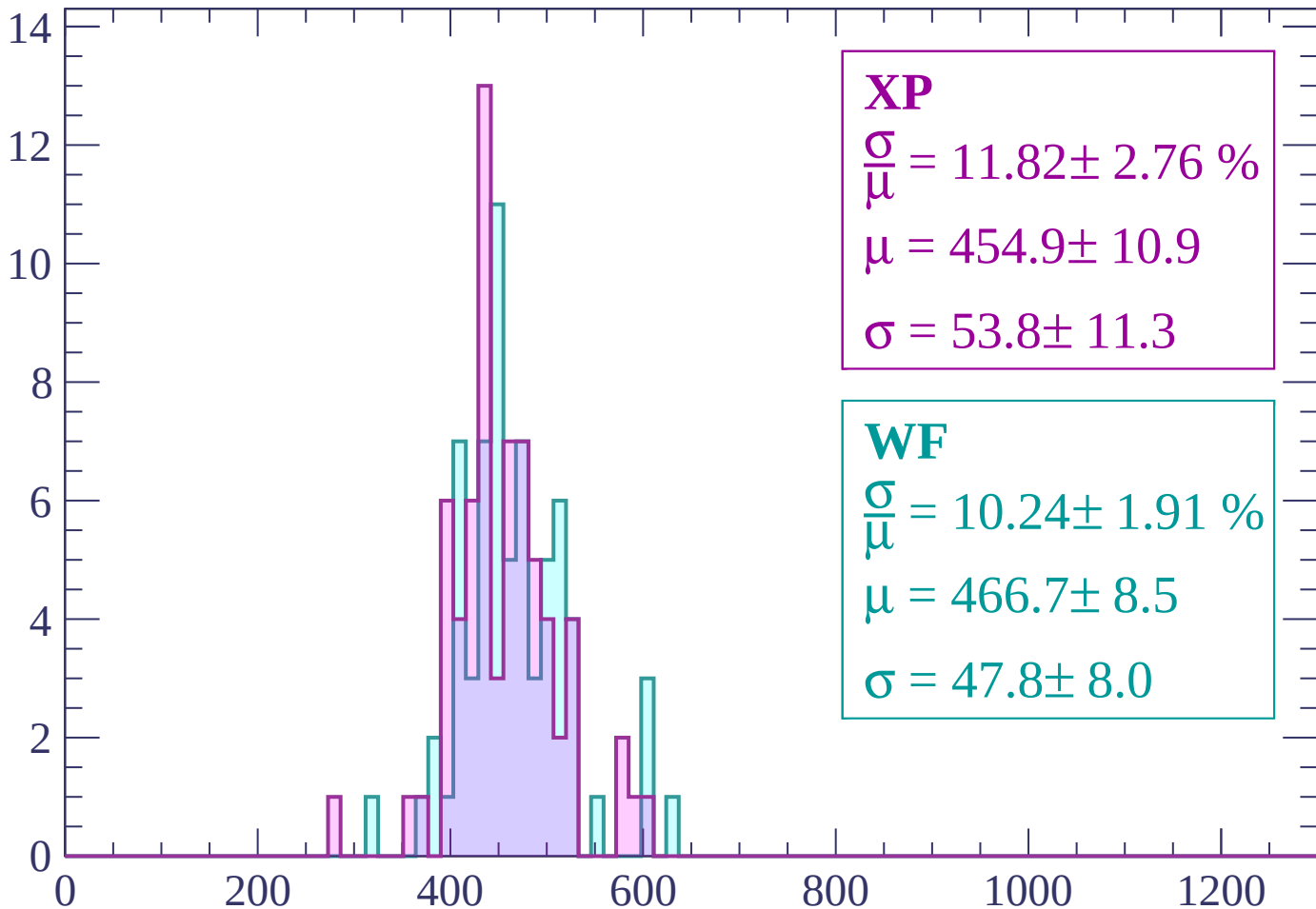
Energy loss |  $2040 < p < 2100$

Count



Energy loss |  $2100 < p < 2160$

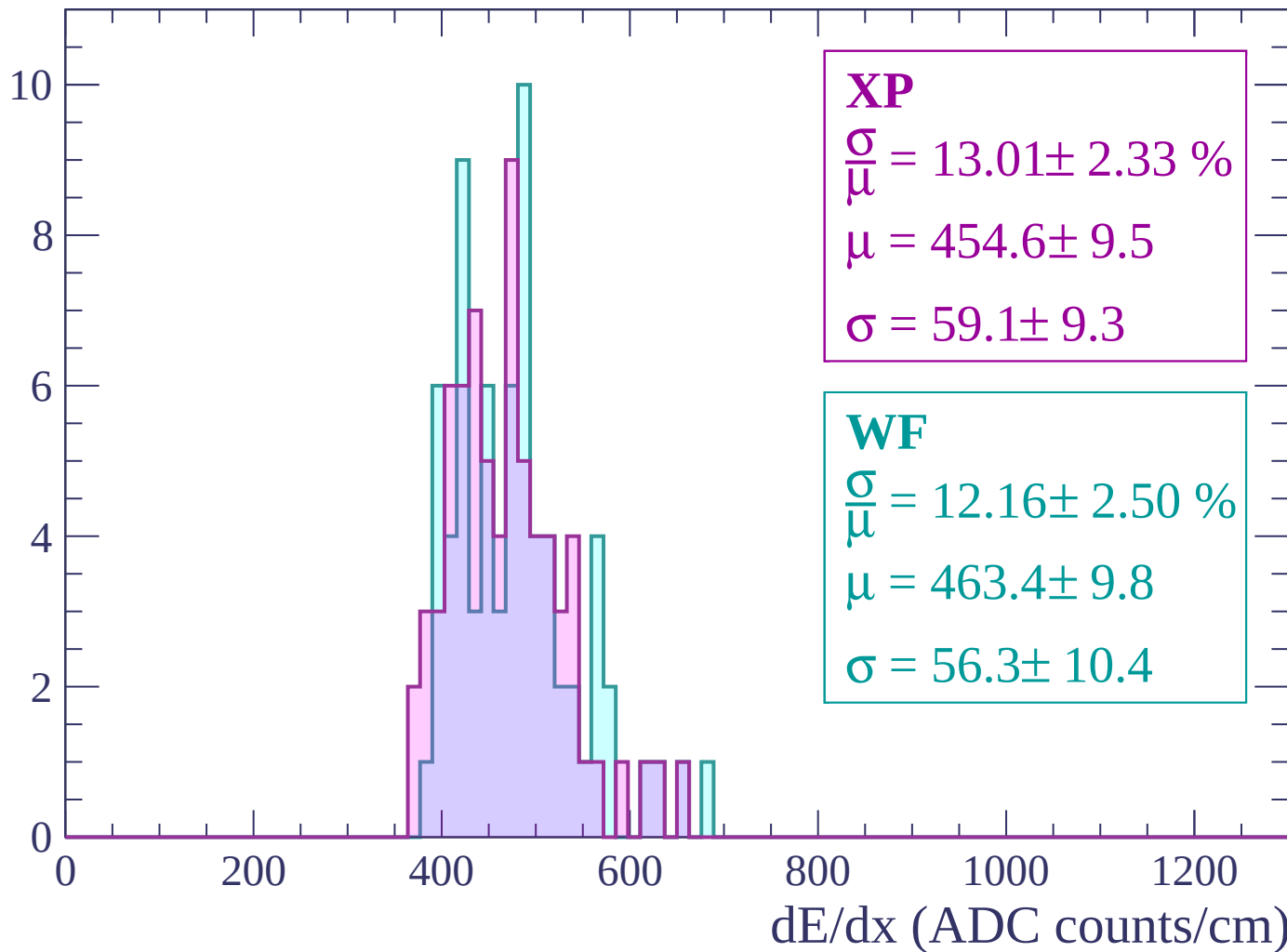
Count



dE/dx (ADC counts/cm)

# Energy loss | $2160 < p < 2220$

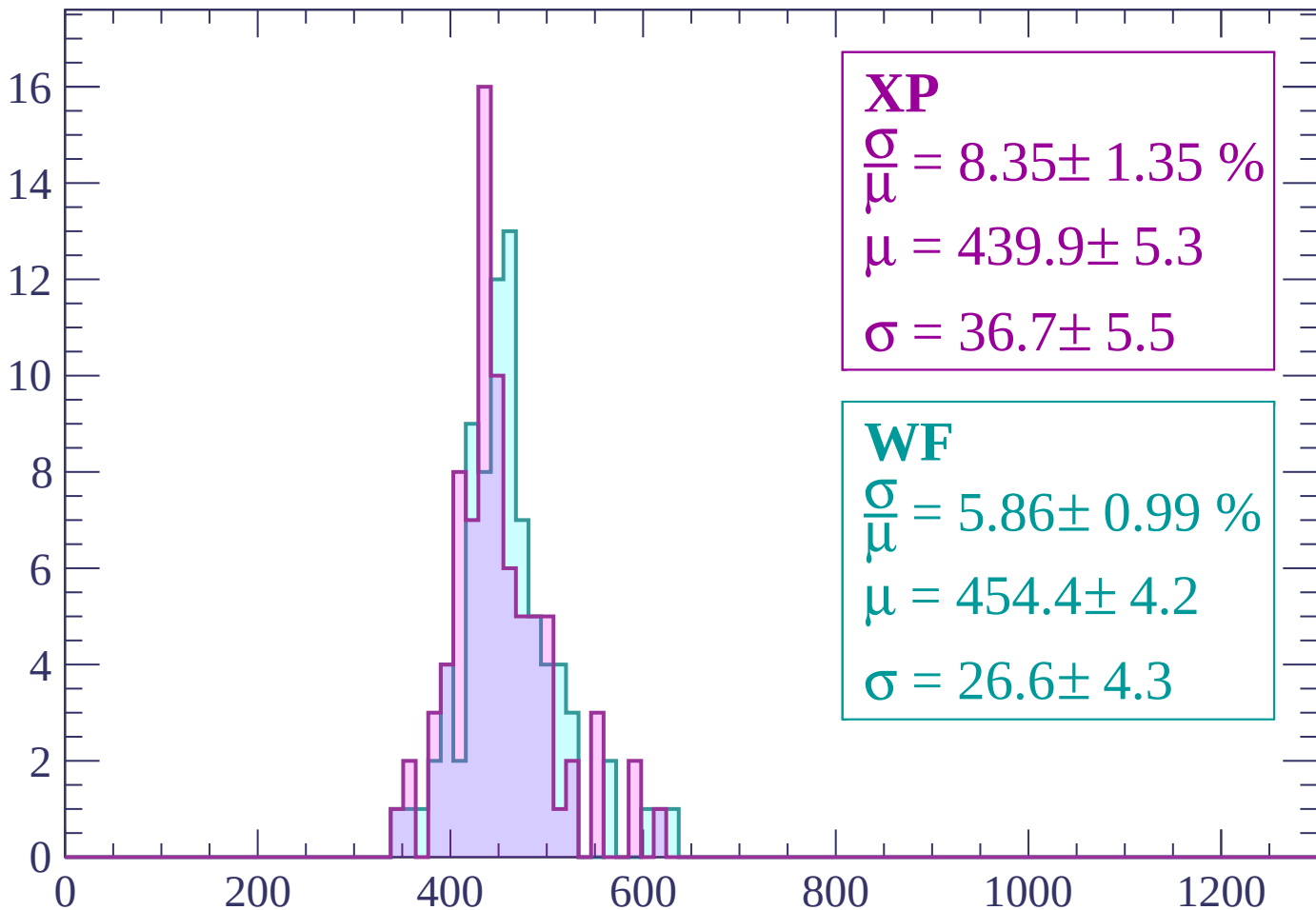
Count





Energy loss |  $2220 < p < 2280$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.35 \pm 1.35 \%$$

$$\mu = 439.9 \pm 5.3$$

$$\sigma = 36.7 \pm 5.5$$

**WF**

$$\frac{\sigma}{\mu} = 5.86 \pm 0.99 \%$$

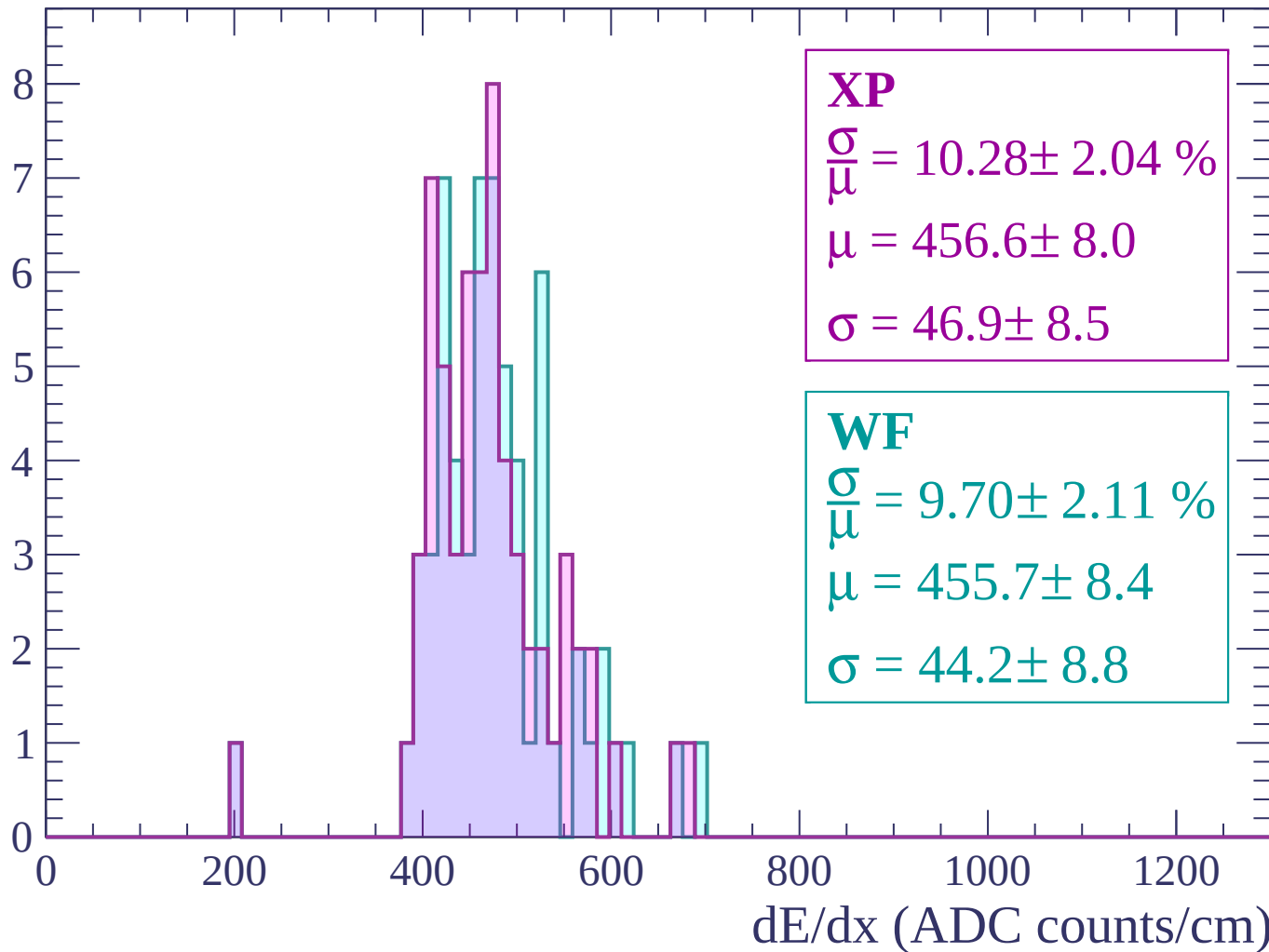
$$\mu = 454.4 \pm 4.2$$

$$\sigma = 26.6 \pm 4.3$$

$dE/dx$  (ADC counts/cm)

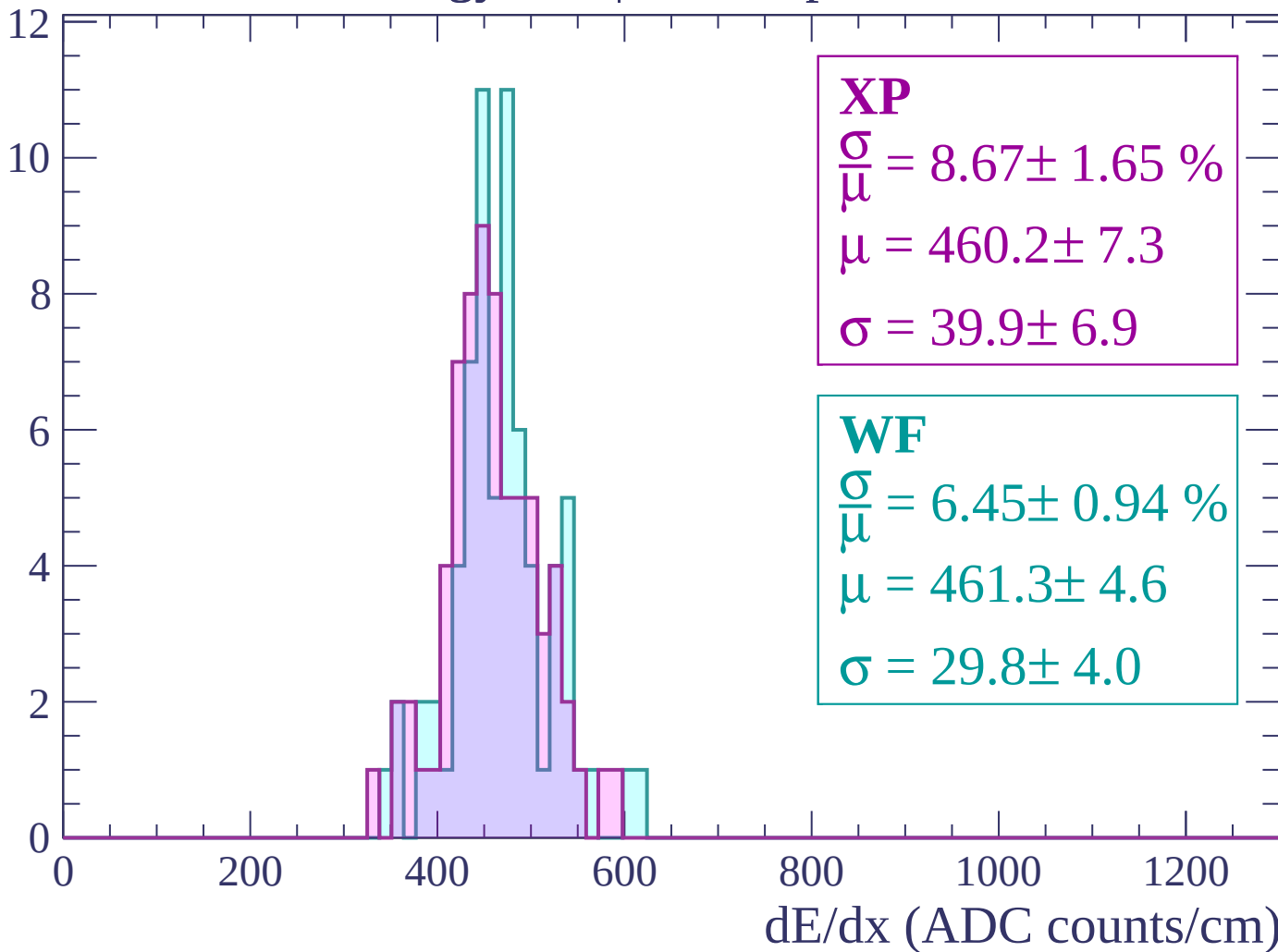
Energy loss |  $2280 < p < 2340$

Count



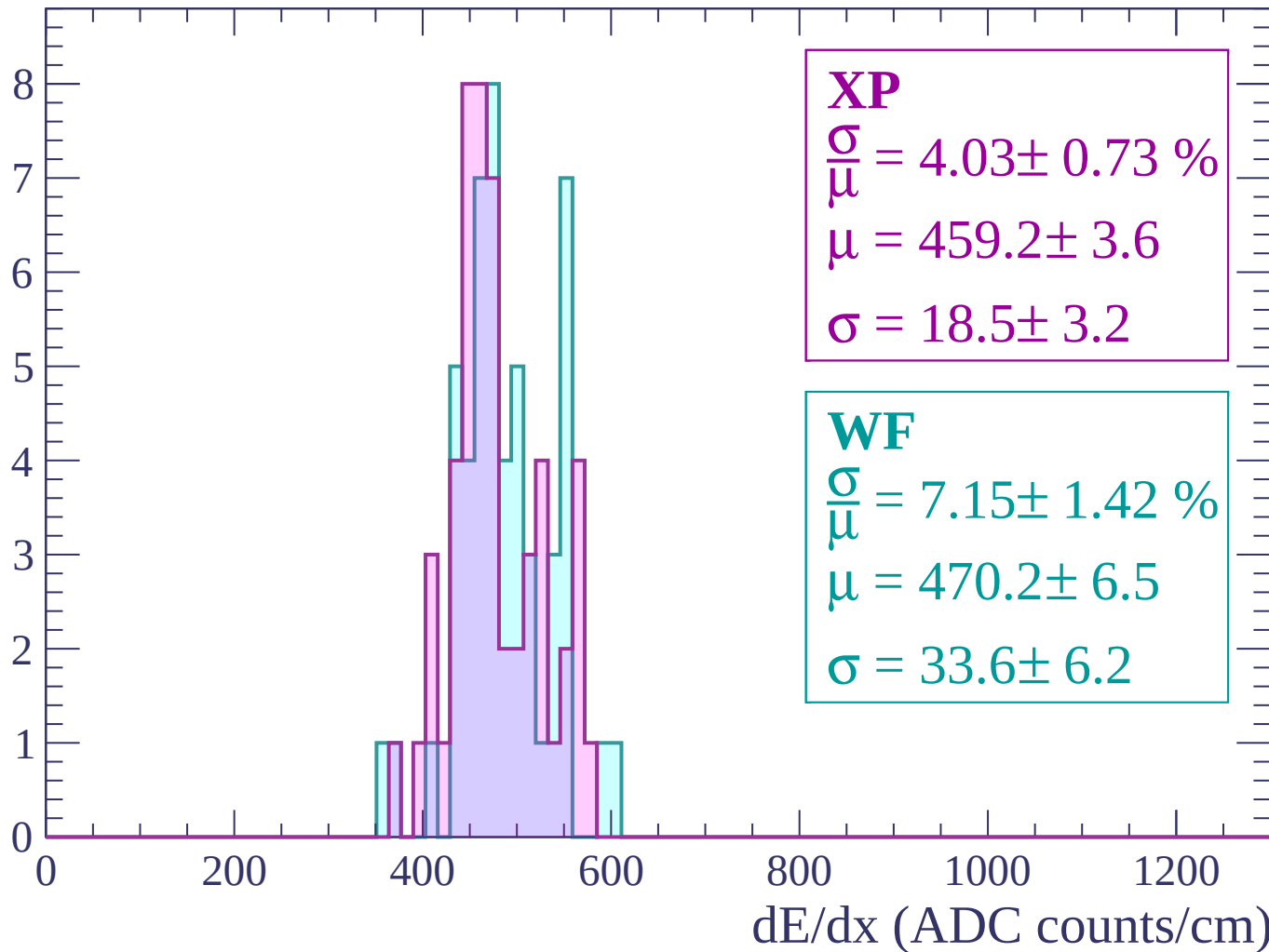
Energy loss |  $2340 < p < 2400$

Count



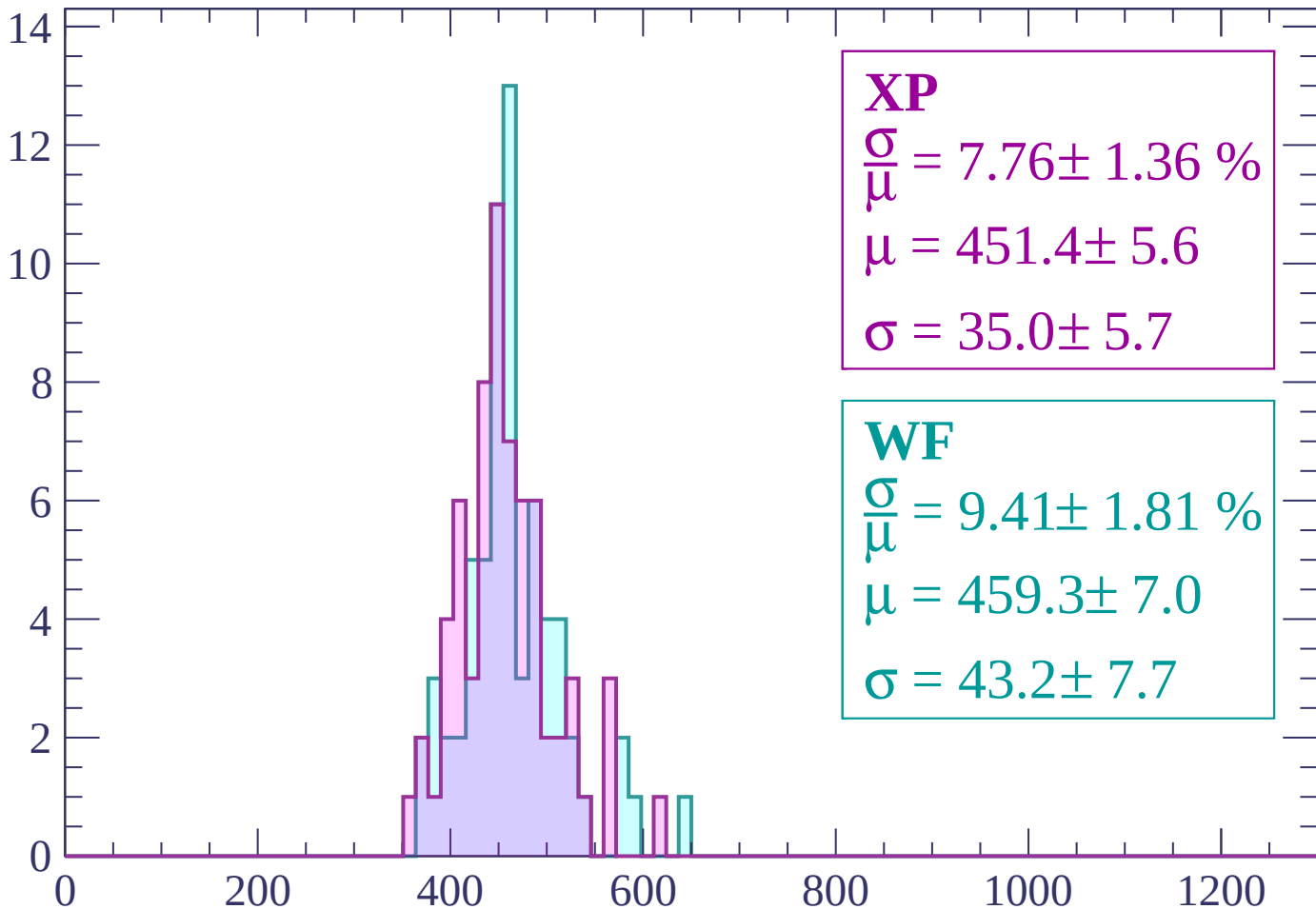
Energy loss |  $2400 < p < 2460$

Count



Energy loss |  $2460 < p < 2520$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.76 \pm 1.36 \%$$

$$\mu = 451.4 \pm 5.6$$

$$\sigma = 35.0 \pm 5.7$$

**WF**

$$\frac{\sigma}{\mu} = 9.41 \pm 1.81 \%$$

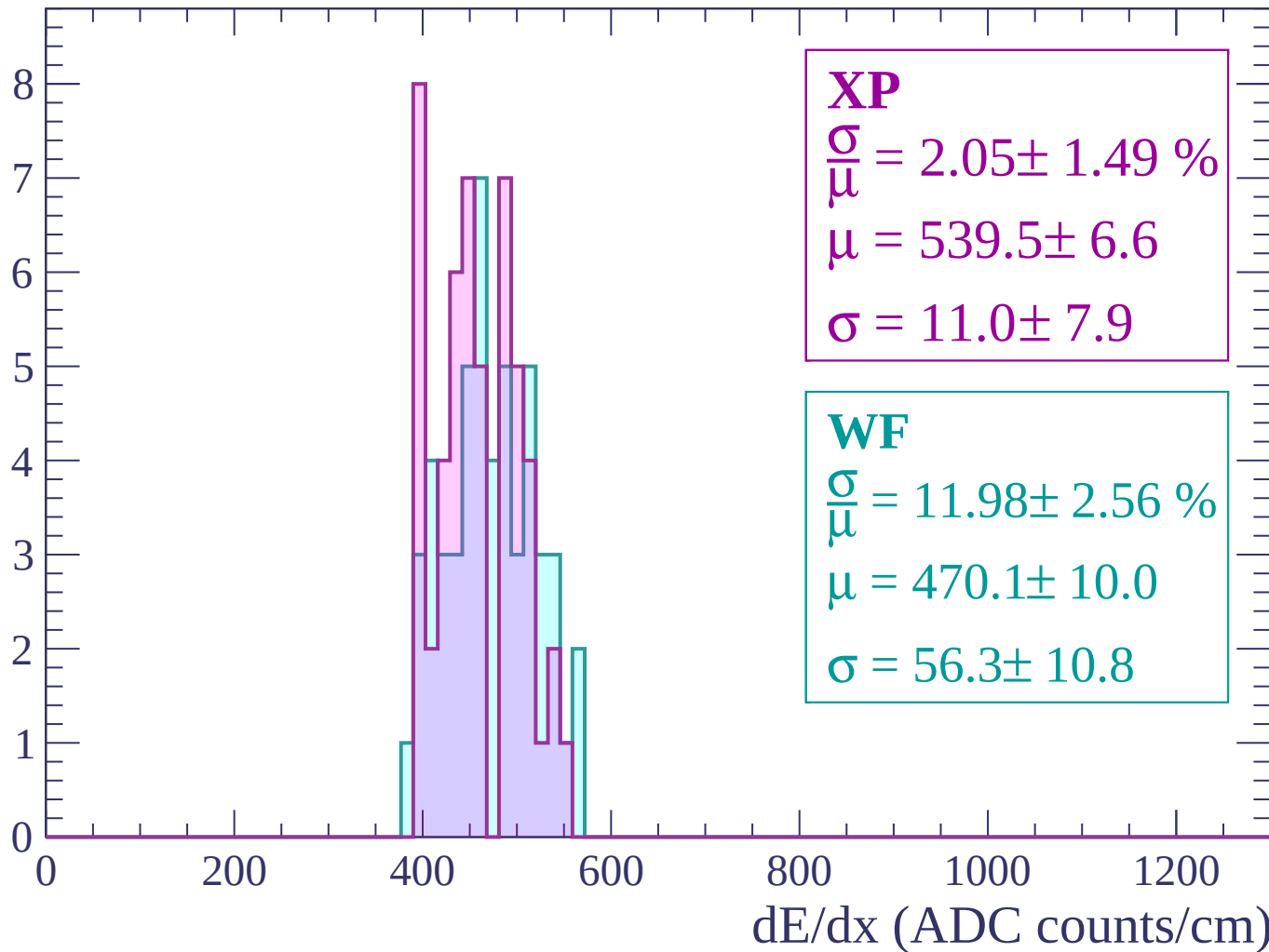
$$\mu = 459.3 \pm 7.0$$

$$\sigma = 43.2 \pm 7.7$$

$dE/dx$  (ADC counts/cm)

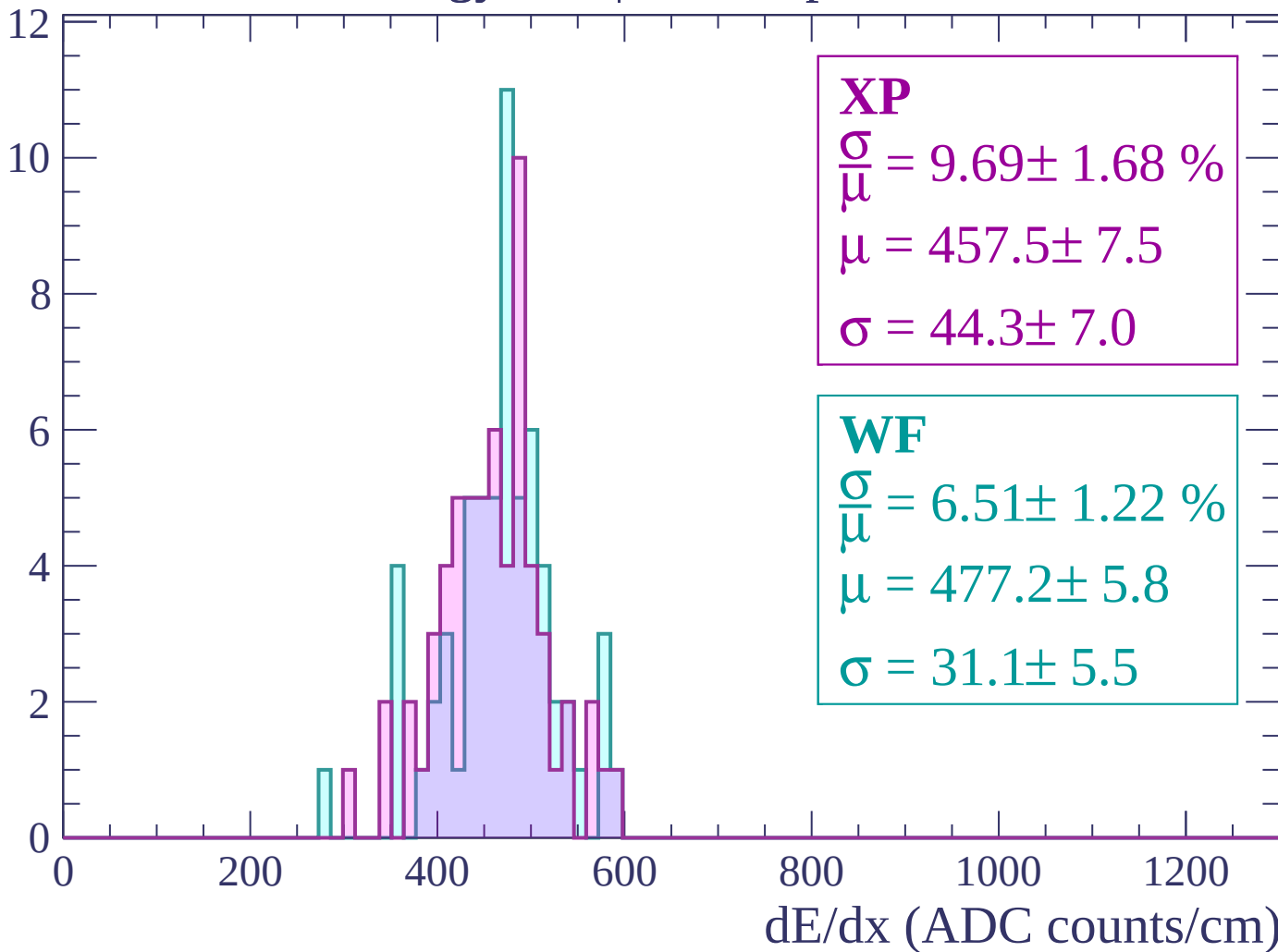
Energy loss |  $2520 < p < 2580$

Count



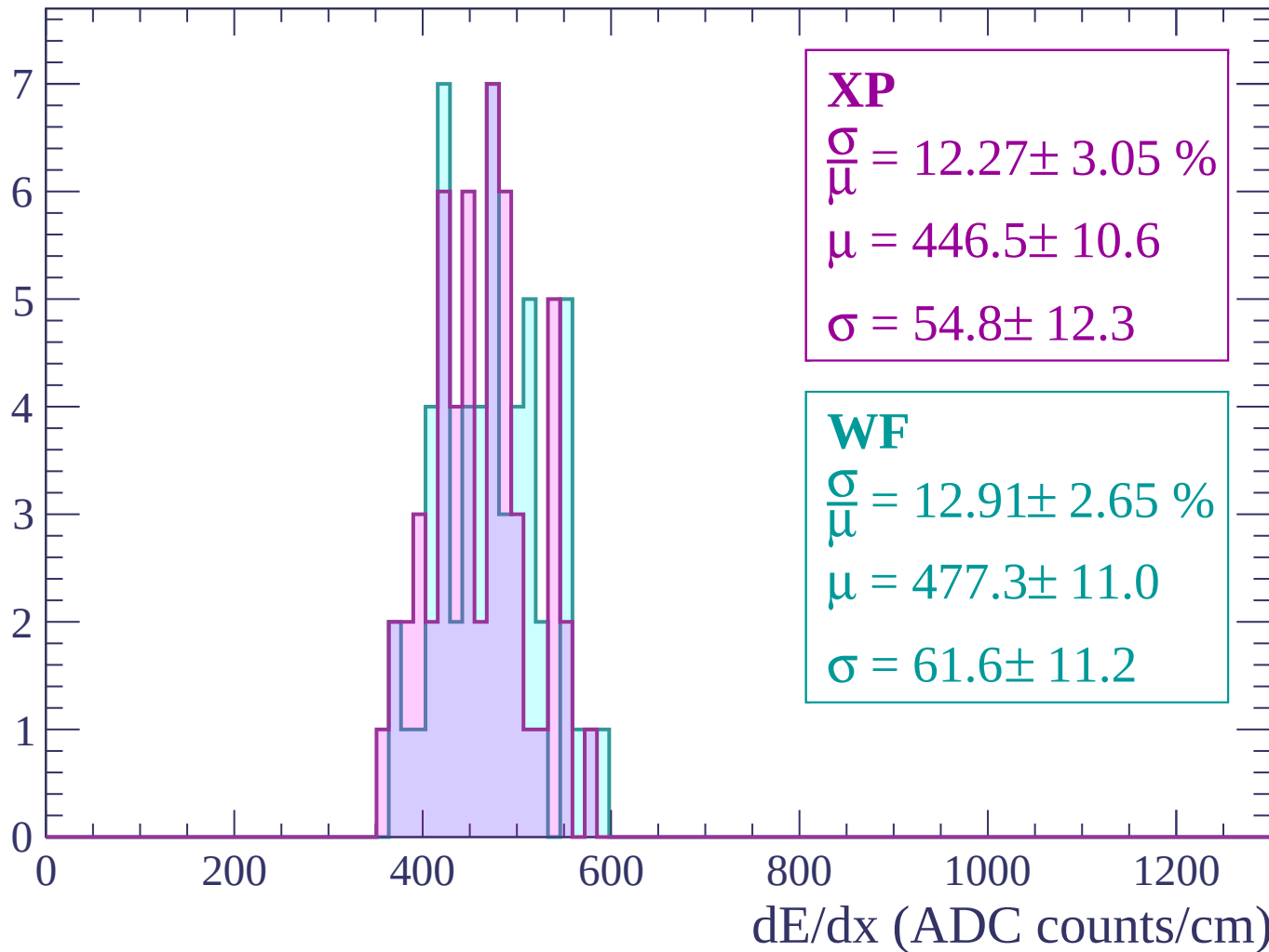
Energy loss |  $2580 < p < 2640$

Count



Energy loss |  $2640 < p < 2700$

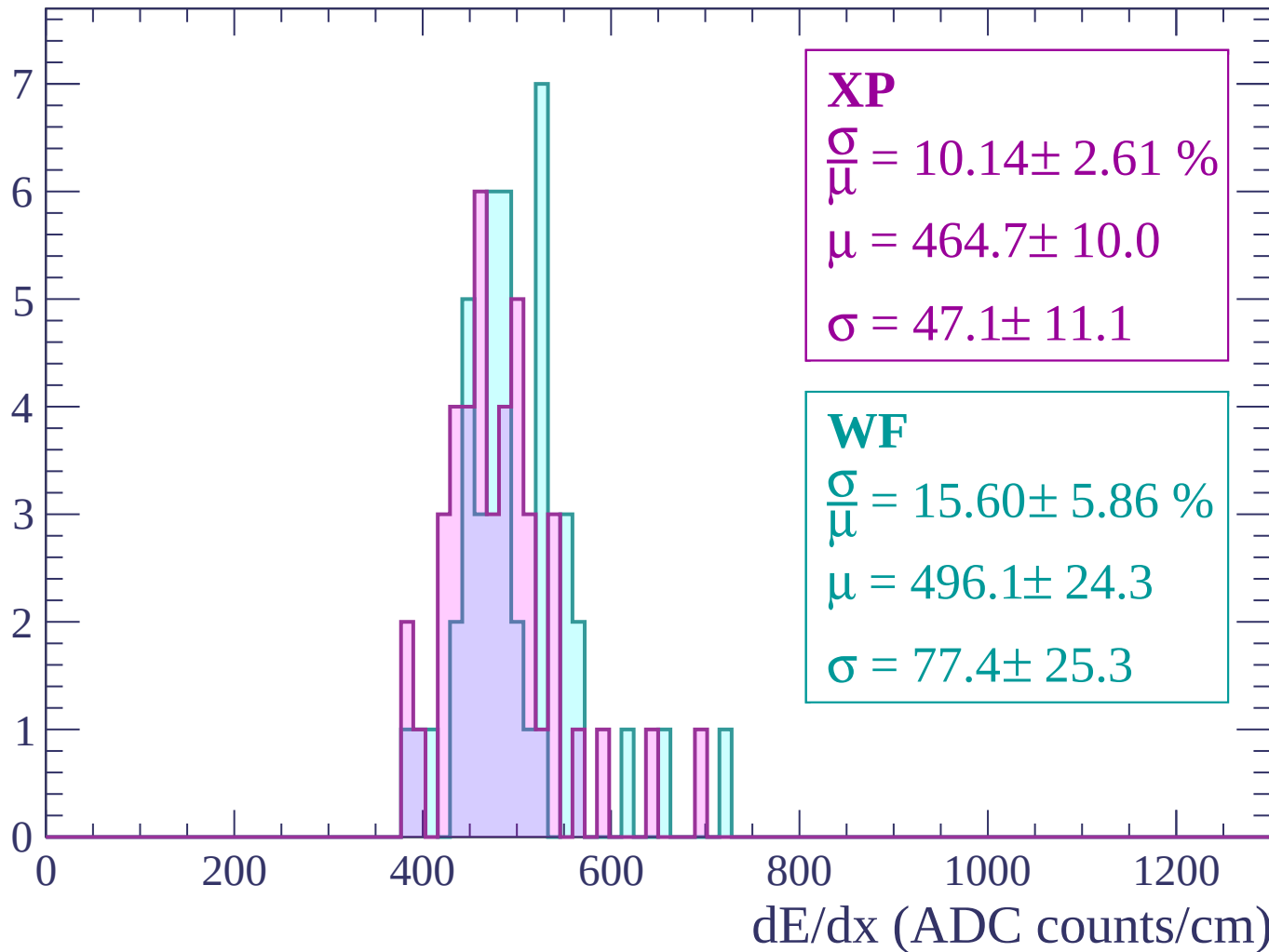
Count





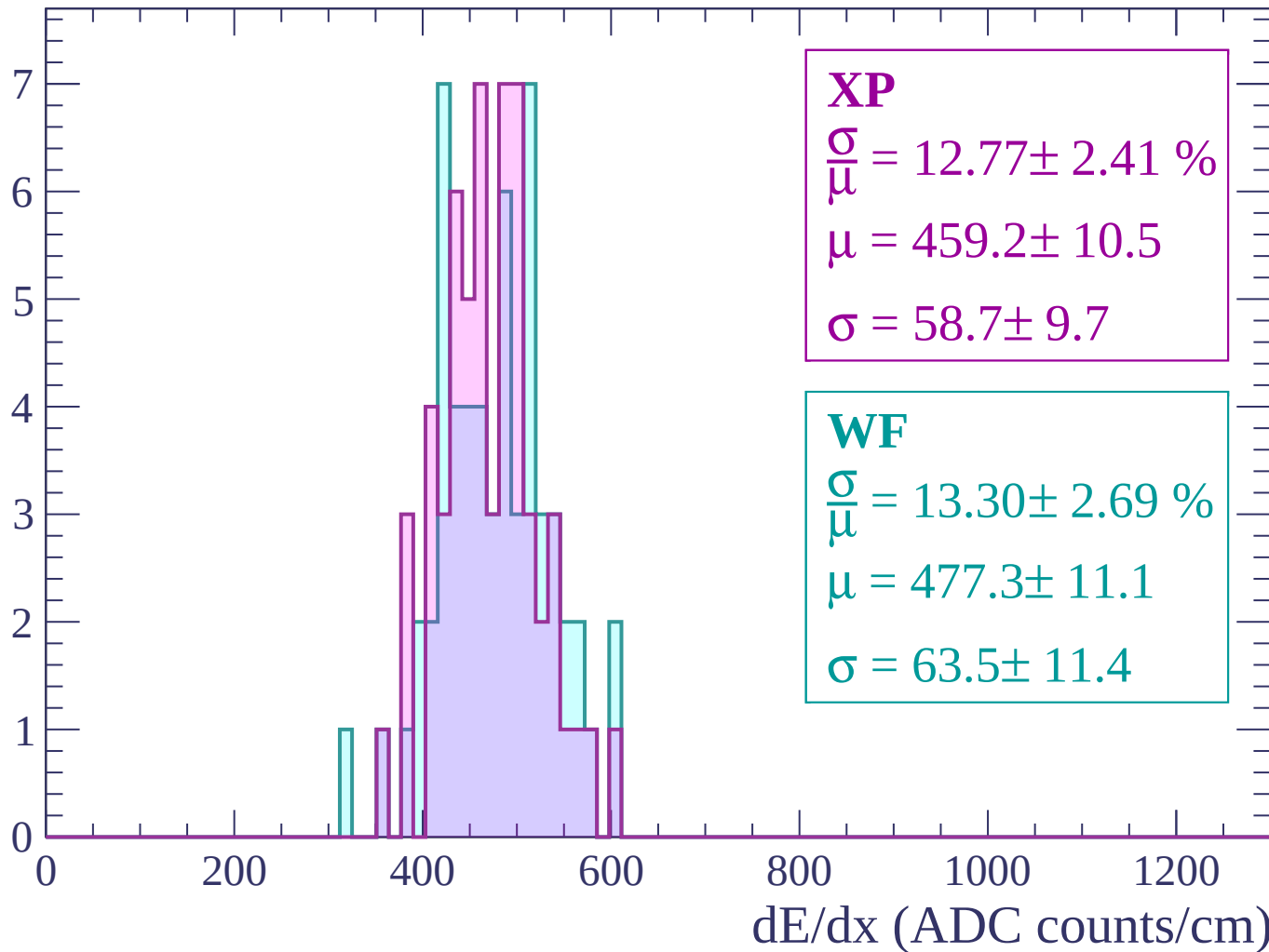
Energy loss |  $2700 < p < 2760$

Count



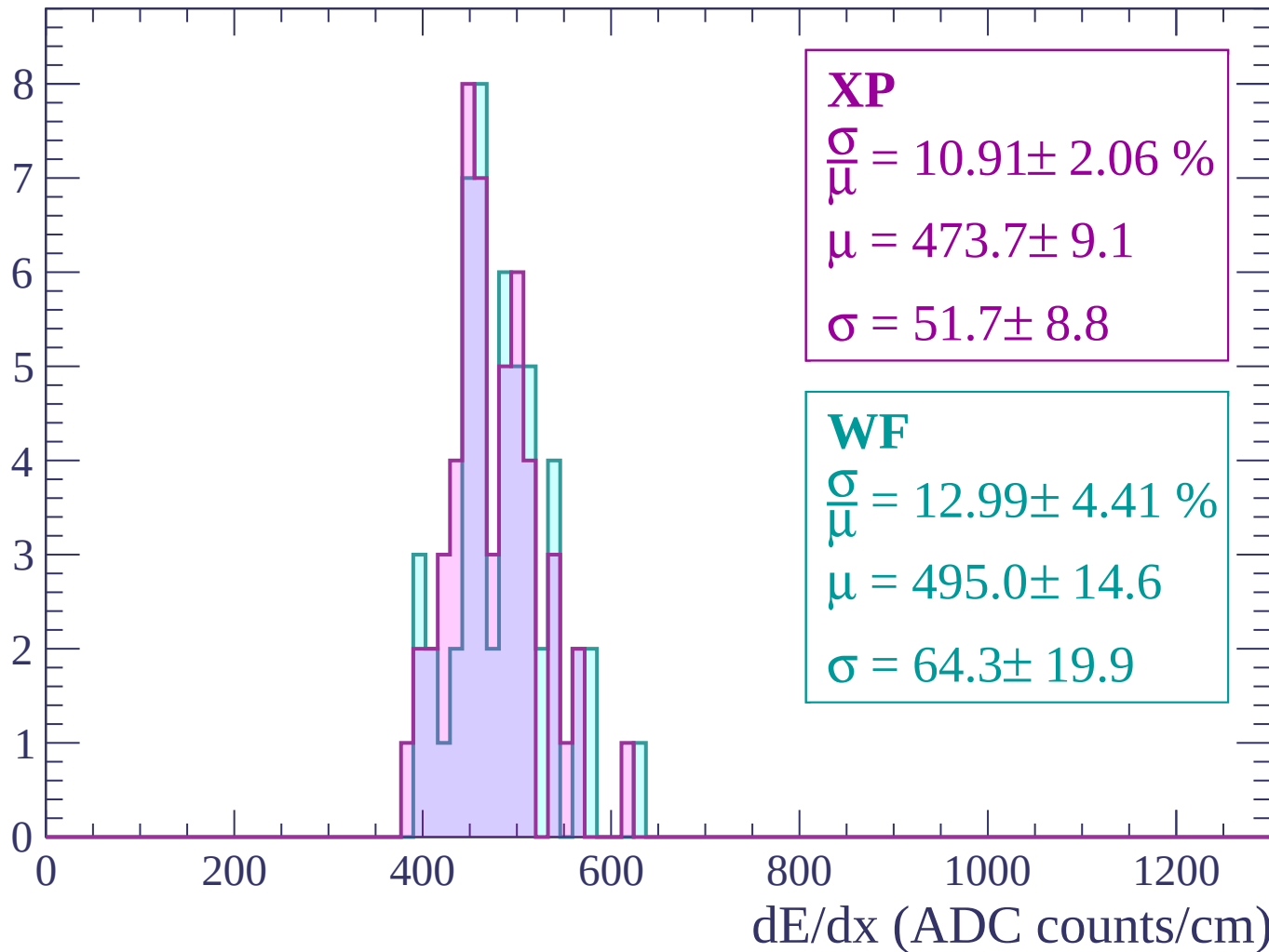
Energy loss |  $2760 < p < 2820$

Count



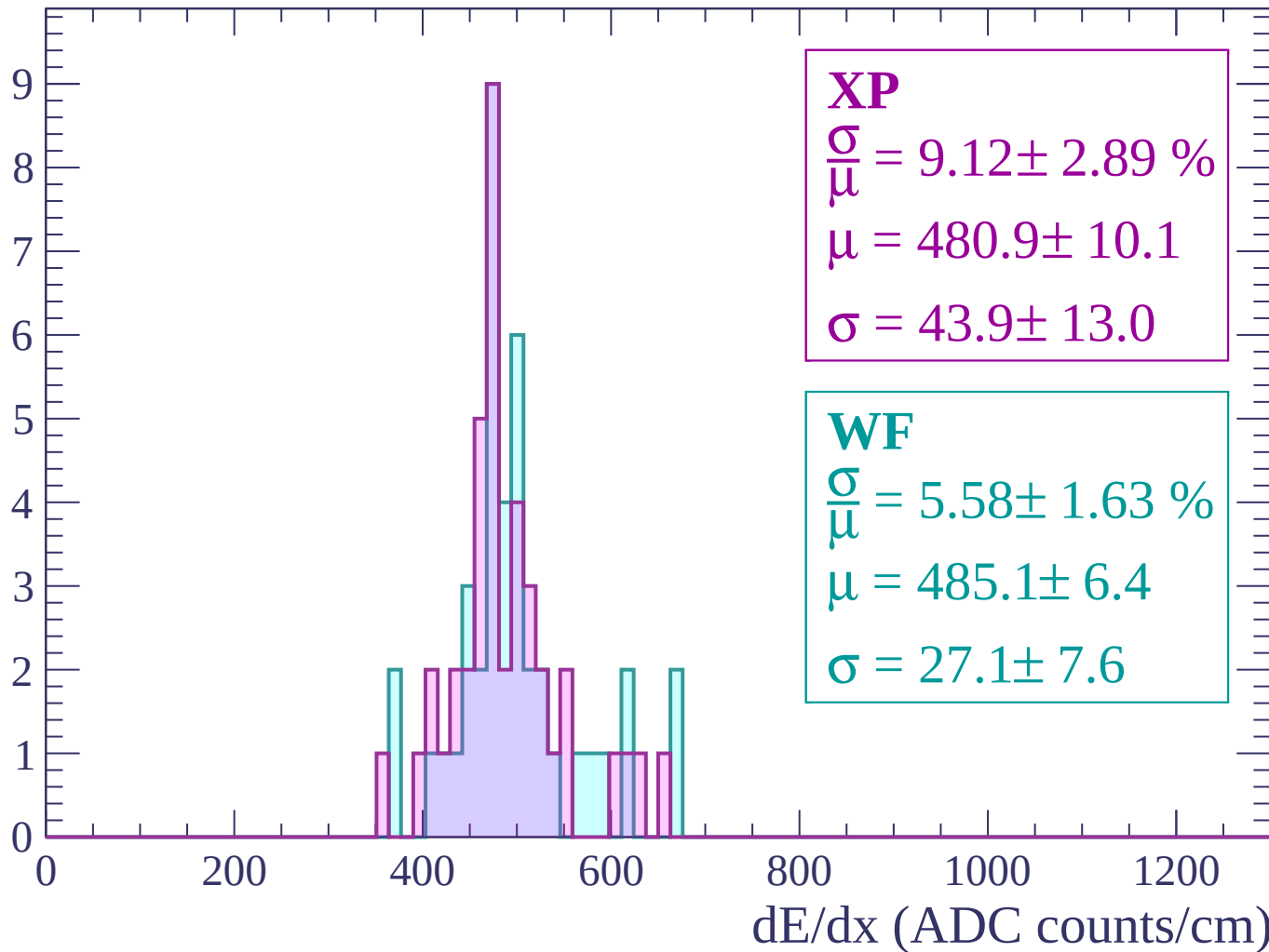
Energy loss |  $2820 < p < 2880$

Count



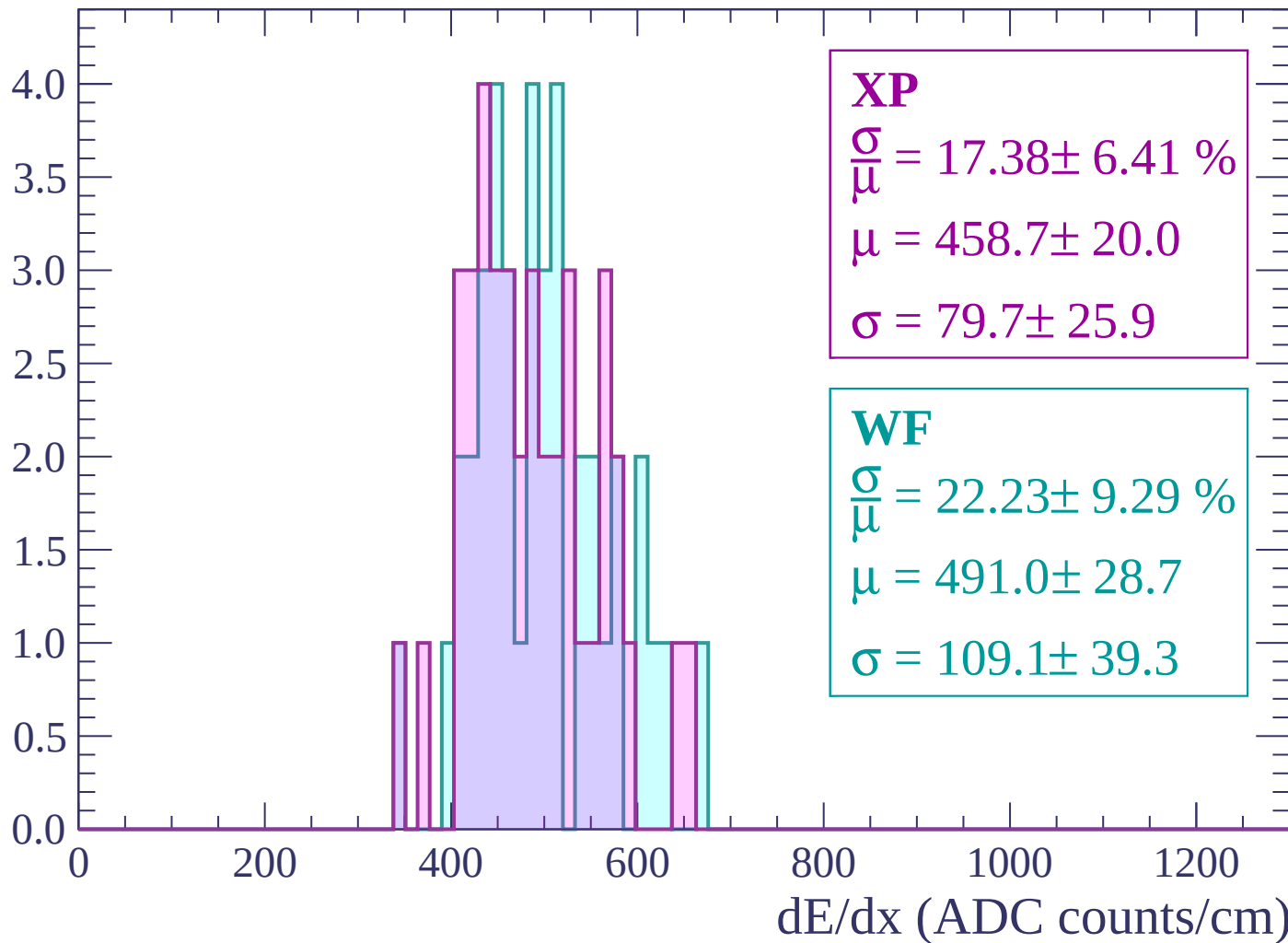
Energy loss |  $2880 < p < 2940$

Count



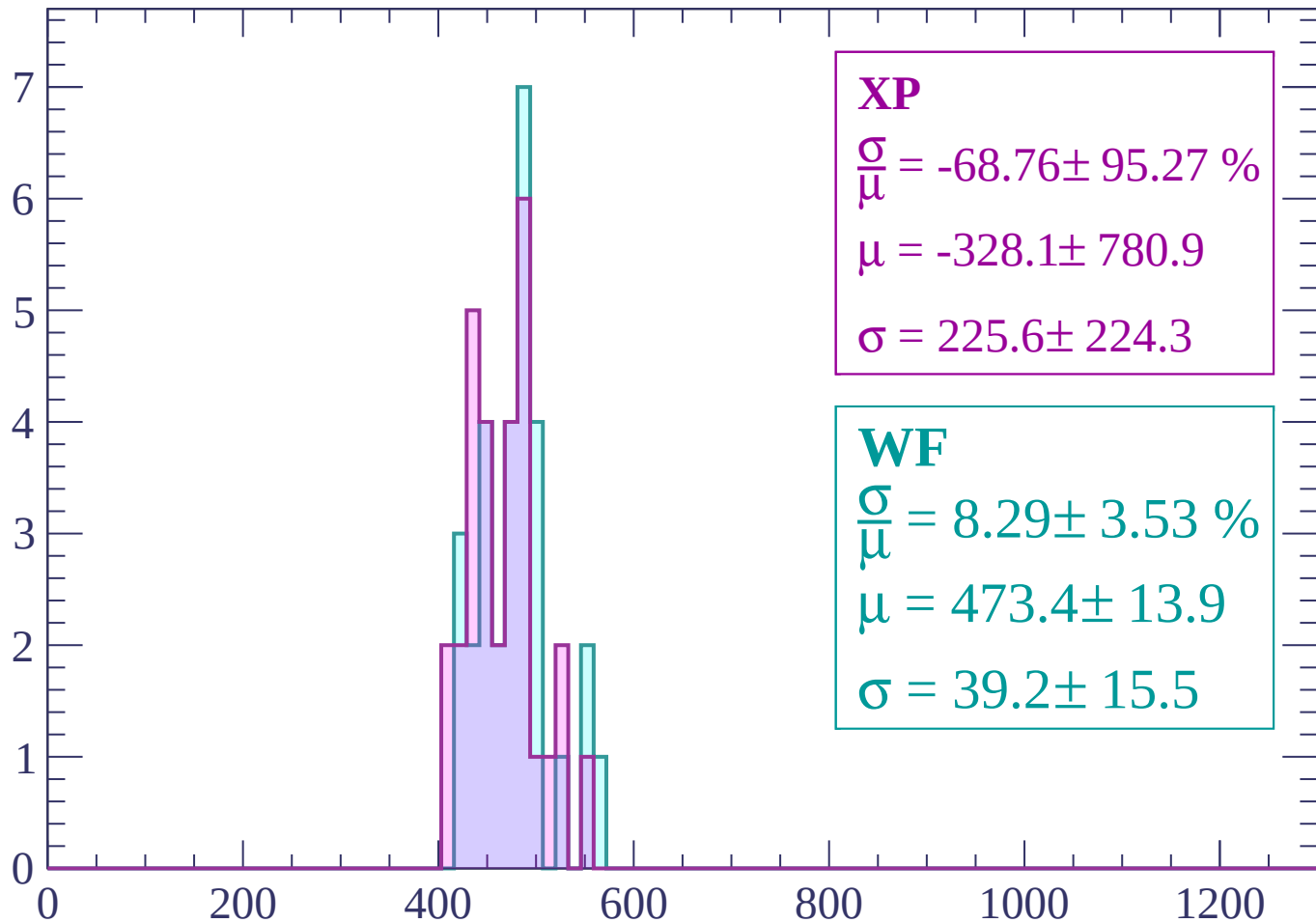
Energy loss |  $2940 < p < 3000$

Count



Energy loss |  $3000 < p < 3060$

Count



**XP**

$$\frac{\sigma}{\mu} = -68.76 \pm 95.27 \%$$

$$\mu = -328.1 \pm 780.9$$

$$\sigma = 225.6 \pm 224.3$$

**WF**

$$\frac{\sigma}{\mu} = 8.29 \pm 3.53 \%$$

$$\mu = 473.4 \pm 13.9$$

$$\sigma = 39.2 \pm 15.5$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $-90 < \theta < -88$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

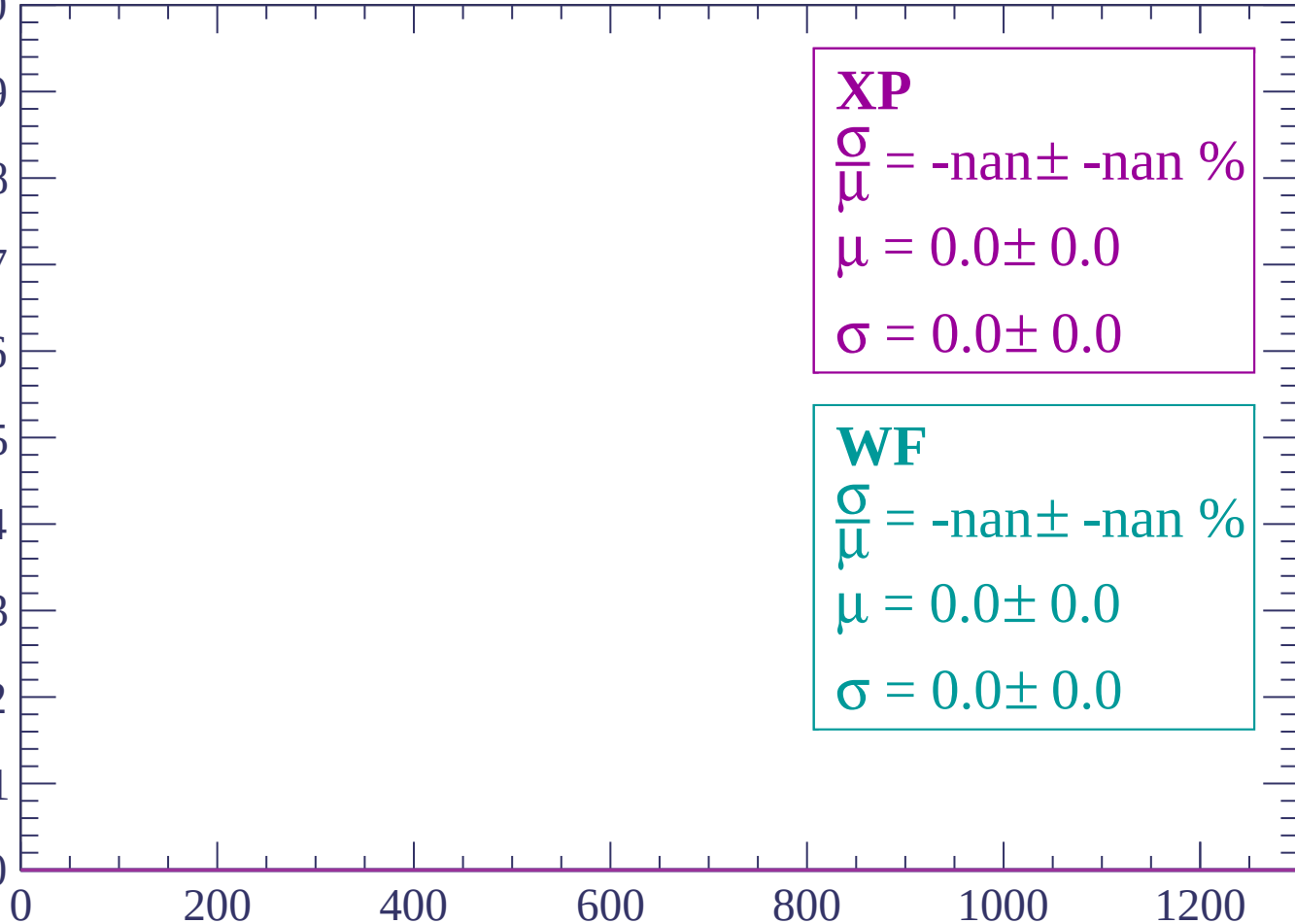
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-88 < \theta < -86$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0



$dE/dx$  (ADC counts/cm)



Energy loss |  $-86 < \theta < -84$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-84 < \theta < -82$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-82 < \theta < -80$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-80 < \theta < -78$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-78 < \theta < -76$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-76 < \theta < -74$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

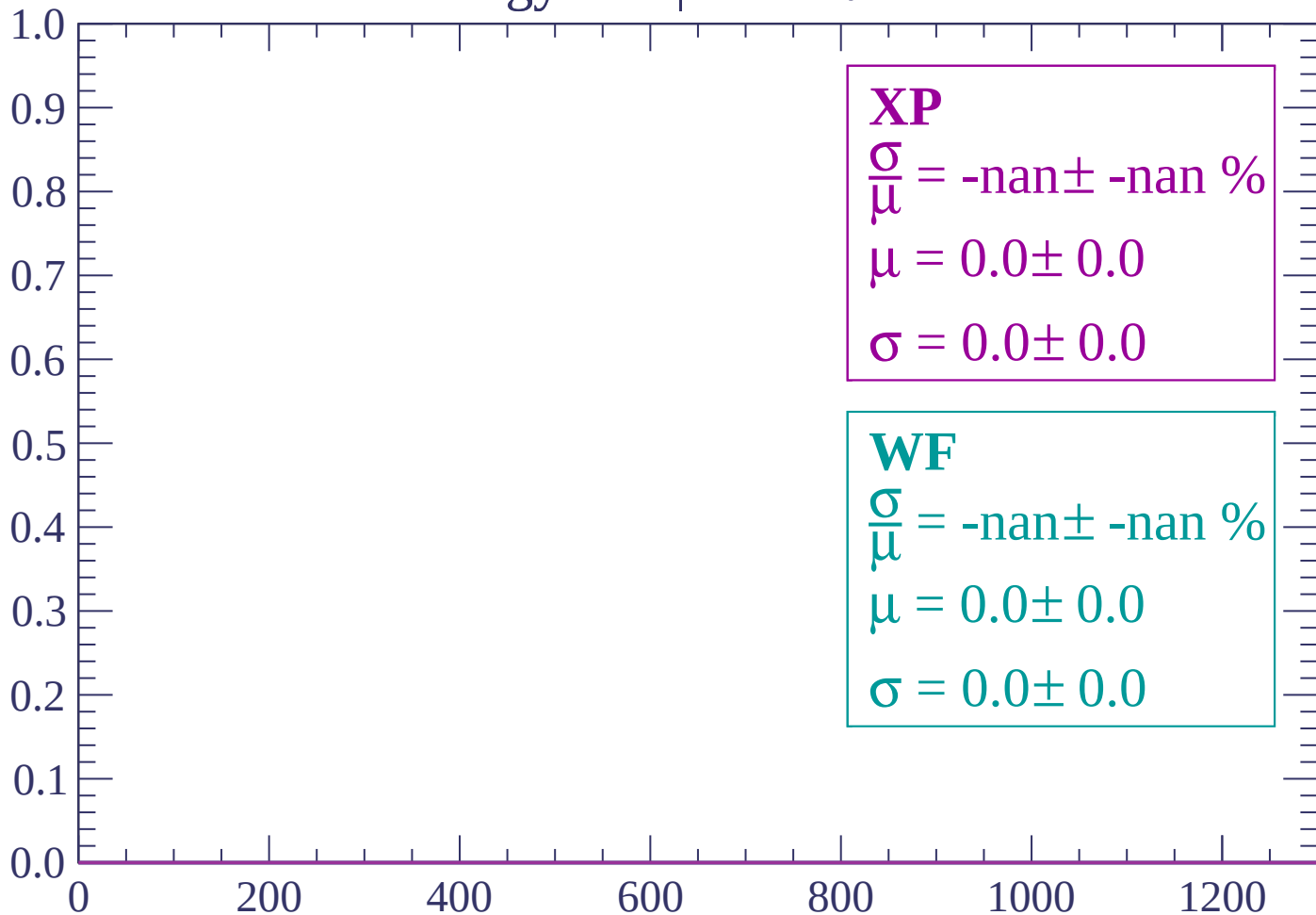
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-74 < \theta < -72$

Count



dE/dx (ADC counts/cm)

Energy loss |  $-72 < \theta < -70$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$



Energy loss |  $-70 < \theta < -68$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-68 < \theta < -66$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

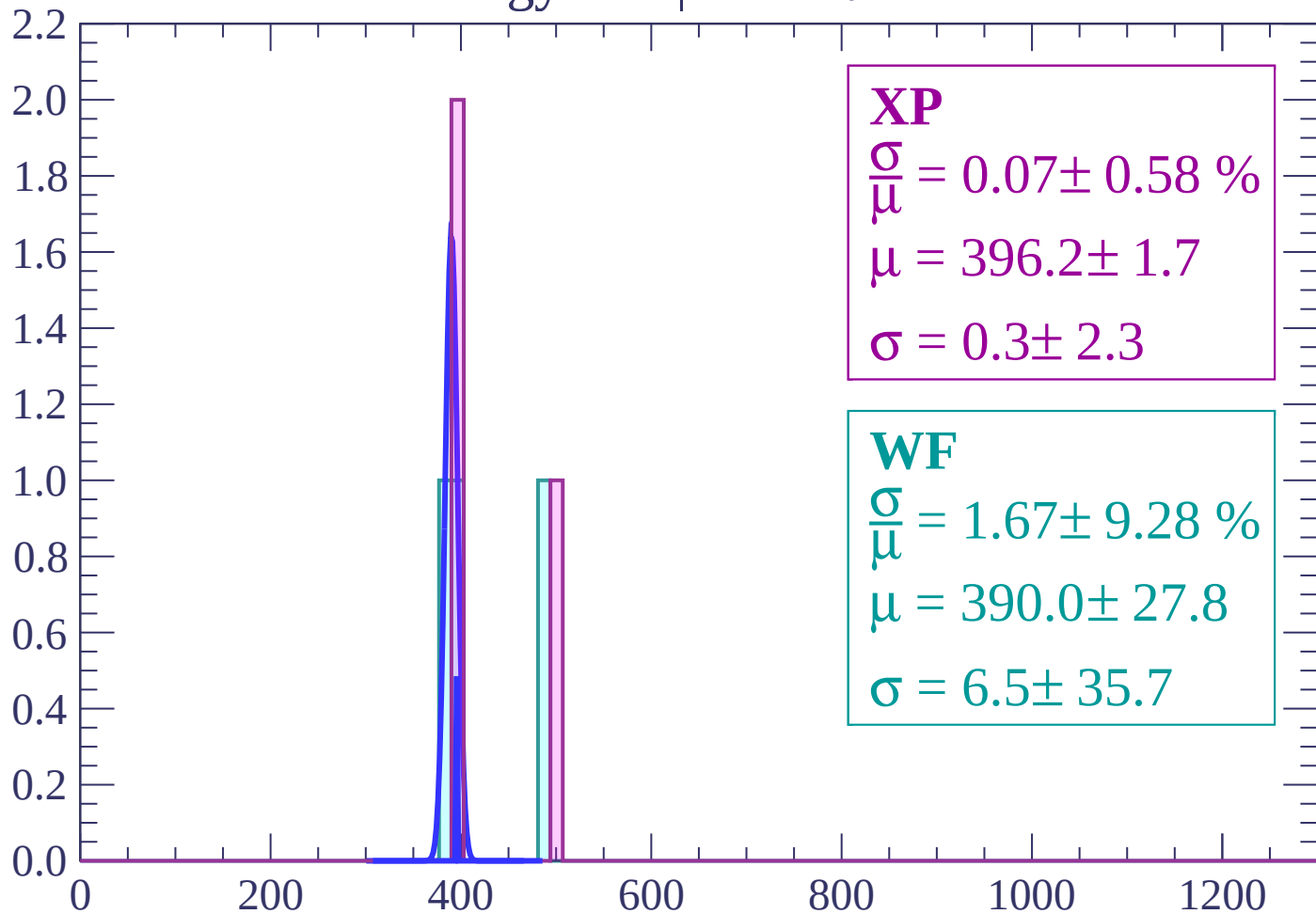
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $-66 < \theta < -64$

Count



**XP**

$$\frac{\sigma}{\mu} = 0.07 \pm 0.58 \%$$

$$\mu = 396.2 \pm 1.7$$

$$\sigma = 0.3 \pm 2.3$$

**WF**

$$\frac{\sigma}{\mu} = 1.67 \pm 9.28 \%$$

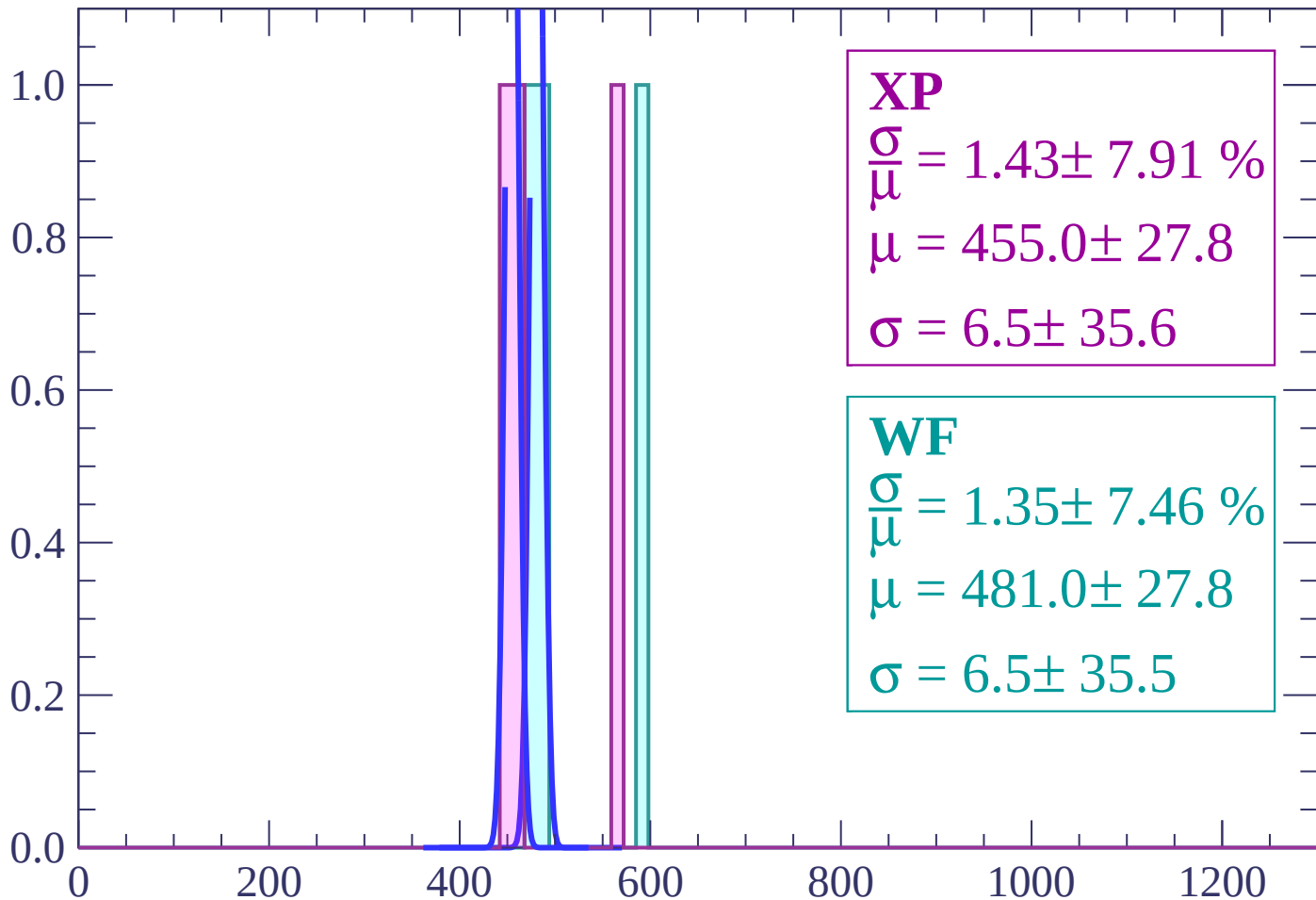
$$\mu = 390.0 \pm 27.8$$

$$\sigma = 6.5 \pm 35.7$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $-64 < \theta < -62$

Count



**XP**

$$\frac{\sigma}{\mu} = 1.43 \pm 7.91 \%$$

$$\mu = 455.0 \pm 27.8$$

$$\sigma = 6.5 \pm 35.6$$

**WF**

$$\frac{\sigma}{\mu} = 1.35 \pm 7.46 \%$$

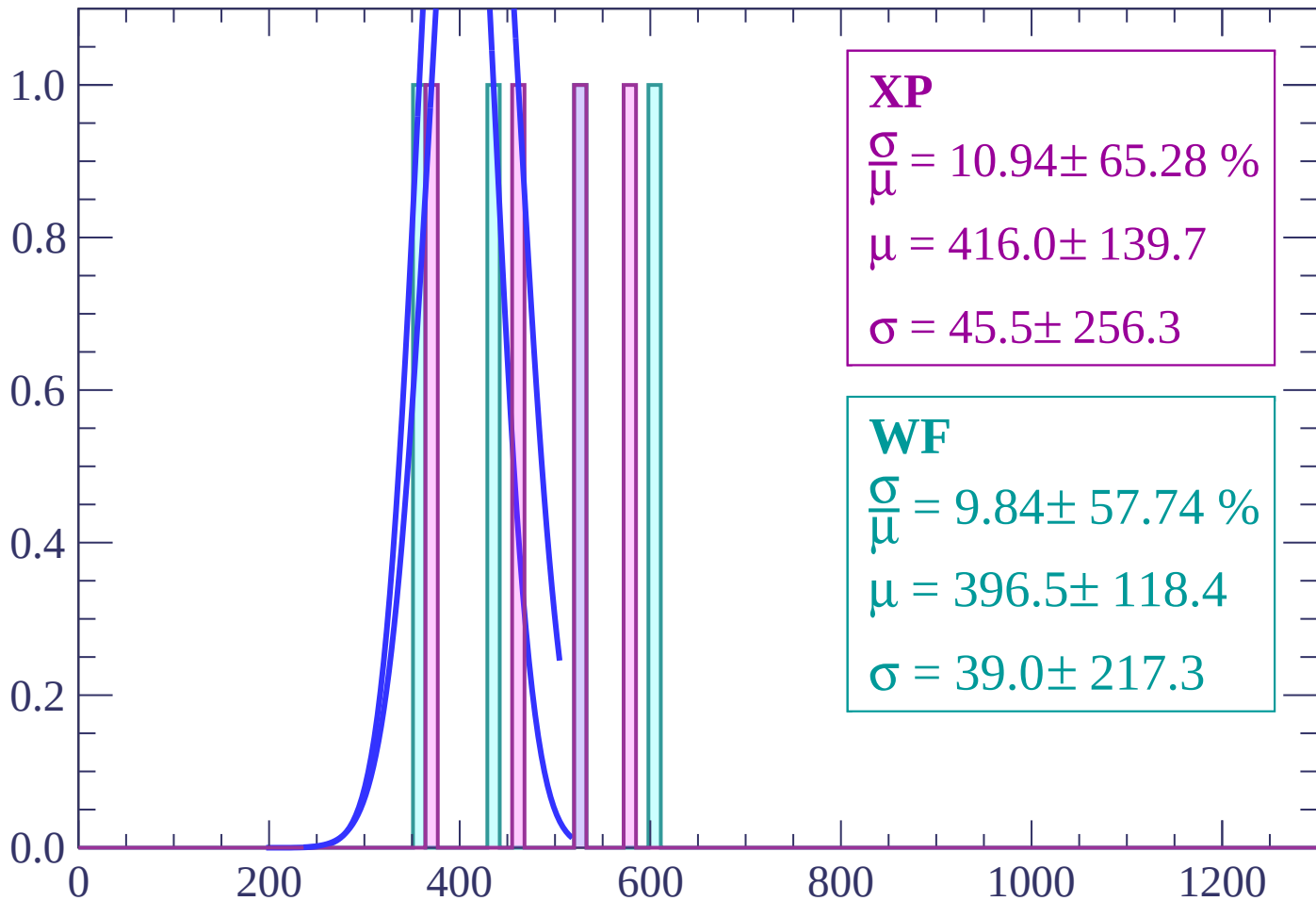
$$\mu = 481.0 \pm 27.8$$

$$\sigma = 6.5 \pm 35.5$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $-62 < \theta < -60$

Count



**XP**

$$\frac{\sigma}{\mu} = 10.94 \pm 65.28 \%$$

$$\mu = 416.0 \pm 139.7$$

$$\sigma = 45.5 \pm 256.3$$

**WF**

$$\frac{\sigma}{\mu} = 9.84 \pm 57.74 \%$$

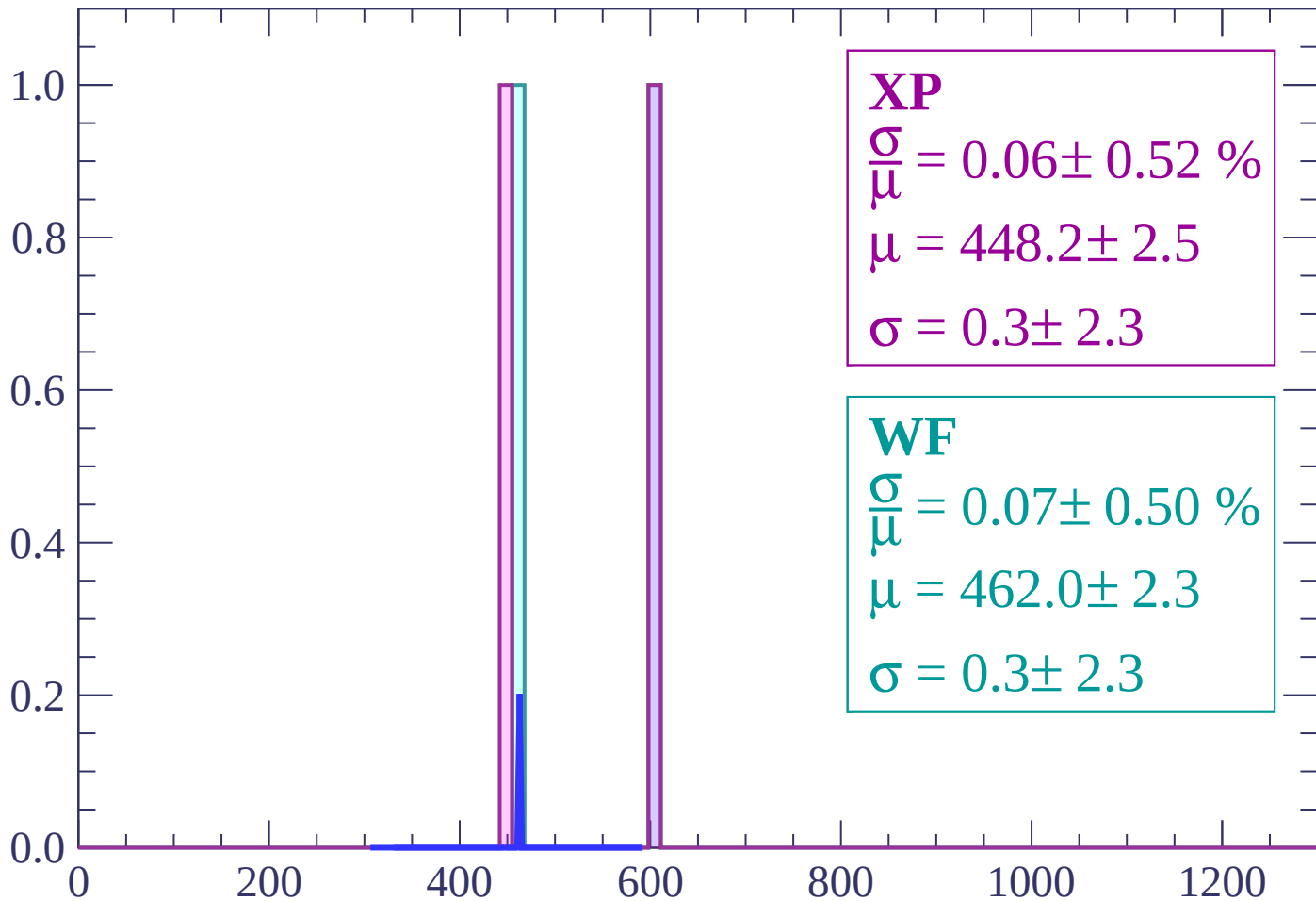
$$\mu = 396.5 \pm 118.4$$

$$\sigma = 39.0 \pm 217.3$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $-60 < \theta < -58$

Count



**XP**

$$\frac{\sigma}{\mu} = 0.06 \pm 0.52 \%$$

$$\mu = 448.2 \pm 2.5$$

$$\sigma = 0.3 \pm 2.3$$

**WF**

$$\frac{\sigma}{\mu} = 0.07 \pm 0.50 \%$$

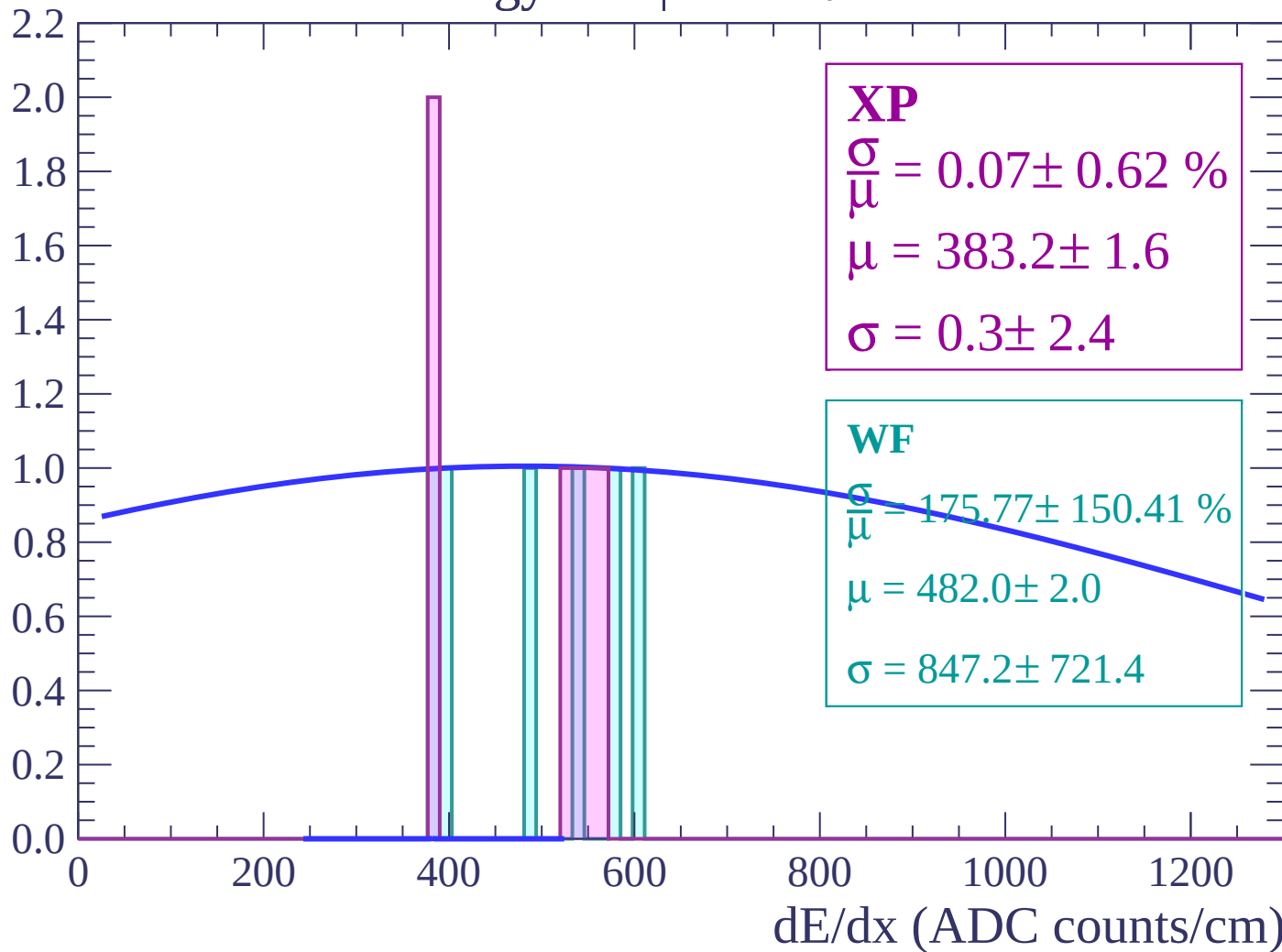
$$\mu = 462.0 \pm 2.3$$

$$\sigma = 0.3 \pm 2.3$$

$dE/dx$  (ADC counts/cm)

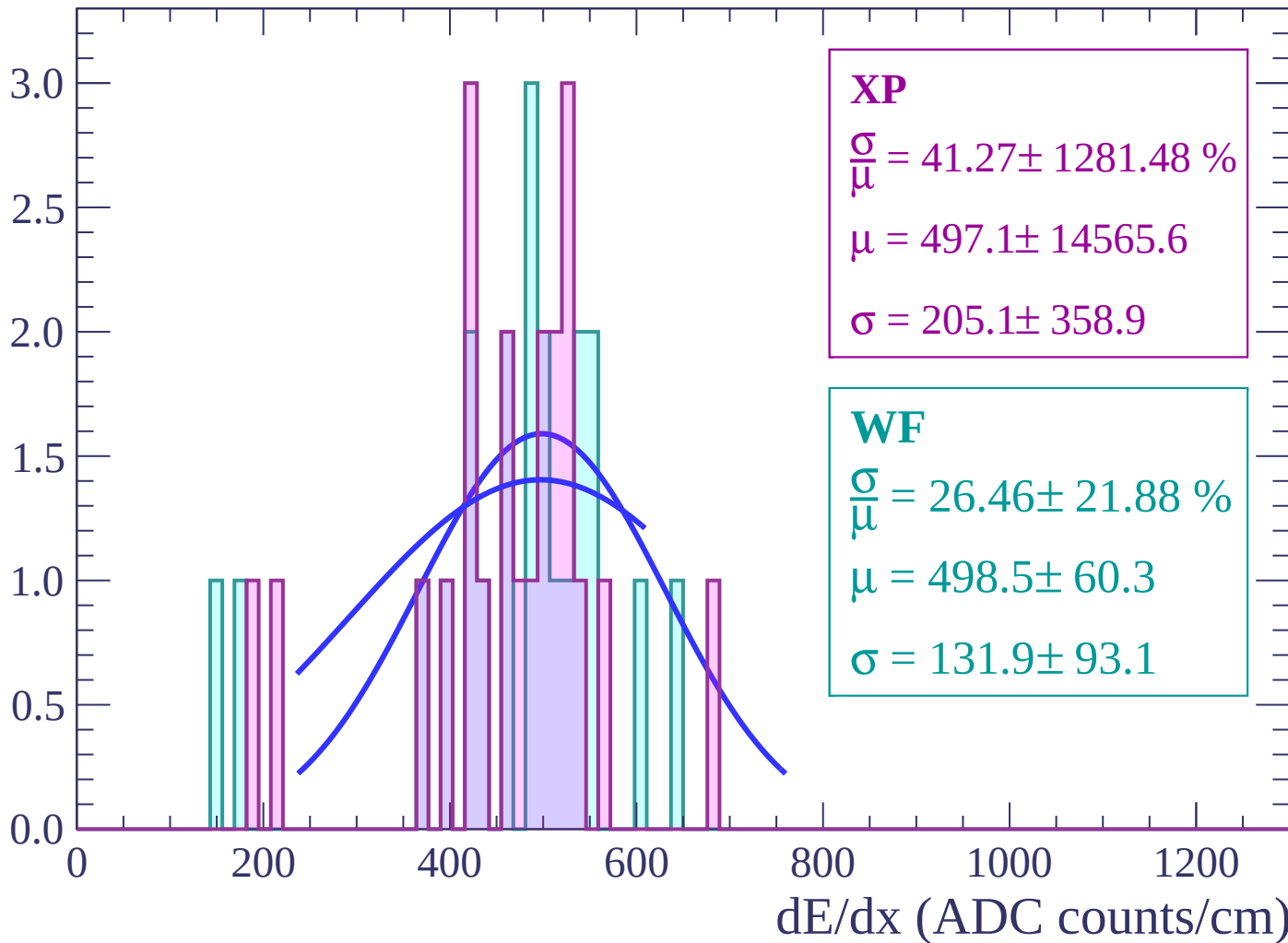
Energy loss |  $-58 < \theta < -56$

Count



Energy loss |  $-56 < \theta < -54$

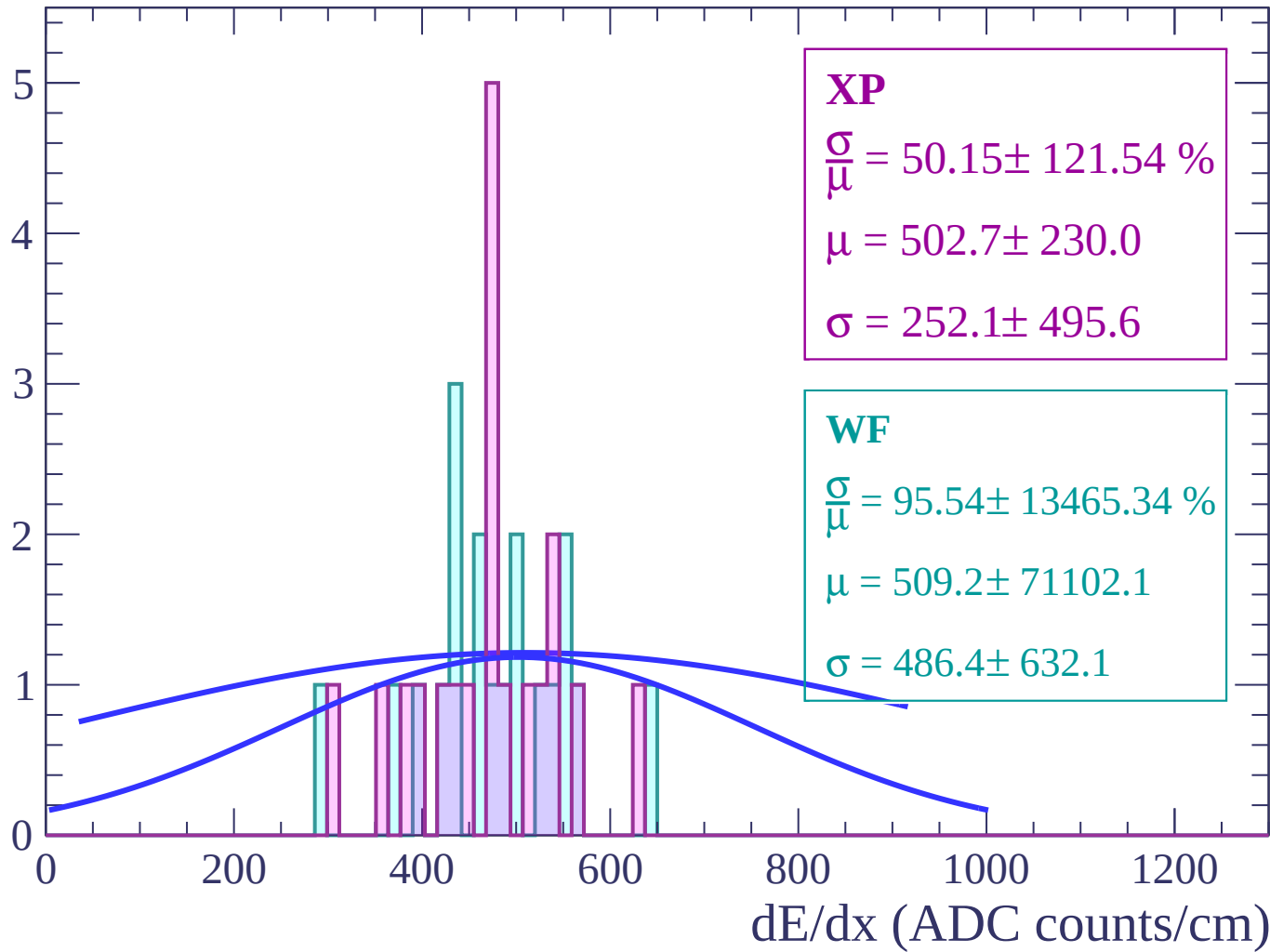
Count





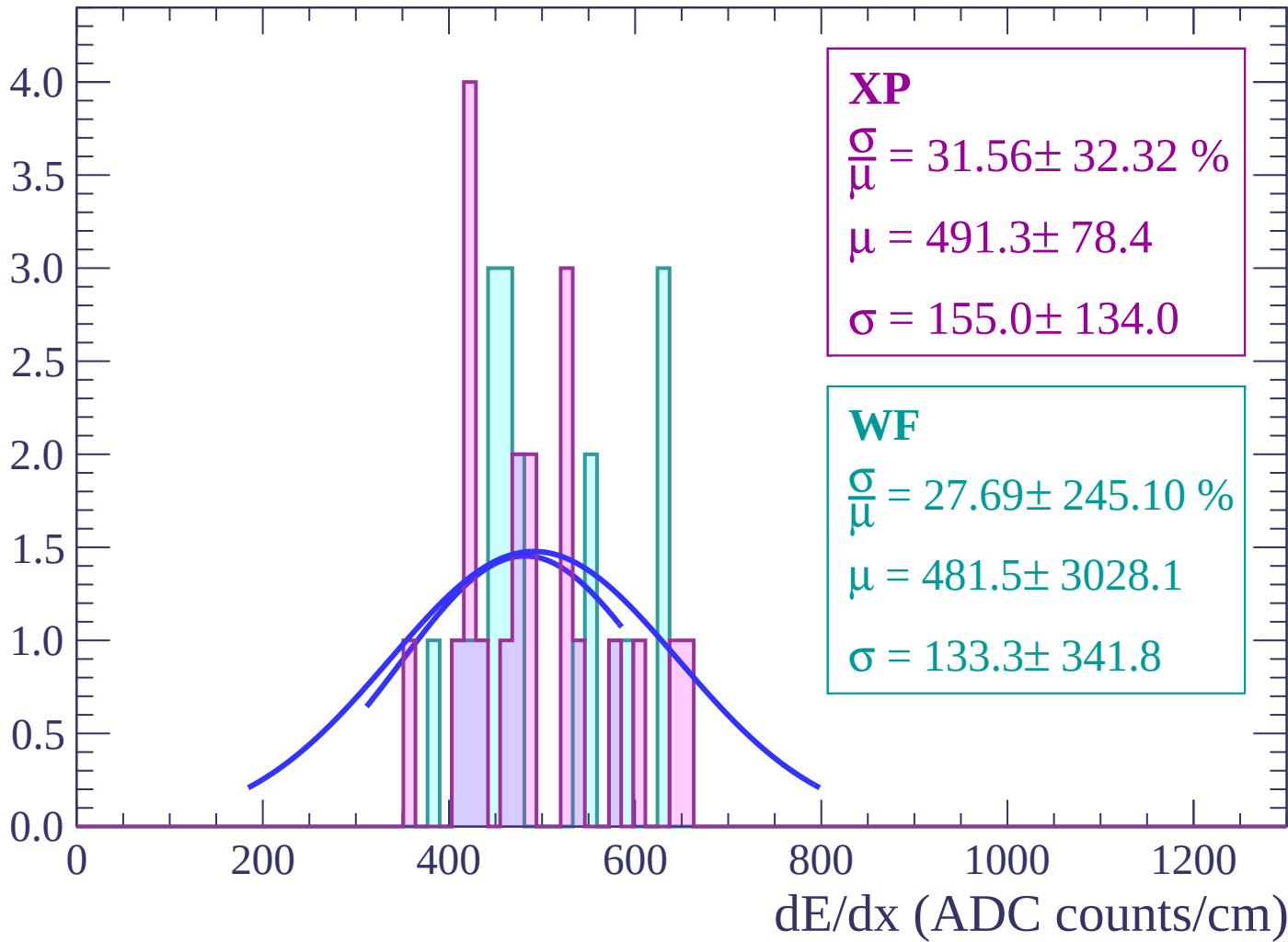
# Energy loss | $-54 < \theta < -52$

Count



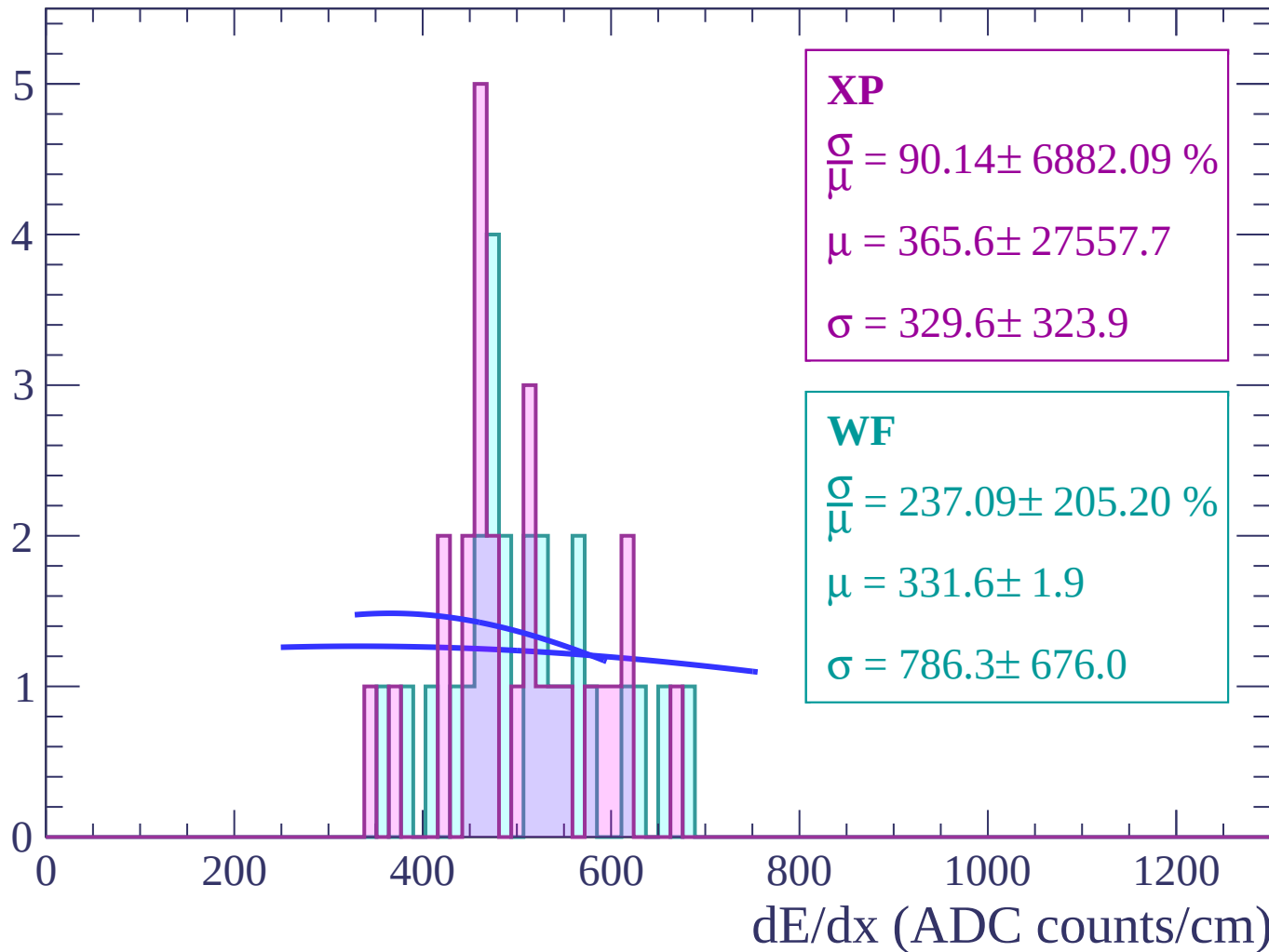
# Energy loss | $-52 < \theta < -50$

Count



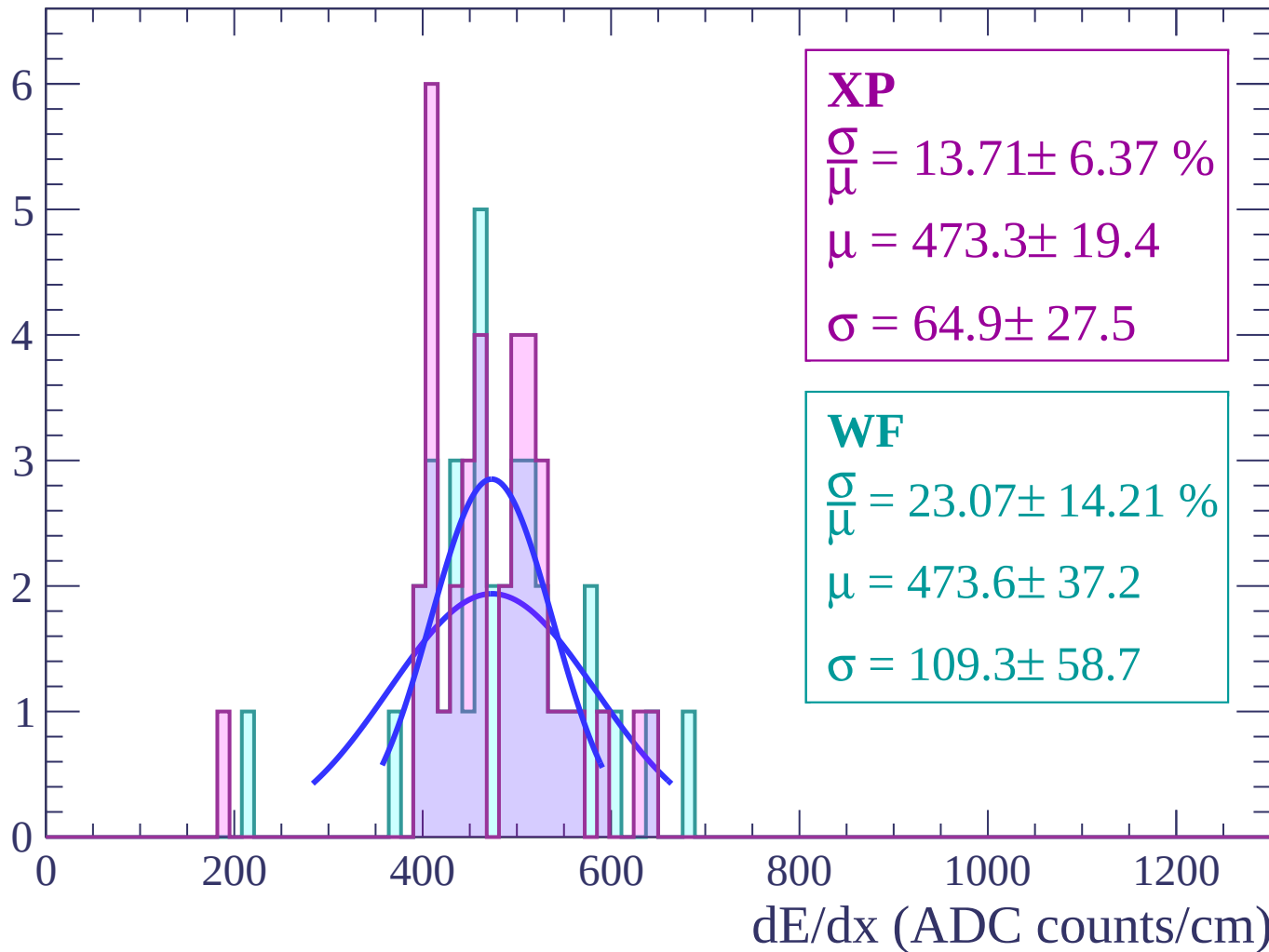
# Energy loss | $-50 < \theta < -48$

Count



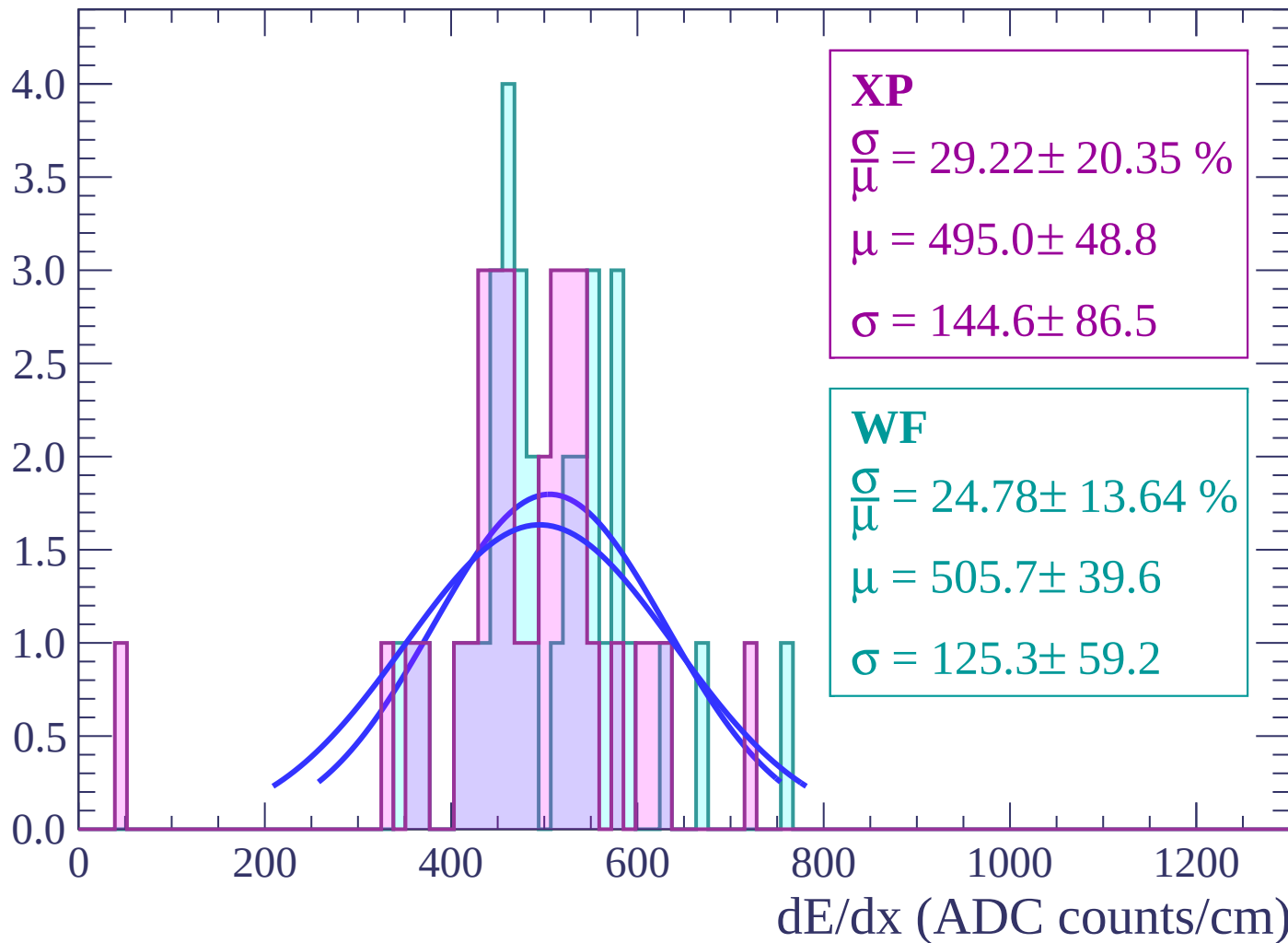
Energy loss |  $-48 < \theta < -46$

Count



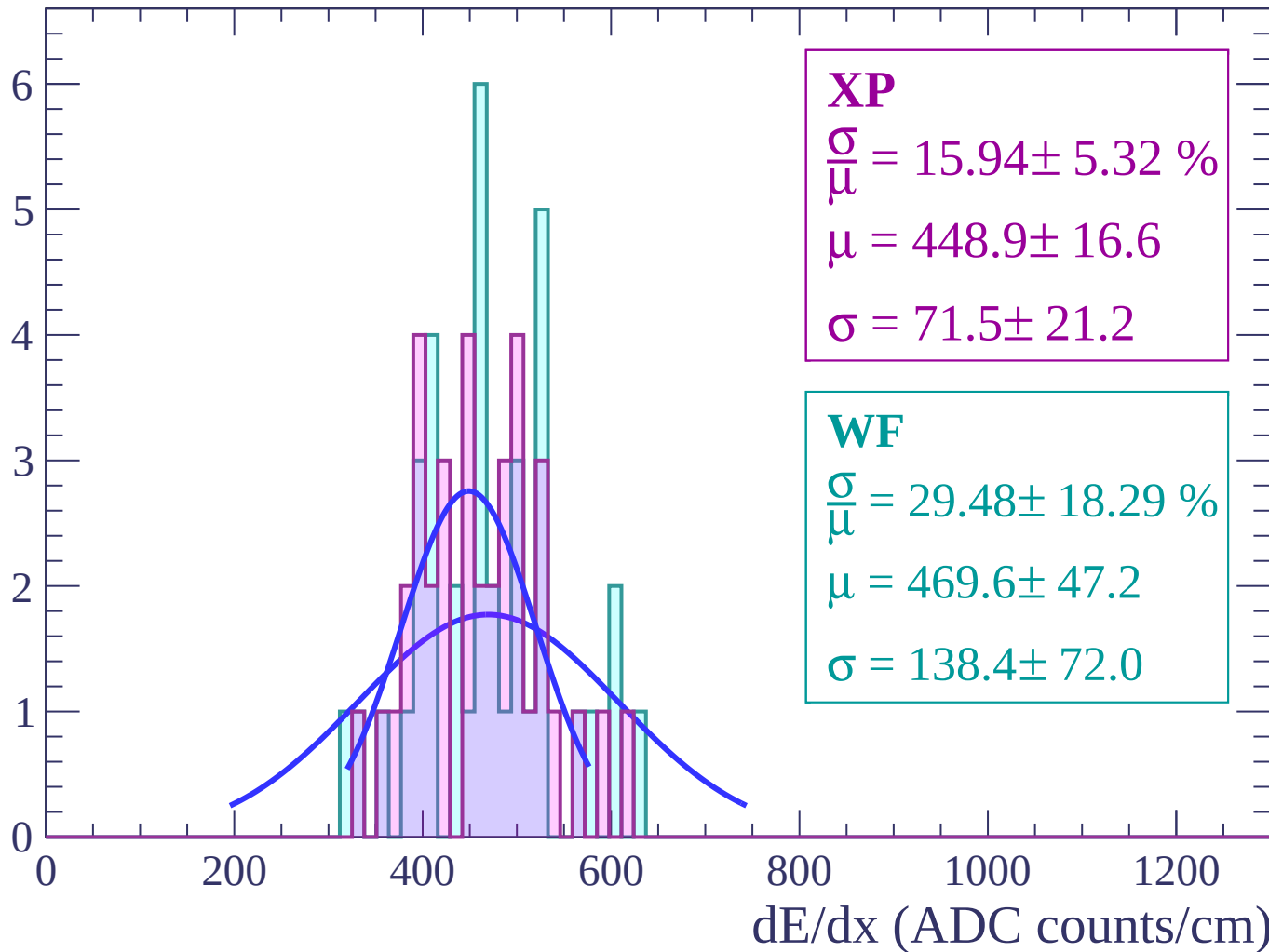
Energy loss |  $-46 < \theta < -44$

Count



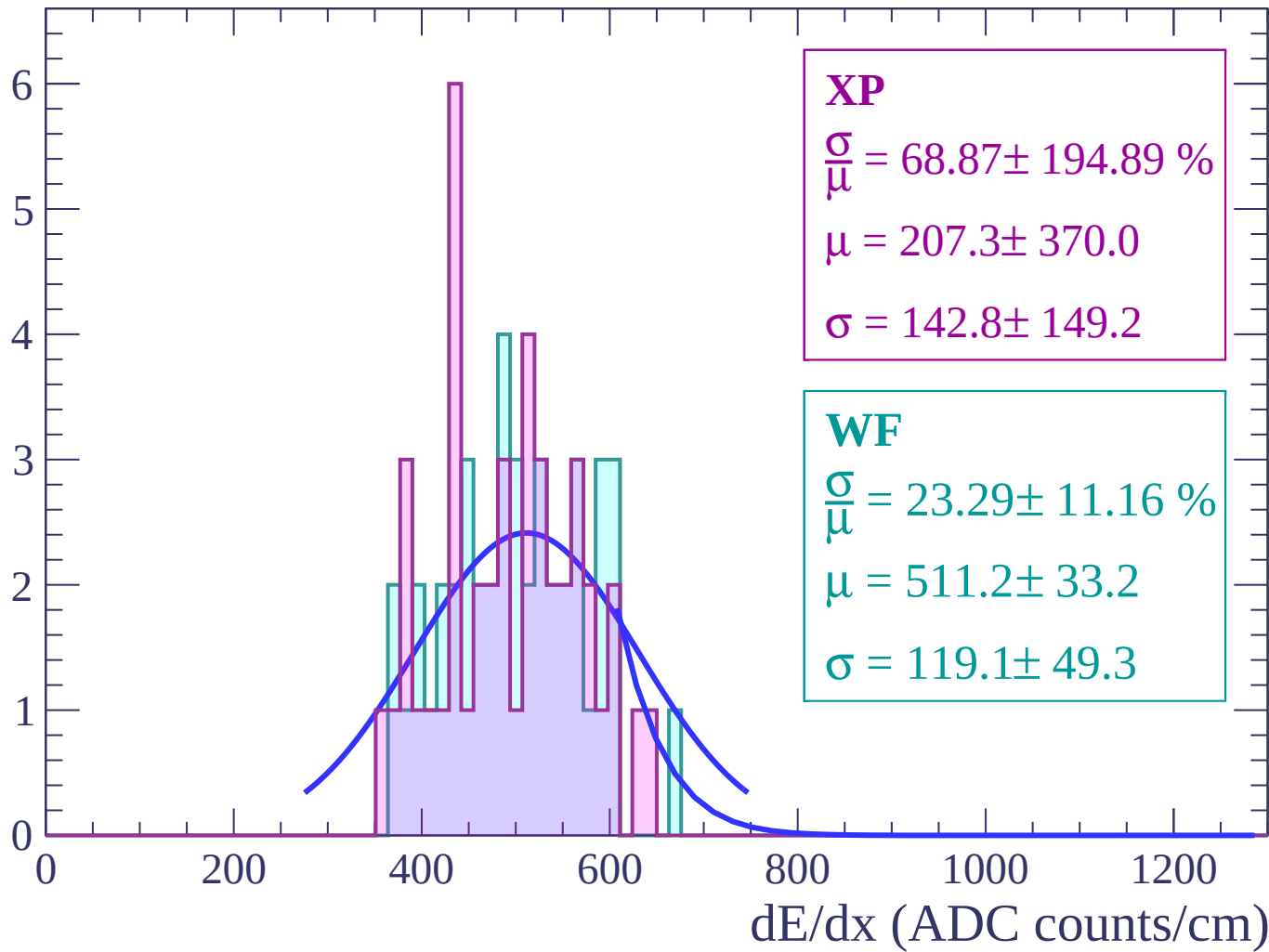
Energy loss |  $-44 < \theta < -42$

Count



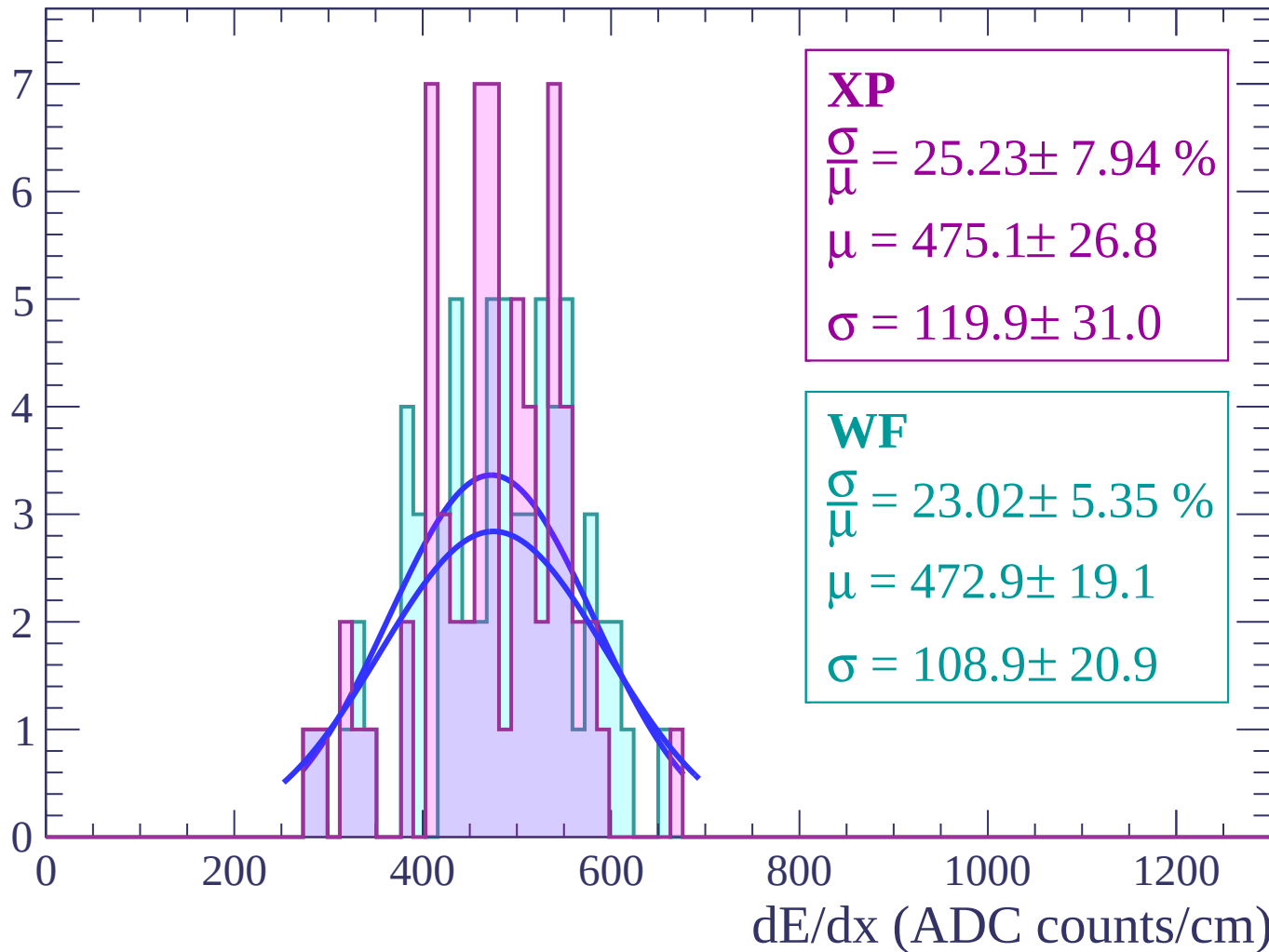
Energy loss |  $-42 < \theta < -40$

Count



Energy loss |  $-40 < \theta < -38$

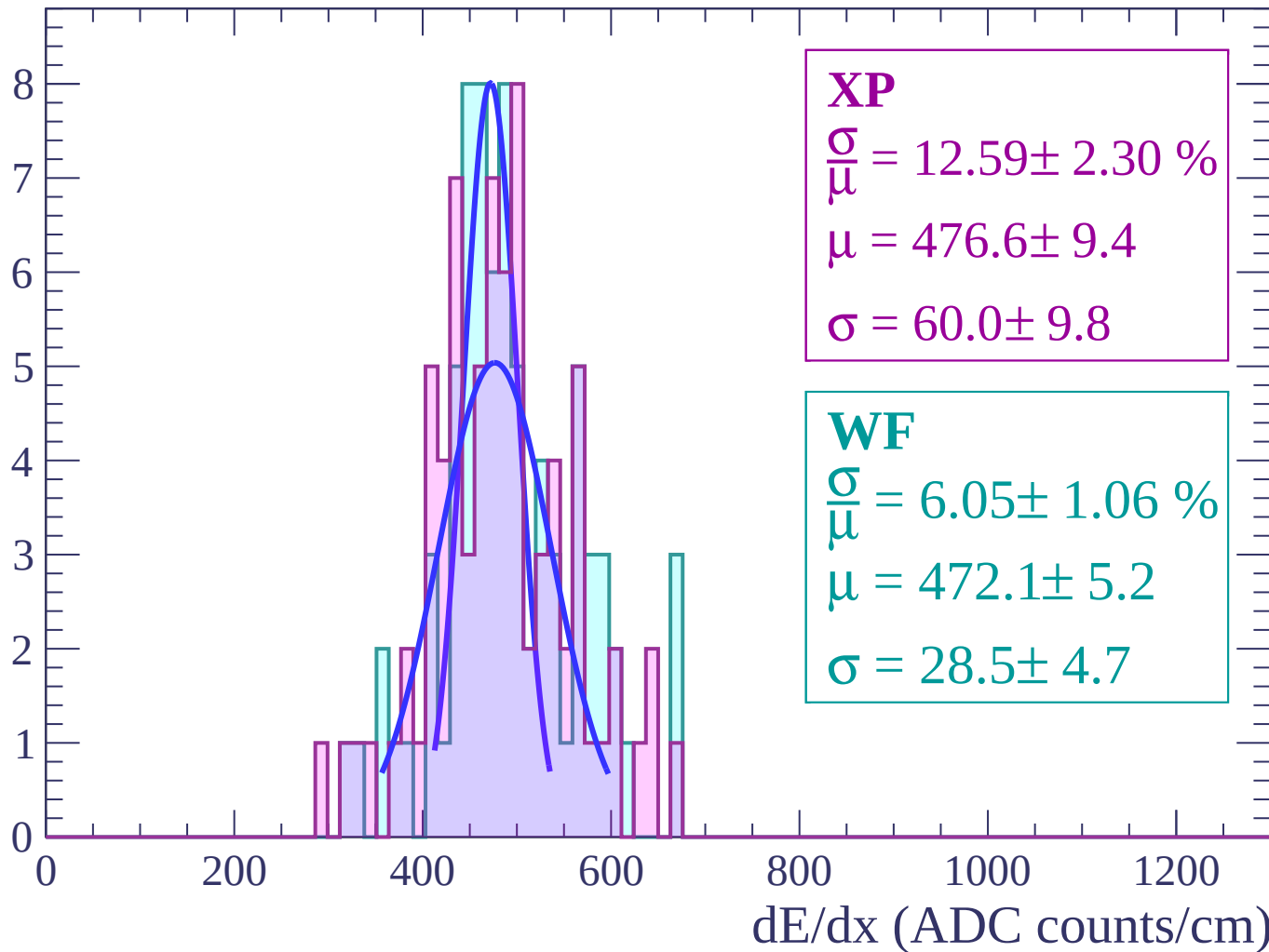
Count





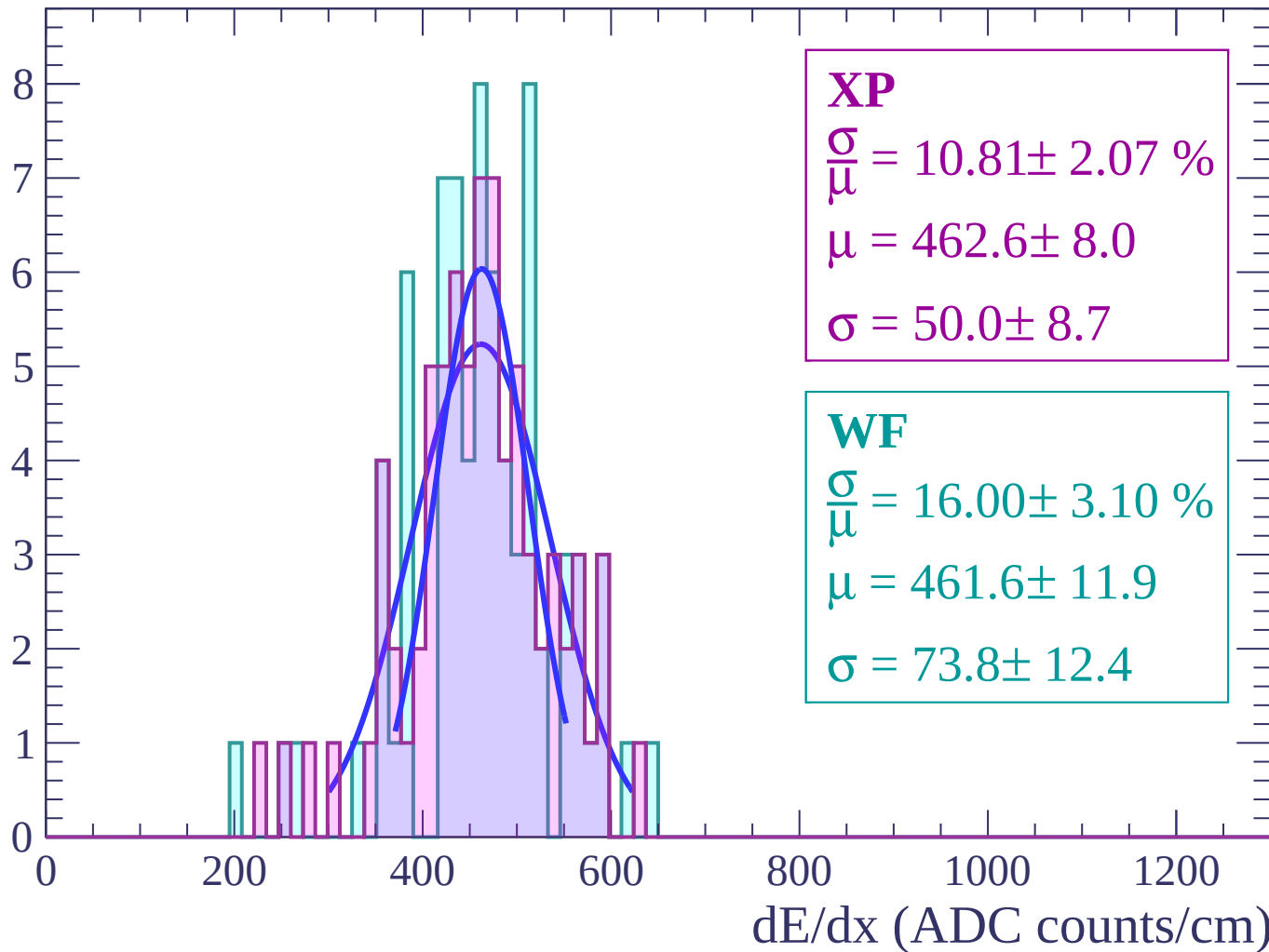
Energy loss |  $-38 < \theta < -36$

Count



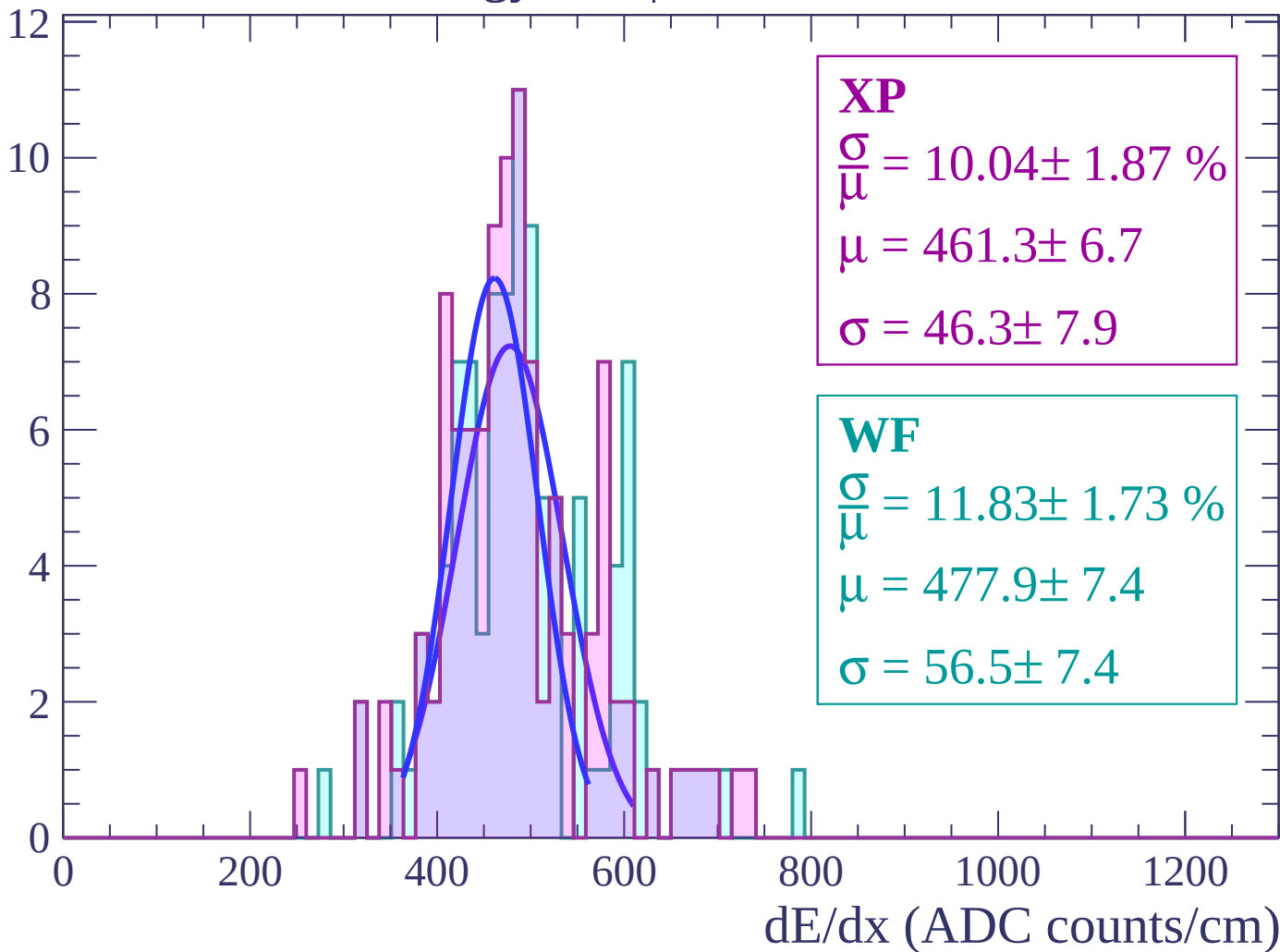
Energy loss |  $-36 < \theta < -34$

Count



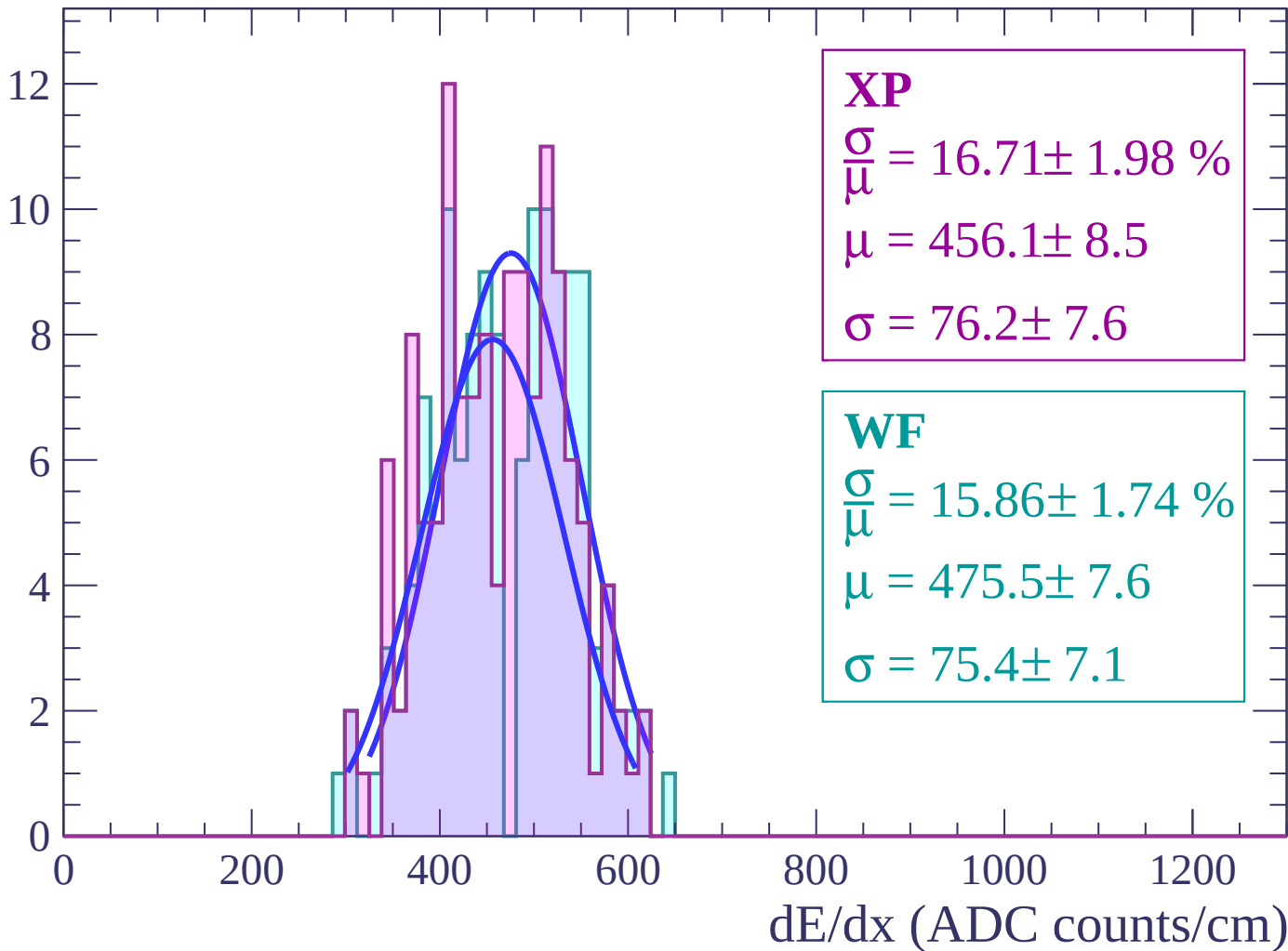
Energy loss |  $-34 < \theta < -32$

Count



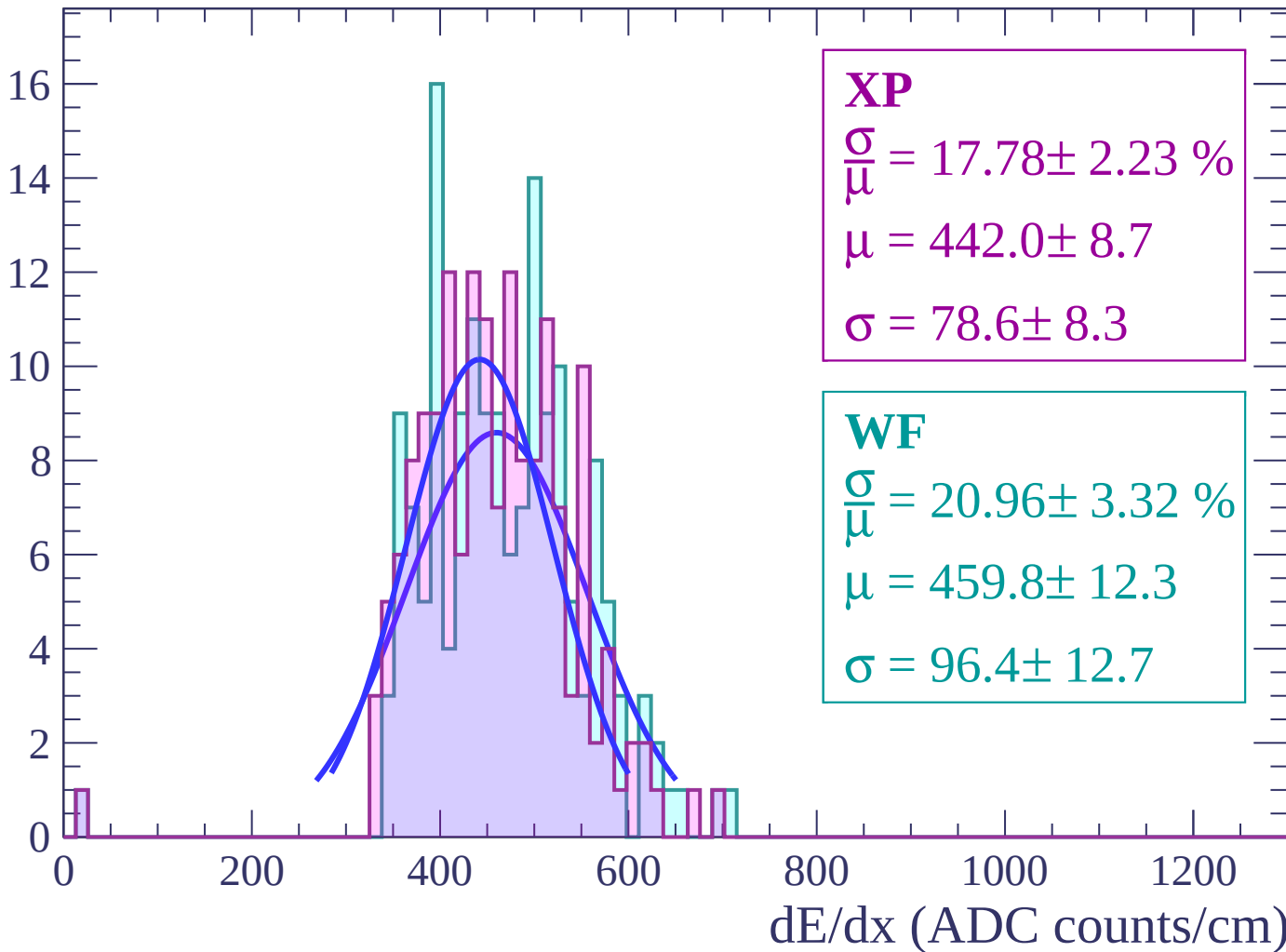
Energy loss |  $-32 < \theta < -30$

Count



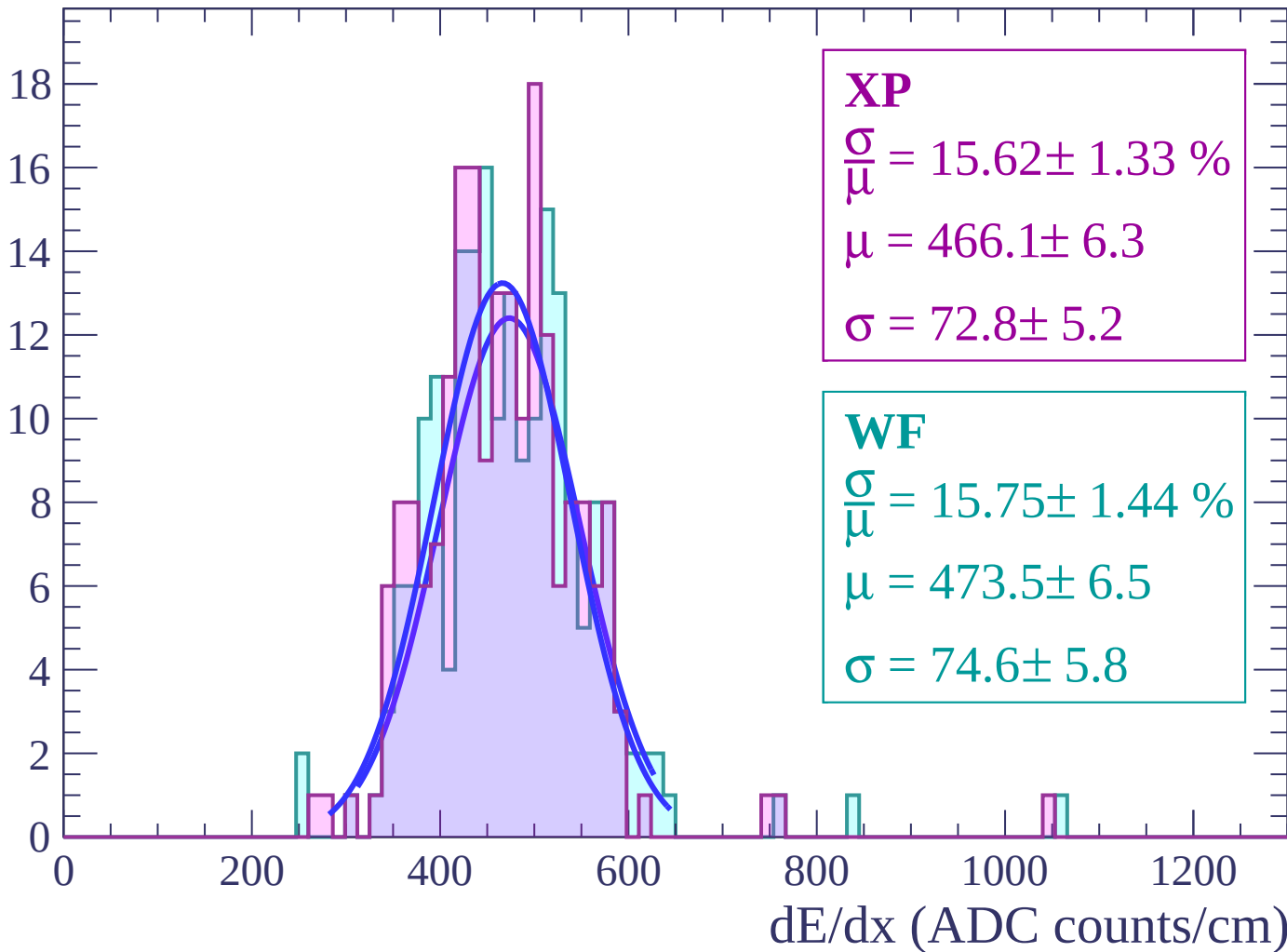
Energy loss |  $-30 < \theta < -28$

Count



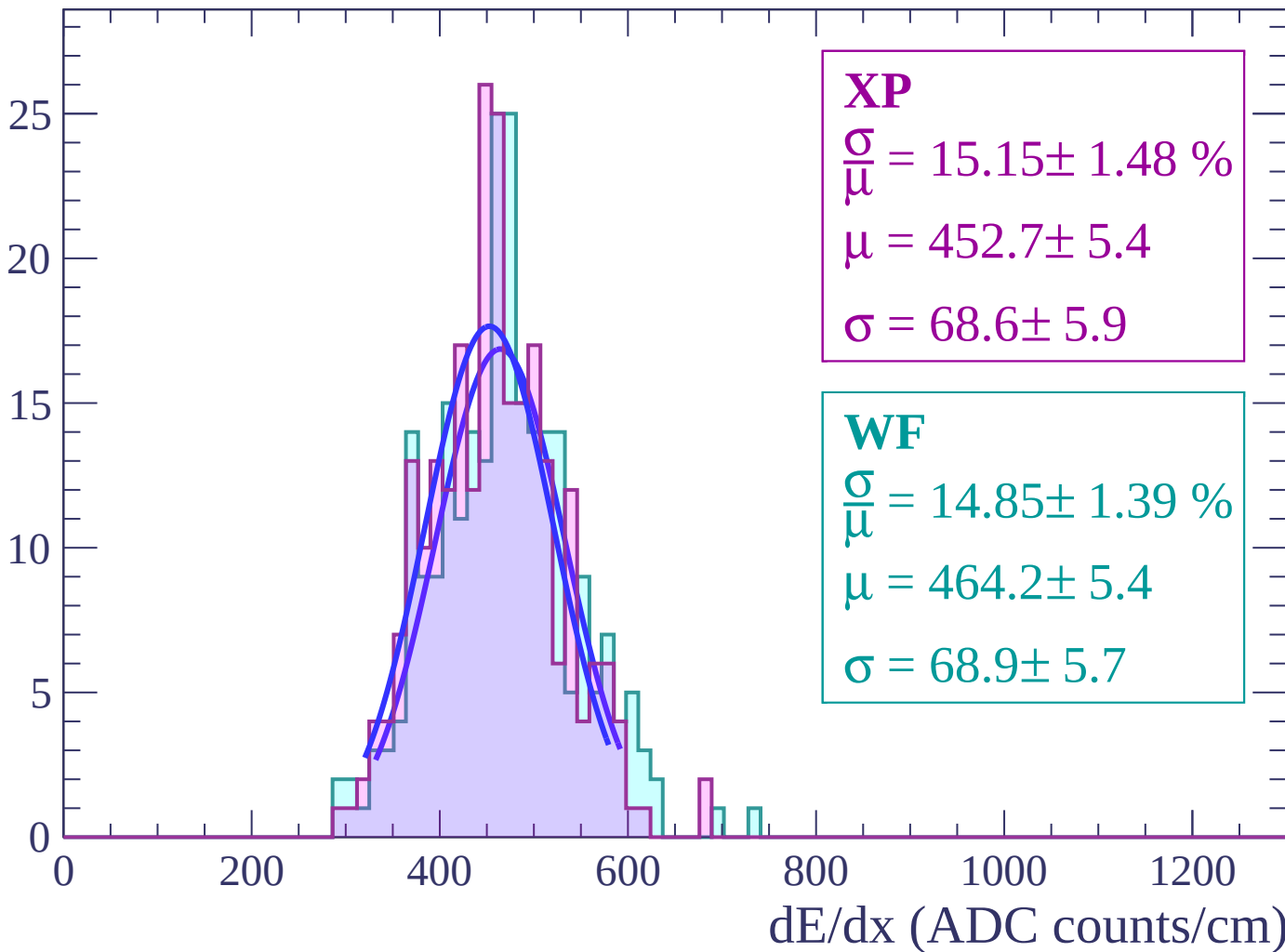
Energy loss |  $-28 < \theta < -26$

Count



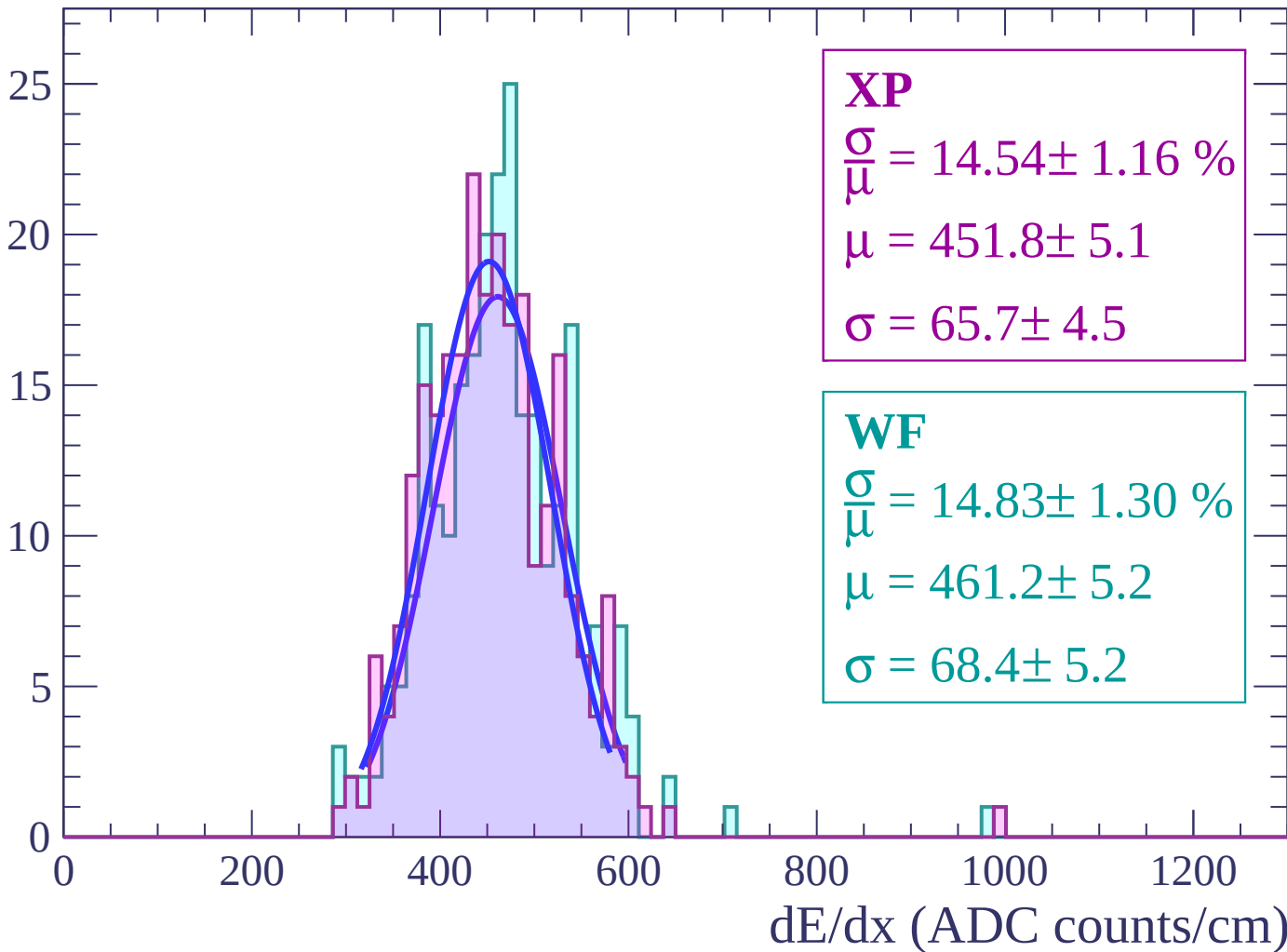
Energy loss |  $-26 < \theta < -24$

Count



Energy loss |  $-24 < \theta < -22$

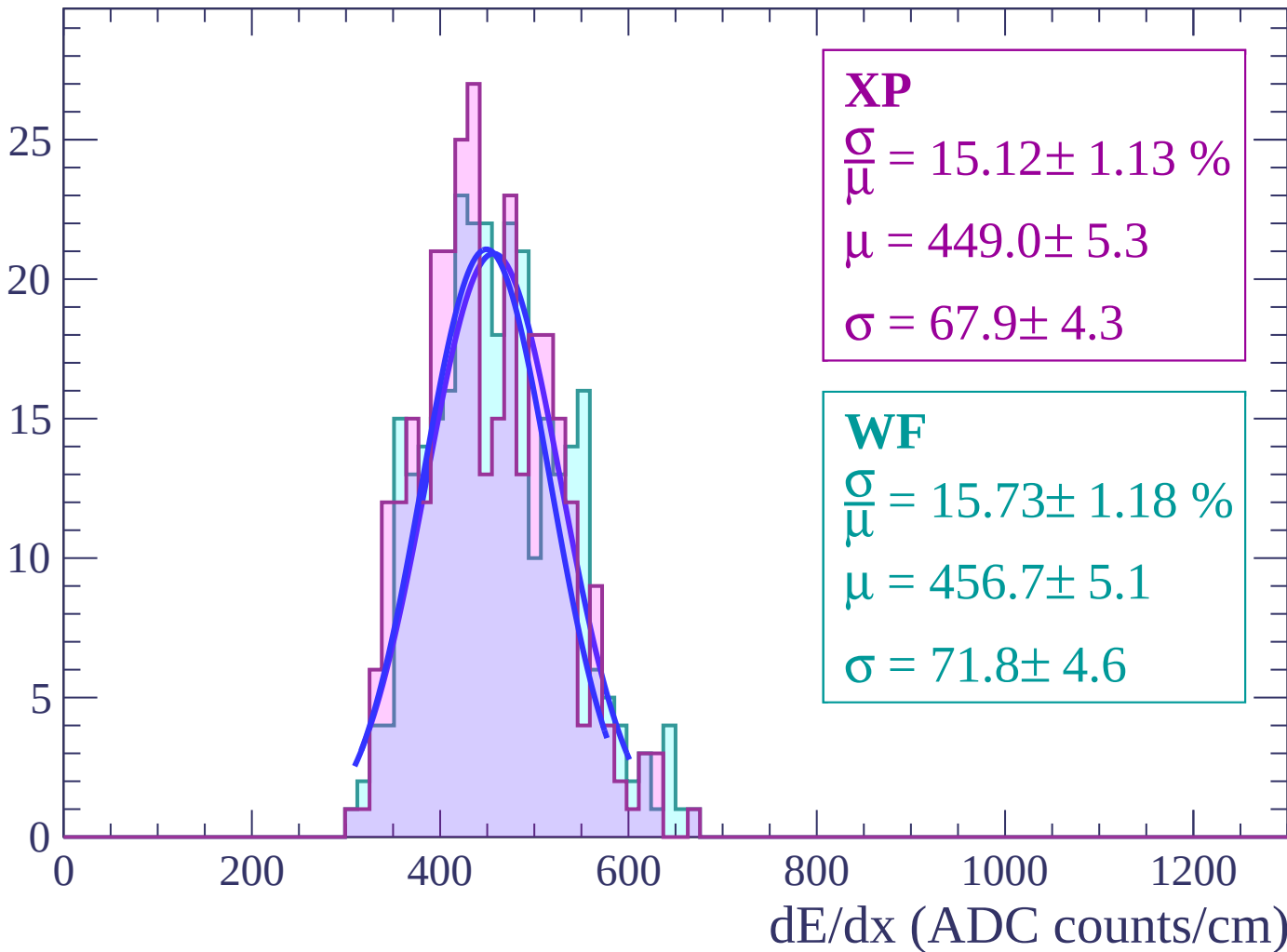
Count





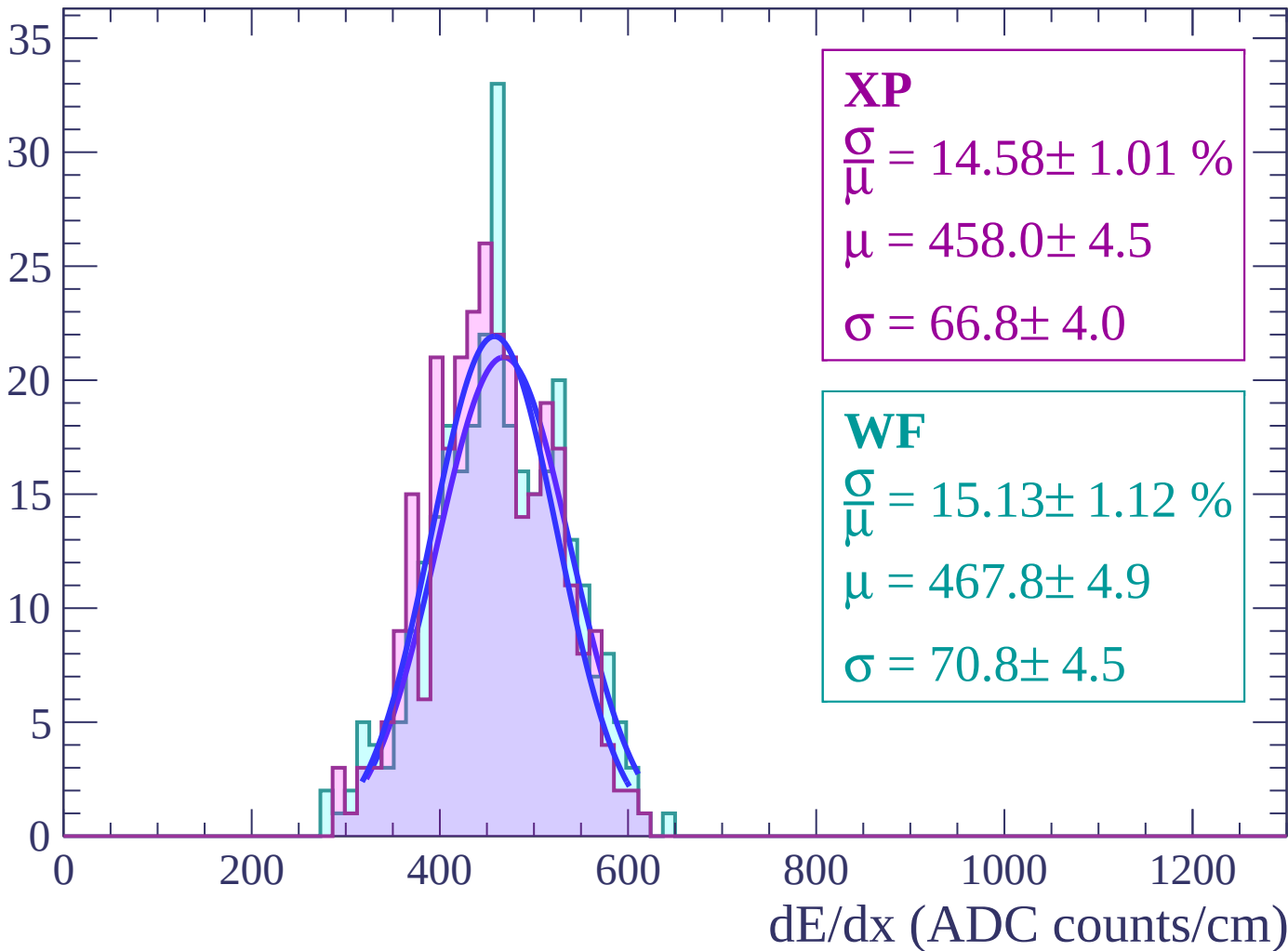
Energy loss |  $-22 < \theta < -20$

Count



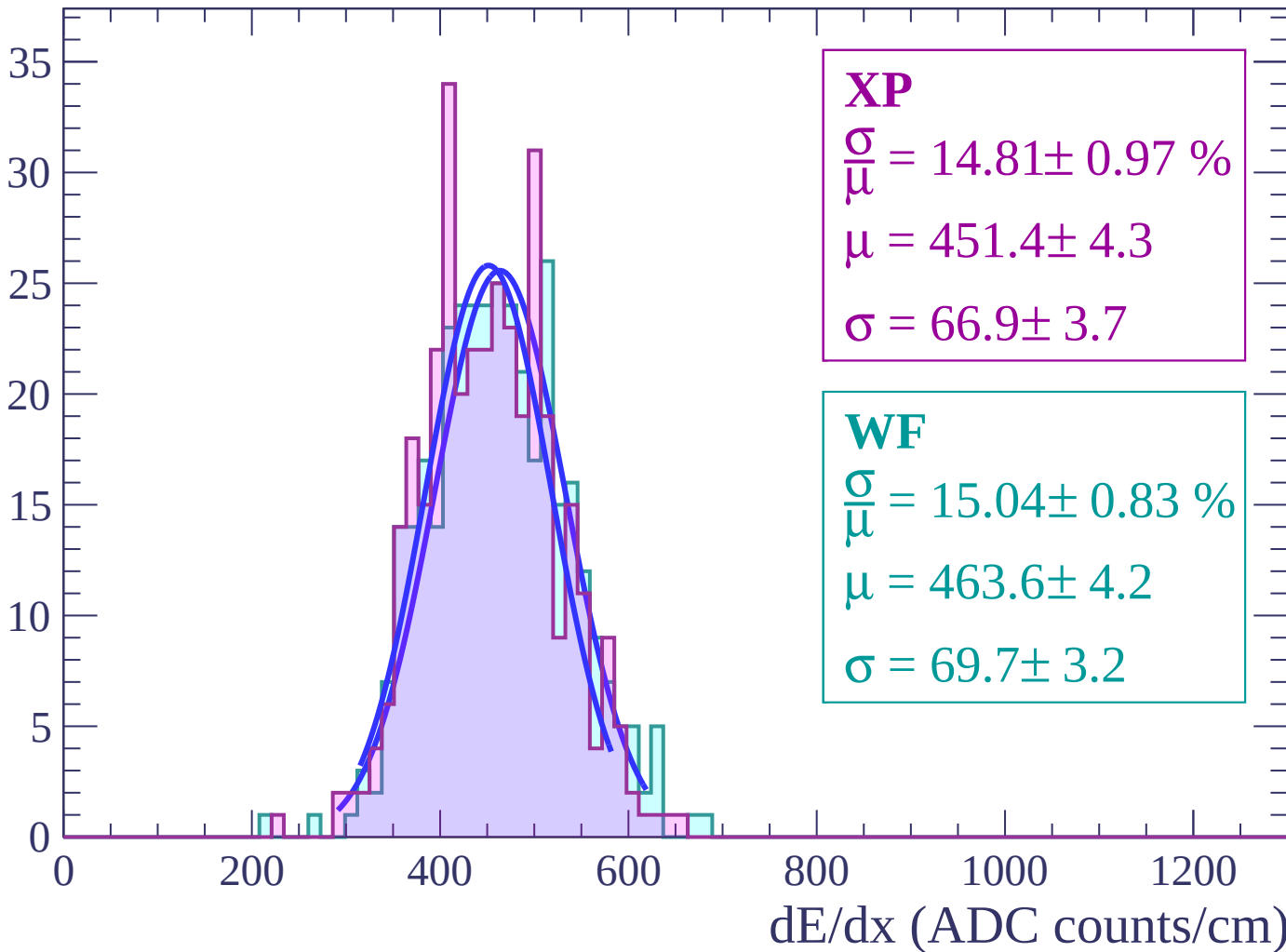
Energy loss |  $-20 < \theta < -18$

Count



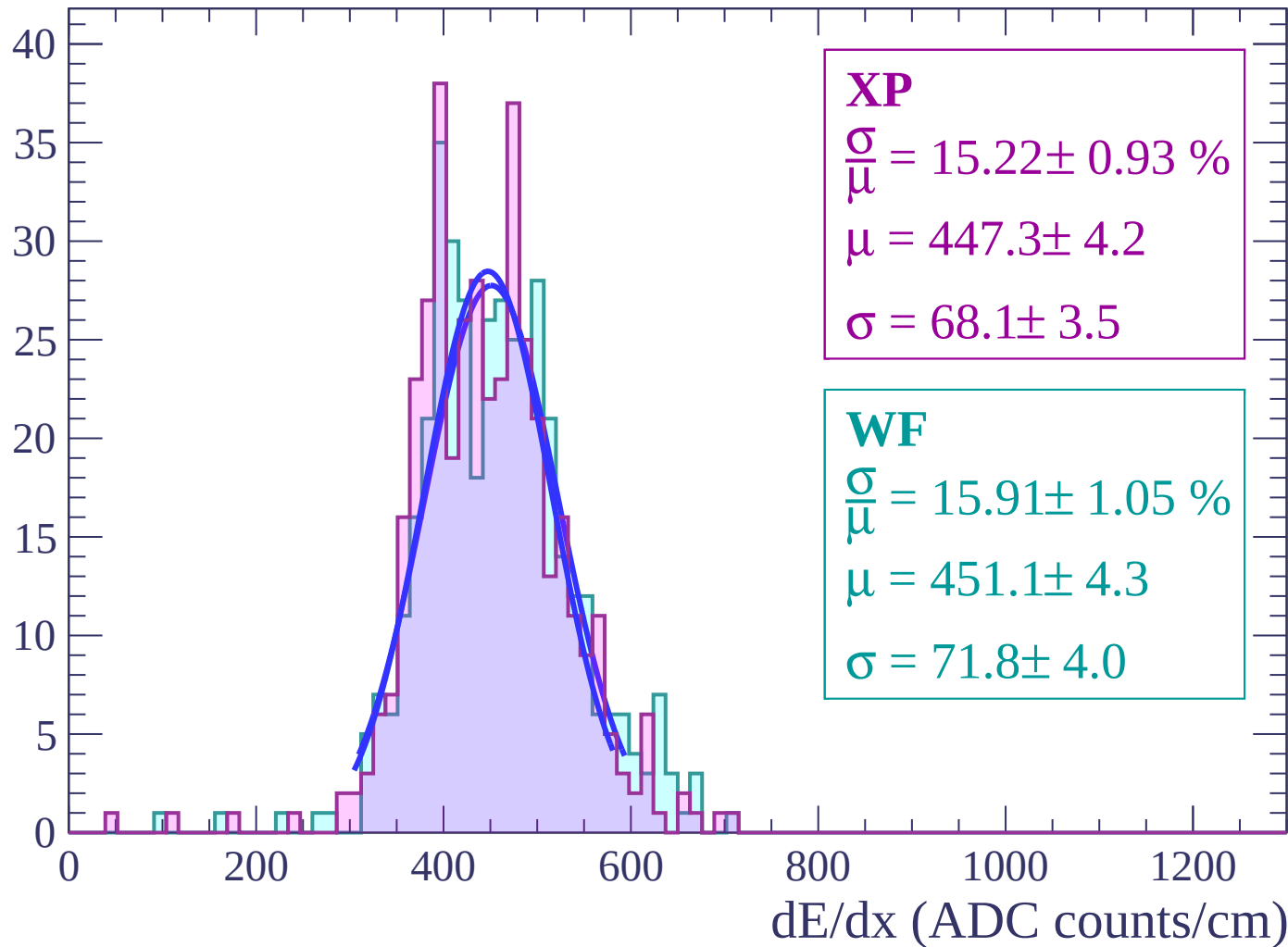
Energy loss |  $-18 < \theta < -16$

Count



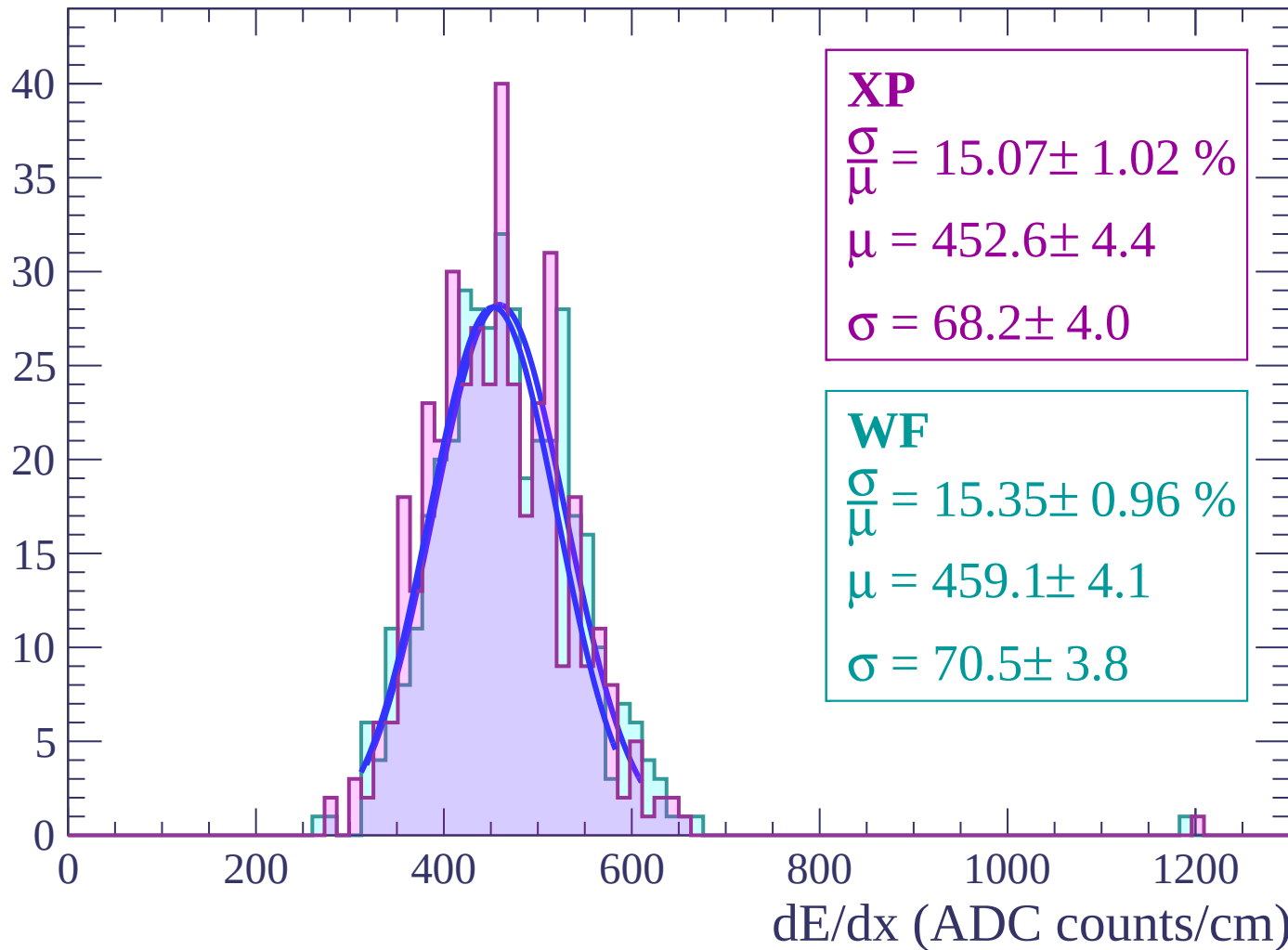
Energy loss |  $-16 < \theta < -14$

Count



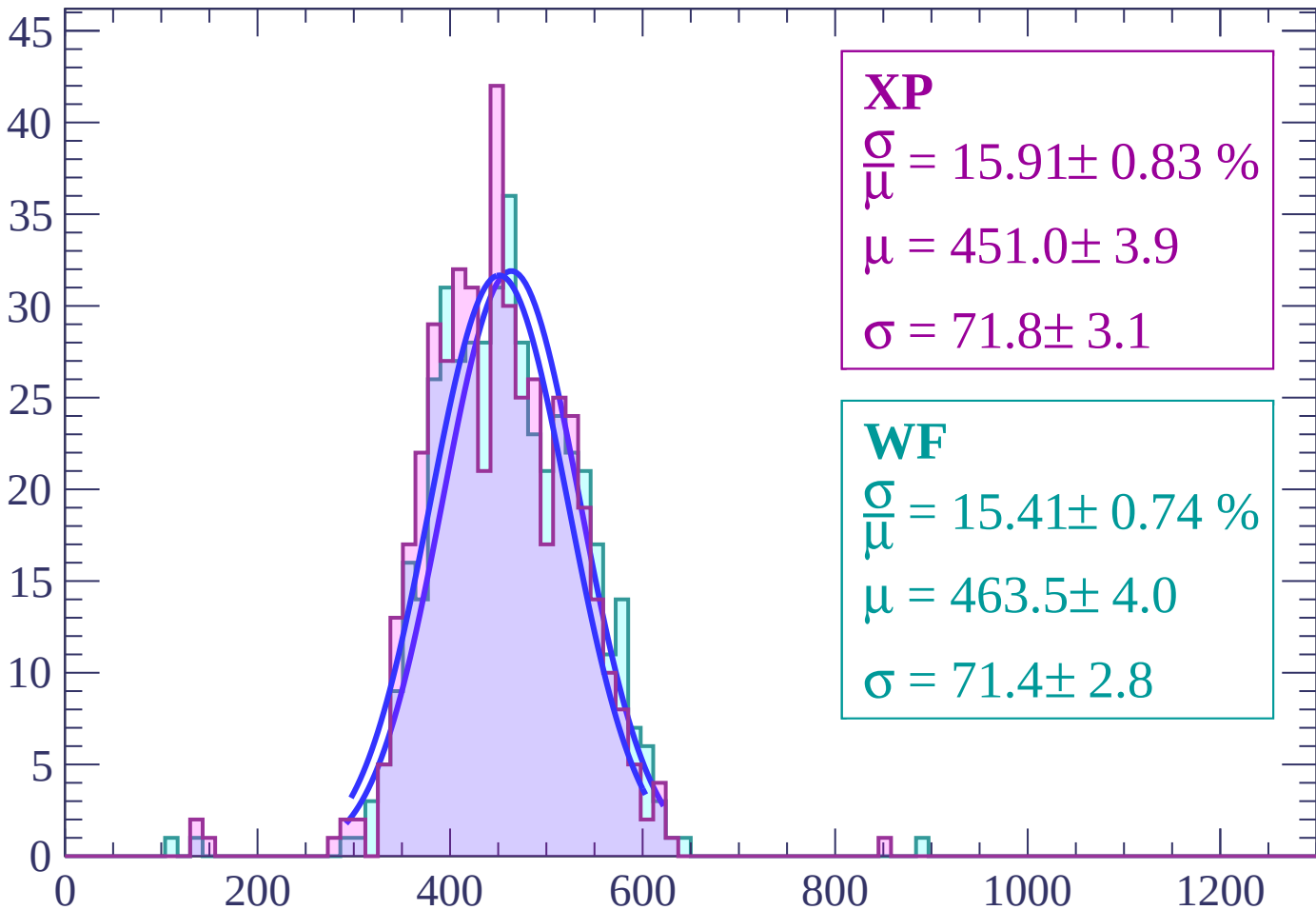
Energy loss |  $-14 < \theta < -12$

Count



Energy loss |  $-12 < \theta < -10$

Count



**XP**

$$\frac{\sigma}{\mu} = 15.91 \pm 0.83 \%$$

$$\mu = 451.0 \pm 3.9$$

$$\sigma = 71.8 \pm 3.1$$

**WF**

$$\frac{\sigma}{\mu} = 15.41 \pm 0.74 \%$$

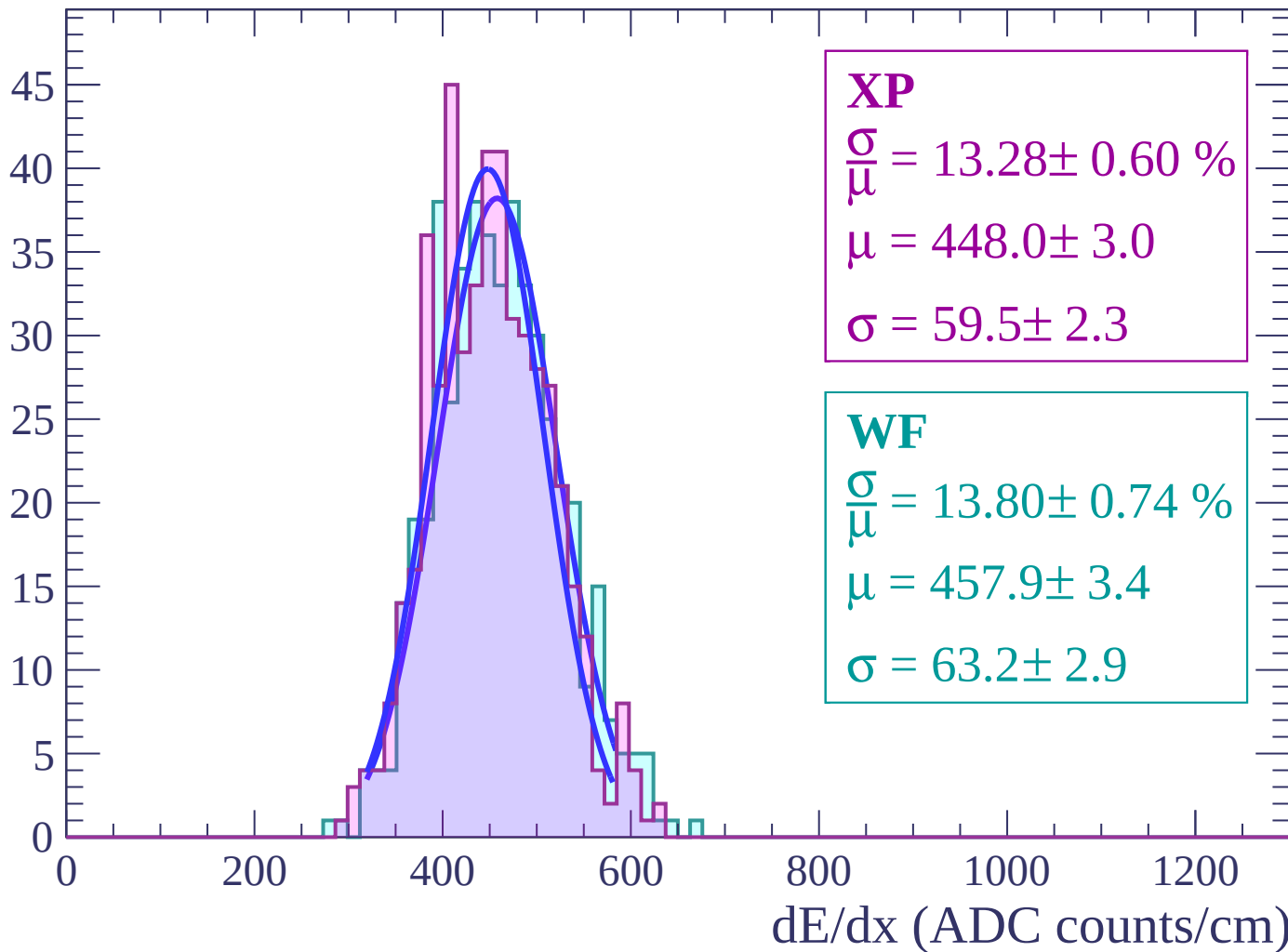
$$\mu = 463.5 \pm 4.0$$

$$\sigma = 71.4 \pm 2.8$$

$dE/dx$  (ADC counts/cm)

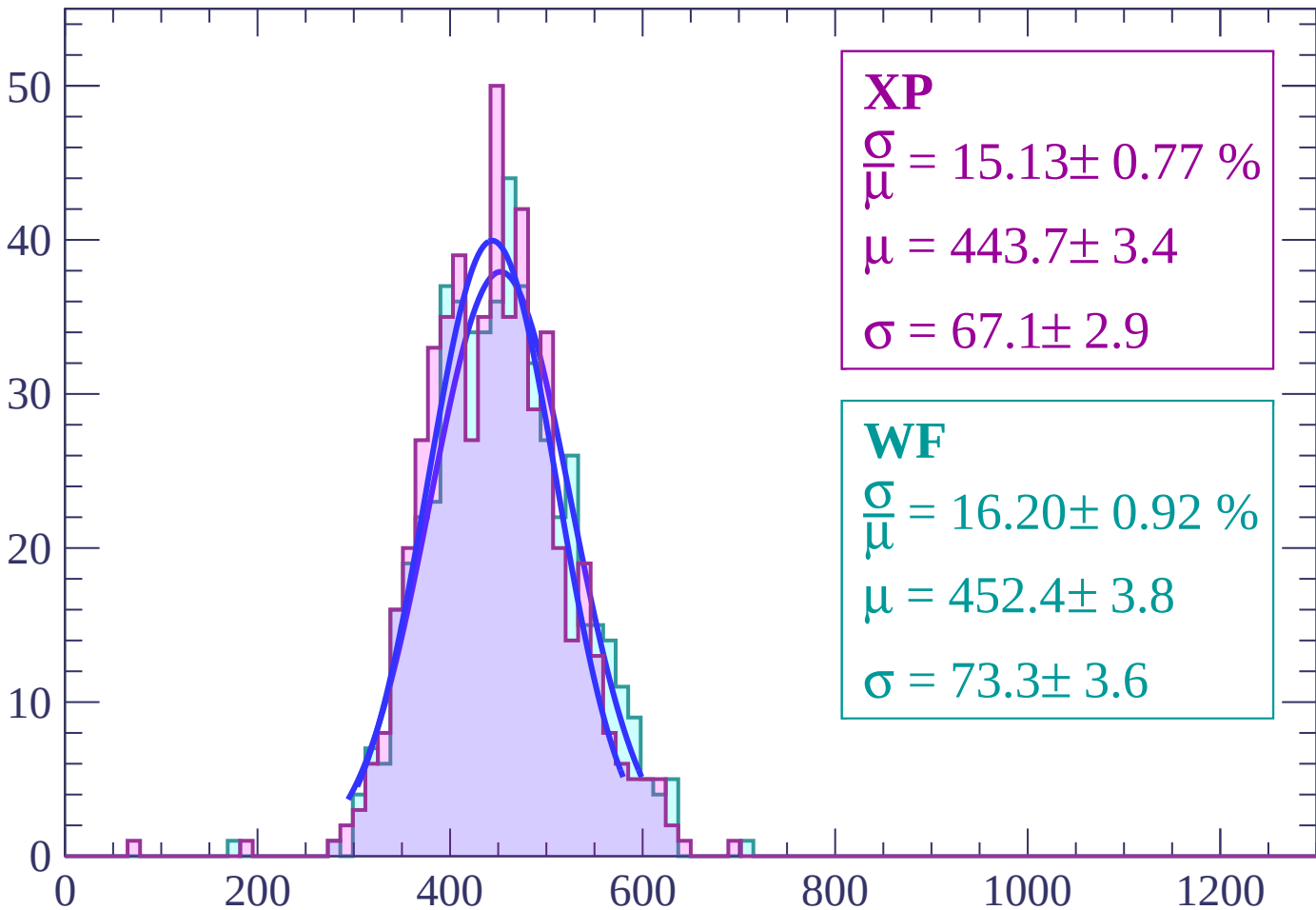
# Energy loss | $-10 < \theta < -8$

Count



Energy loss |  $-8 < \theta < -6$

Count



**XP**

$$\frac{\sigma}{\mu} = 15.13 \pm 0.77 \%$$

$$\mu = 443.7 \pm 3.4$$

$$\sigma = 67.1 \pm 2.9$$

**WF**

$$\frac{\sigma}{\mu} = 16.20 \pm 0.92 \%$$

$$\mu = 452.4 \pm 3.8$$

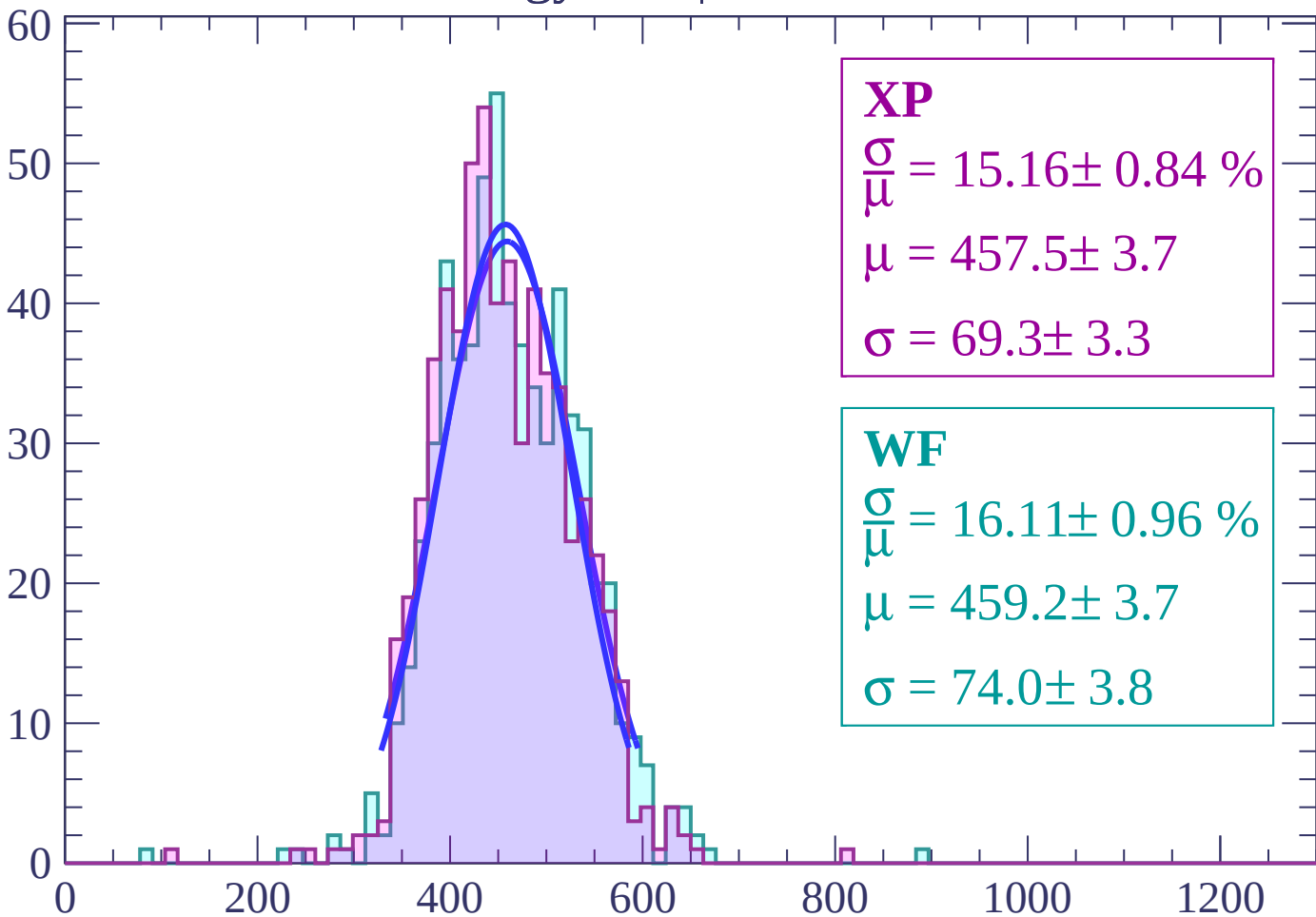
$$\sigma = 73.3 \pm 3.6$$

$dE/dx$  (ADC counts/cm)



Energy loss |  $-6 < \theta < -4$

Count



**XP**

$$\frac{\sigma}{\mu} = 15.16 \pm 0.84 \%$$

$$\mu = 457.5 \pm 3.7$$

$$\sigma = 69.3 \pm 3.3$$

**WF**

$$\frac{\sigma}{\mu} = 16.11 \pm 0.96 \%$$

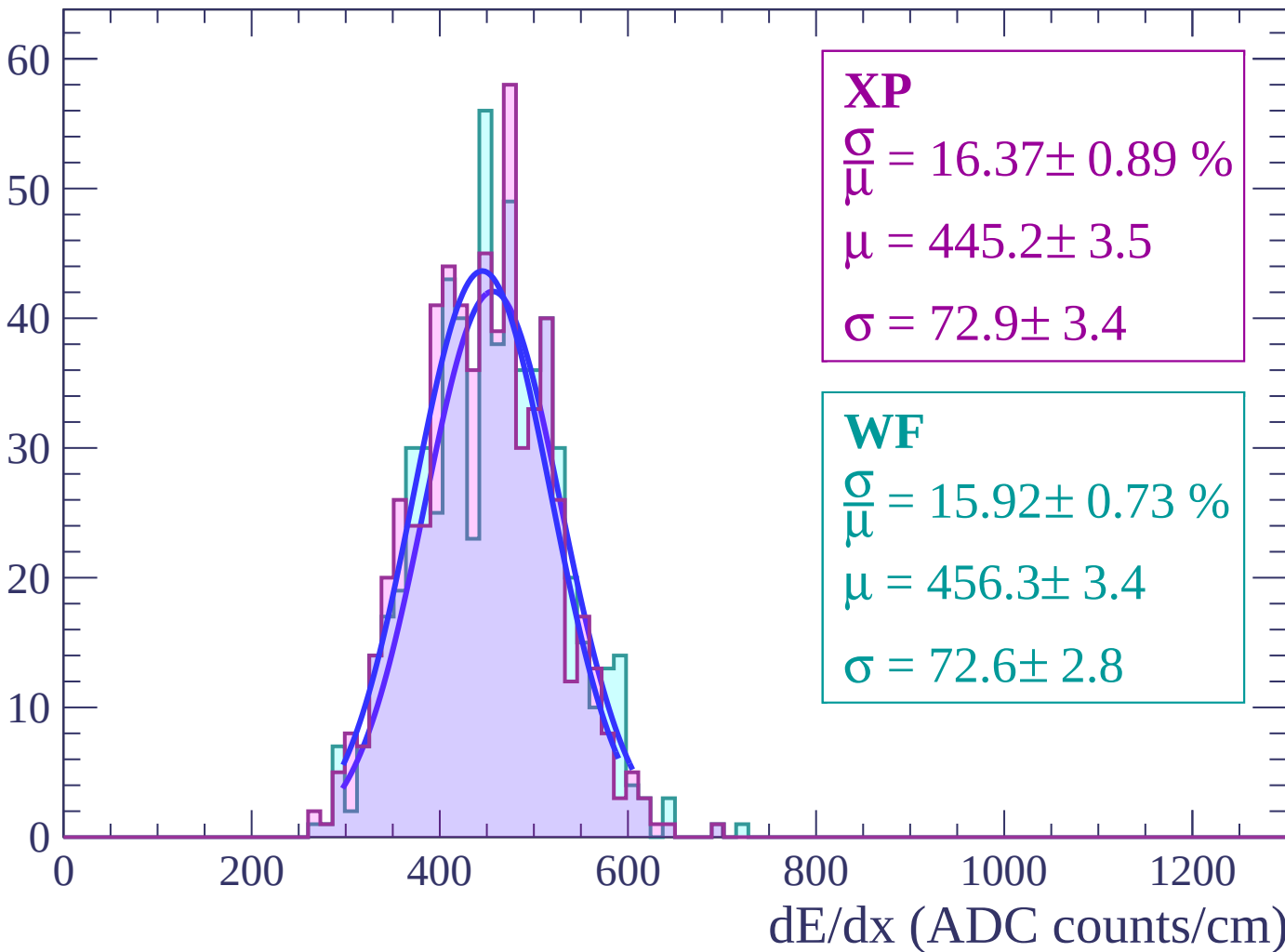
$$\mu = 459.2 \pm 3.7$$

$$\sigma = 74.0 \pm 3.8$$

$dE/dx$  (ADC counts/cm)

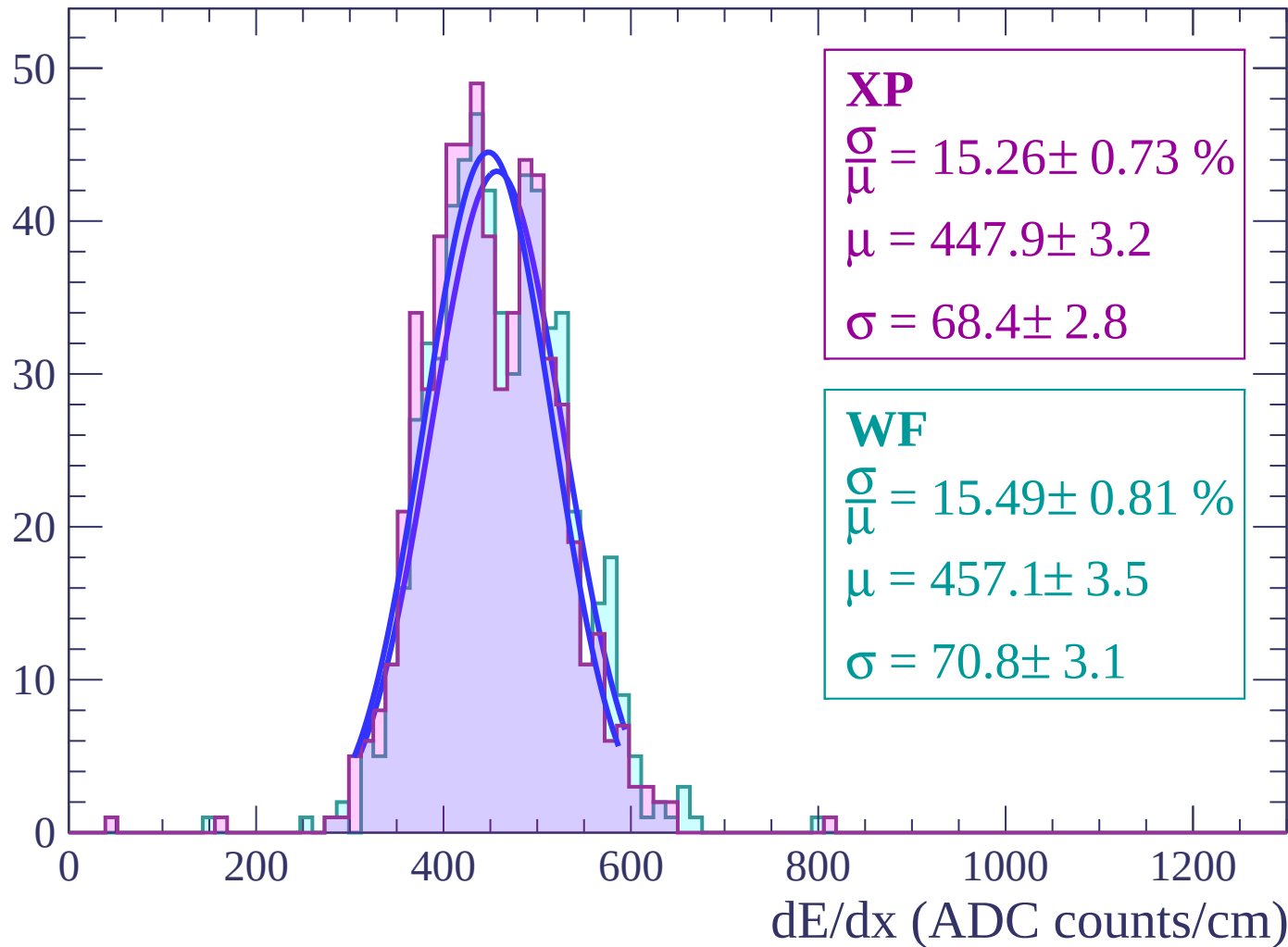
Energy loss |  $-4 < \theta < -2$

Count



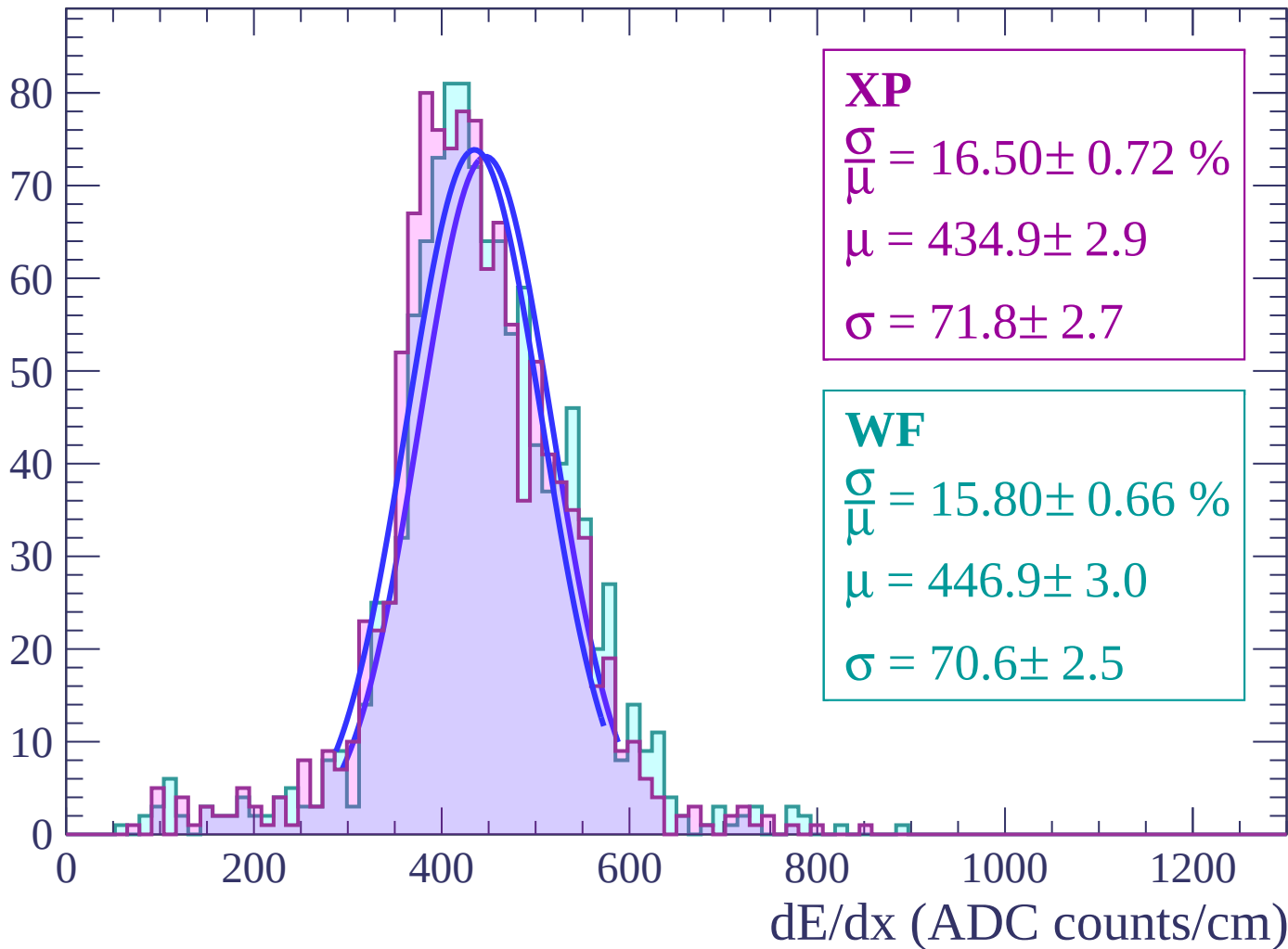
# Energy loss | $-2 < \theta < 0$

Count



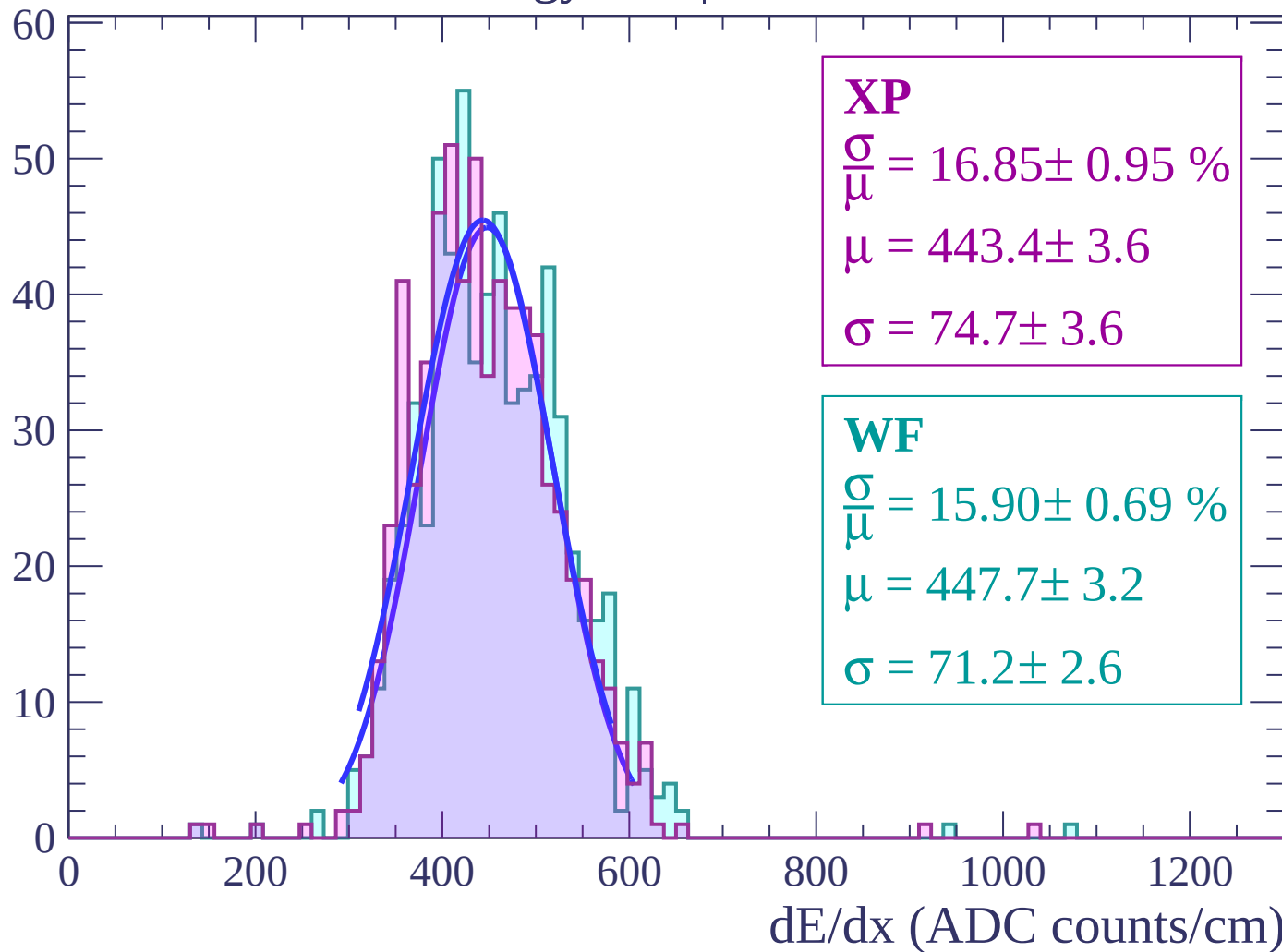
# Energy loss | $0 < \theta < 2$

Count



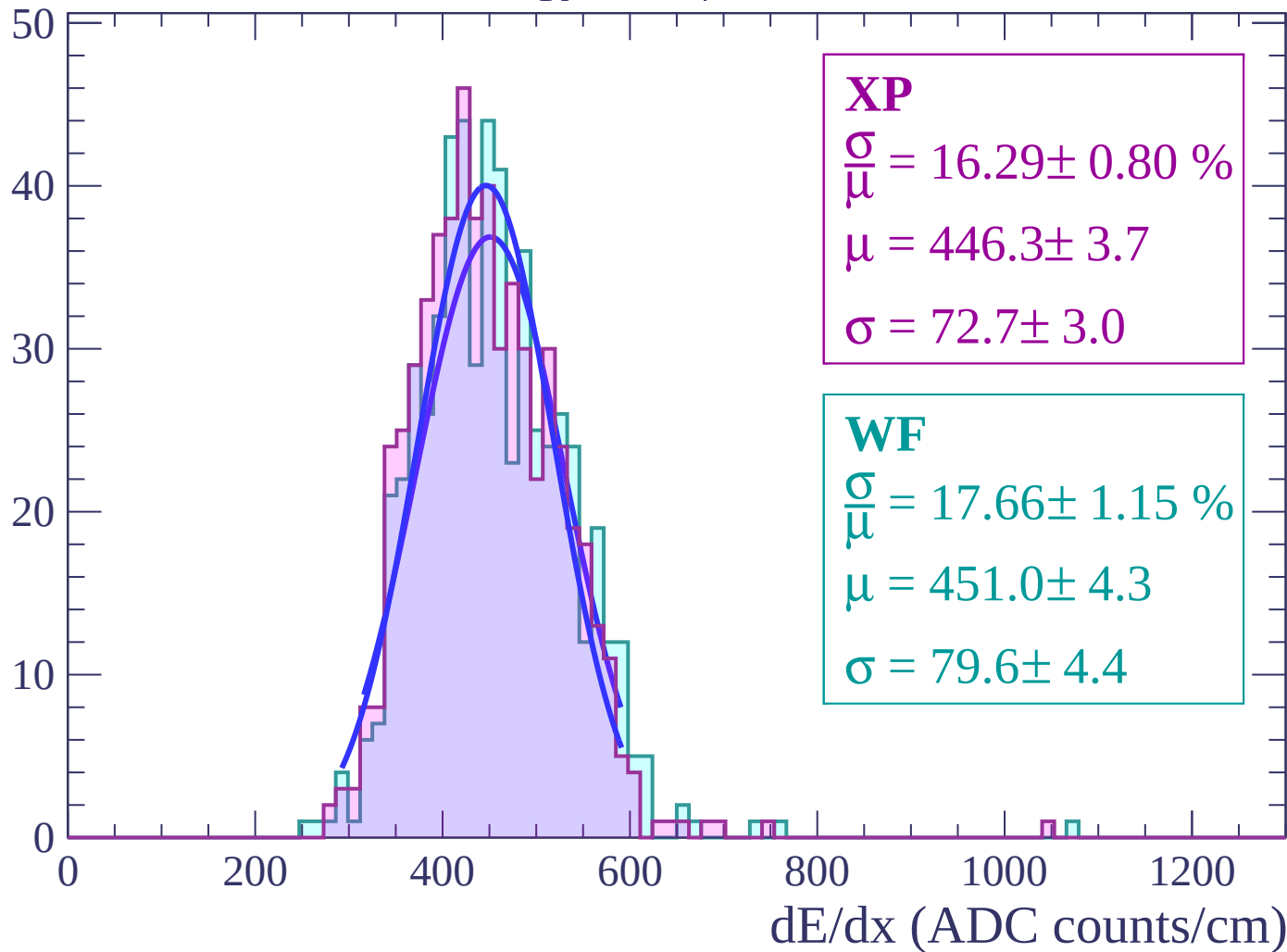
# Energy loss | $2 < \theta < 4$

Count



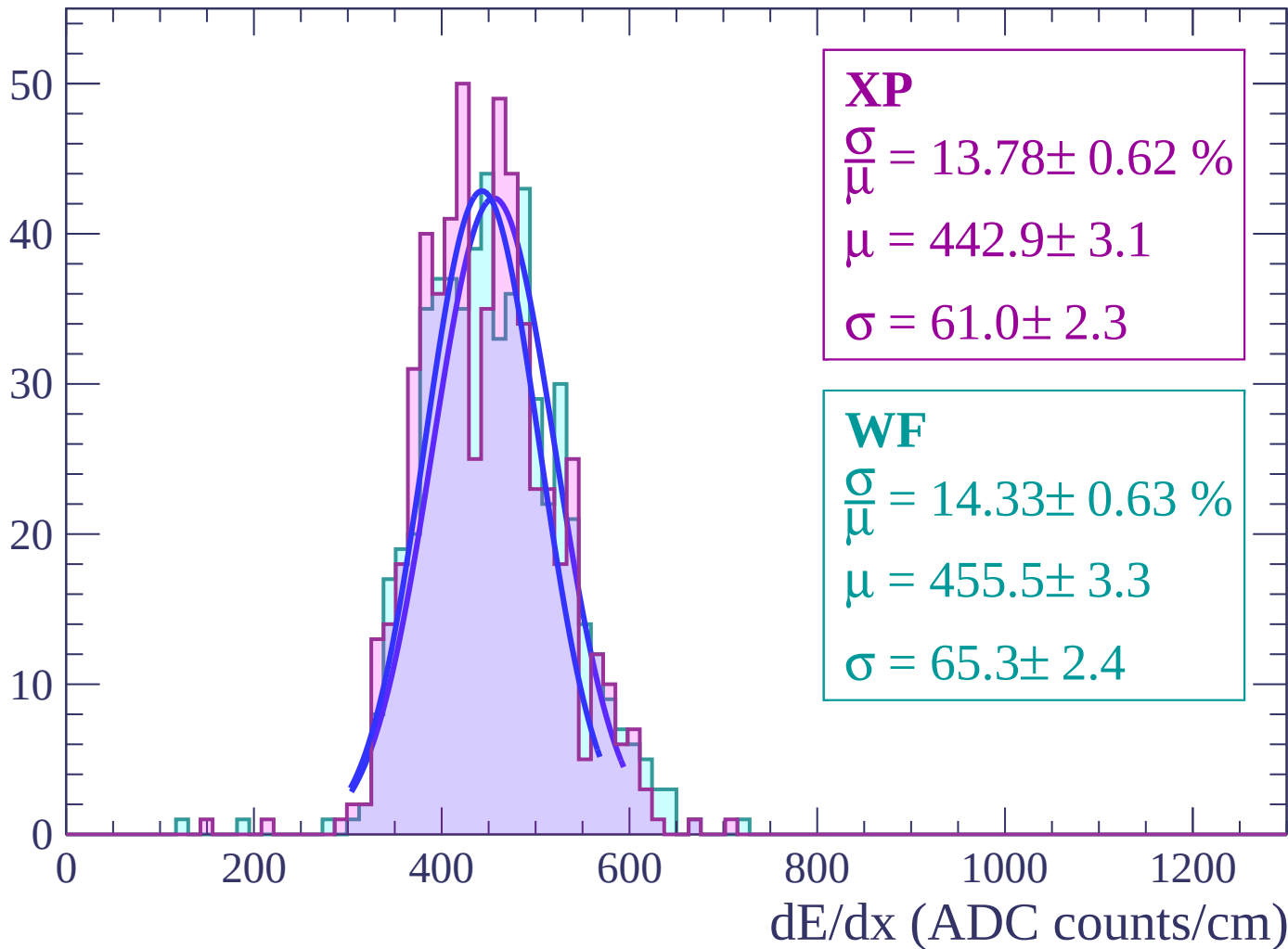
Energy loss |  $4 < \theta < 6$

Count



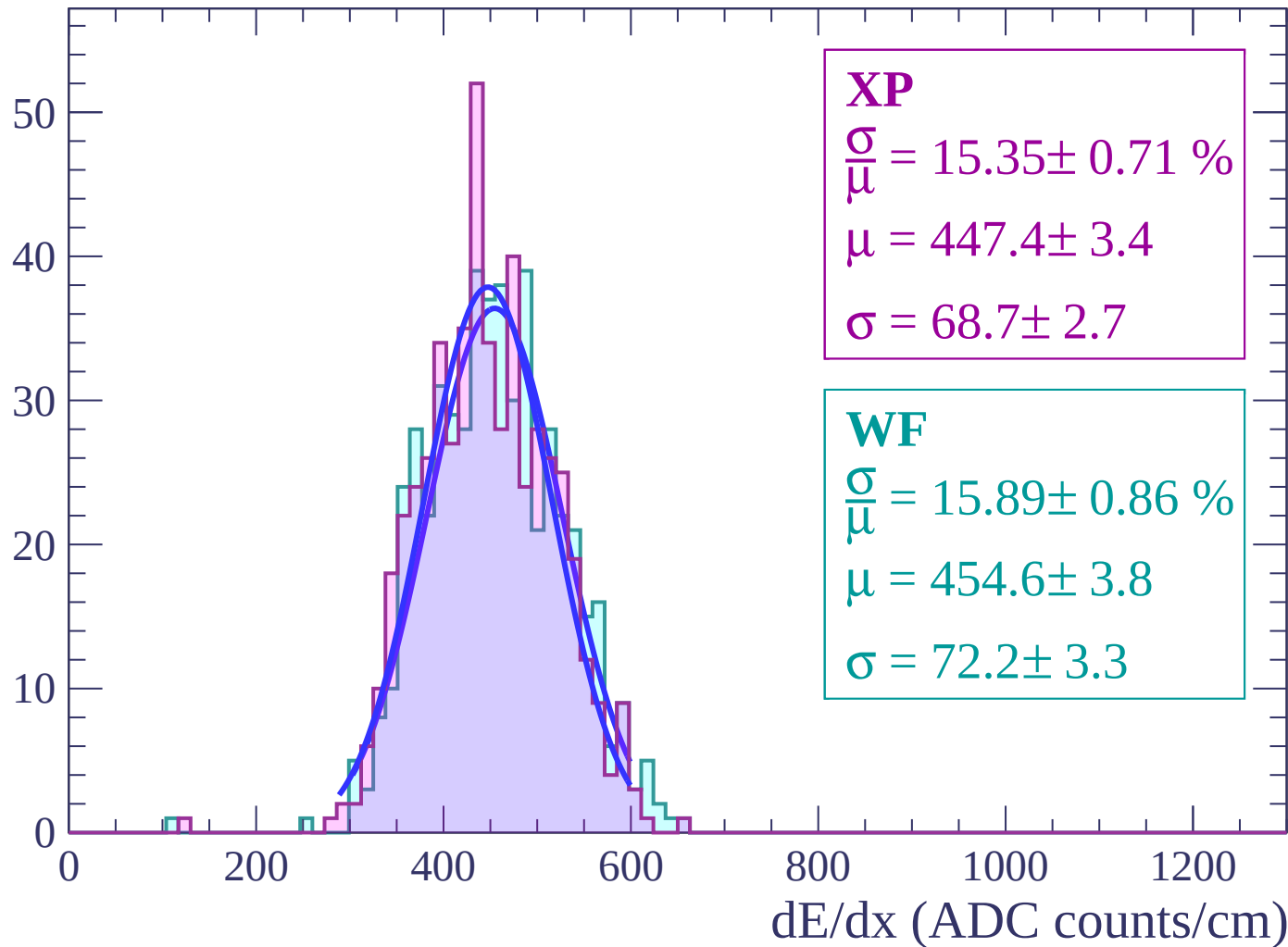
# Energy loss | $6 < \theta < 8$

Count



Energy loss |  $8 < \theta < 10$

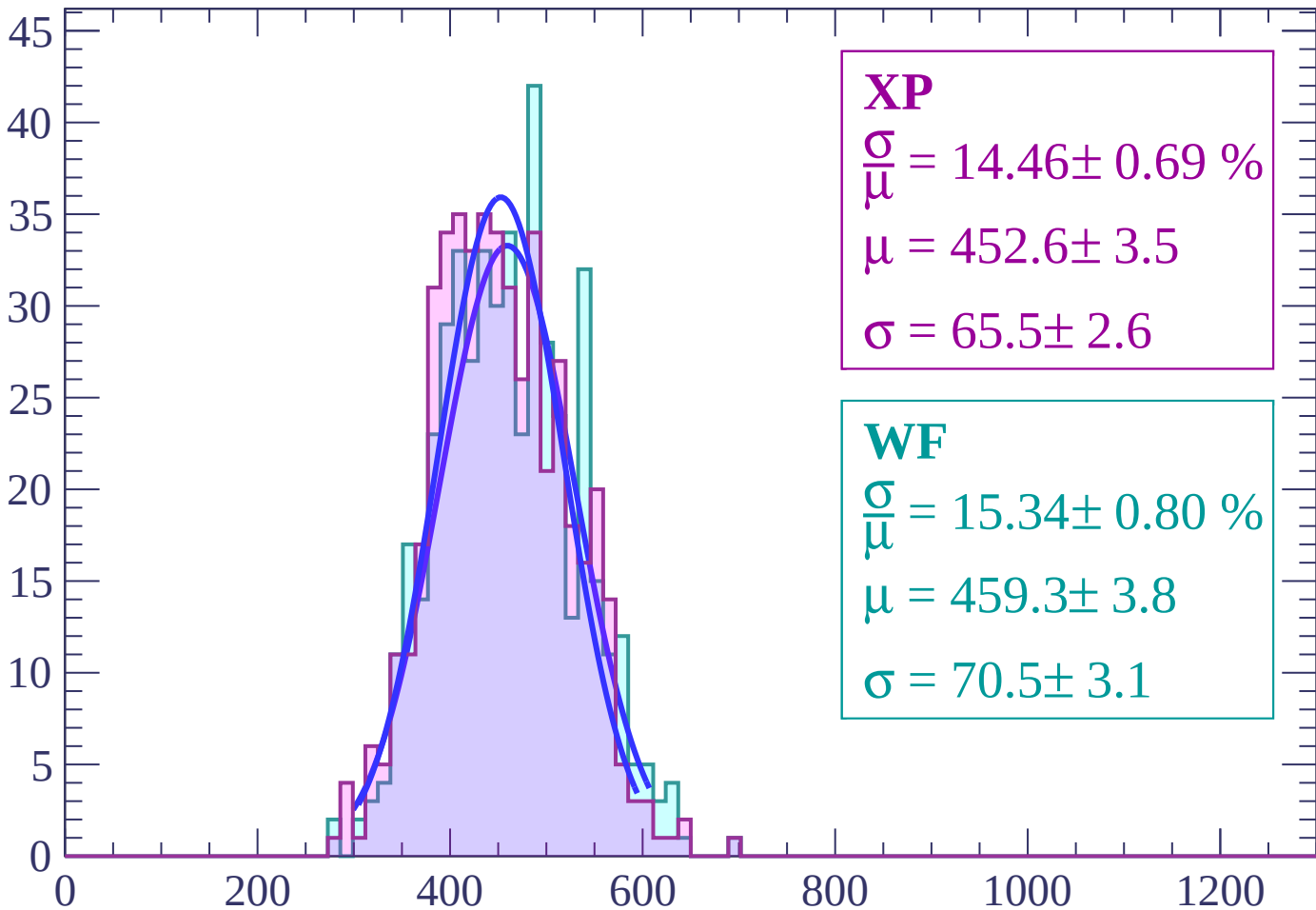
Count





Energy loss |  $10 < \theta < 12$

Count



**XP**

$$\frac{\sigma}{\mu} = 14.46 \pm 0.69 \%$$

$$\mu = 452.6 \pm 3.5$$

$$\sigma = 65.5 \pm 2.6$$

**WF**

$$\frac{\sigma}{\mu} = 15.34 \pm 0.80 \%$$

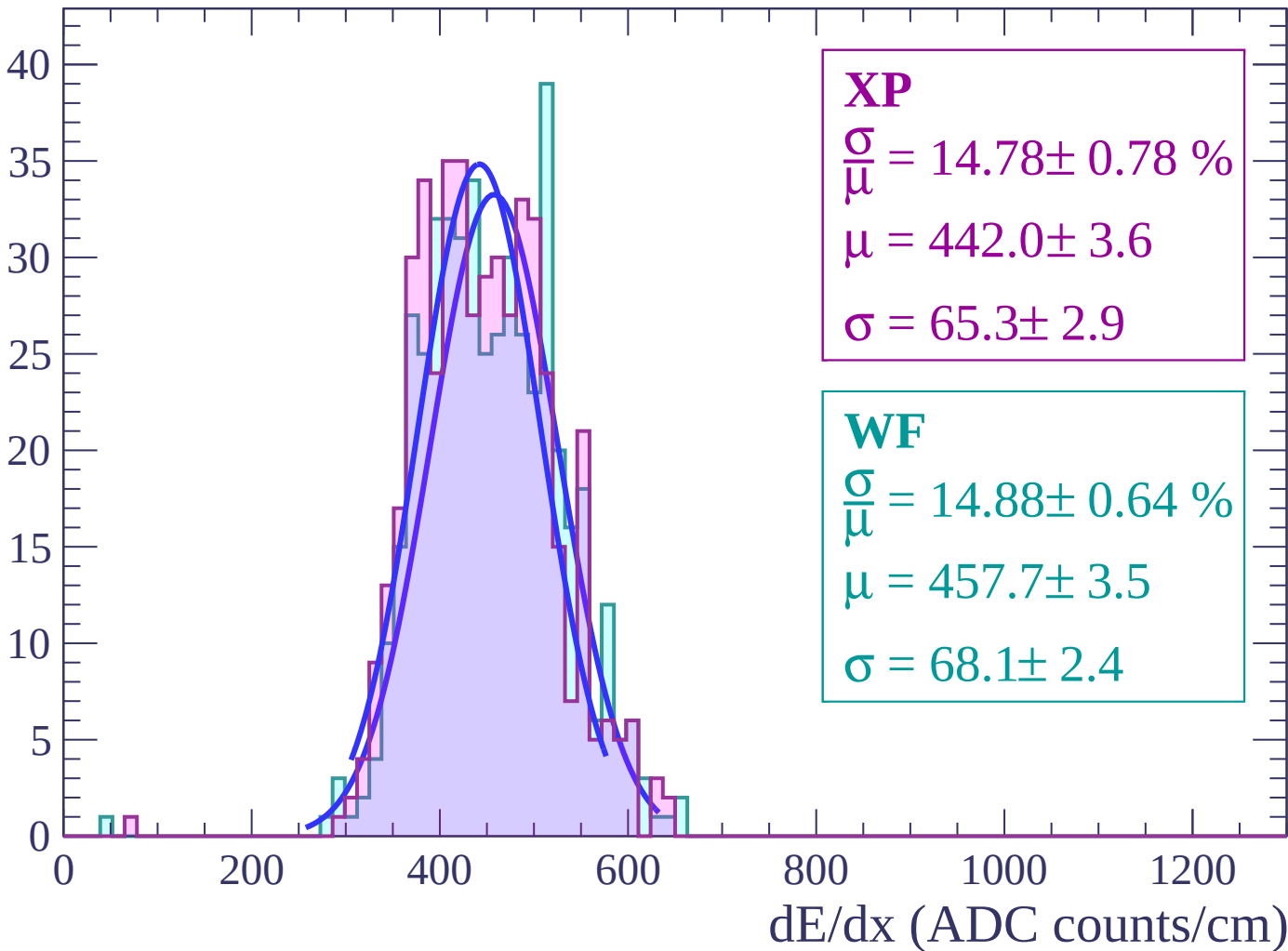
$$\mu = 459.3 \pm 3.8$$

$$\sigma = 70.5 \pm 3.1$$

$dE/dx$  (ADC counts/cm)

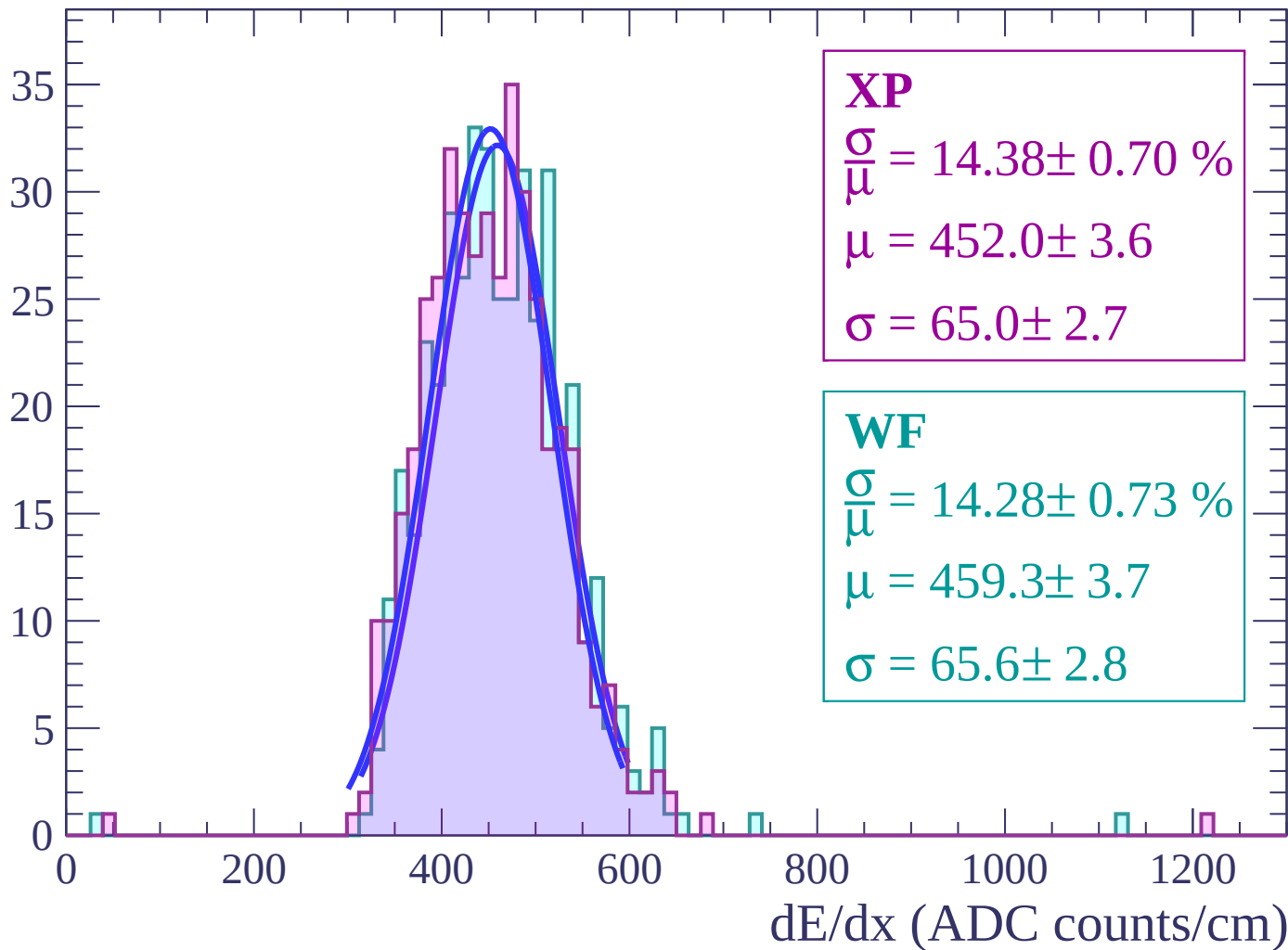
Energy loss |  $12 < \theta < 14$

Count



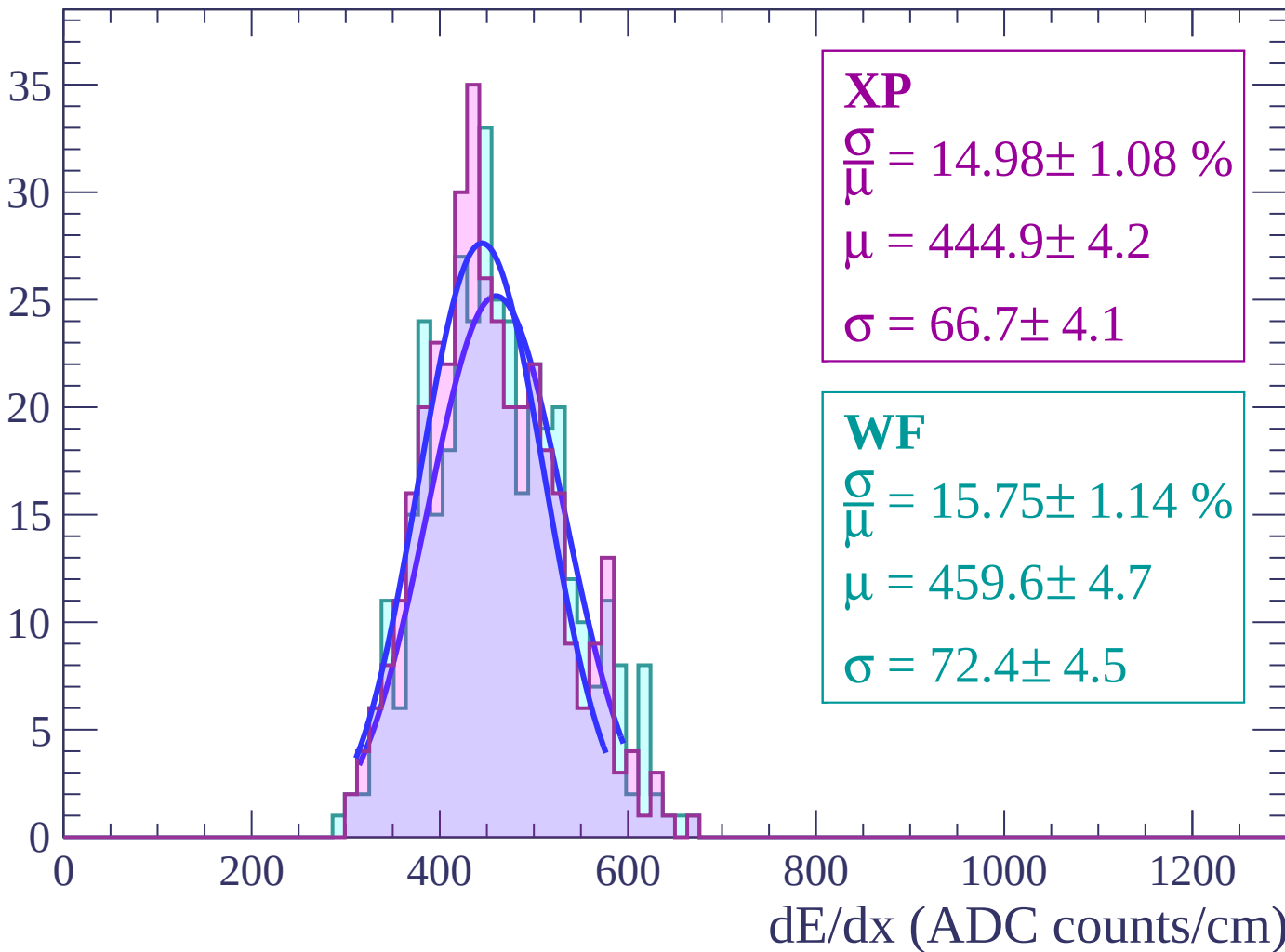
# Energy loss | $14 < \theta < 16$

Count



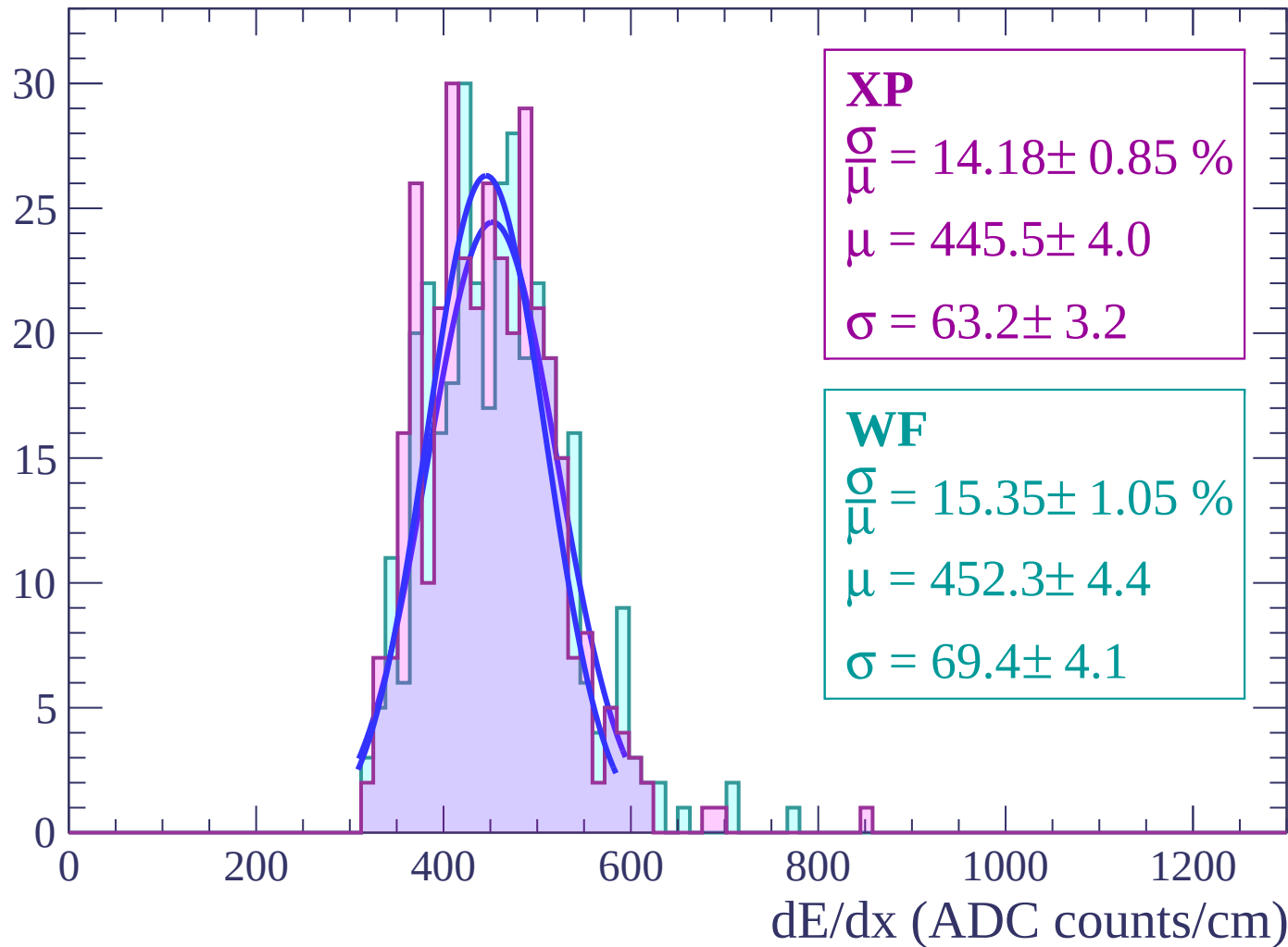
Energy loss |  $16 < \theta < 18$

Count



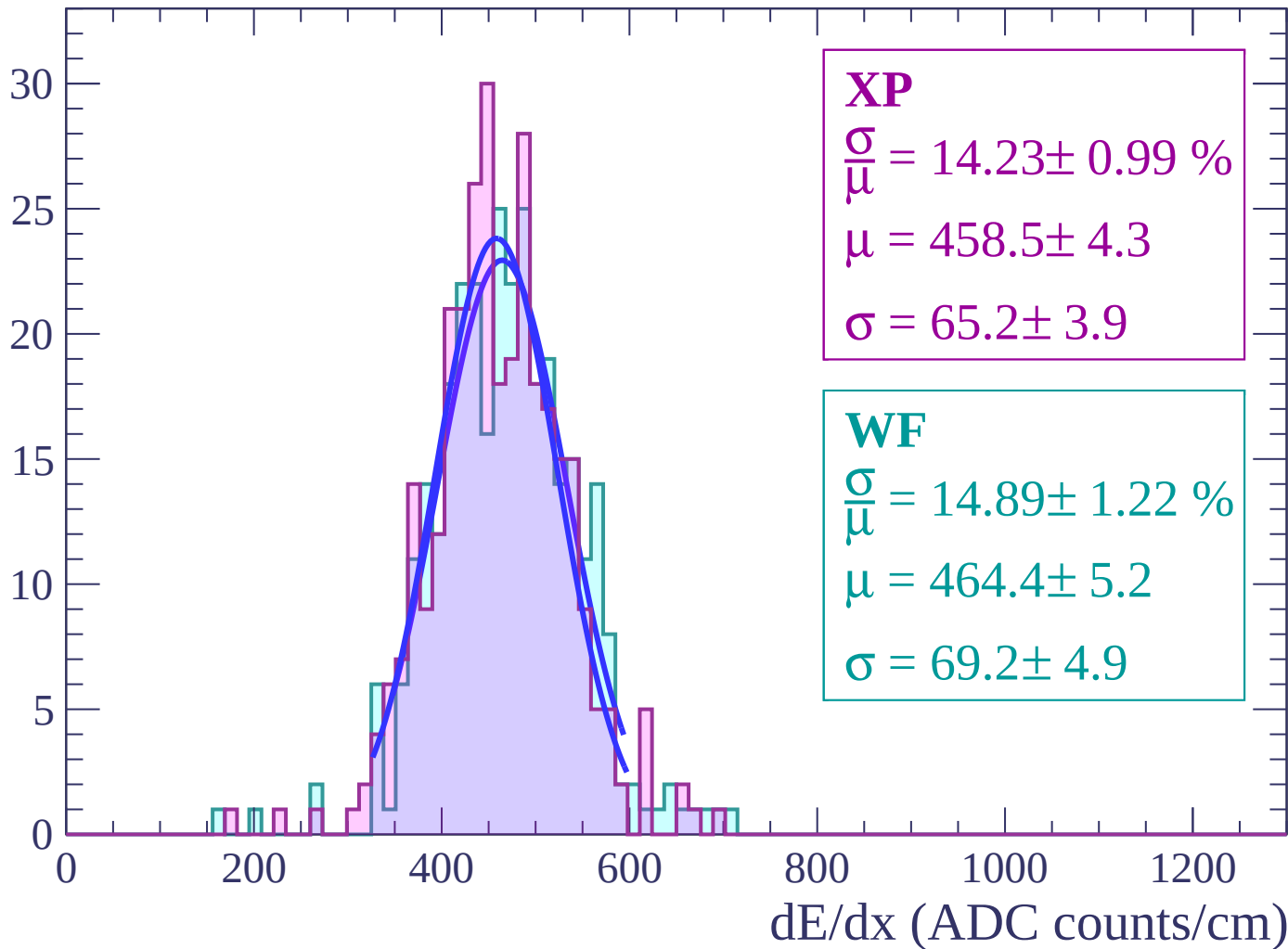
# Energy loss | $18 < \theta < 20$

Count



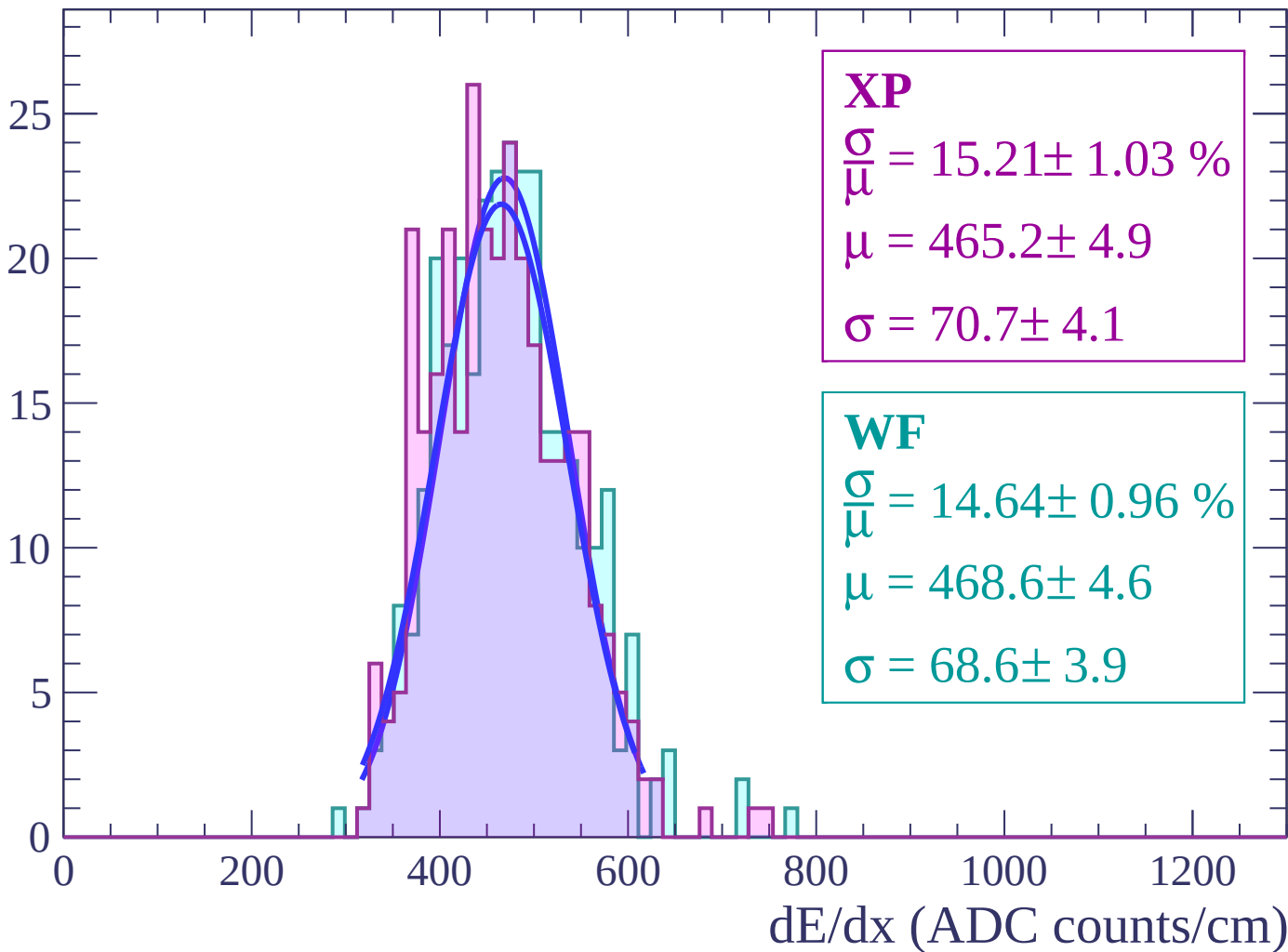
Energy loss |  $20 < \theta < 22$

Count



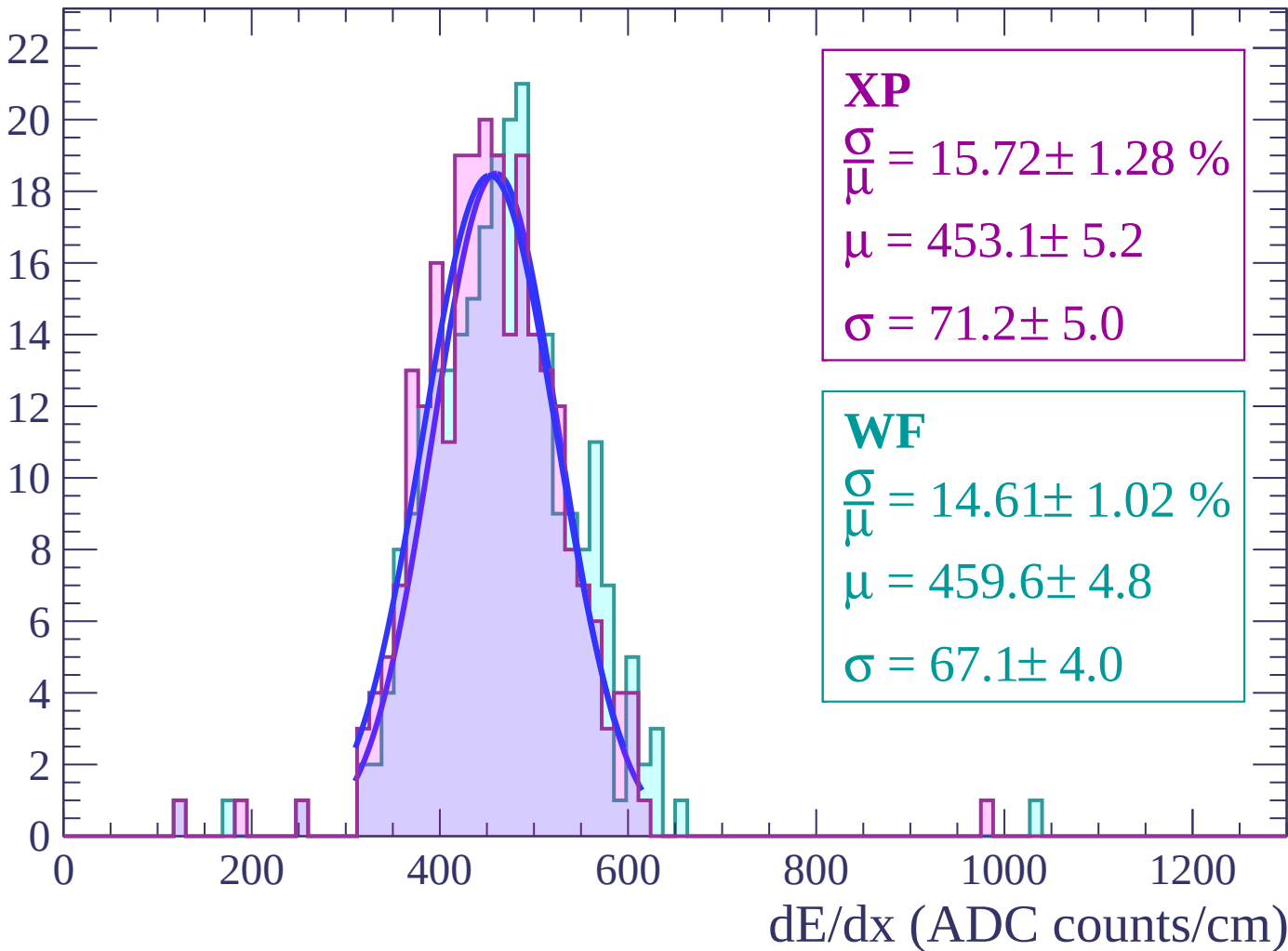
Energy loss |  $22 < \theta < 24$

Count



Energy loss |  $24 < \theta < 26$

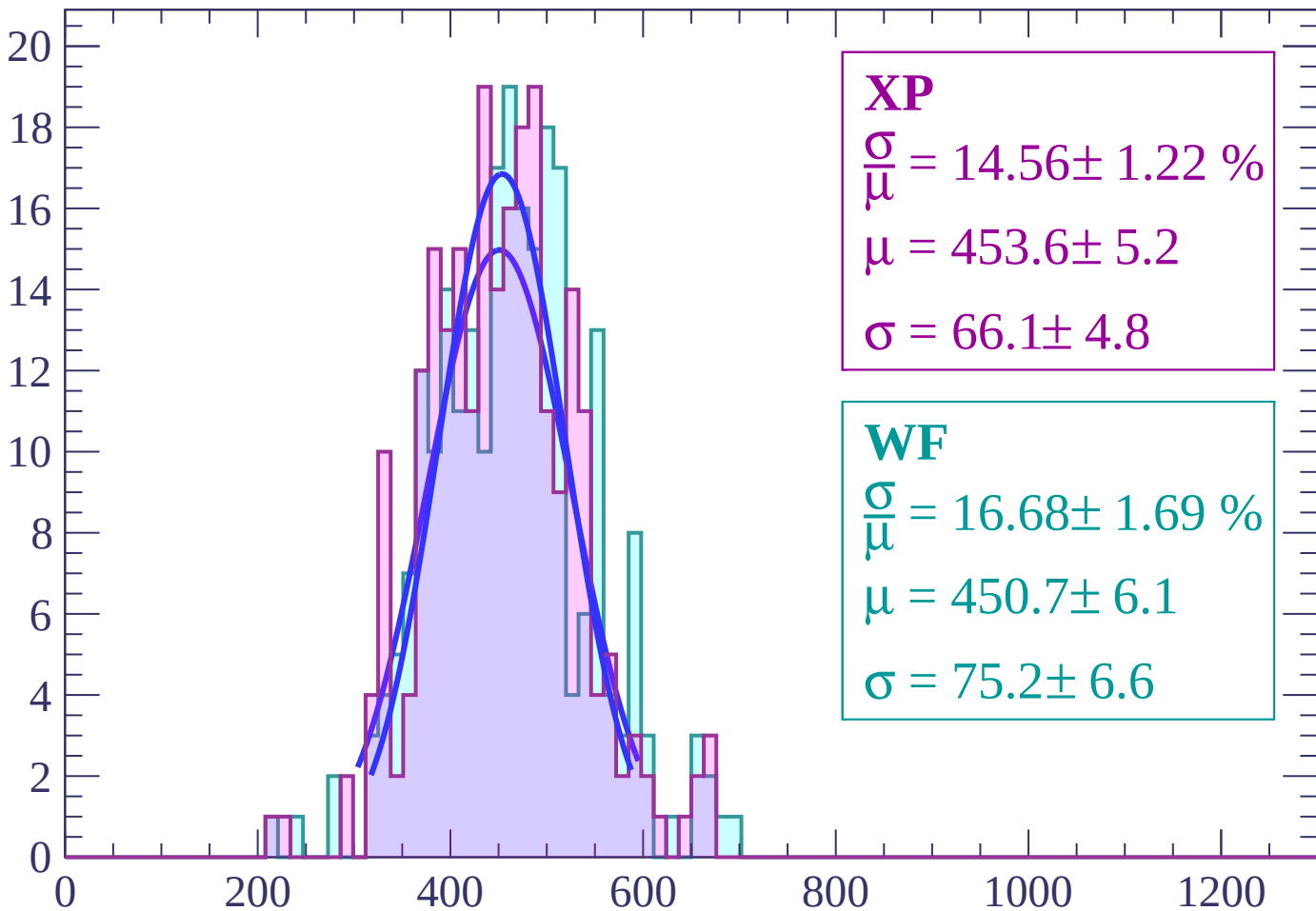
Count





Energy loss |  $26 < \theta < 28$

Count



**XP**

$$\frac{\sigma}{\mu} = 14.56 \pm 1.22 \%$$

$$\mu = 453.6 \pm 5.2$$

$$\sigma = 66.1 \pm 4.8$$

**WF**

$$\frac{\sigma}{\mu} = 16.68 \pm 1.69 \%$$

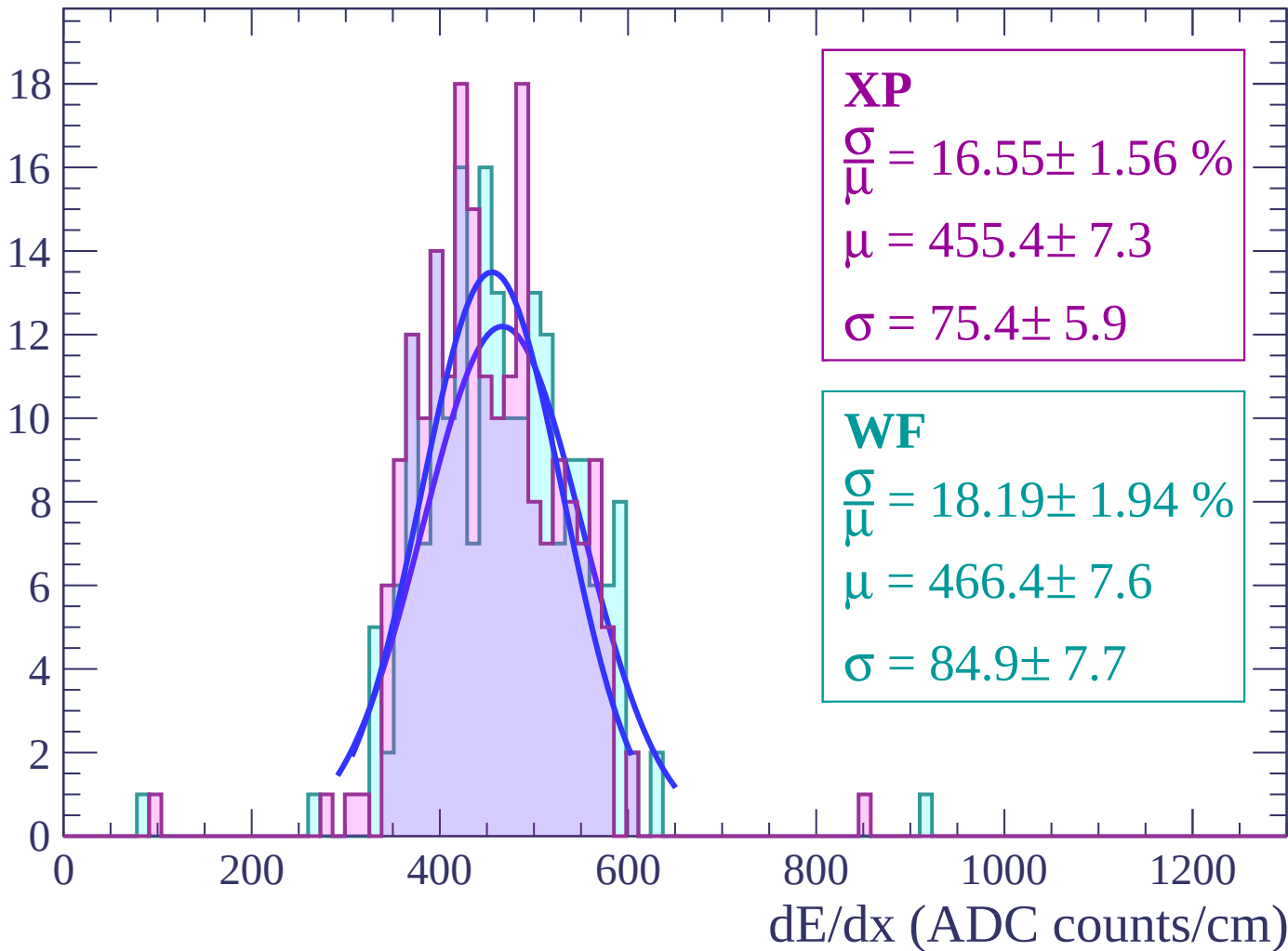
$$\mu = 450.7 \pm 6.1$$

$$\sigma = 75.2 \pm 6.6$$

$dE/dx$  (ADC counts/cm)

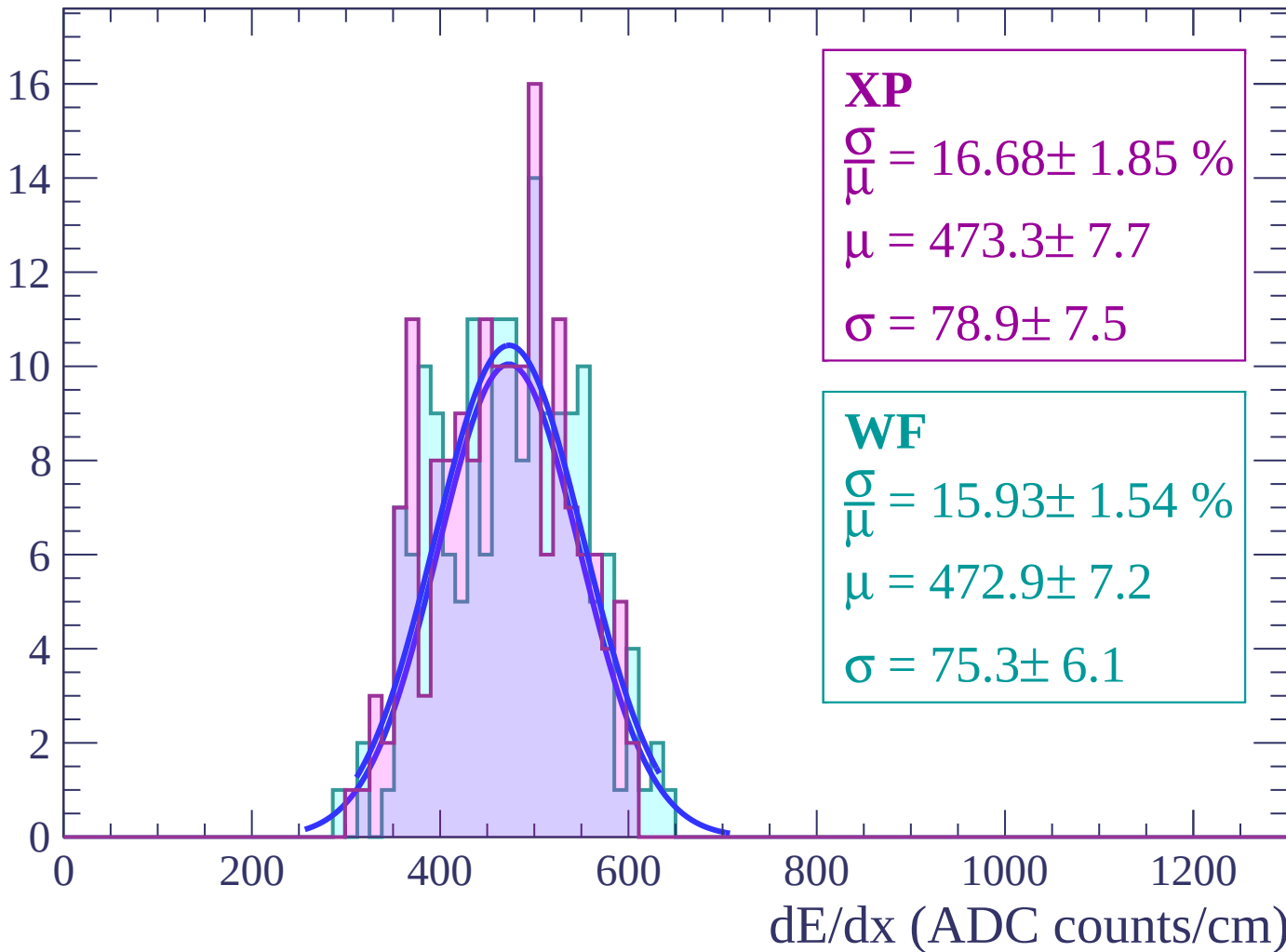
Energy loss |  $28 < \theta < 30$

Count



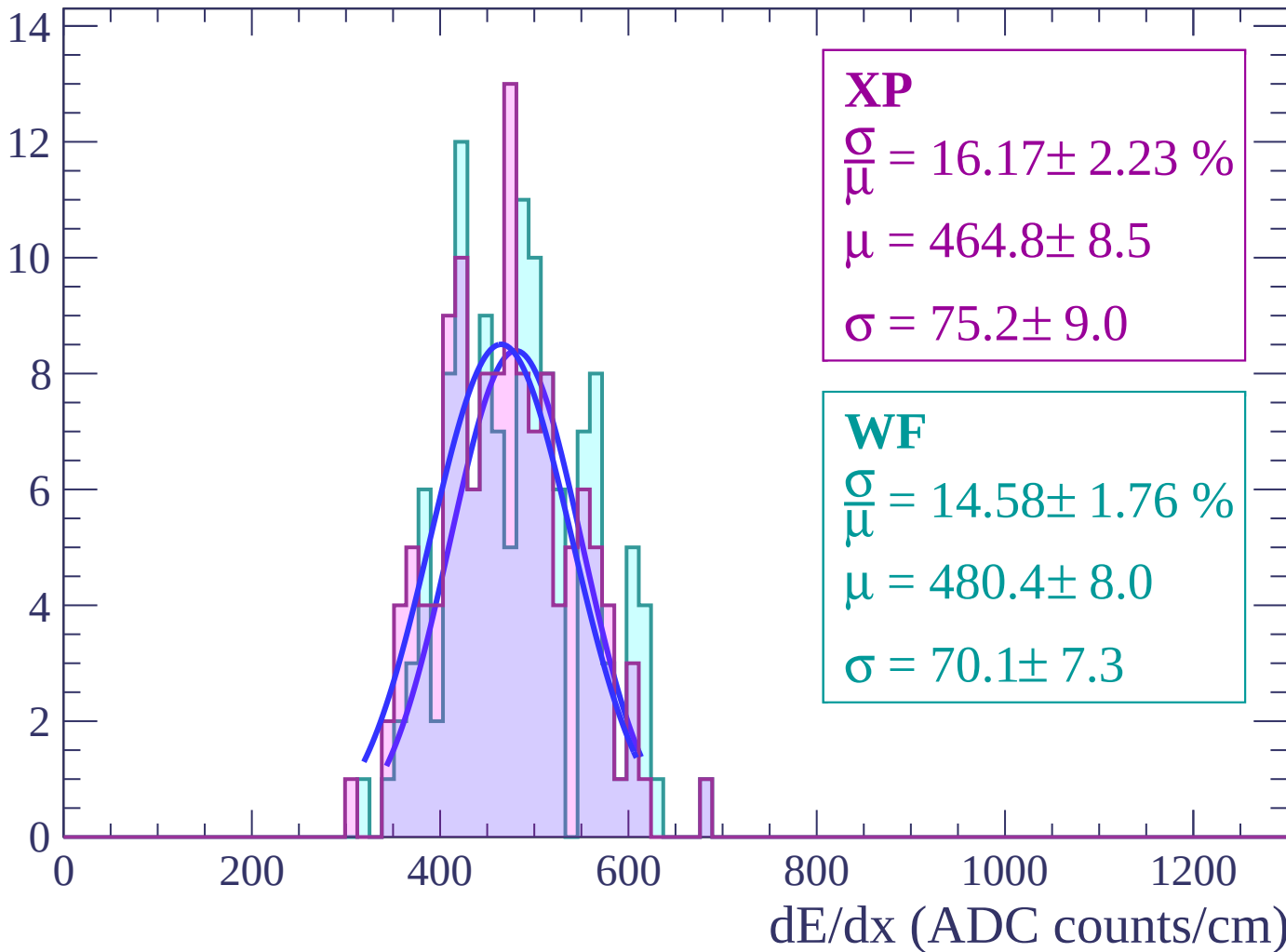
Energy loss |  $30 < \theta < 32$

Count



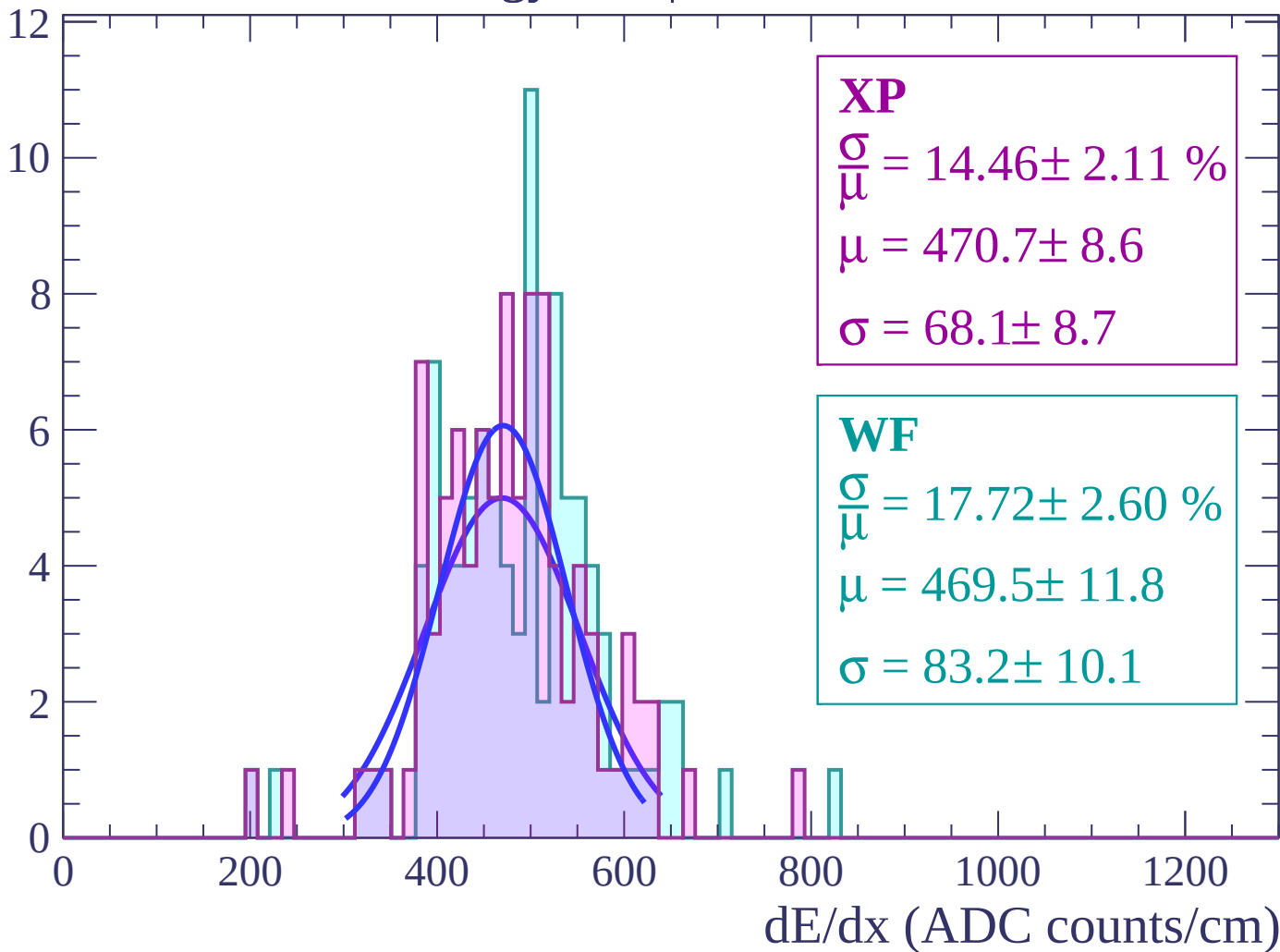
Energy loss |  $32 < \theta < 34$

Count



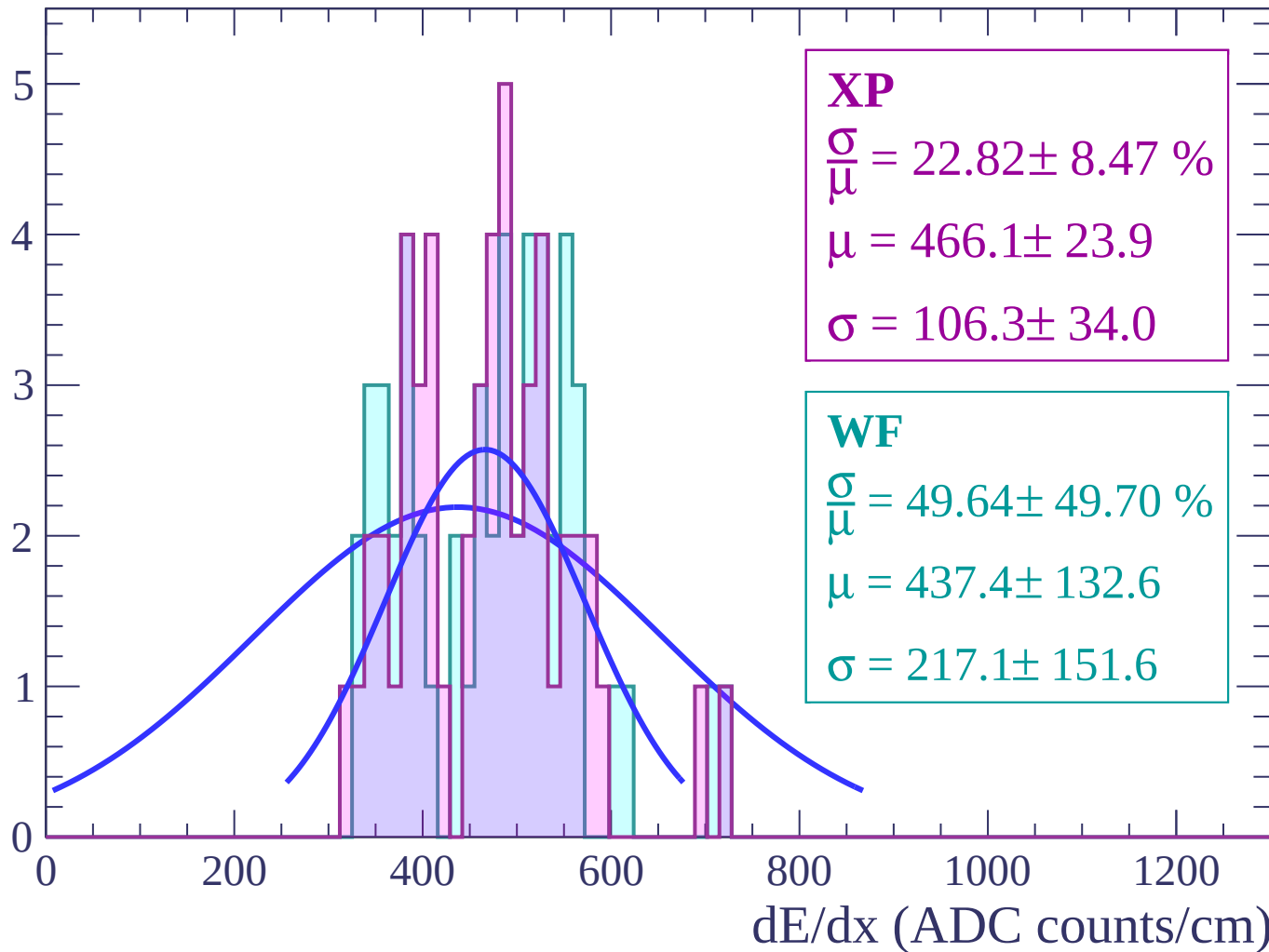
# Energy loss | $34 < \theta < 36$

Count



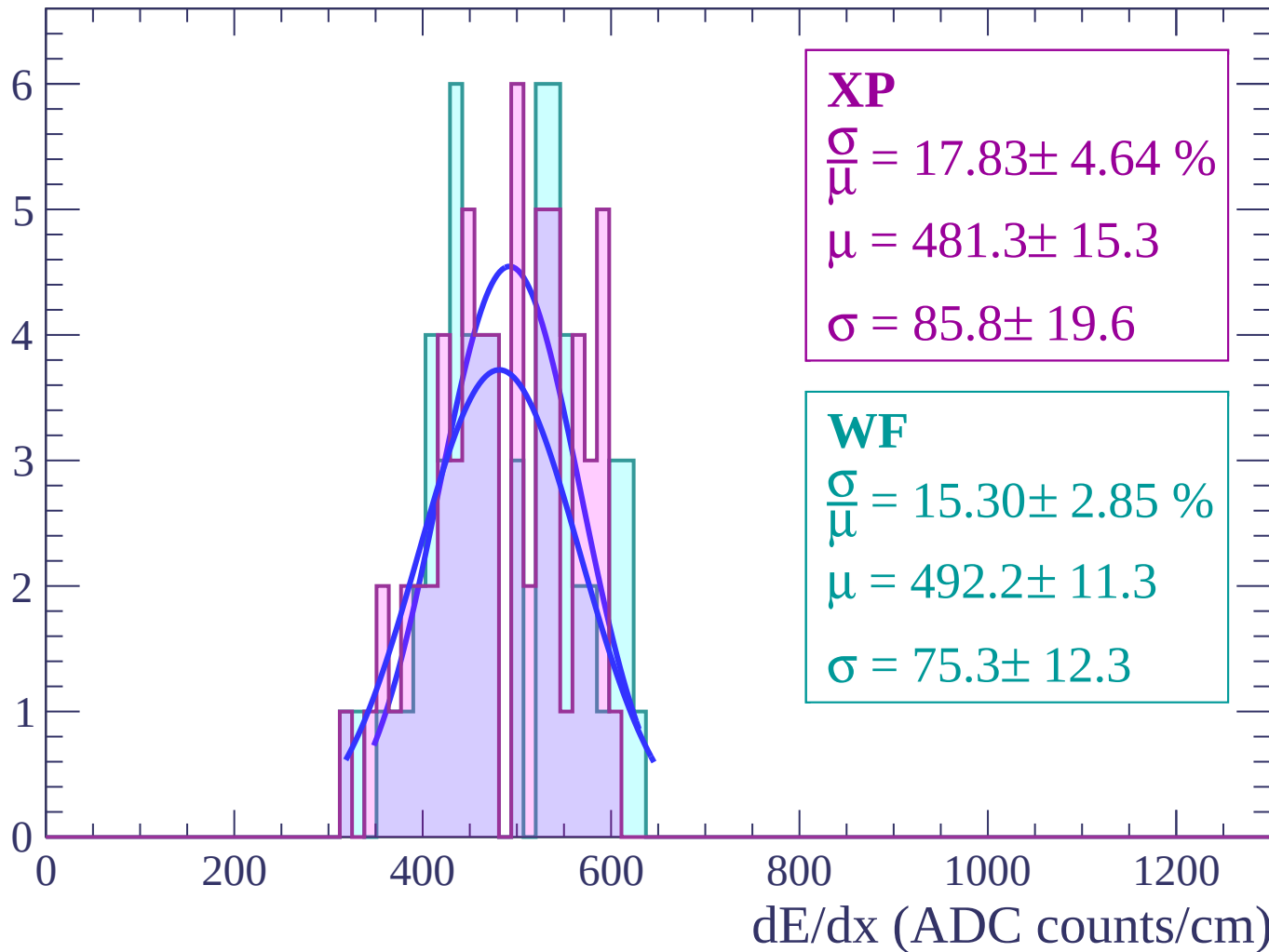
Energy loss |  $36 < \theta < 38$

Count



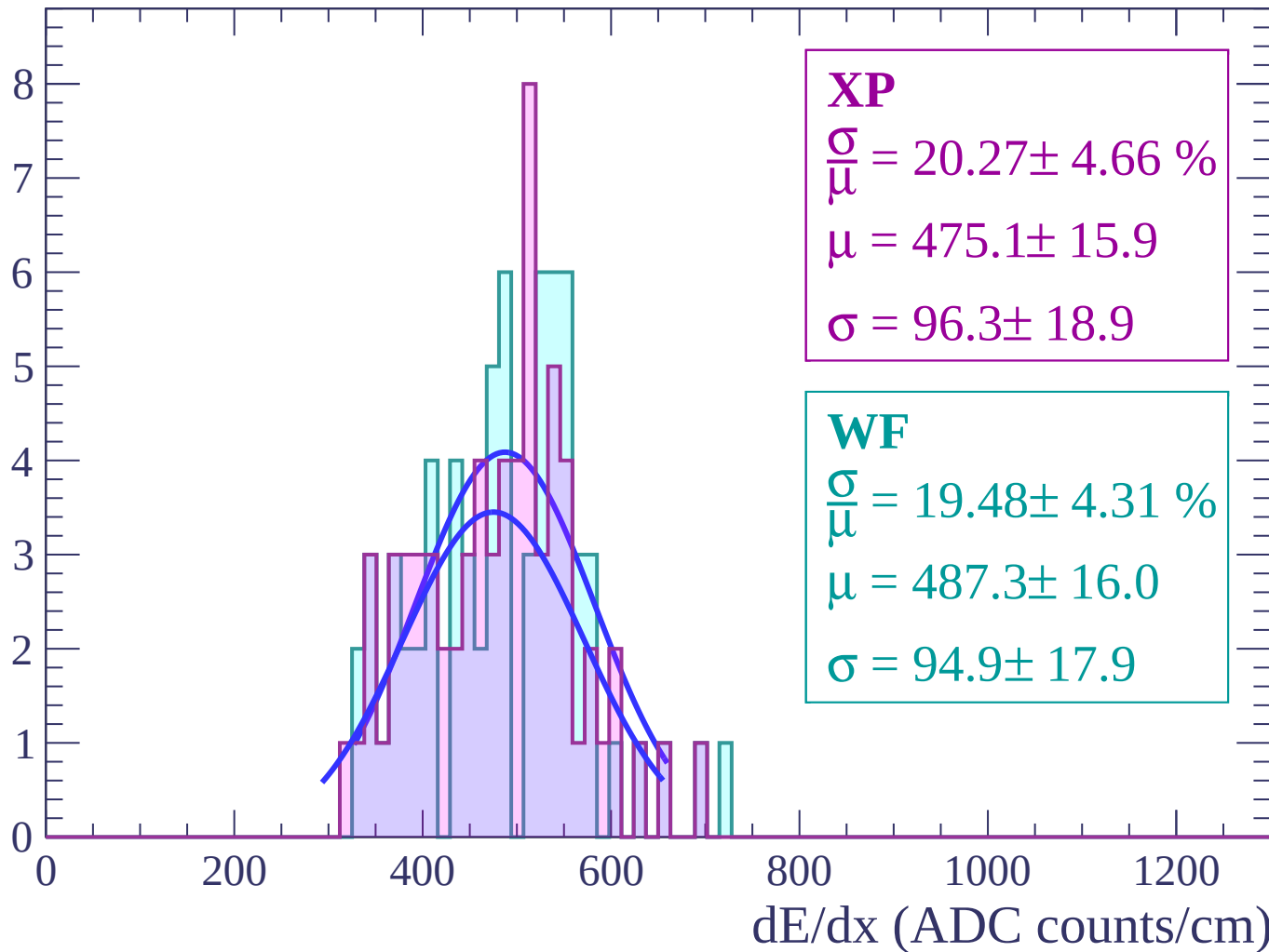
Energy loss |  $38 < \theta < 40$

Count



Energy loss |  $40 < \theta < 42$

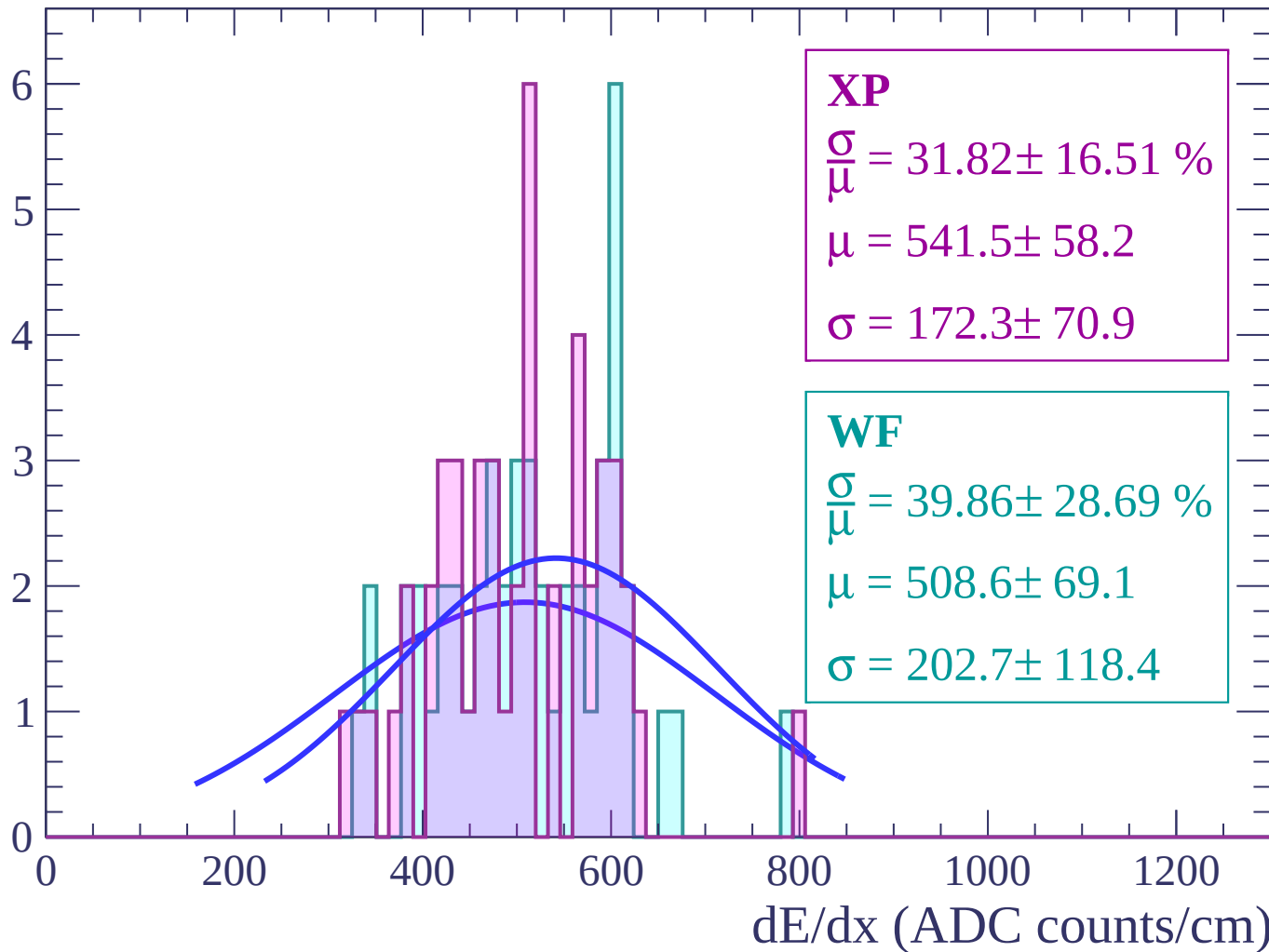
Count





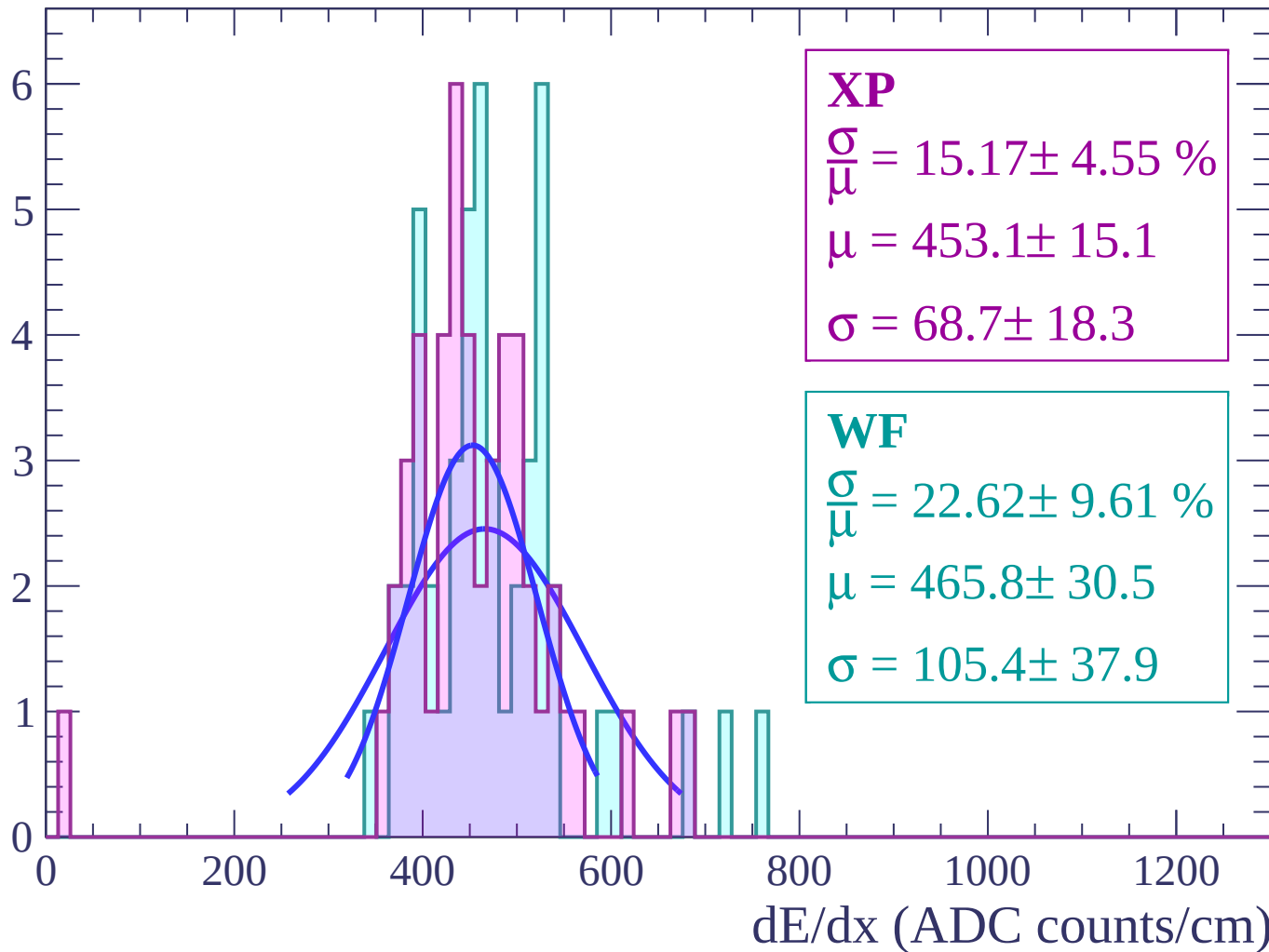
Energy loss |  $42 < \theta < 44$

Count



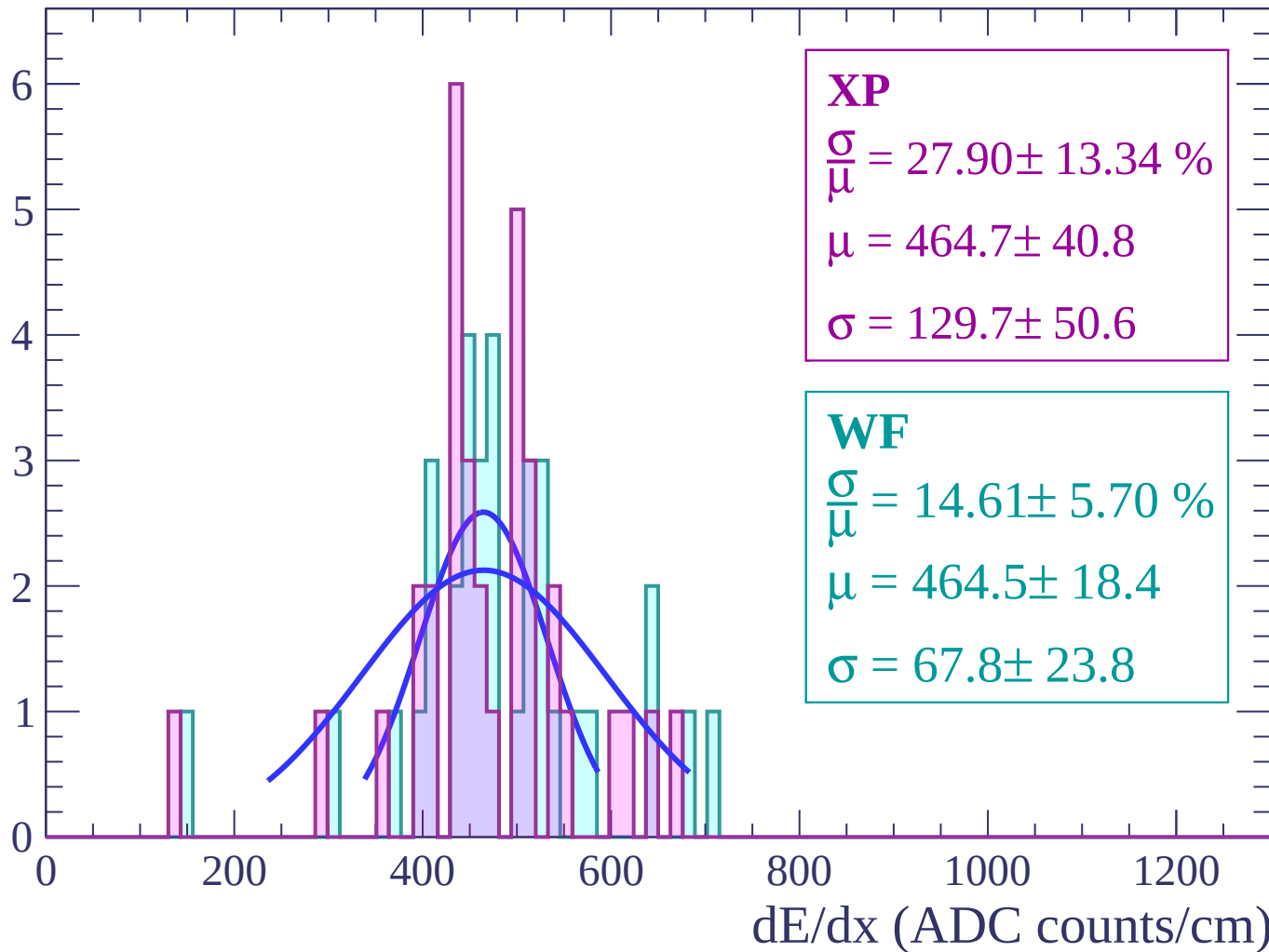
Energy loss |  $44 < \theta < 46$

Count



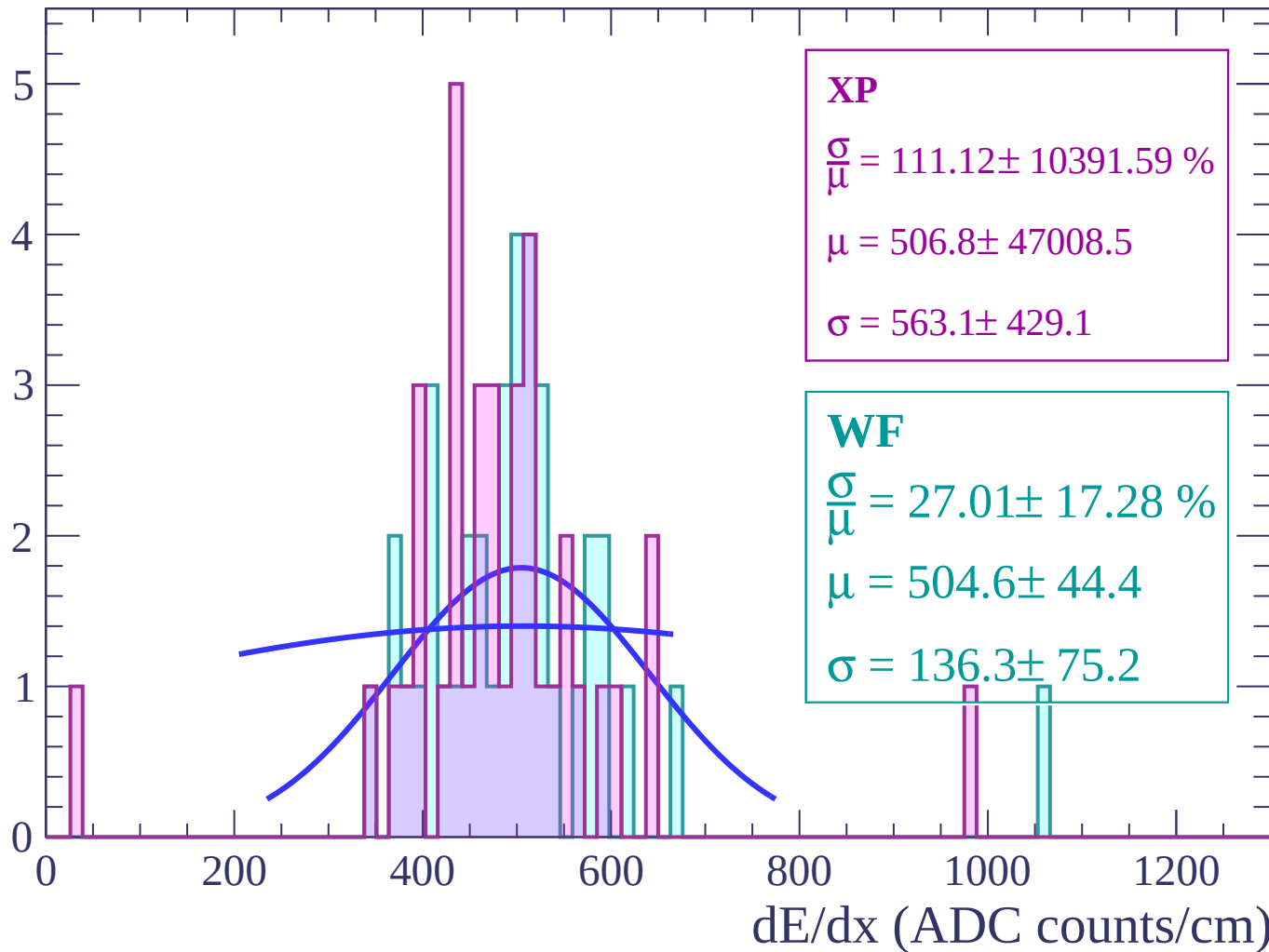
Energy loss |  $46 < \theta < 48$

Count



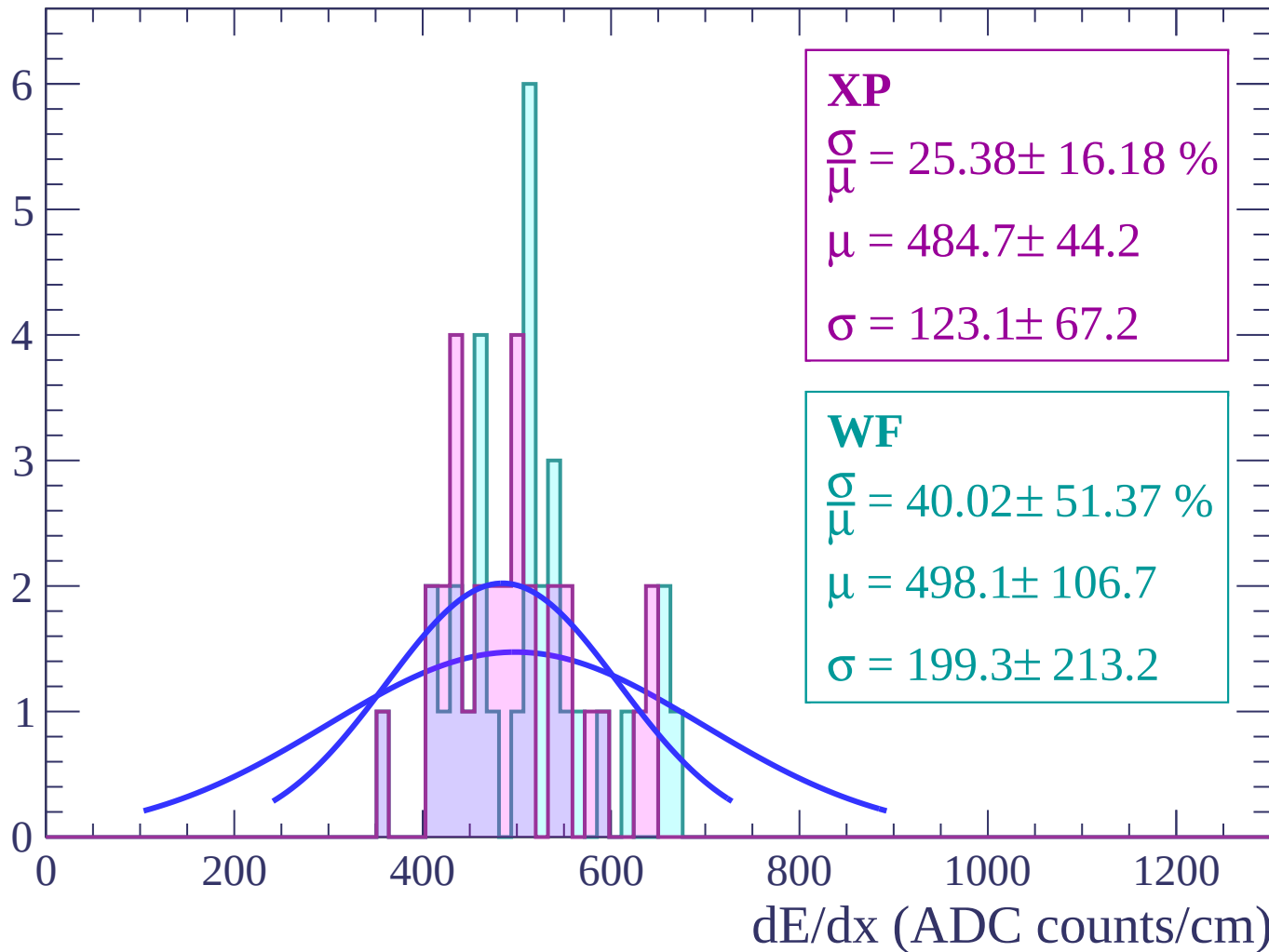
Energy loss |  $48 < \theta < 50$

Count



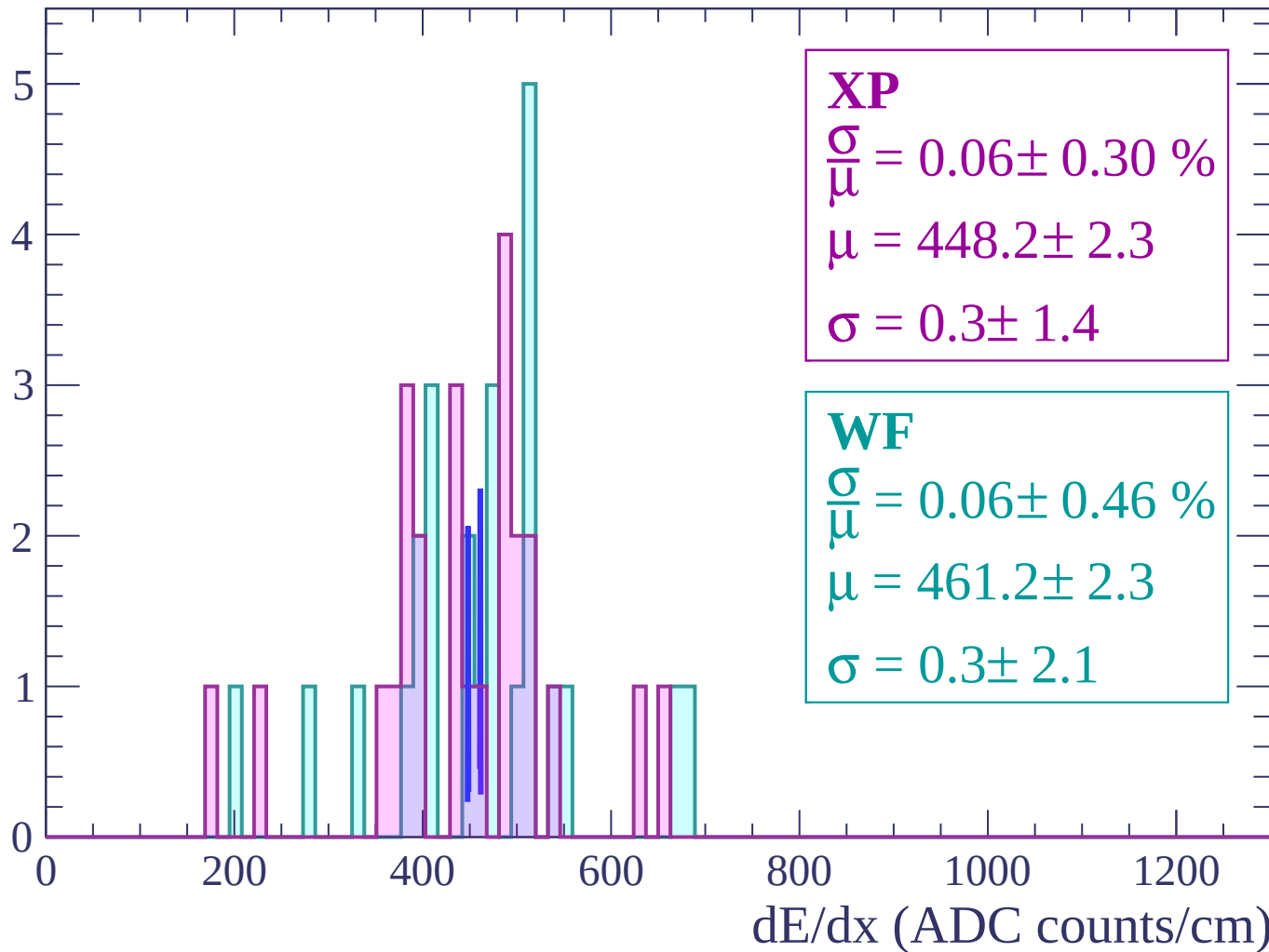
Energy loss |  $50 < \theta < 52$

Count



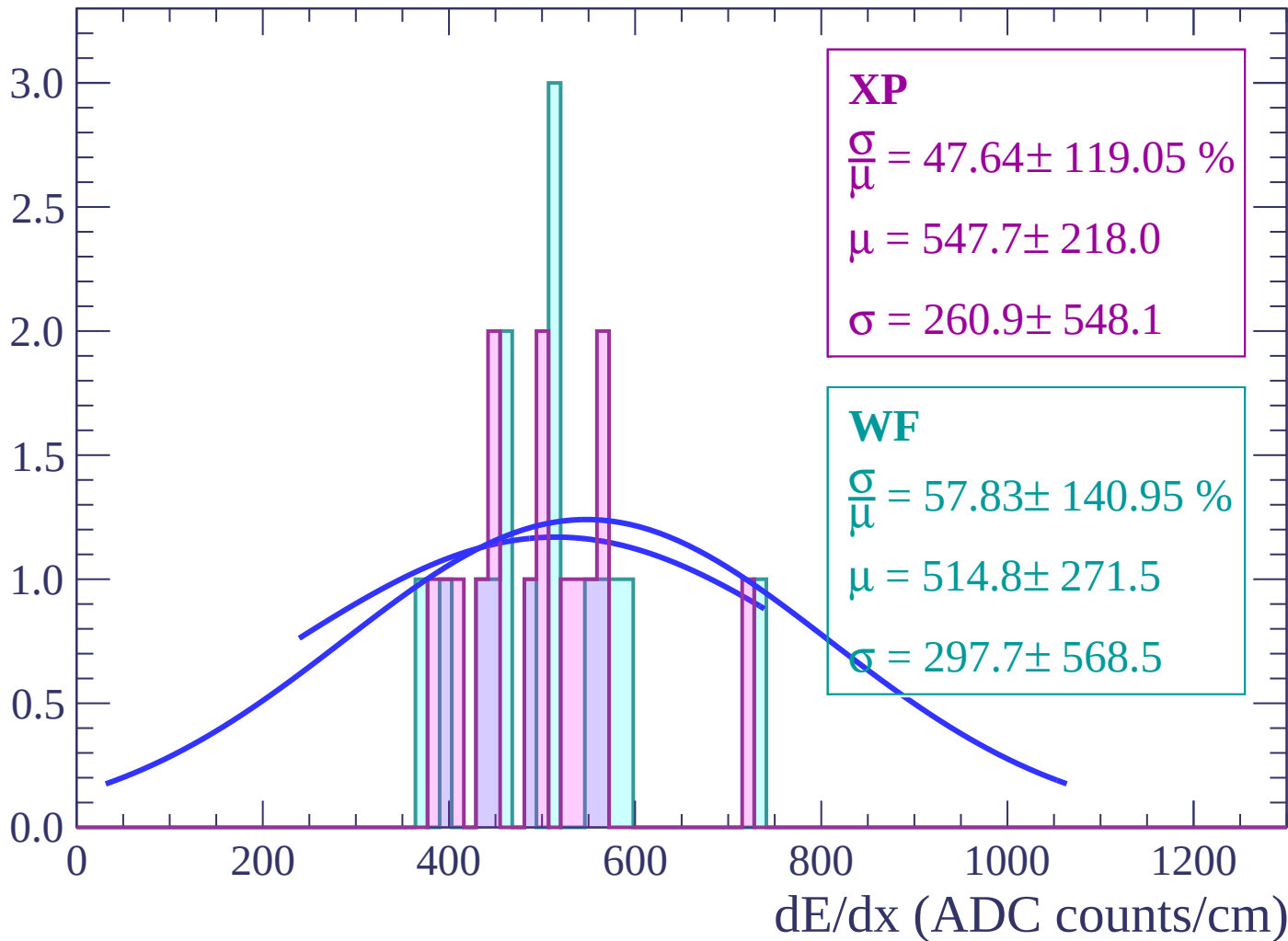
Energy loss |  $52 < \theta < 54$

Count



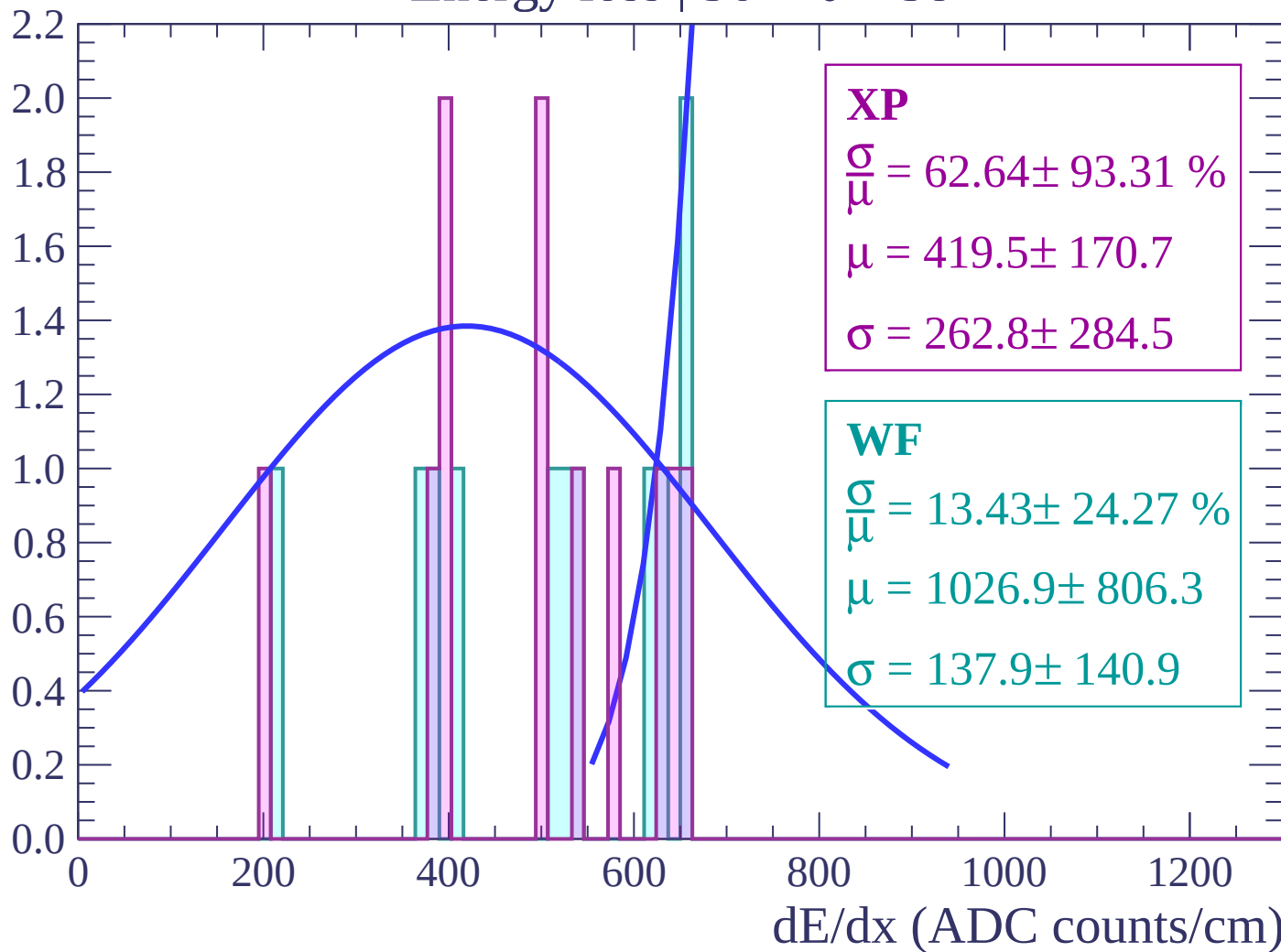
# Energy loss | $54 < \theta < 56$

Count



Energy loss |  $56 < \theta < 58$

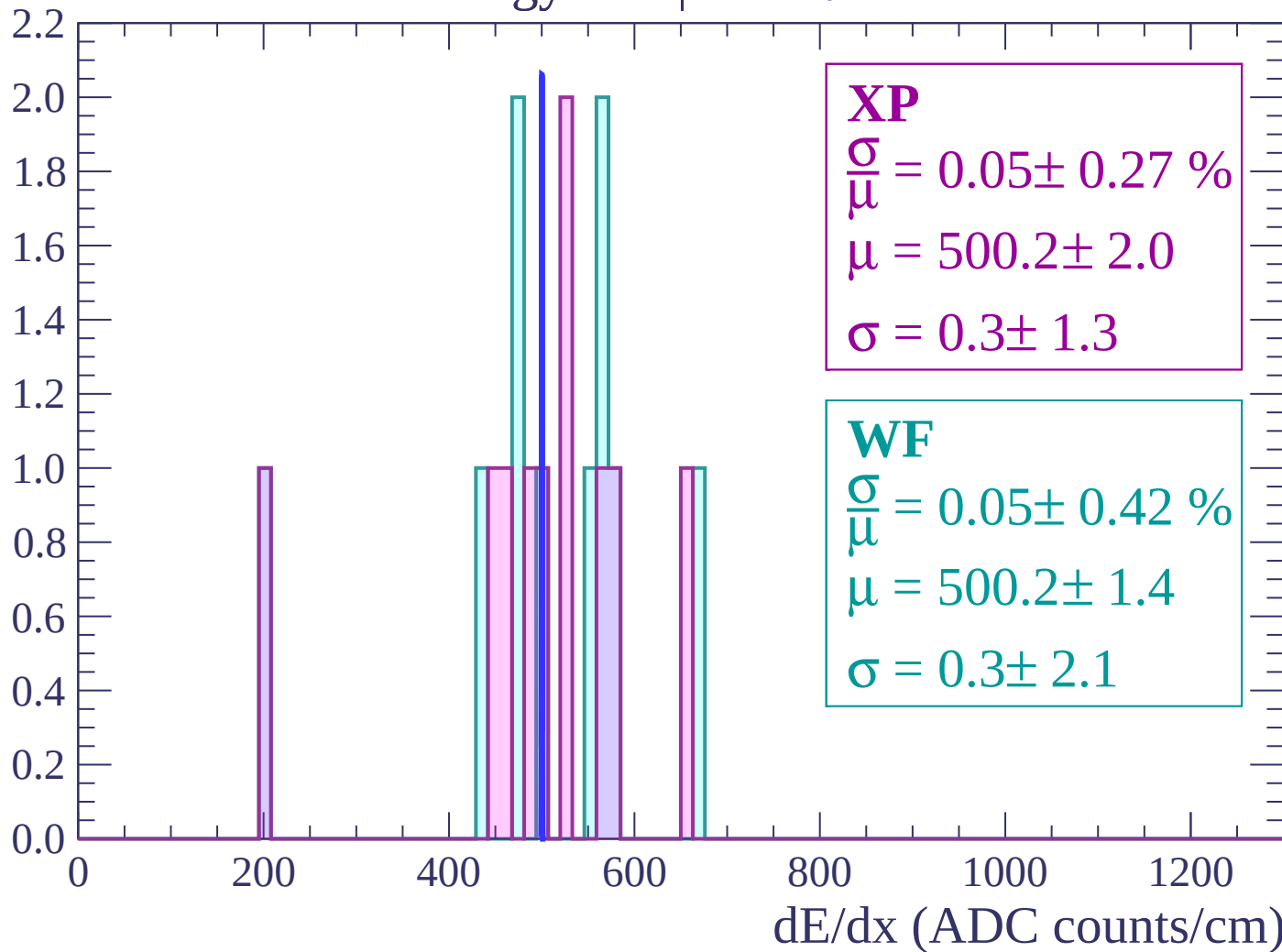
Count





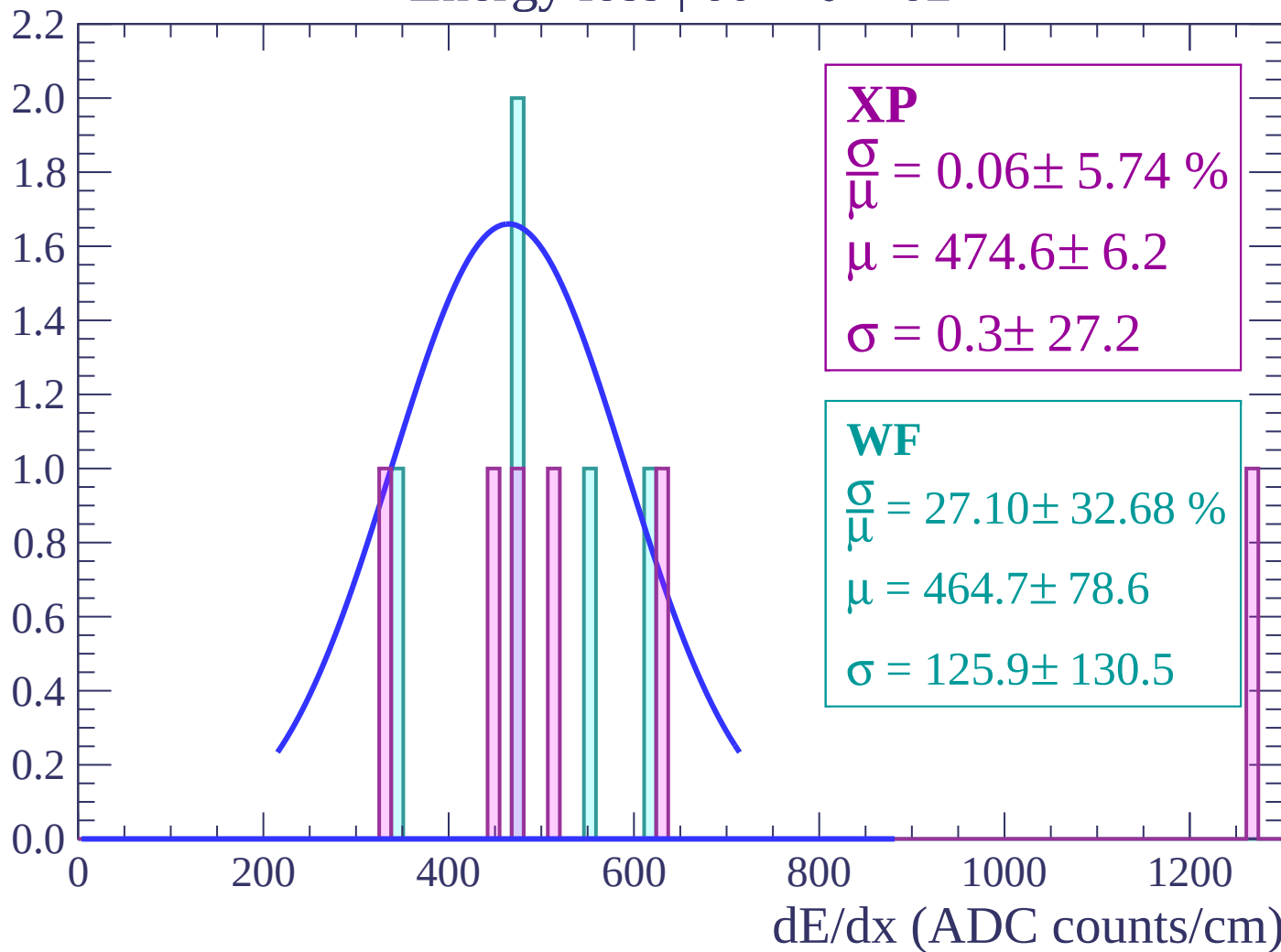
Energy loss |  $58 < \theta < 60$

Count



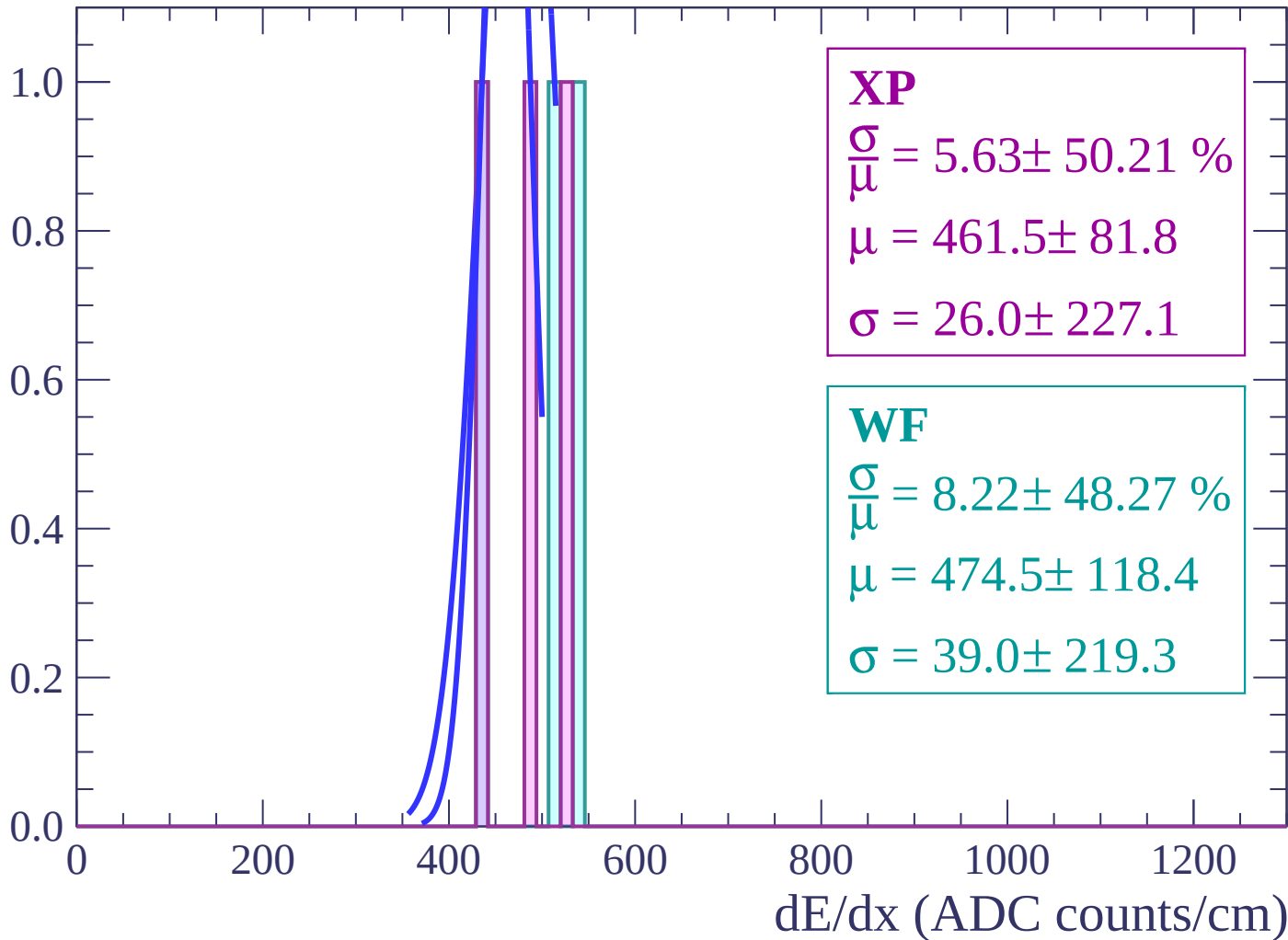
Energy loss |  $60 < \theta < 62$

Count



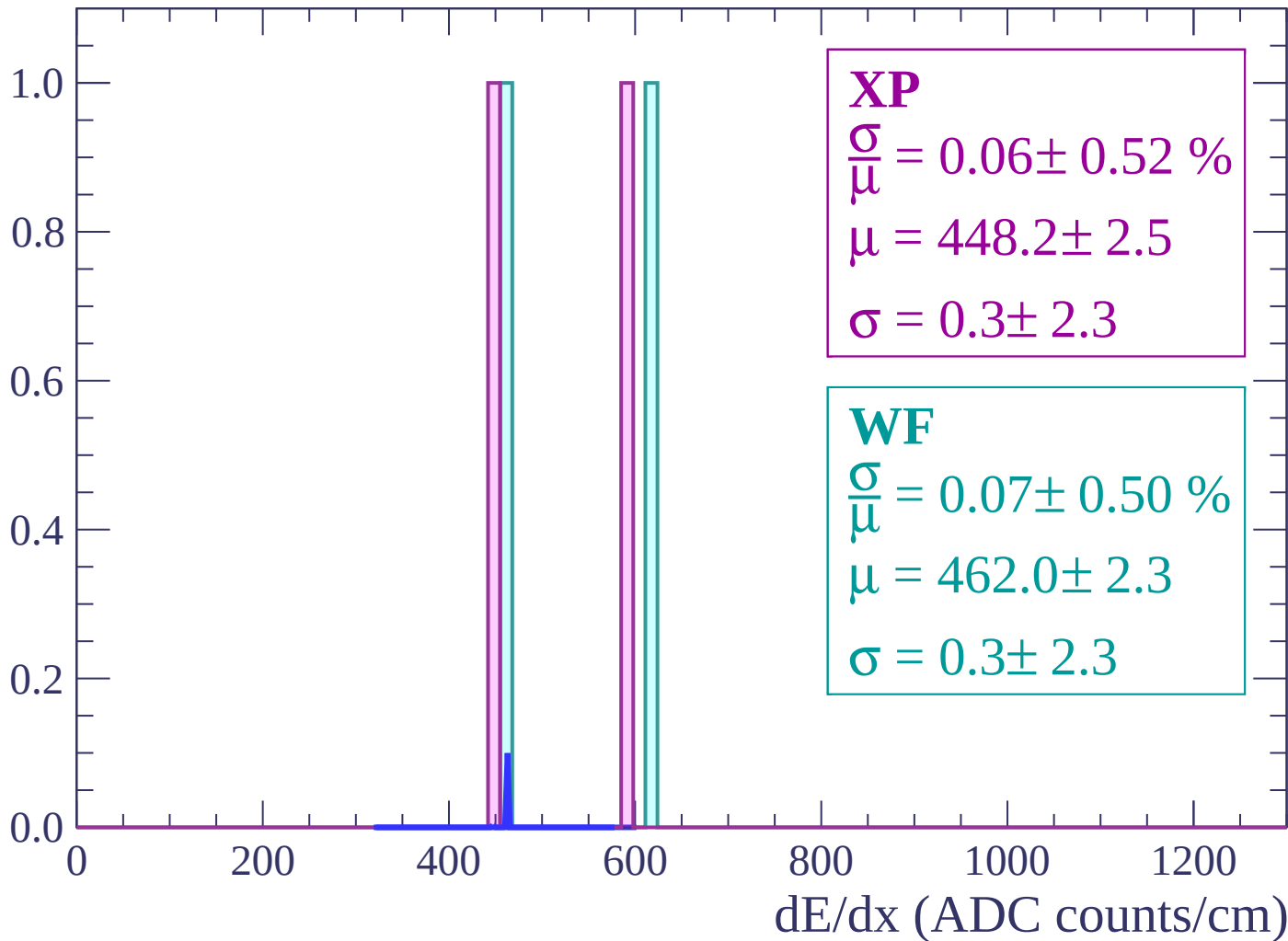
Energy loss |  $62 < \theta < 64$

Count



Energy loss |  $64 < \theta < 66$

Count



Energy loss |  $66 < \theta < 68$

Count

1.0

0.8

0.6

0.4

0.2

0.0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $68 < \theta < 70$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $70 < \theta < 72$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $72 < \theta < 74$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

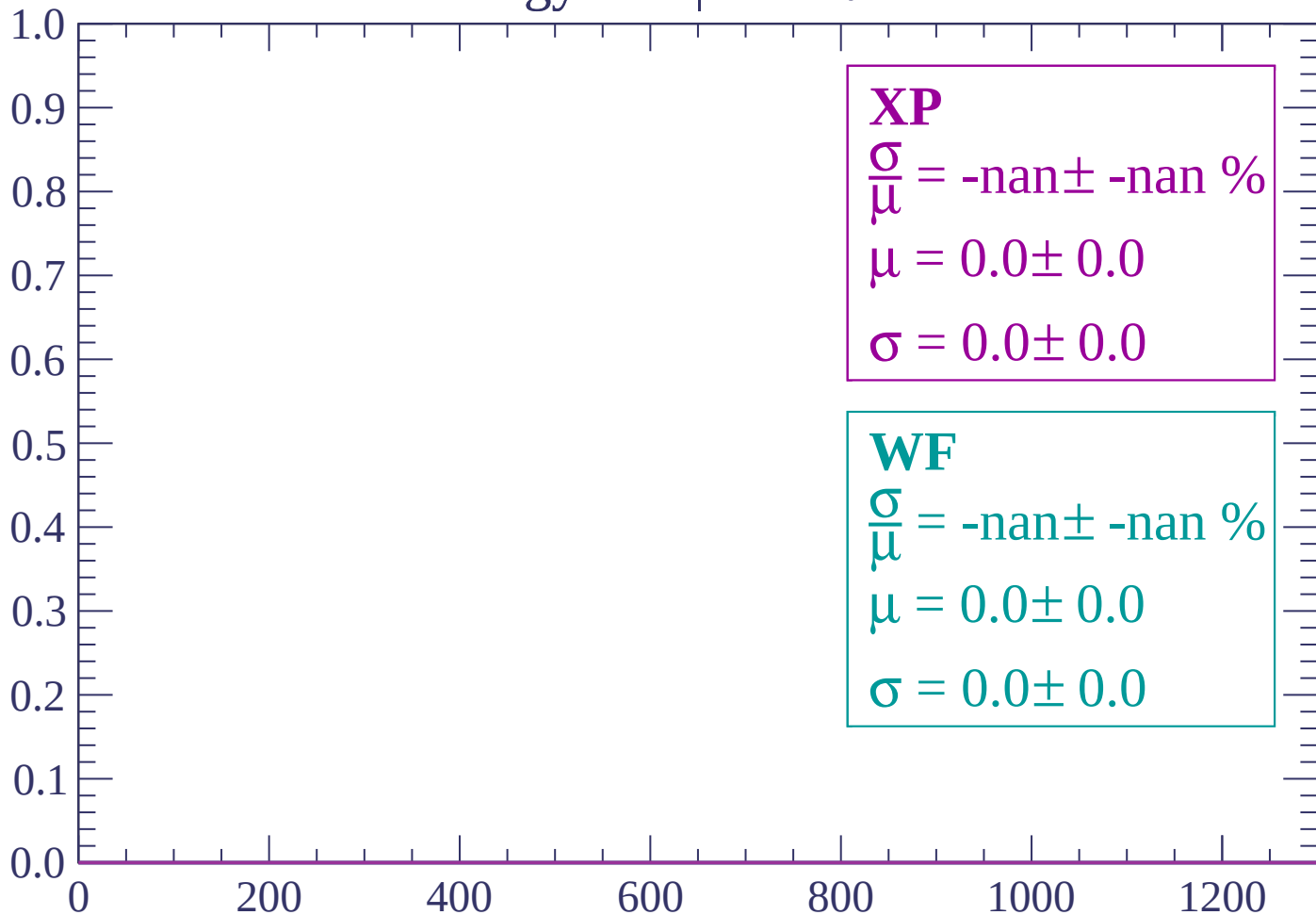
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$



Energy loss |  $74 < \theta < 76$

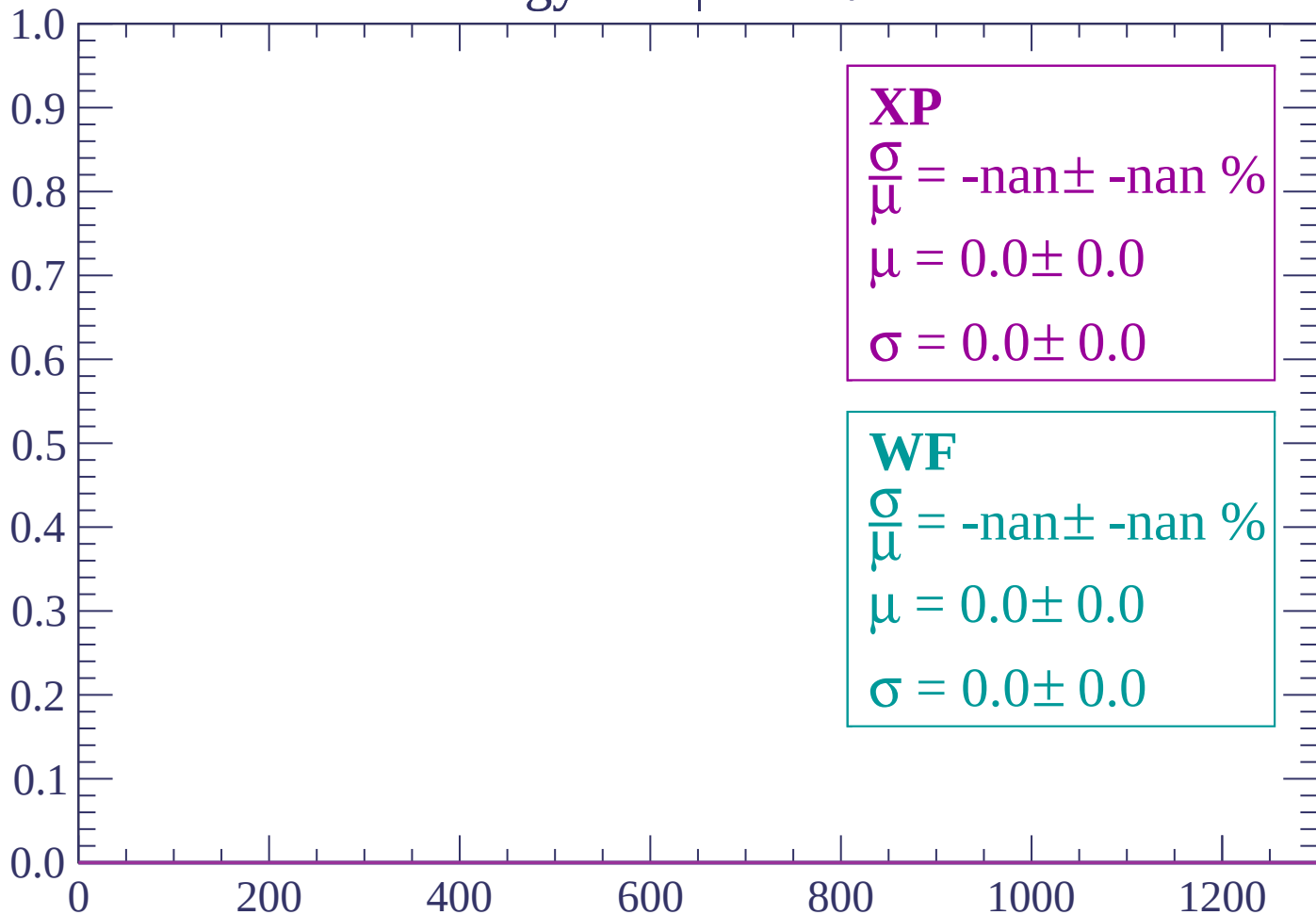
Count



dE/dx (ADC counts/cm)

Energy loss |  $76 < \theta < 78$

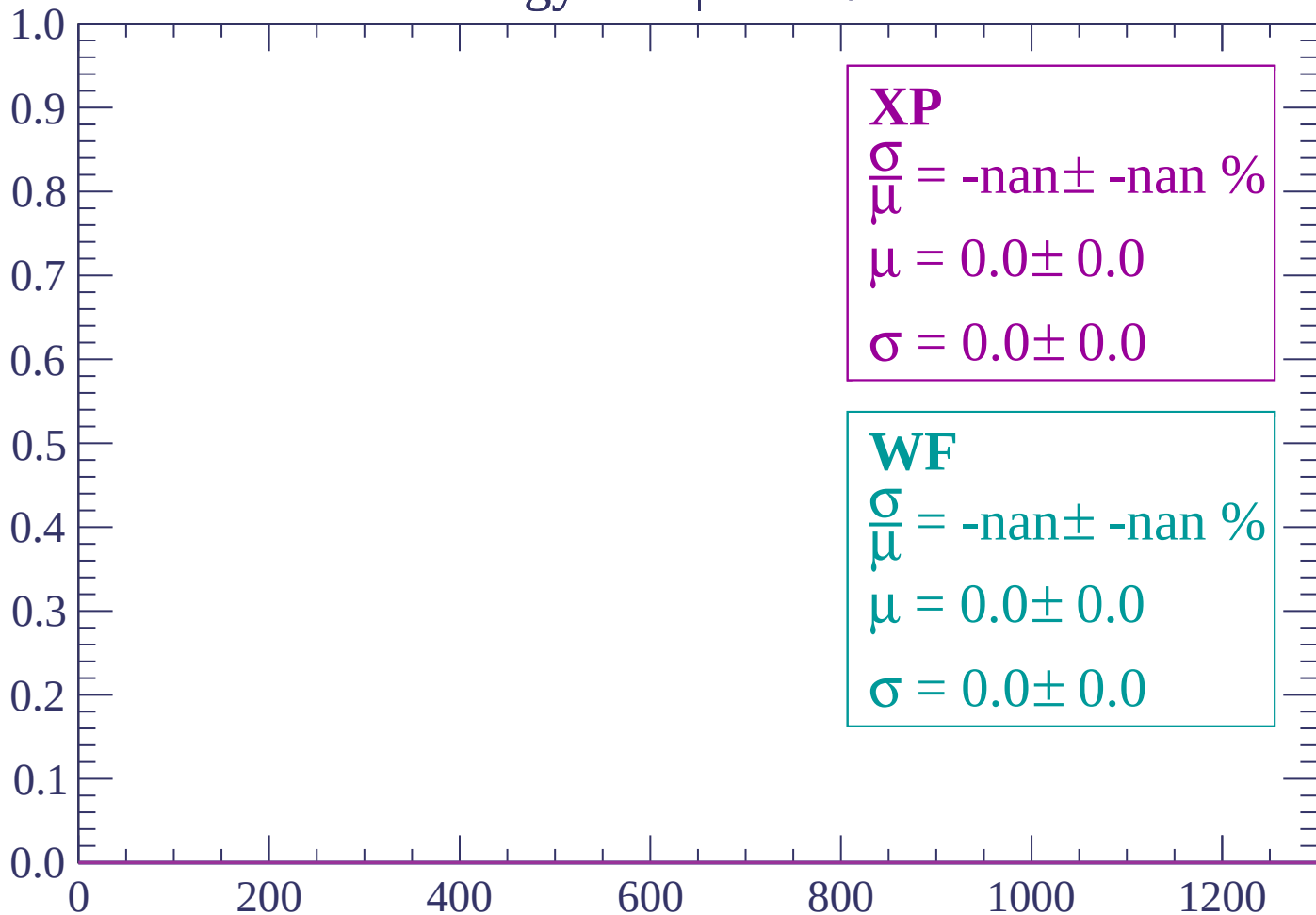
Count



dE/dx (ADC counts/cm)

Energy loss |  $78 < \theta < 80$

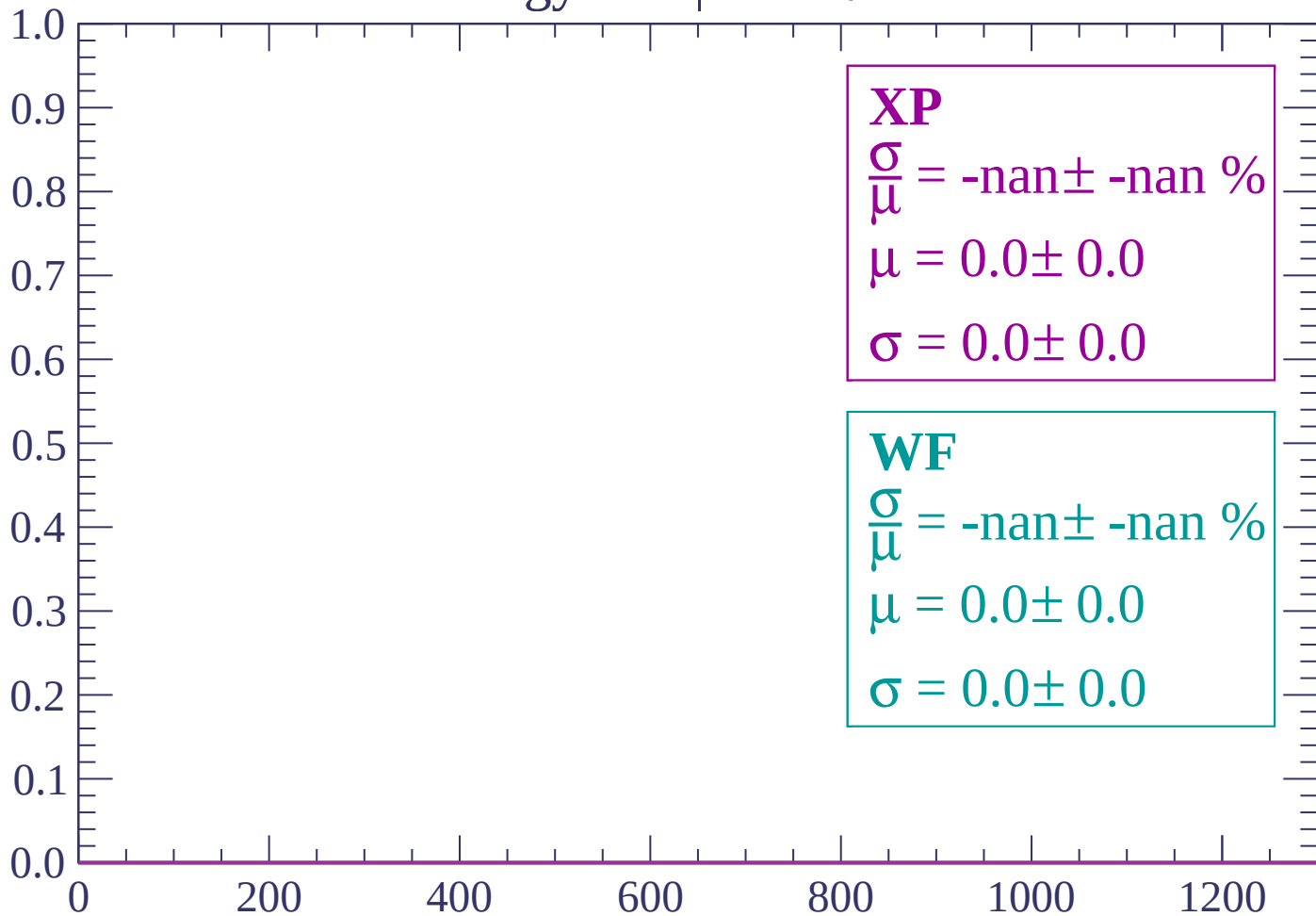
Count



dE/dx (ADC counts/cm)

Energy loss |  $80 < \theta < 82$

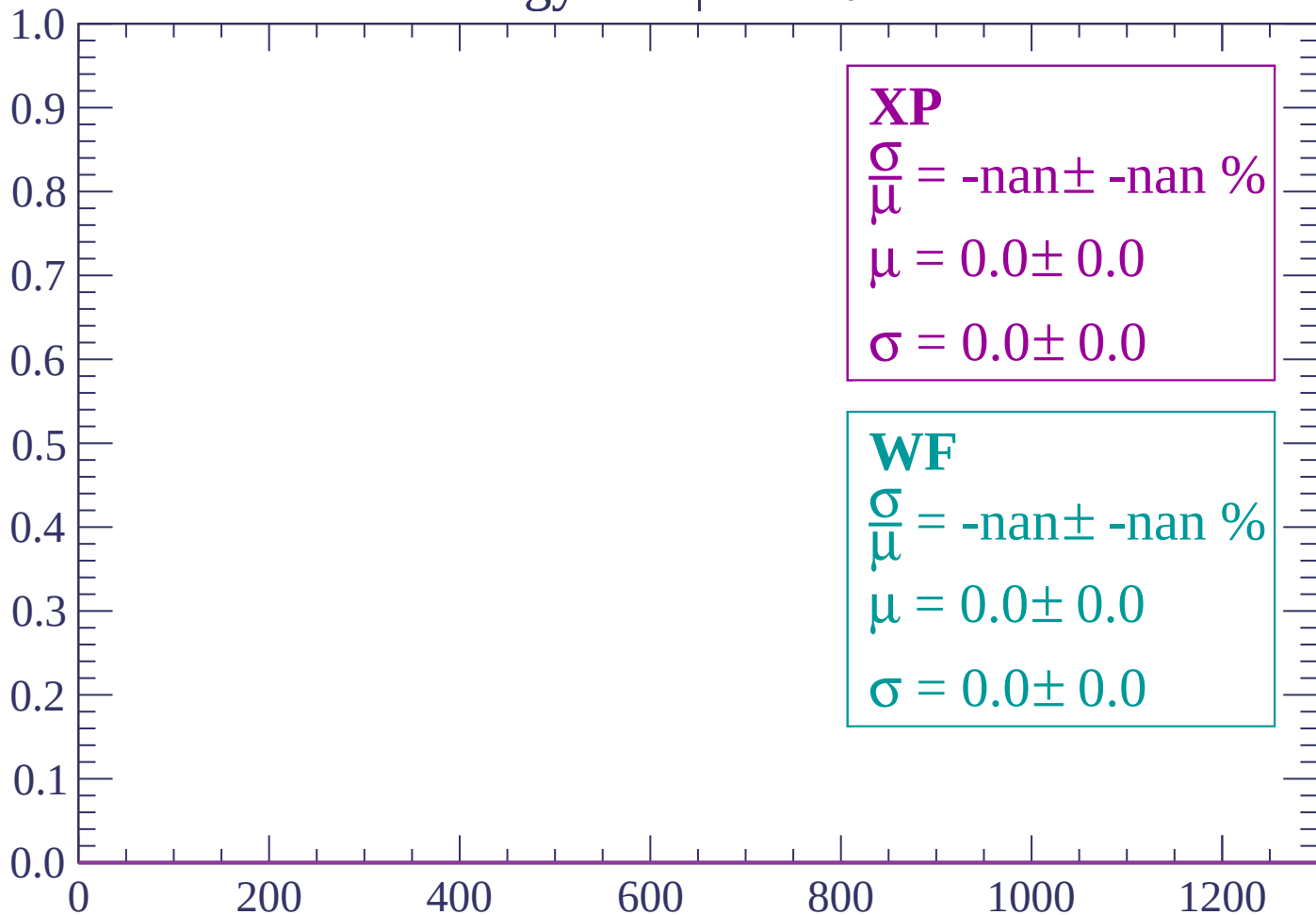
Count



dE/dx (ADC counts/cm)

Energy loss |  $82 < \theta < 84$

Count



dE/dx (ADC counts/cm)

Energy loss |  $84 < \theta < 86$

Count

1.0  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

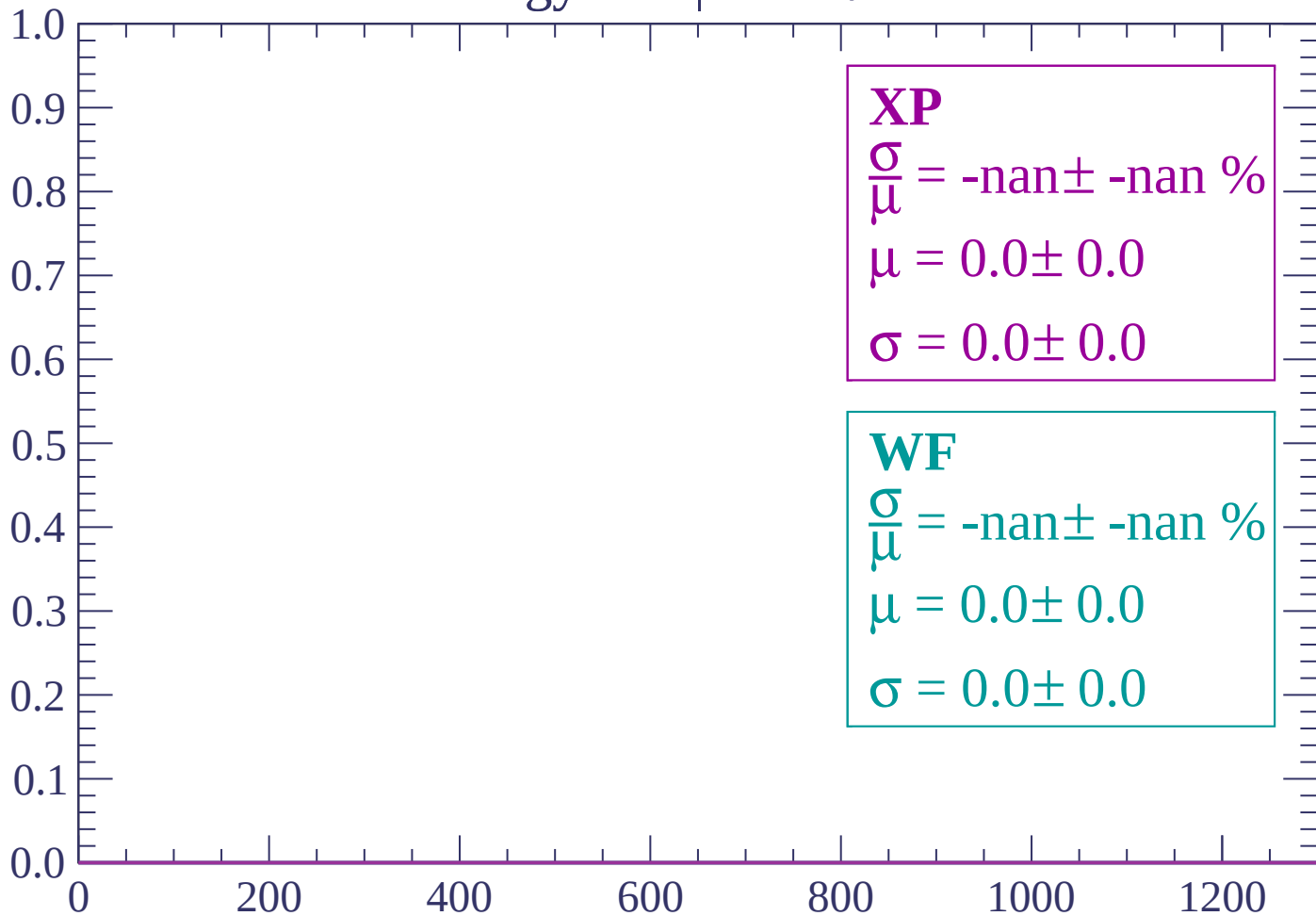
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss |  $86 < \theta < 88$

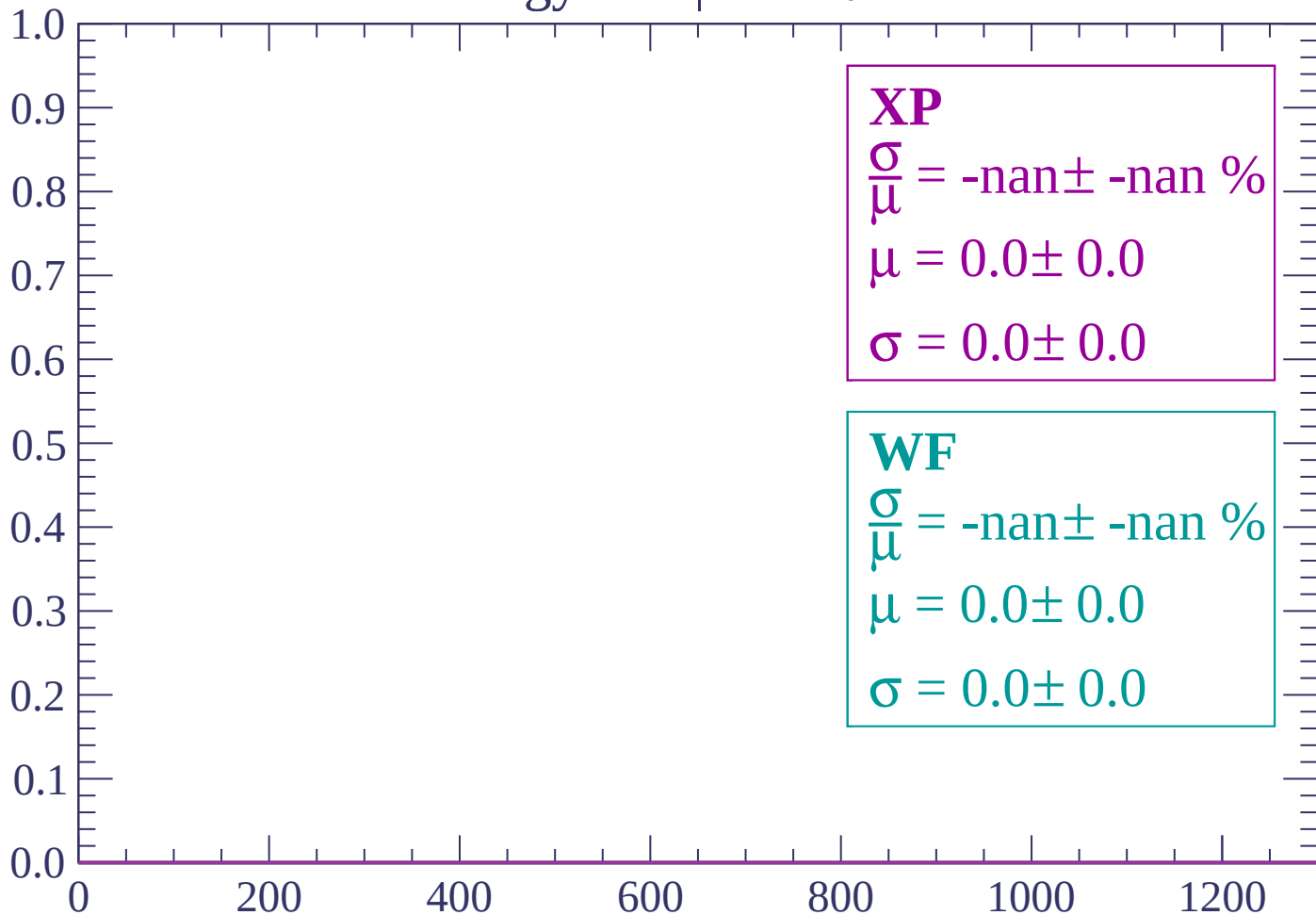
Count



dE/dx (ADC counts/cm)

Energy loss |  $88 < \theta < 90$

Count

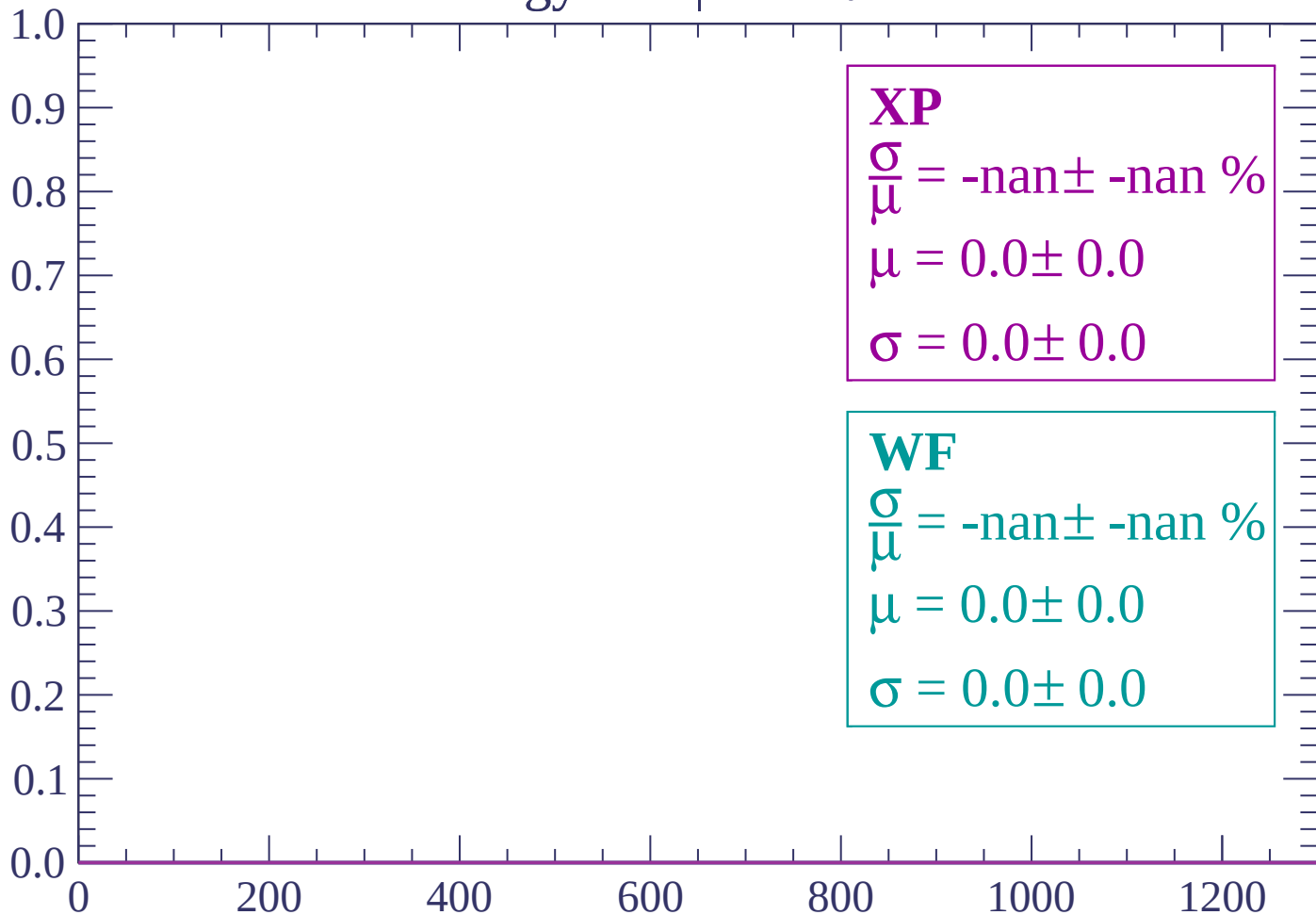


dE/dx (ADC counts/cm)

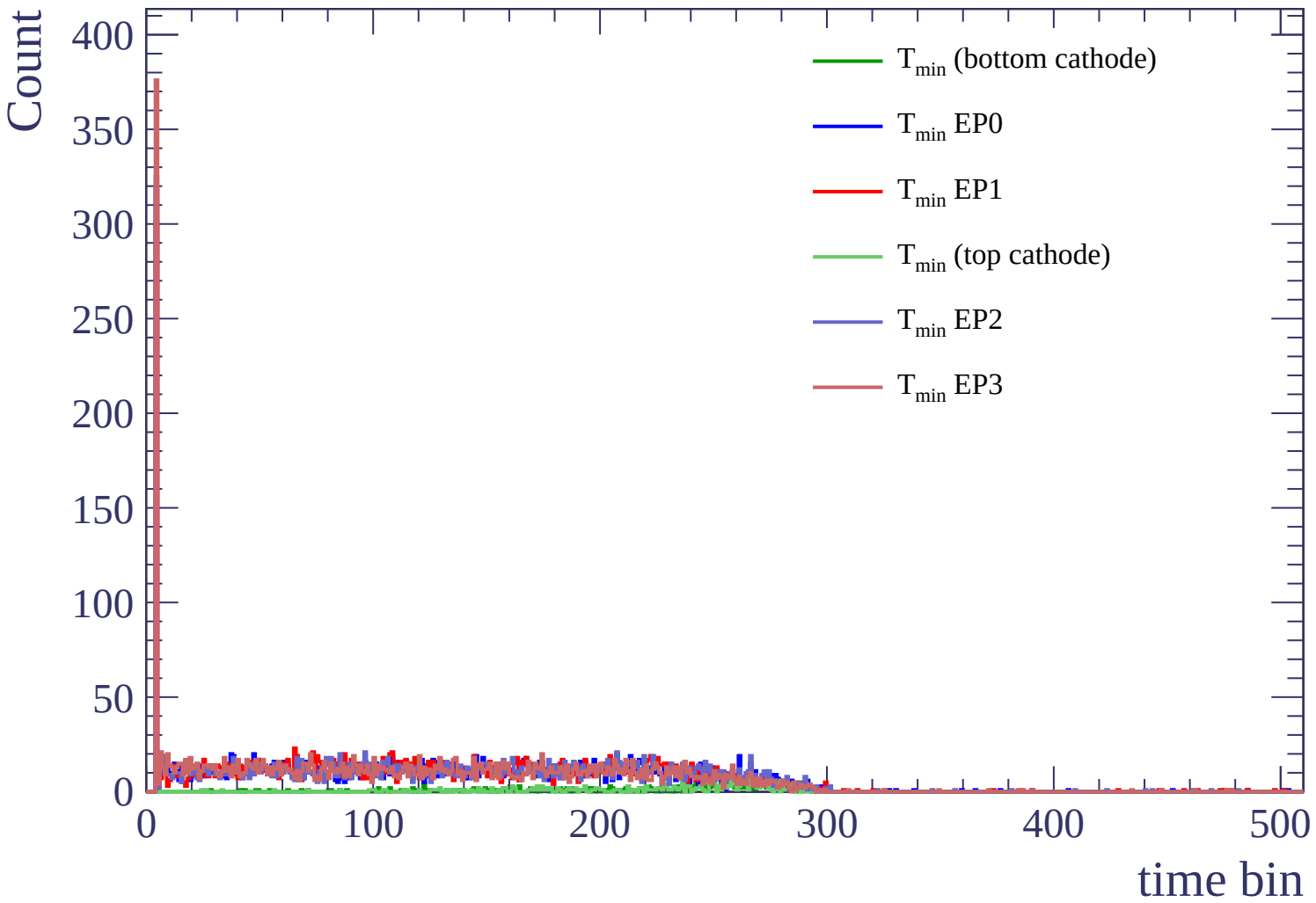


Energy loss |  $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)



Count

